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(54) **DRY POWDER FORMULATIONS OF
EPINEPHRINE AND ASSOCIATED
METHODS**

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None

See application file for complete search history.

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(57) **ABSTRACT**

Provided herein are dry powder formulations comprising
epinephrine alone or in combination with at least one
enabling agent suitable for nasal application. Also provided
are unit dose forms and devices comprising such formula-
tions and methods of using such formulations for the treat-
ment of various conditions including anaphylaxis, anaphy-
lactoid reaction, respiratory conditions, hemodynamic
collapse, and for administration during cardiopulmonary
arrest and other life-threatening conditions.

15 Claims, 74 Drawing Sheets

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Formulation			Spray Dryer Parameters				Particle Size Distribution Results		
Formulation feed strength, (% w/v)	Composition (% w/w)	Solvent mix	Inlet temp (°C)	Aspirator Power (%)	Pump Rate (%)	Nozzle power (%)		d(4,3) μm	d(0.9) μm
2.5%	Lactose/API 90/10	EtOH/H ₂ O, 20/80 v/v	110	100	10	40	non-sonicated	18.060	36.741
							sonicated	13.144	26.649
2.5%	Lactose/API/Leucine 80/10/10	EtOH/H ₂ O, 20/80 v/v	110	100	10	40	non-sonicated	9.956	18.007
							sonicated	9.700	17.798
2.5%	Trehalose/API/Leucine 80/10/10	EtOH/H ₂ O, 20/80 v/v	110	100	10	40	non-sonicated	12.135	18.829
							sonicated	12.075	18.793
2.5%	Lactose/API/Leucine 80/10/10	EtOH/H ₂ O, 20/80 v/v	110	100	10	35	non-sonicated	11.237	19.006
							sonicated	10.201	19.473
2.5%	Trehalose/API/Leucine 80/10/10	EtOH/H ₂ O, 20/80 v/v	110	100	15	30	non-sonicated	11.820	18.417
							sonicated	34.783	24.682
1%	Na-CMC/API/Leucine 80/10/10	EtOH/H ₂ O, 20/80 v/v	130	100	15	30	non-sonicated	31.638	64.248
							sonicated	17.281	33.140
5%	Lactose/API/Leucine 80/10/10	EtOH/H ₂ O, 20/80 v/v	110	100	10	35	non-sonicated	13.809	23.552
							sonicated	12.729	23.591
2.5%	Lactose/API/Leucine/NaCl 78/10/10/	EtOH/H ₂ O, 20/80 v/v	120	100	10	35	non-sonicated	9.819	18.285
							sonicated	-	-
2.5%	Lactose/API/chitosan 70/10/20	EtOH/H ₂ O, 20/80 v/v	-	-	-	-	Insoluble in the solvent mix		
2.5%	Lactose/API/chitosan 70/10/20	EtOH/H ₂ O, 10/90 v/v	-	-	-	-	Insoluble in the solvent mix		
2.5%	Lactose/API/chitosan 70/10/20	EtOH/H ₂ O, 10/90 v/v (acidified)	-	-	-	-	Viscosity too high to spray dry		
2.5%	Lactose/API/chitosan 80/10/10	EtOH/H ₂ O, 10/90 v/v (acidified)	-	-	-	-	Insoluble in the solvent mix		
1%	Lactose/API/Carbomer 934 85/10/5	EtOH/H ₂ O, 20/80 v/v	130	100	15	45	non-sonicated	26.103	60.283
							sonicated	14.574	31.981
2.5%	Lactose/API/Tween 80 89.9/10/0.1/	EtOH/H ₂ O, 20/80 v/v	130	100	10	40	non-sonicated	19.191	42.749
							sonicated	13.824	28.463

FIG. 1

Formulation		Spray Dryer Parameters				Particle Size Distribution Results	
Formulation feed strength, (% w/v)	Composition (% w/w)	Solvent mix	Inlet temp (°C)	Aspirator Power (%)	Pump Rate (%)	Nozzle power (%)	
1%	Na-CMC/API 90/10	EtOH/H ₂ O, 20/80 v/v	130	100	15	40	
							non-sonicated 14.471 24.979
							sonicated 12.324 21.007
2.5%	Lactose/API/NaCl 88/10/2	EtOH/H ₂ O, 20/80 v/v	130	100	15	35	
							non-sonicated 18.727 37.747
							sonicated 20.220 41.416
1%	Na-CMC/API/NaCl 88/10/2	EtOH/H ₂ O, 20/80 v/v	130	100	15	37	
							non-sonicated 13.084 21.962
							sonicated 11.605 18.723
2.5%	Lactose/API/Tween 80 89.5/10/0.5	EtOH/H ₂ O, 20/80 v/v	130	100	23	35	
							non-sonicated 42.106 83.484
							sonicated 32.413 61.869
1%	Na-CMC/API/Tween 80 89.5/10/0.5	EtOH/H ₂ O, 20/80 v/v	130	100	25	40	
							non-sonicated 14.769 25.036
							sonicated 12.622 21.299
1%	Na-CMC/API/Leucine/NaCl 78/10/10/2	EtOH/H ₂ O, 20/80 v/v	130	100	15	45	
							non-sonicated 17.229 34.355
							sonicated 12.139 20.724
1%	Na-CMC/API/Carbomer 934 85/10/5	EtOH/H ₂ O, 20/80 v/v	130	100	20	45	
							non-sonicated 11.714 18.123
							sonicated 9.423 14.518
5%	Lactose/API/Nicotinic acid 88/10/2	EtOH/H ₂ O, 20/80 v/v	130	100	23	35	
							non-sonicated 17.376 34.458
							sonicated 18.476 35.815
1.5%	Na-CMC/API/Nicotinic acid 88/10/2	EtOH/H ₂ O, 20/80 v/v	130	100	25	47	
							non-sonicated 17.436 30.728
							sonicated 13.701 23.134
5%	Lactose/API/Hyaluronate 89.5/10/0.5	EtOH/H ₂ O, 20/80 v/v	130	100	15	45	
							non-sonicated 20.479 46.885
							sonicated 18.419 41.286
2%	Na-CMC/API/Hyaluronate 89.5/10/0.5	EtOH/H ₂ O, 20/80 v/v	130	100	25	55	
							non-sonicated 18.111 32.800
							sonicated 14.827 25.977
5%	Lactose/API/Caffeine 85/10/5	EtOH/H ₂ O, 20/80 v/v	130	100	22	34	
							non-sonicated 74.973 190.149
							sonicated 53.744 118.603
5%	Lactose/API/Caffeine 85/10/5	EtOH/H ₂ O, 20/80 v/v	130	100	15	45	
							non-sonicated 26.829 52.352
							sonicated 24.960 48.269
5%	Lactose/API/Caffeine 85/10/5	EtOH/H ₂ O, 20/80 v/v	140	100	15	48	
							non-sonicated 28.430 55.336
							sonicated 25.269 50.206
2%	Na-CMC/API/Caffeine 85/10/5	EtOH/H ₂ O, 20/80 v/v	130	100	20	50	
							non-sonicated 12.995 212.626
							sonicated 11.206 18.457

FIG. 2

Formulation		Spray Dryer Parameters				Particle Size Distribution Results		
Formulation feed strength, (% w/v)	Composition (% w/w)	Solvent mix	Inlet temp (°C)	Aspirator Power (%)	Pump Rate (%)	Nozzle power (%)		
2.5%	Lactose/API 90/10	EtOH/H ₂ O, 20/80 v/v	110	100	10	40	non-sonicated	d(4,3) μ m
							sonicated	d(0.9) μ m
2.5%	Lactose/API 90/10	EtOH/H ₂ O, 20/80 v/v	110	100	10	40	non-sonicated	19.721
							sonicated	14.486
2.5%	Lactose/API/NaCl 88/10/2	EtOH/H ₂ O, 20/80 v/v	140	100	20	35	non-sonicated	16.648
							sonicated	34.541
5%	Lactose/API/Nicotinic acid 88/10/2	EtOH/H ₂ O, 20/80 v/v	130	100	23	35	n/a	152.013
							sonicated	24.604
2%	Na-CMC/API/Nicotinic acid 88/10/2	EtOH/H ₂ O, 20/80 v/v	130	100	16	46	n/a	49.635
							sonicated	38.188
1%	Na-CMC/API/Leucine 80/10/10	EtOH/H ₂ O, 20/80 v/v	130	100	15	45	sonicated	76.434
							n/a	14.624
2%	Na-CMC/API 90/10	EtOH/H ₂ O, 20/80 v/v	130	100	15	47	sonicated	24.057
							n/a	14.624
2.5%	Lactose/API/Tween 80 89.5/10/0.5	EtOH/H ₂ O, 20/80 v/v	130	100	18	45	sonicated	24.057
							n/a	14.757
5%	Lactose/API/Nicotinic acid 88/10/2	EtOH/H ₂ O, 20/80 v/v	130	100	20	36	sonicated	24.665
							n/a	22.480
							sonicated	41.142
							n/a	24.589
							sonicated	48.277

FIG. 3

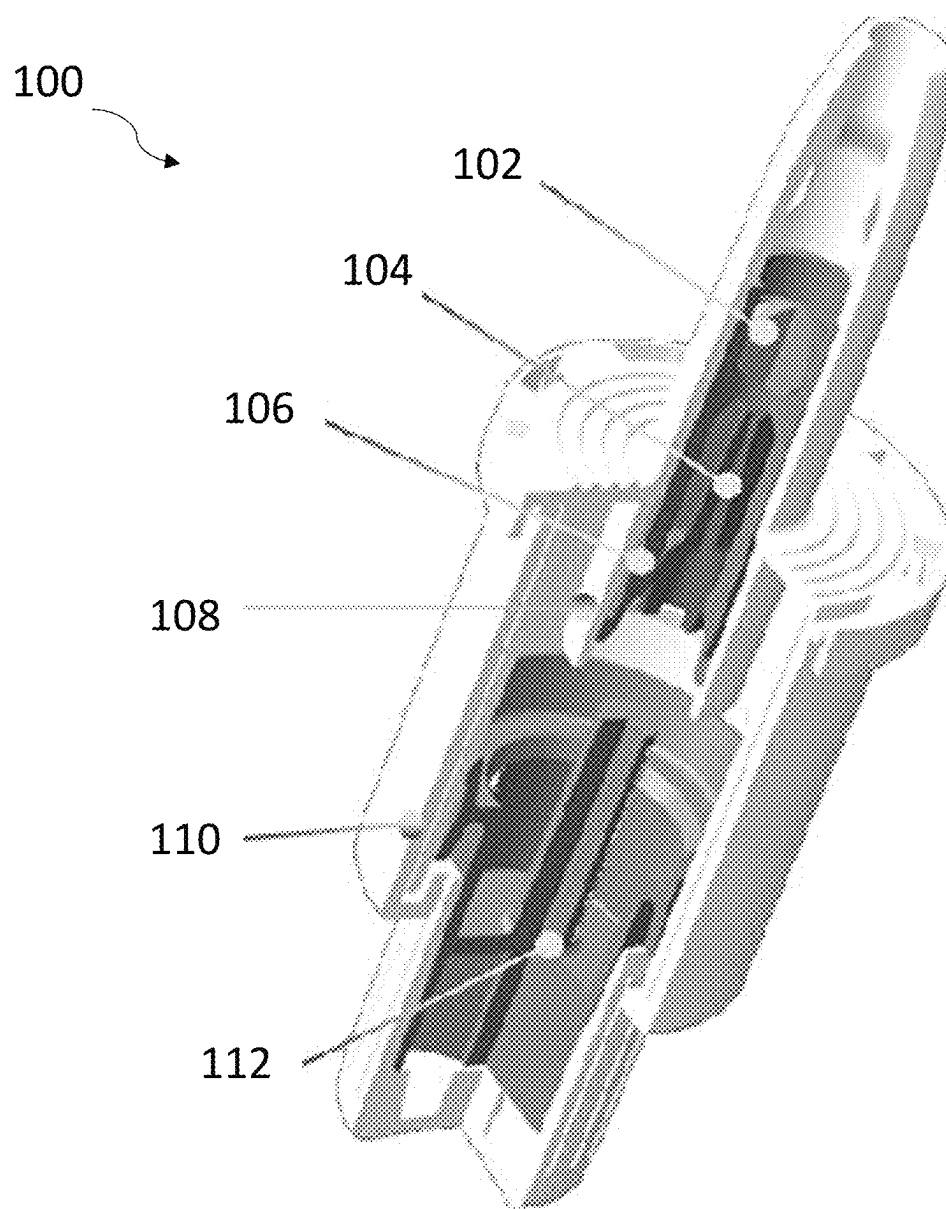


FIG. 4

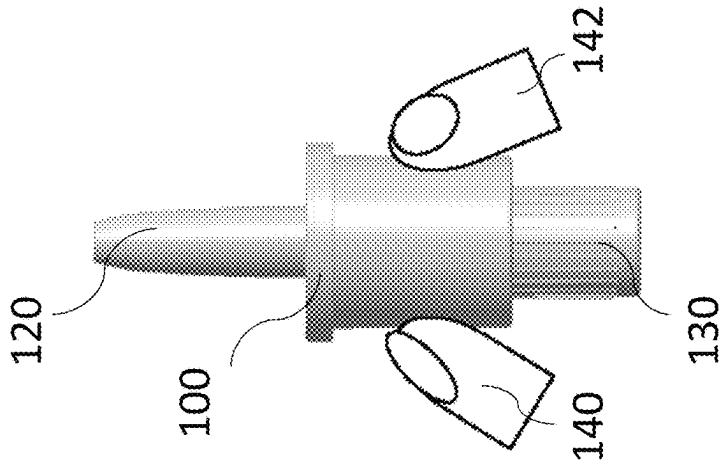


FIG. 5A

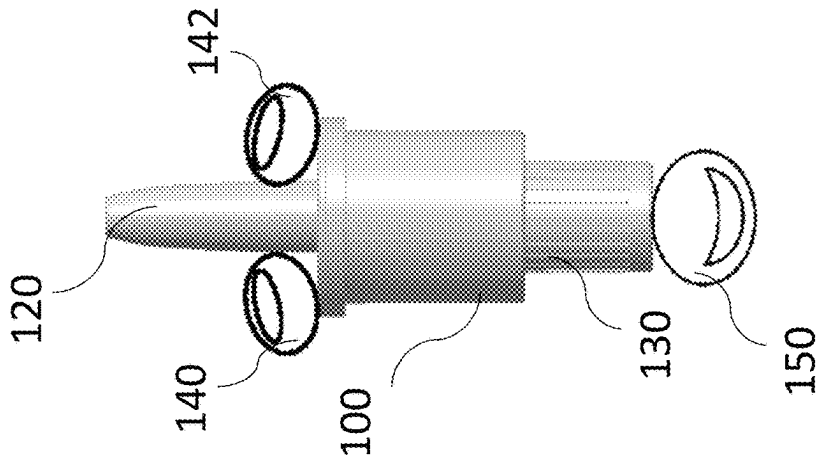


FIG. 5B

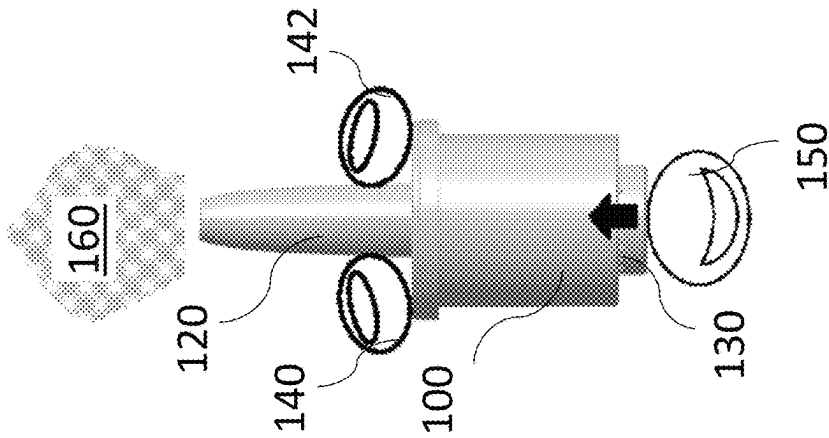


FIG. 5C

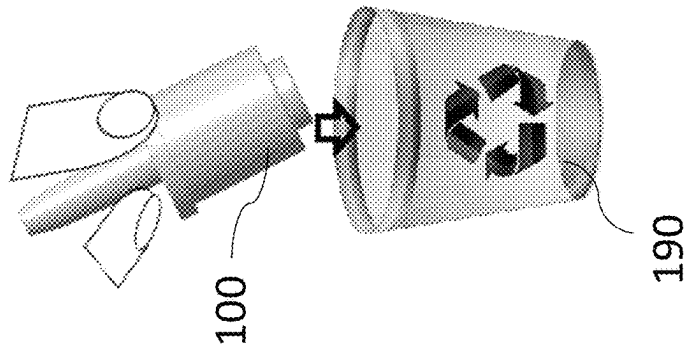


FIG. 5D

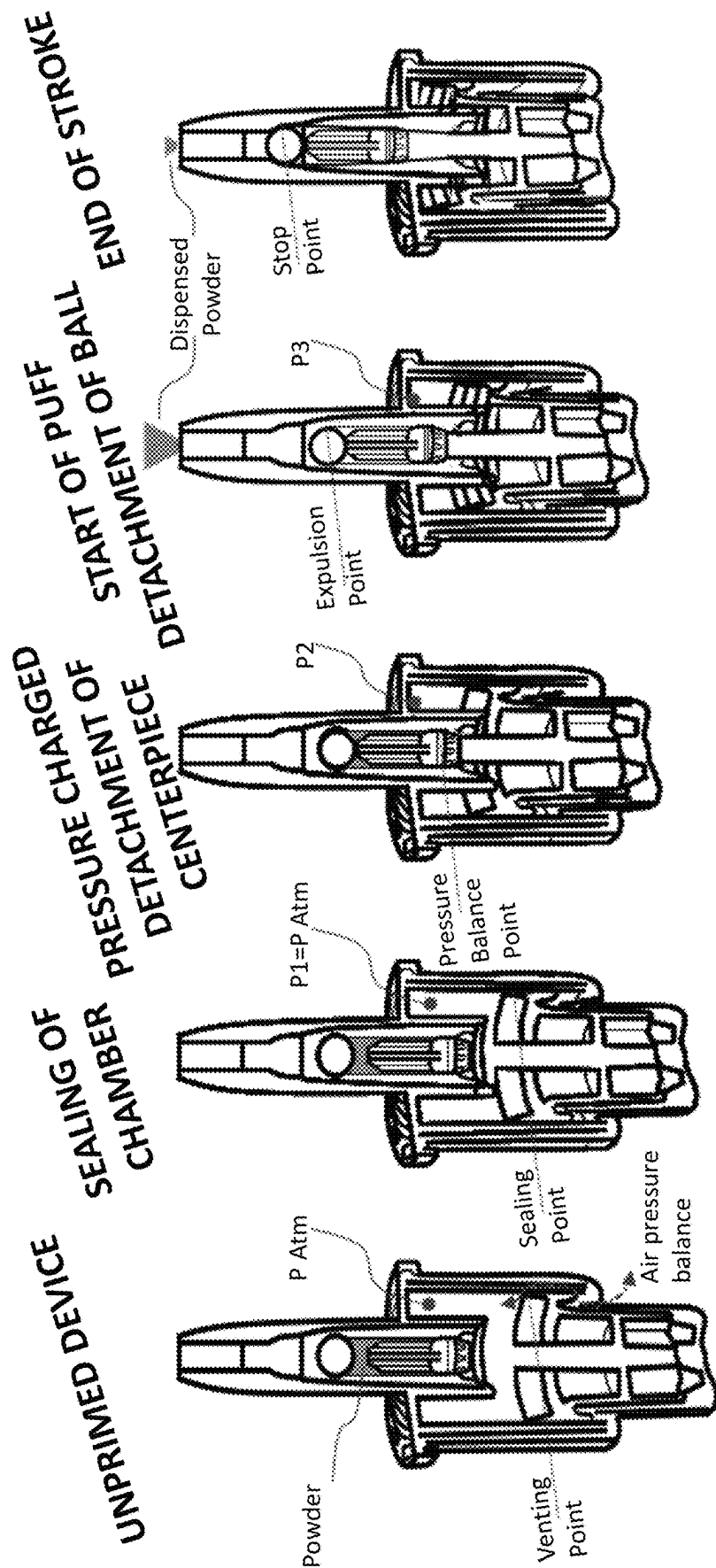


FIG. 6

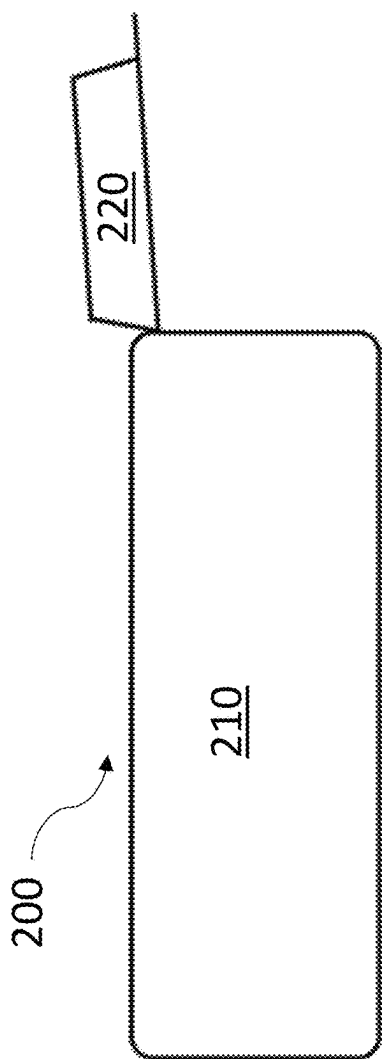
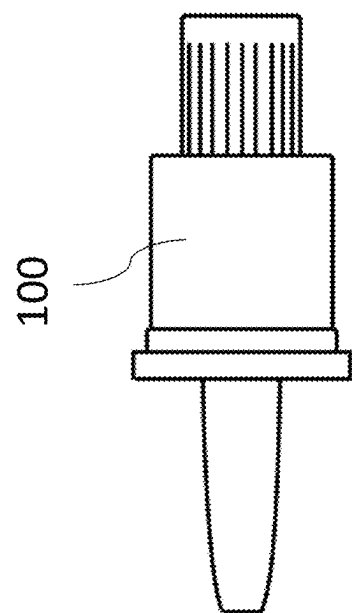


FIG. 7

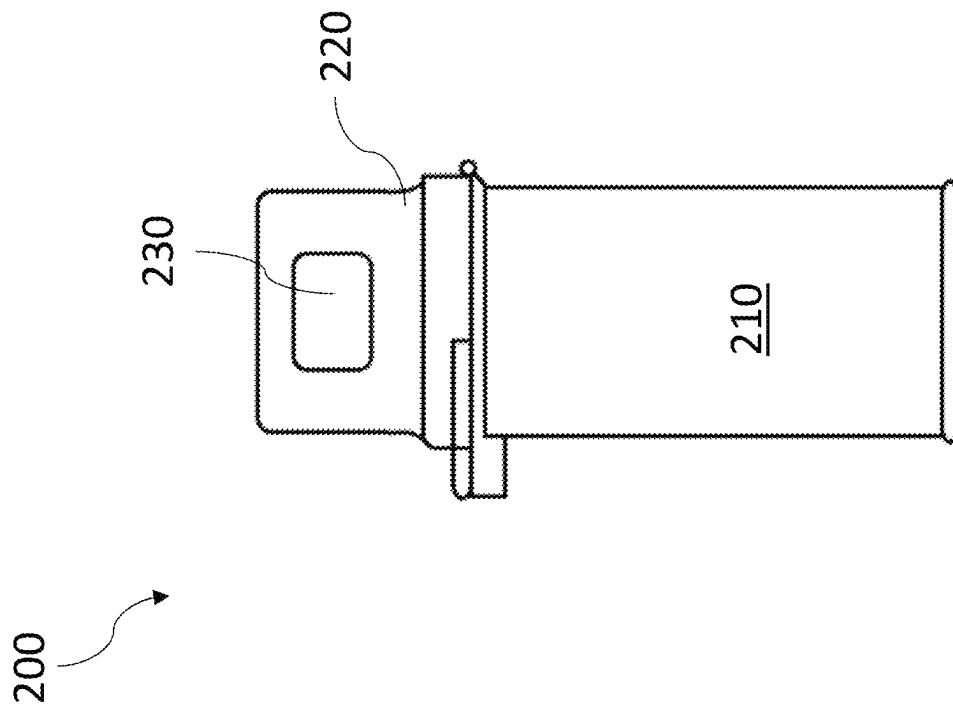


FIG. 8

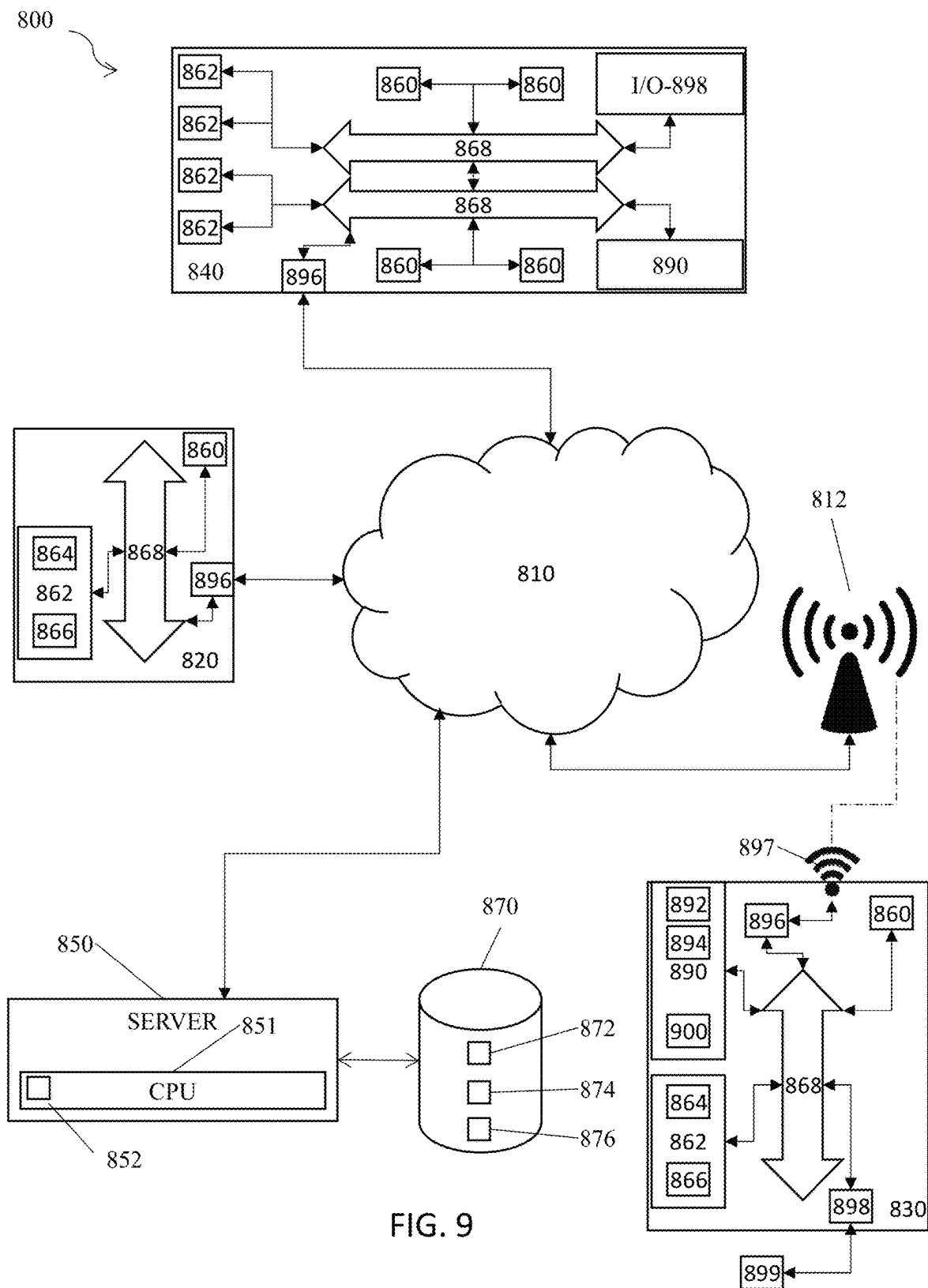


FIG. 9

FIG. 10

			Time (h)								
			0.0	1.0	3.0	5.0	10.0	15.0	30.0	60.0	
Test Article	Carrier	Animal	Epinephrine (ng/mL)								
Epinephrine	Lactose	1	0.00	32.2	46.3	64.7	41.8	38.5	9.87	0.00	
		2	0.00	0.00	28.0	18.5	16.4	10.1	0.00	0.00	
		3	0.00	111	153	67.6	38.3	14.8	0.00	0.00	
N			3	3	3	3	3	3	3	3	
Mean			0.00	47.7	75.8	50.3	32.2	21.1	3.29	0.00	
CV%			NC	119.6	89.1	54.8	42.8	72.0	173.2	NC	

FIG. 11

			Time (h)								
			0.0	1.0	3.0	5.0	10.0	15.0	30.0	60.0	
Test Article	Carrier	Animal	Epinephrine (ng/mL)								
Epinephrine	Sodium CMC	22	0.00	0.00	2.90	0.00	5.28	13.3	6.17	4.36	
		23	0.00	0.00	2.95	14.6	19.6	19.6	10.1	7.65	
		24	0.00	11.1	5.71	9.25	22.5	23.6	39.2	5.59	
N		3	3	3	3	3	3	3	3	3	
Mean		0.00	3.70	3.85	7.95	15.8	18.8	18.5	5.87	5.87	
CV%		NC	173.2	41.7	92.9	58.4	27.6	97.6	28.3	28.3	

FIG. 12

Test Article			Time (h)									
			Carrier	Animal	0.0	1.0	3.0	5.0	10.0	15.0	30.0	60.0
Epinephrine + Caffeine (5% w/w)					Epinephrine (ng/mL)							
			Lactose	19	0.00	87.3	62.3	23.6	16.0	4.15	2.69	0.00
				20	0.00	106	95.8	86.2	36.0	13.1	7.85	6.20
				21	0.00	43.2	88.4	71.8	23.3	19.3	10.4	3.53
N				3	3	3	3	3	3	3	3	3
Mean					0.00	78.8	82.2	60.5	25.1	12.2	6.98	3.24
CV%					NC	40.9	21.4	54.2	40.3	62.5	56.3	95.9

FIG. 13

Test Article			Time (h)									
			Carrier	Animal	0.0	1.0	3.0	5.0	10.0	15.0	30.0	60.0
Epinephrine + Caffeine (5% w/w)					Epinephrine (ng/mL)							
			Sodium CMC	40	0.00	7.80	18.9	16.0	15.2	16.3	20.5	14.4
				41	0.00	18.4	23.1	27.7	29.0	36.3	43.8	0.00
				42	0.00	16.3	16.1	24.0	16.3	20.9	10.1	4.36
N				3	3	3	3	3	3	3	3	3
Mean					0.00	14.2	19.4	22.6	20.2	24.5	24.8	6.25
CV%					NC	39.6	18.2	26.5	38.0	42.8	69.4	118.1

FIG. 14

Test Article			Time (h)									
			Carrier	Animal	0.0	1.0	3.0	5.0	10.0	15.0	30.0	60.0
Epinephrine + Hyaluronate (0.5% w/w)					Epinephrine (ng/mL)							
			Lactose	16	0.00	3.57	3.45	4.45	3.50	0.00	0.00	0.00
				17	0.00	43.7	13.7	11.3	2.88	0.00	2.72	0.00
				18	0.00	11.9	5.29	0.00	2.92	0.00	0.00	0.00
N				3	3	3	3	3	3	3	3	3
Mean					0.00	19.7	7.48	5.25	3.10	0.00	0.907	0.00
CV%					NC	107.4	73.1	108.4	11.2	NC	173.2	NC

FIG. 15

			Time (h)									
			0.0	1.0	3.0	5.0	10.0	15.0	30.0	60.0		
Epinephrine + Hyaluronate (0.5% w/w)	Test Article	Carrier	Animal	Epinephrine (ng/mL)								
				0.00	4.78	14.3	16.7	31.8	26.1	28.9	46.1	
				0.00	3.47	10.2	20.2	ISV	28.2	15.9	14.3	
				0.00	4.53	13.4	14.6	16.8	20.1	19.8	12.6	
				N	3	3	3	3	2	3	3	3
			Mean	0.00	4.26	12.6	17.2	24.3	24.8	21.5	24.3	
			CV%	NC	16.3	17.1	16.5	43.6	16.9	31.0	77.5	

FIG. 16

[illegible]

FIG. 17

Test Article			Time (h)							
			0.0	1.0	3.0	5.0	10.0	15.0	30.0	60.0
			Epinephrine (ng/mL)							
Epinephrine + Leucine (10% w/w)	Sodium CMC	Animal								
		25	0.00	0.00	6.00	4.08	7.56	6.80	8.47	7.98
		26	0.00	3.43	7.58	14.2	36.1	18.8	8.61	7.12
		27	0.00	0.00	4.62	10.6	14.9	11.6	8.65	9.92
N			3	3	3	3	3	3	3	3
Mean			0.00	1.14	6.07	9.63	19.5	12.4	8.58	8.34
CV%			NC	173.2	24.4	53.3	75.9	48.7	1.1	17.2

FIG. 18

Test Article			Time (h)							
			0.0	1.0	3.0	5.0	10.0	15.0	30.0	60.0
			Epinephrine (ng/mL)							
Epinephrine + NaCl (2% w/w)	Lactose	Animal								
		10	0.00	6.57	26.4	27.6	26.7	21.0	12.3	11.2
		11	0.00	22.9	21.7	22.4	91.1	16.3	16.2	6.31
		12	0.00	42.1	61.5	46.6	20.1	7.31	7.68	0.00
N			3	3	3	3	3	3	3	3
Mean			0.00	23.9	36.5	32.2	46.0	14.9	12.1	5.84
CV%			NC	74.5	59.5	39.6	85.3	46.8	35.4	96.2

FIG. 19

			Time (h)							
			0.0	1.0	3.0	5.0	10.0	15.0	30.0	60.0
Test Article	Carrier	Animal	Epinephrine (ng/mL)							
Epinephrine + NaCl (2% w/w)	Sodium CMC	31	0.00	0.00	4.49	7.32	13.8	9.65	10.2	7.32
		32	0.00	3.40	7.33	15.3	14.0	15.4	15.8	8.10
		33	0.00	4.57	19.8	54.2	177	117	74.6	39.9
N			3	3	3	3	3	3	3	3
Mean			0.00	2.66	10.5	25.6	68.3	47.4	33.5	18.4
CV%			NC	89.4	77.3	98.0	137.9	127.5	106.4	100.8

FIG. 20

Test Article			Time (h)												
			0.0	1.0	3.0	5.0	10.0	15.0	30.0	60.0					
			Epinephrine (ng/mL)												
Epinephrine + Niacin (2% w/w)			Animal	Carrier	7	Lactose	0.00	82.8	78.1	47.1	32.3	55.1	Not received	NC	
					8		0.00	97.4	135	107	28.2	21.0	0.00	0.00	
					9		0.00	575	683	NC	NC	NC	NC	NC	
					46		0.00	61.2	83.8	35.2	24.9	16.1	7.90	0.00	
					47		0.00	116	180	43.6	13.4	10.2	0.00	0.00	
			N				5	5	5	4	4	4	3	3	
			Mean				0.00	186	232	58.2	24.7	25.6	2.63	0.00	
			CV%					117.0	110.1	56.5	32.9	78.7	173.2		

FIG. 21

Test Article			Time (h)							
			0.0	1.0	3.0	5.0	10.0	15.0	30.0	60.0
			Epinephrine (ng/mL)							
Epinephrine + Niacin (2% w/w)	Sodium CMC	Animal								
		28	0.00	5.96	7.30	12.8	21.0	15.4	11.3	2.90
		29	0.00	4.25	13.8	18.9	17.0	16.1	13.7	11.7
		30	0.00	5.24	10.8	11.3	23.1	27.6	28.7	18.0
N			3	3	3	3	3	3	3	3
Mean			0.00	5.15	10.6	14.3	20.4	19.7	17.9	10.9
CV%			NC	16.7	30.6	28.1	15.2	34.8	52.7	69.8

FIG. 22

Test Article			Time (h)							
			0.0	1.0	3.0	5.0	10.0	15.0	30.0	60.0
			Epinephrine (ng/mL)							
Epinephrine + Polysorbate (0.5% w/w)	Lactose	Animal								
		13	0.00	52.3	79.5	67.9	32.3	34.0	14.7	13.5
		14	0.00	124	40.0	51.3	23.4	9.07	8.07	5.91
		15	0.00	88.4	32.8	100	18.2	7.18	5.57	7.55
N			3	3	3	3	3	3	3	3
Mean			0.00	88.2	50.8	73.1	24.6	16.8	9.45	8.99
CV%			NC	40.6	49.5	33.9	28.9	89.4	49.9	44.4

FIG. 23

Test Article			Time (h)							
			0.0	1.0	3.0	5.0	10.0	15.0	30.0	60.0
Epinephrine + Polysorbate (0.5% w/w)			Epinephrine (ng/mL)							
Carrier	Animal		0.00	0.00	15.2	16.2	18.5	12.0	12.9	9.70
Sodium CMC	34		0.00	0.00	7.28	8.05	10.7	12.1	11.3	10.8
	35		0.00	0.00	7.72	34.0	13.7	14.2	24.4	15.1
	36		0.00	2.57	7.72	34.0	13.7	14.2	24.4	15.1
	N		3	3	3	3	3	3	3	3
	Mean		0.00	0.857	10.1	19.4	14.3	12.8	16.2	11.9
	CV%		NC	173.2	44.2	68.3	27.5	9.7	44.1	24.0

FIG. 24

Test Article			Time (h)							
			0.0	1.0	3.0	5.0	10.0	15.0	30.0	60.0
Epinephrine intramuscular injection at 0.7 mg/kg			Epinephrine (ng/mL)							
Carrier	Animal		0.00	165	62.1	36.9	44.5	296	-	-
Liquid	43		0.00	933	464	-	-	-	-	-
	44		0.00	5.66	54.2	20.4	55.7	37.1	23.2	10.3
	45		0.00	5.66	54.2	20.4	55.7	37.1	23.2	10.3
	N		3	3	3	2	2	2	1	1
	Mean		0.00	368	193	28.7	50.1	167	23.2	10.3
	CV%		NC	134.8	121.2	40.7	15.8	109.9	NC	NC

FIG. 25

Test Article	Carrier	Animal	Dose (mg)	Cmax (ng/mL)	Tmax (min)	AUClast (min*ng/mL)	t½ (min)	AUCinf (min*ng/mL)
Epinephrine	Lactose	1	0.45	64.7	5.0	984	9.07	1110
		2	0.48	28.0	3.0	226	11.5	393
		3	0.45	153	3.0	910	4.56	1010
N			3	3	3	3	3	3
Mean			81.9	3.67	707	8.36	838	
CV%			78.4	31.5	59.1	41.8	46.4	

FIG. 26

Test Article	Carrier	Animal	Dose (mg)	Cmax (ng/mL)	Tmax (min)	AUClast (min*ng/mL)	t½ (min)	AUCinf (min*ng/mL)
Epinephrine	Sodium CMC	22	0.42	13.3	15.0	361	-	-
		23	0.47	19.6	10.0	683	36.0	1080
		24	0.60	39.2	30.0	1220	-	-
		N		3	3	3	1	1
		Mean		24.0	18.3	755	36.0	1080
		CV%		56.2	56.8	57.5	NC	NC

FIG. 27

Test Article	Carrier	Animal	Dose (mg)	Cmax (ng/mL)	Tmax (min)	AUClast (min*ng/mL)	t½ (min)	AUCinf (min*ng/mL)
Epinephrine + Caffeine (5% w/w)	Lactose	19	0.46	87.3	1.0	464	6.41	489
		20	0.51	106	1.0	1200	45.1	1600
		21	0.48	88.4	3.0	1040	18.3	1130
N			3	3	3	3	3	3
Mean			93.9	1.67	902	23.3	1080	
CV%			11.2	69.3	43.0	85.1	52.1	

FIG. 28

Test Article	Carrier	Animal	Dose (mg)	Cmax (ng/mL)	Tmax (min)	AUClast (min*ng/mL)	t _{1/2} (min)	AUCinf (min*ng/mL)
Epinephrine + Caffeine (5% w/w)	Sodium CMC	40	0.52	20.6	30.0	1020	-	-
		41	0.51	43.8	30.0	1010	-	-
		42	0.48	24.0	5.0	701	20.5	830
		N		3	3	3	1	1
		Mean		29.5	21.7	909	20.5	830
		CV%		42.5	66.6	19.8	NC	NC

FIG. 29

Test Article	Carrier	Animal	Dose (mg)	Cmax (ng/mL)	Tmax (min)	AUClast (min*ng/mL)	t _{1/2} (min)	AUCinf (min*ng/mL)
Epinephrine + Hyaluronate (0.5% w/w)	Lactose	16	0.22	4.45	5.0	36.5	-	-
		17	0.14	43.7	1.0	157	12.9	207
		18	0.08	11.9	1.0	34.8	-	-
		N		3	3	3	1	1
		Mean		20.0	2.33	76.1	12.9	207
		CV%		104.1	99.0	92.0	NC	NC

FIG. 30

Test Article	Carrier	Animal	Dose (mg)	Cmax (ng/mL)	Tmax (min)	AUClast (min*ng/mL)	t _{1/2} (min)	AUCinf (min*ng/mL)
Epinephrine + Hyaluronate (0.5% w/w)	Sodium CMC	37	0.65	46.1	60.0	1860	-	-
		38	0.47	28.2	15.0	1060	-	-
		39	0.48	20.1	15.0	996	-	-
		N		3	3	3	0	0
		Mean		31.5	30.0	1300	NC	NC
		CV%		42.3	86.6	36.7	NC	NC

FIG. 31

Test Article	Carrier	Animal	Dose (mg)	Cmax (ng/mL)	Tmax (min)	AUClast (min*ng/mL)	t½ (min)	AUCinf (min*ng/mL)
Epinephrine + Leucine (10% w/w)	Lactose	4	0.37	0.00	0.0	-	-	-
		5	0.33	0.00	0.0	-	-	-
		6	0.16	0.00	0.0	-	-	-
N			3	3	0	0	0	0
Mean			0.00	0.00	0.00	-	-	-
CV%			NC	NC	NC	-	-	-

FIG. 32

Test Article	Carrier	Animal	Dose (mg)	Cmax (ng/mL)	Tmax (min)	AUClast (min*ng/mL)	t½ (min)	AUCinf (min*ng/mL)
Epinephrine + Leucine (10% w/w)	Sodium CMC	25	0.52	8.47	30.0	442	-	-
		26	0.48	36.1	10.0	724	35.7	1090
		27	0.50	14.9	10.0	579	298	4840
		N		3	3	3	2	2
		Mean		19.8	16.7	582	167	2970
		CV%		72.9	69.3	24.2	111.1	89.4

FIG. 33

Test Article	Carrier	Animal	Dose (mg)	Cmax (ng/mL)	Tmax (min)	AUClast (min*ng/mL)	t½ (min)	AUCinf (min*ng/mL)
Epinephrine + NaCl (2% w/w)	Lactose	10	0.45	27.6	5.0	941	42.4	1630
		11	0.43	91.1	10.0	1160	30.7	1440
		12	0.44	61.5	3.0	565	10.6	683
N			3	3	3	3	3	3
Mean			60.1	6.00	889	27.9	1250	
CV%			52.9	60.1	33.8	57.6	40.0	

FIG. 34

Test Article	Carrier	Animal	Dose (mg)	Cmax (ng/mL)	Tmax (min)	AUClast (min*ng/mL)	t½ (min)	AUCInf (min*ng/mL)
Epinephrine + NaCl (2% w/w)	Sodium CMC	31	0.40	13.8	10.0	536	101	1610
		32	0.54	15.8	30.0	761	-	-
		33	0.46	177	10.0	4480	29.5	6180
			N	3	3	3	2	2
			Mean	68.9	16.7	1930	65.4	3890
			CV%	136.0	69.3	115.0	77.6	83.1

FIG. 35

Test Article	Carrier	Animal	Dose (mg)	Cmax (ng/mL)	Tmax (min)	AUClast (min*ng/mL)	t½ (min)	AUCinf (min*ng/mL)
Epinephrine + Niacin (2% w/w)	Lactose	7	9.61	82.8	1.0	740	26.7	2860
		8	0.29	135	3.0	940	4.26	1070
		9	0.50	683	3.0	1550	-	-
		46	0.00	83.8	3.0	710	11.7	844
		47	0.00	180	3.0	733	4.77	803
			N	5	5	5	4	4
			Mean	233	2.60	934	11.9	1390
			CV%	109.4	34.4	38.0	88.1	70.6

FIG. 36

Test Article	Carrier	Animal	Dose (mg)	Cmax (ng/mL)	Tmax (min)	AUClast (min*ng/mL)	t½ (min)	AUCinf (min*ng/mL)
Epinephrine + Niacin (2% w/w)	Sodium CMC	28	0.50	21.0	10.0	595	18.1	671
		29	0.45	18.9	5.0	828	94.0	2420
		30	0.51	28.7	30.0	1360	-	-
N			3	3	3	3	2	2
Mean			Mean	22.9	15.0	929	56.1	1540
CV%			CV%	22.6	88.2	42.4	95.7	79.9

FIG. 37

Test Article	Carrier	Animal	Dose (mg)	Cmax (ng/mL)	Tmax (min)	AUClast (min*ng/mL)	t½ (min)	AUCInf (min*ng/mL)
Epinephrine + Polysorbate (0.5% w/w)	Lactose	13	0.34	79.5	3.0	1480	26.9	2000
		14	0.47	124	1.0	892	71.9	1500
		15	0.44	100	5.0	880	65.0	1590
		N		3	3	3	3	3
		Mean		101	3.00	1080	54.6	1700
		CV%		22.0	66.7	31.6	44.3	15.7

FIG. 38

Test Article	Carrier	Animal	Dose (mg)	Cmax (ng/mL)	Tmax (min)	AUClast (min*ng/mL)	t½ (min)	AUCInf (min*ng/mL)
Epinephrine + Polysorbate (0.5% w/w)	Sodium CMC	34	0.42	18.5	10.0	732	128	2520
		35	0.42	12.1	15.0	633	-	-
		36	0.51	34.0	5.0	1110	622	14700
		N		3	3	3	2	2
		Mean		21.5	10.0	824	375	8590
		CV%		52.3	50.0	30.2	93.1	99.8

FIG. 39

Test Article	Carrier	Animal	Dose (mg)	Cmax (ng/mL)	Tmax (min)	AUClast (min*ng/mL)	t½ (min)	AUCInf (min*ng/mL)
Epinephrine intramuscular injection at 0.7 mg/kg	Liquid	43	0.00	296	15.0	1440	-	-
		44	0.00	933	1.0	1810	-	-
		45	0.00	55.7	10.0	1470	24.5	1840
	N		3	3	3	1	1	
Mean		428	8.67	1580	24.5	1840		
CV%		105.9	81.9	12.9	NC	NC		

Test_Article=Epinephrine, Carrier=Lactose

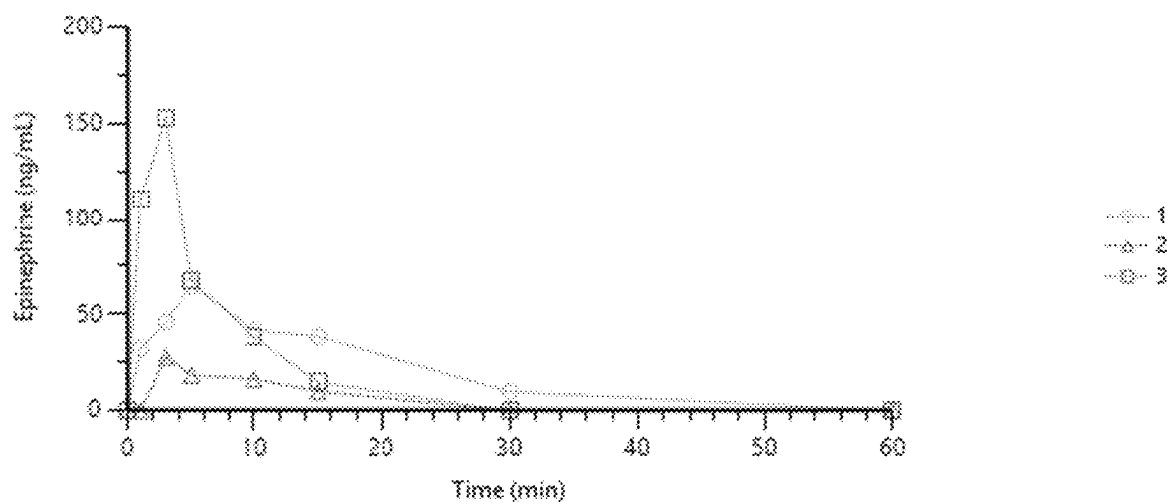


FIG. 40

Test_Article=Epinephrine, Carrier=Lactose

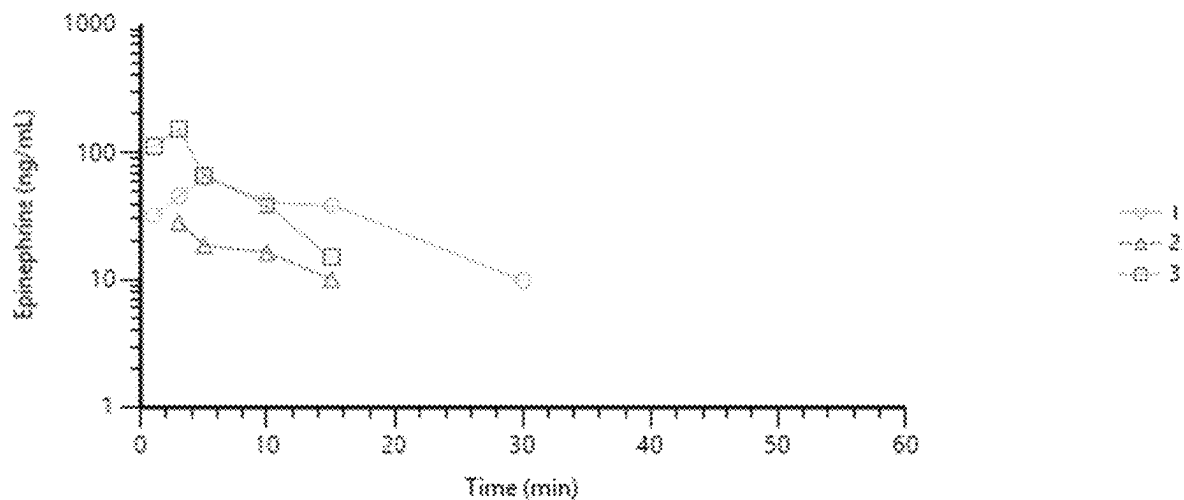


FIG. 41

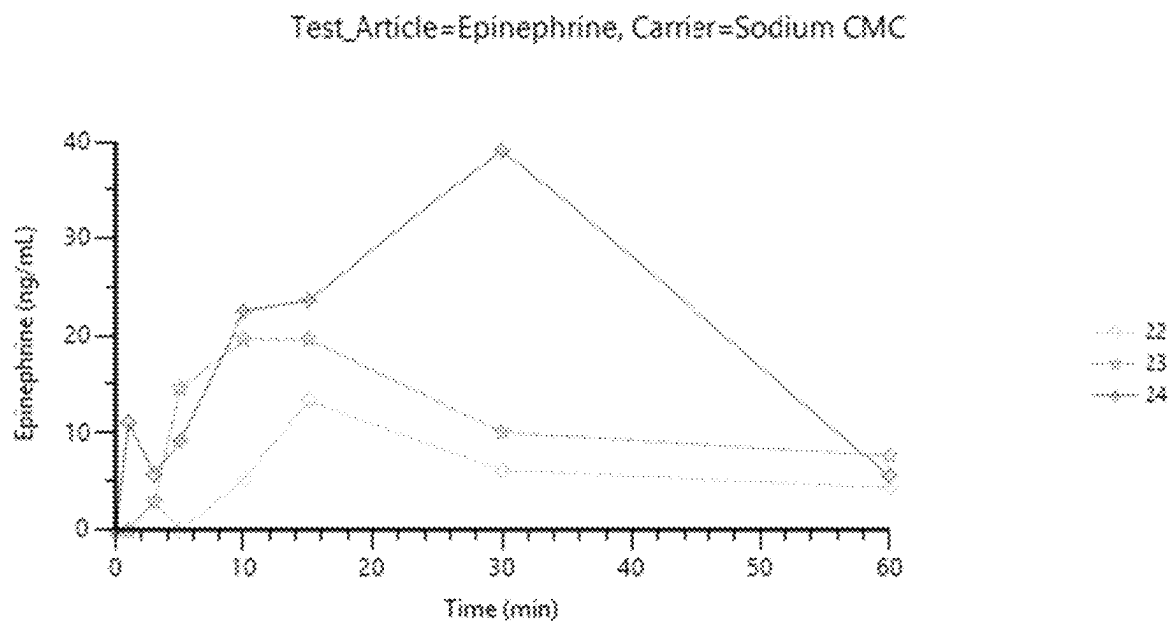


FIG. 42

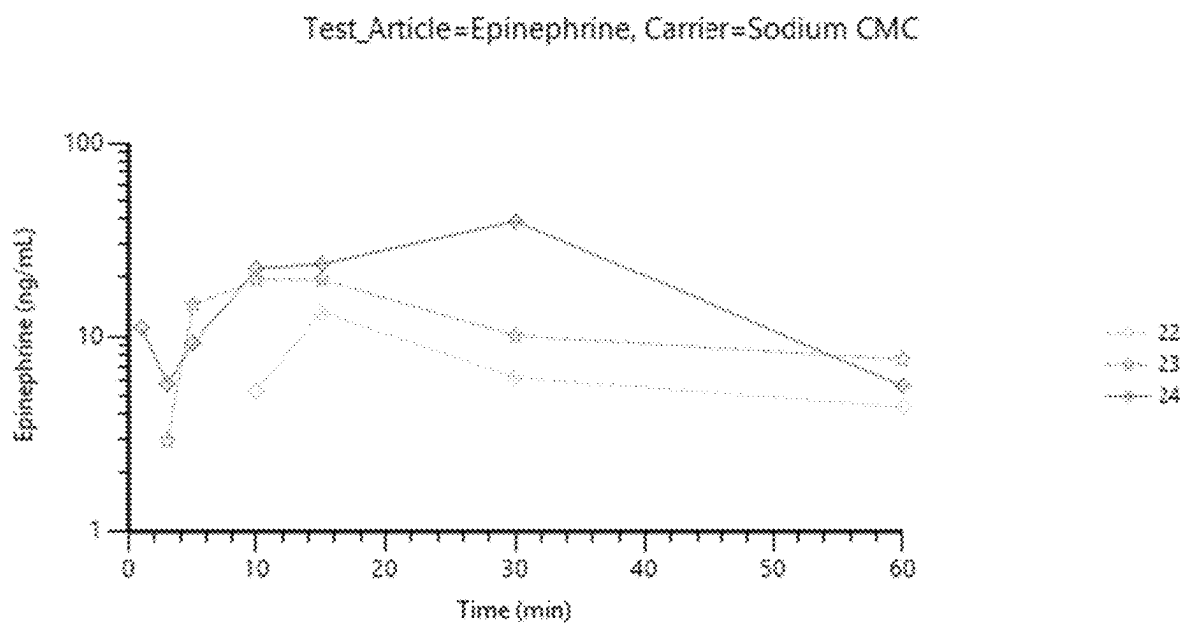


FIG. 43

Test_Article=Epinephrine + Caffeine (5% w/w), Carrier=Lactose

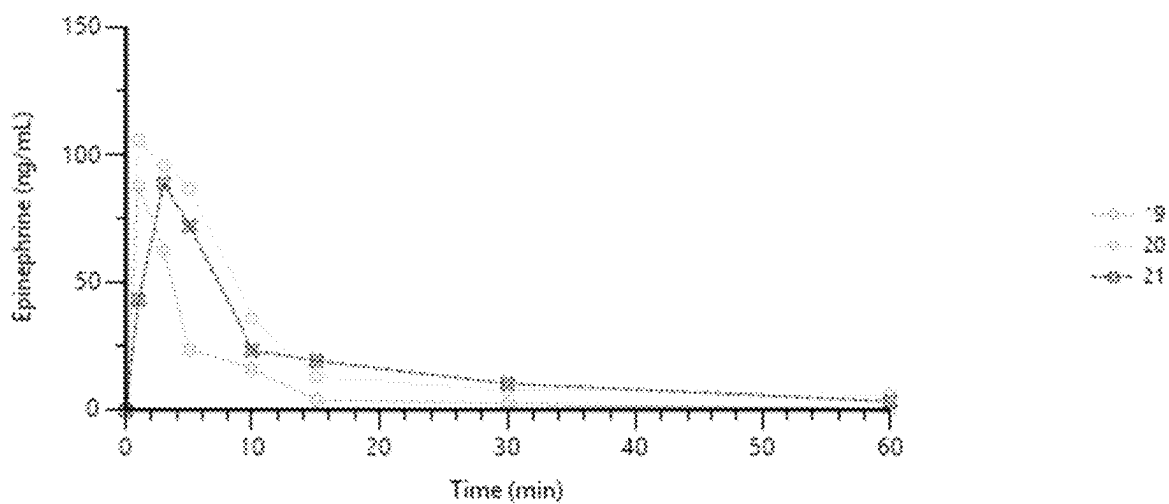


FIG. 44

Test_Article=Epinephrine + Caffeine (5% w/w), Carrier=Lactose

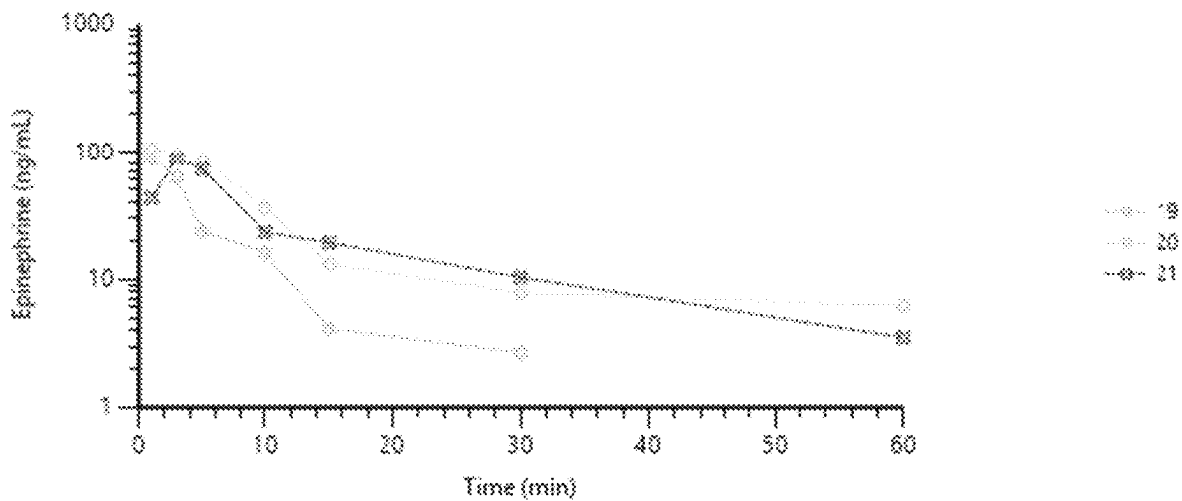


FIG. 45

Test_Article=Epinephrine + Caffeine (5% w/w), Carrier=Sodium CMC

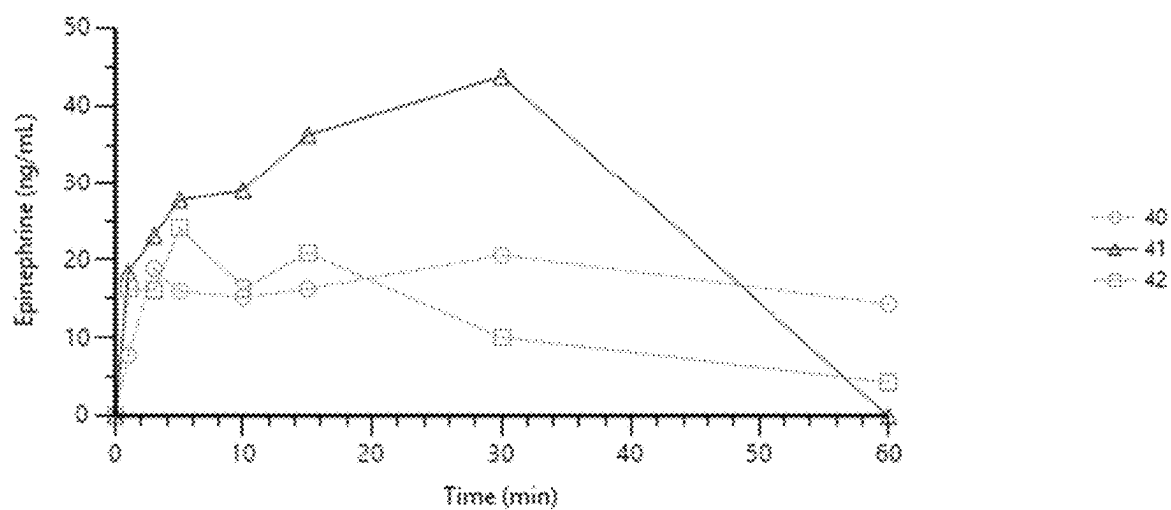


FIG. 46

Test_Article=Epinephrine + Caffeine (5% w/w), Carrier=Sodium CMC

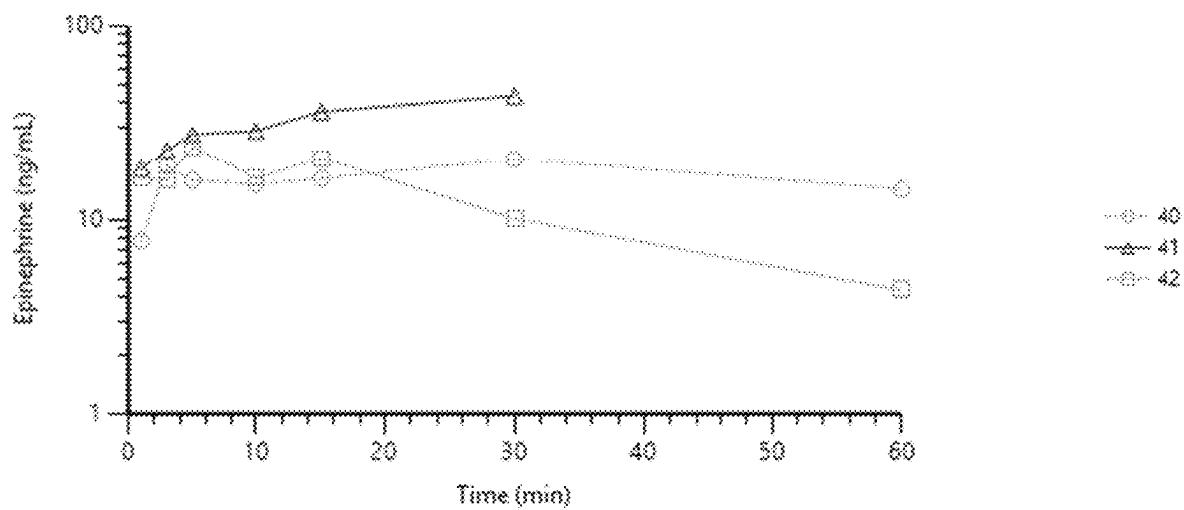


FIG. 47

Test_Article=Epinephrine + Hyaluronate 0.5% w/w, Carrier=Lactose

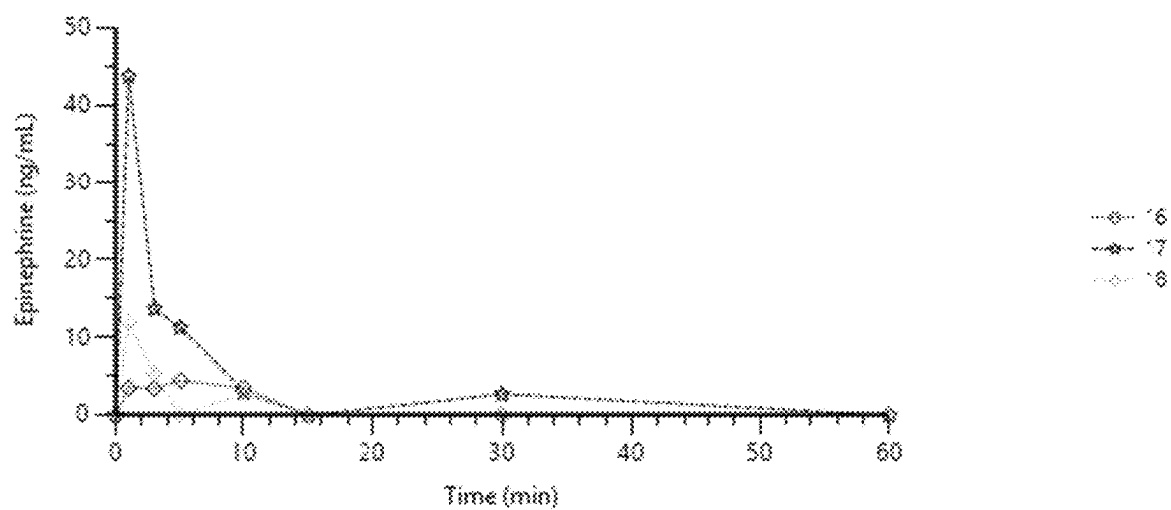


FIG. 48

Test_Article=Epinephrine + Hyaluronate 0.5% w/w, Carrier=Lactose

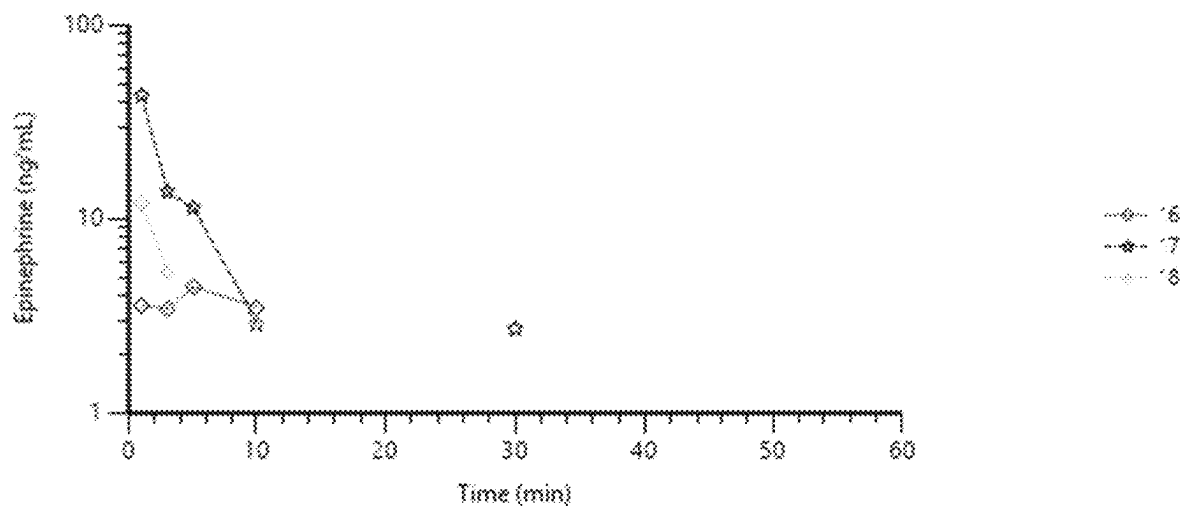


FIG. 49

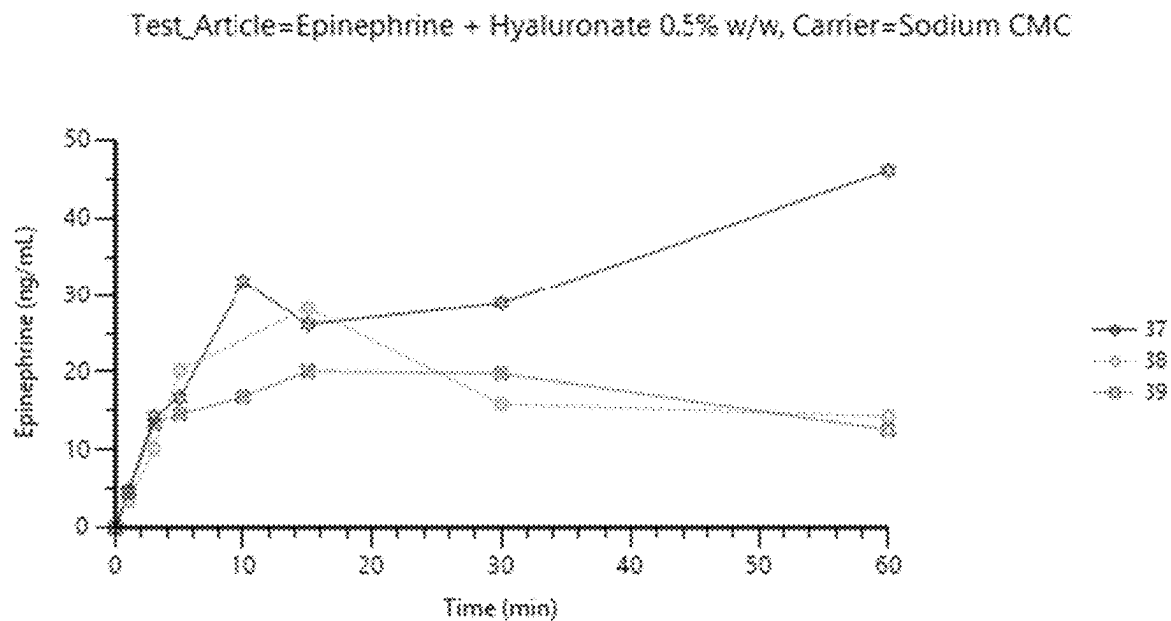


FIG. 50

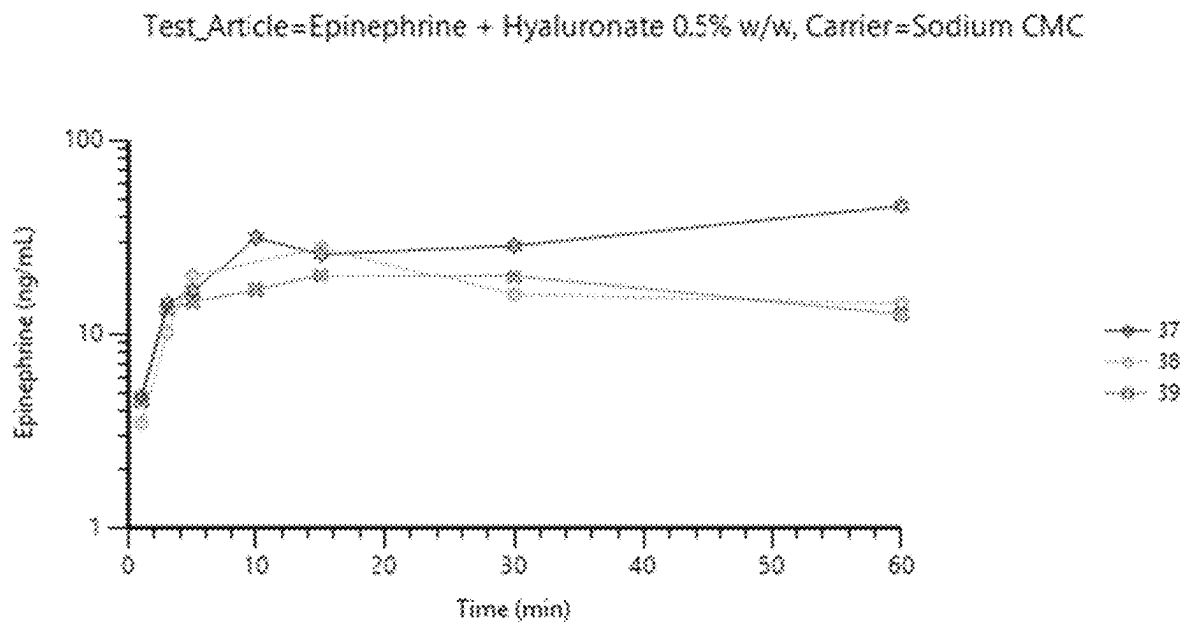


FIG. 51

Test_Article=Epinephrine + Leucine (10% w/w), Carrier=Lactose

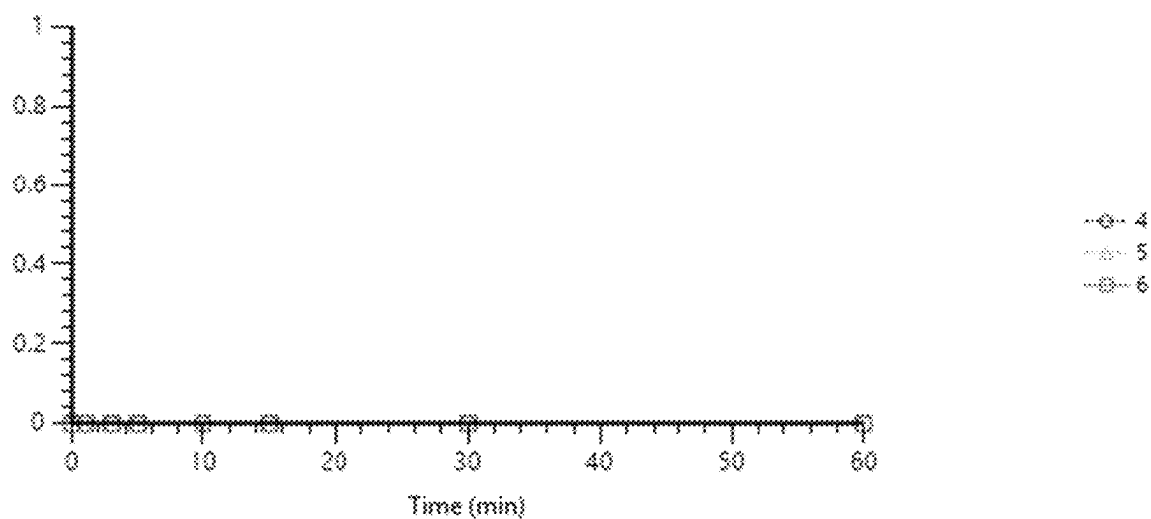


FIG. 52

Test_Article=Epinephrine + Leucine (10% w/w), Carrier=Lactose

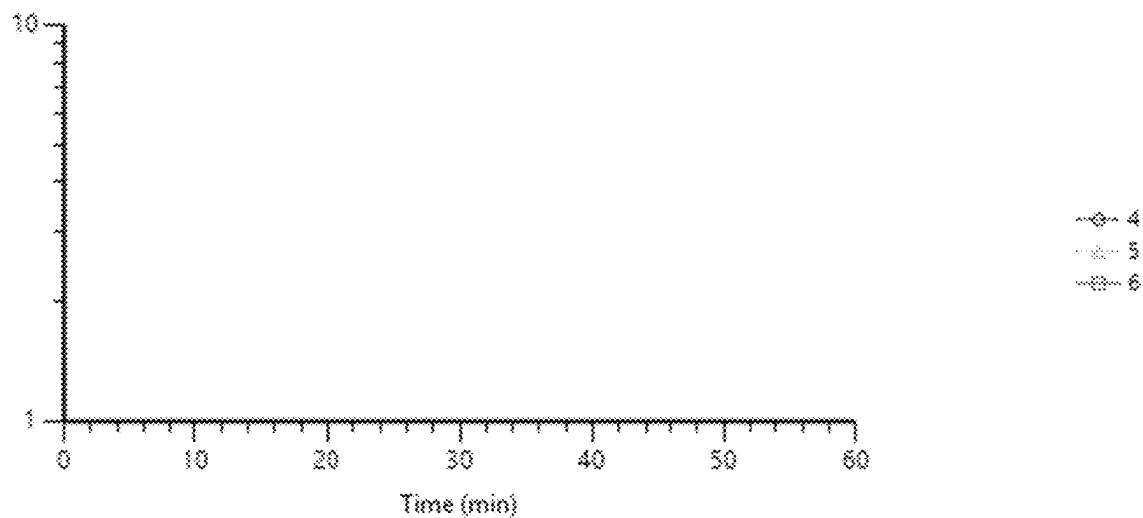


FIG. 53

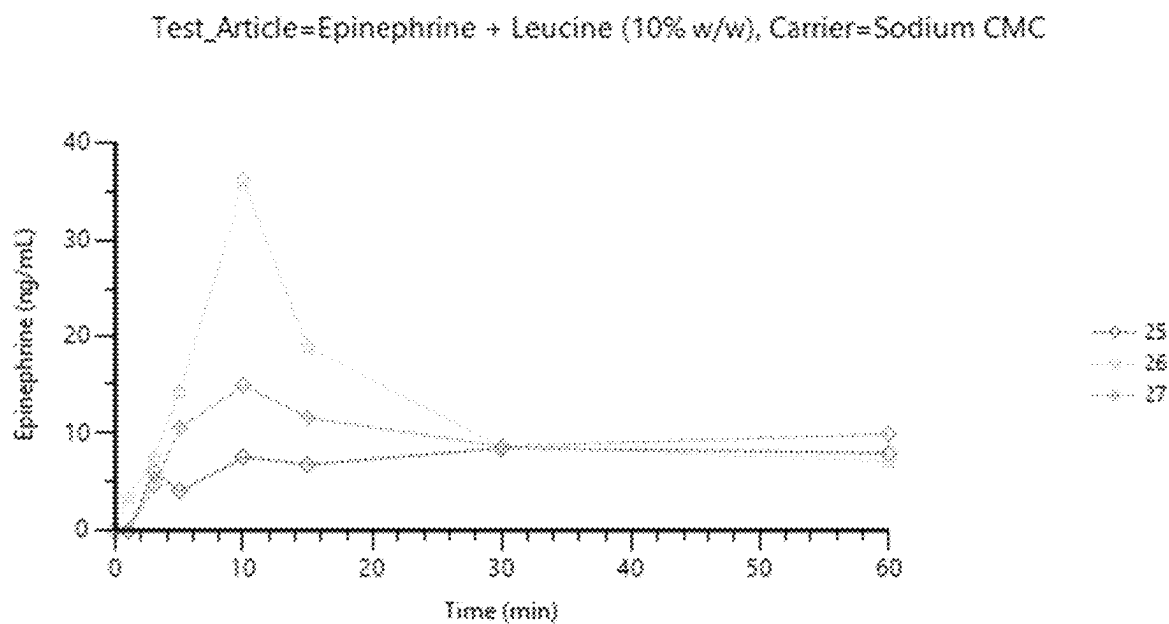


FIG. 54

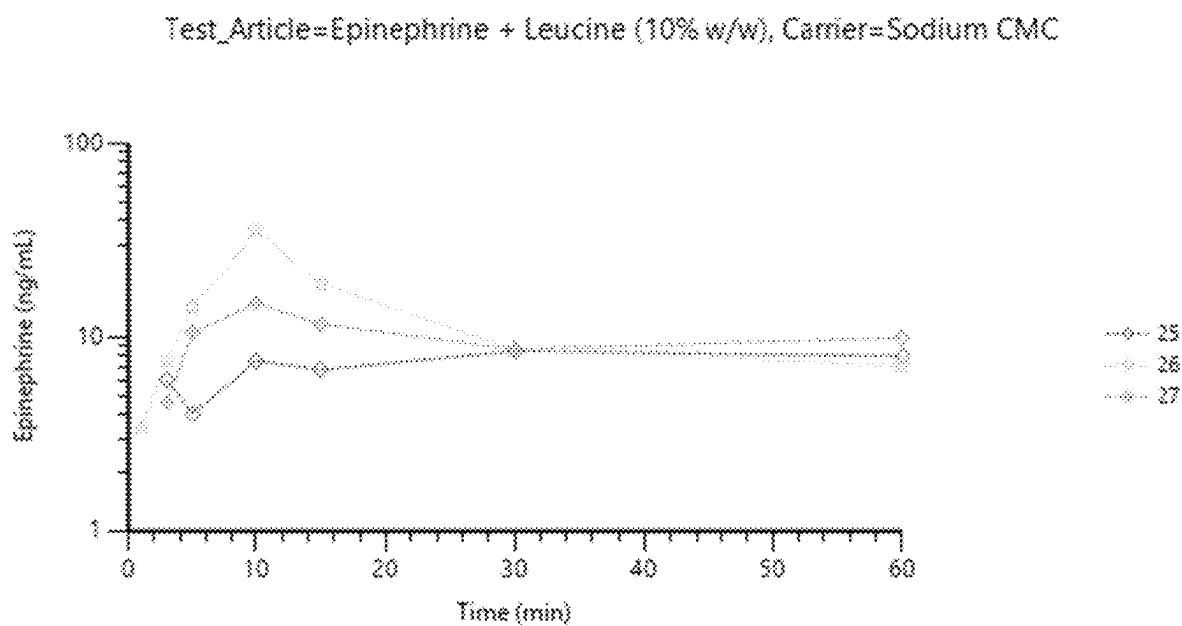


FIG. 55

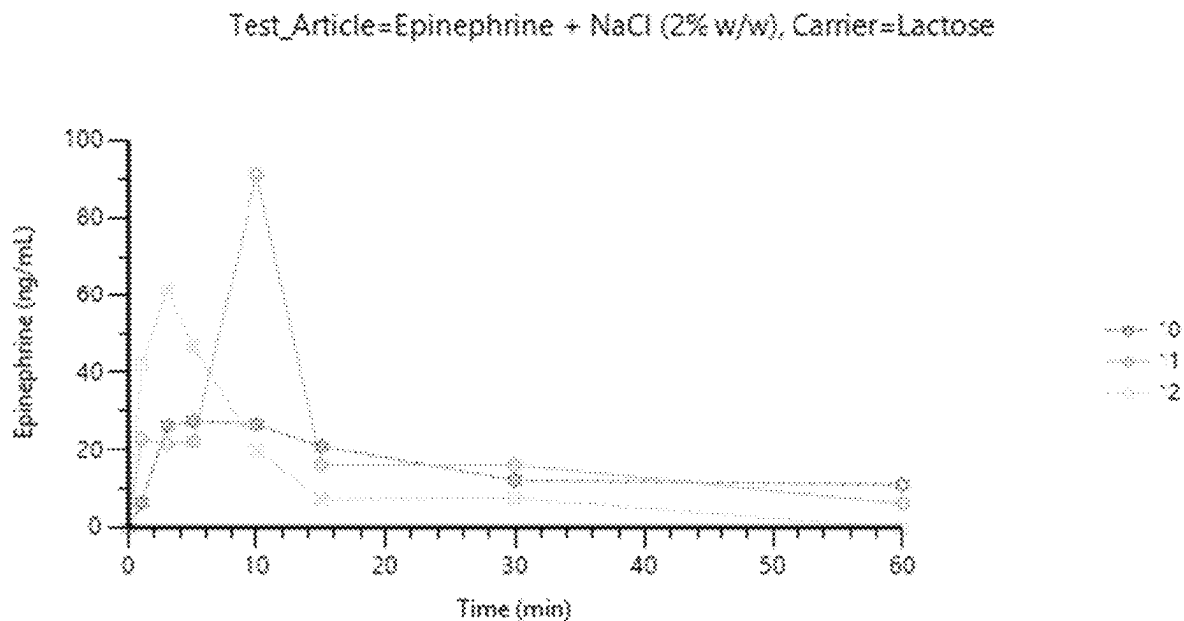


FIG. 56

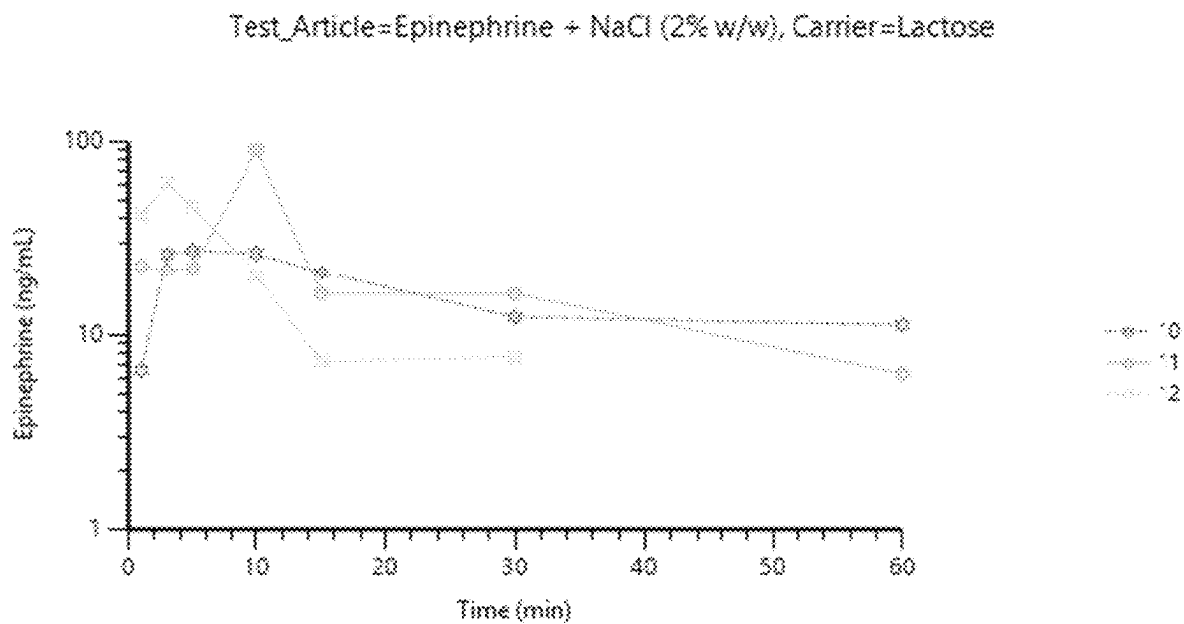


FIG. 57

Test_Article=Epinephrine + NaCl (2% w/w), Carrier=Sodium CMC

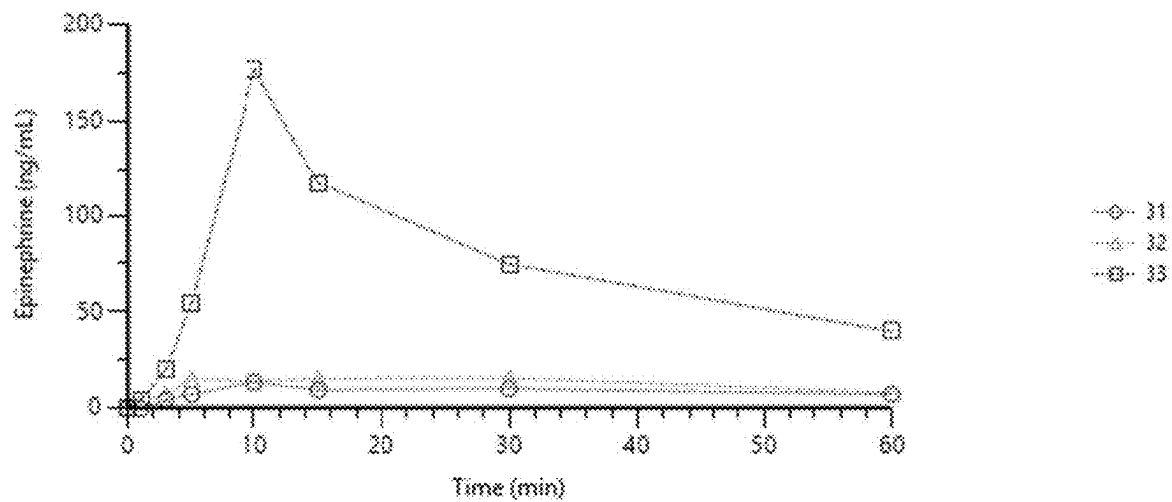


FIG. 58

Test_Article=Epinephrine + NaCl (2% w/w), Carrier=Sodium CMC

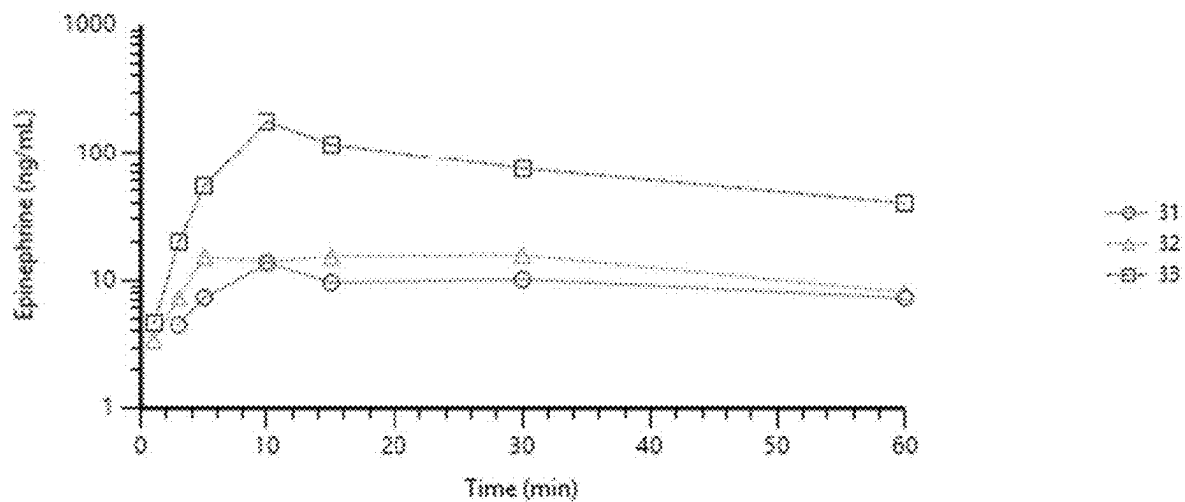


FIG. 59

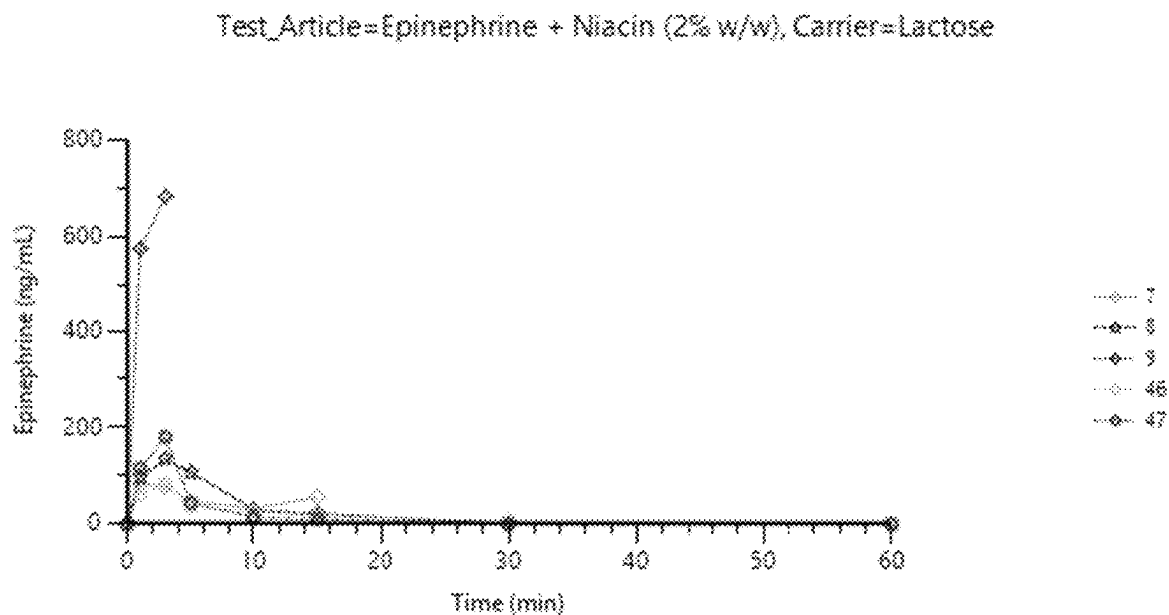


FIG. 60

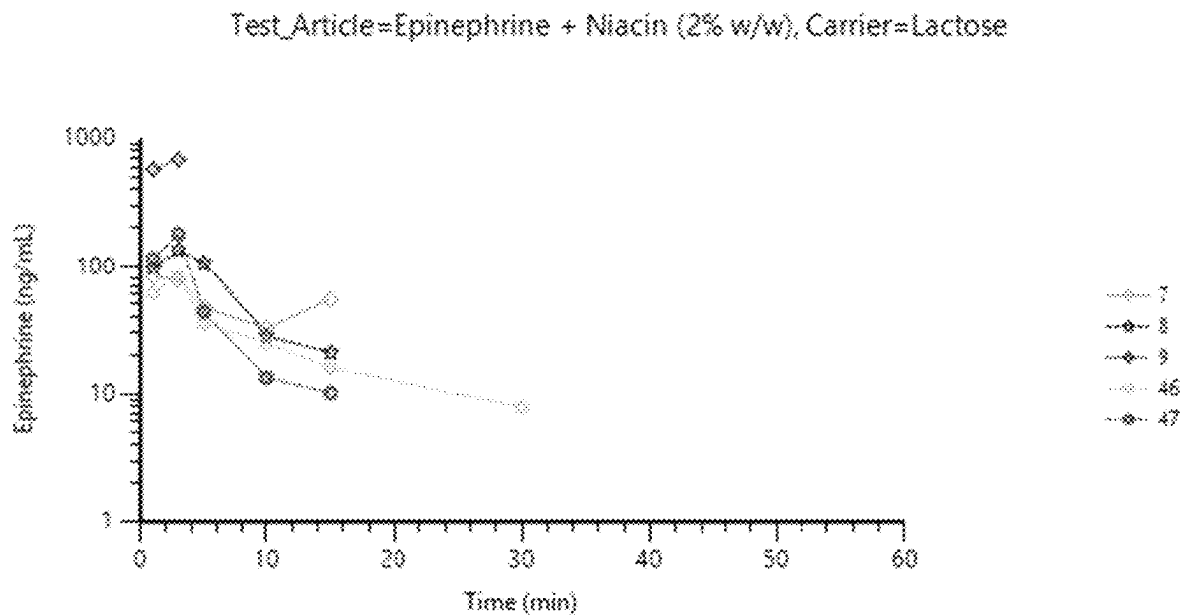


FIG. 61

Test_Article=Epinephrine + Niacin (2% w/w), Carrier=Sodium CMC

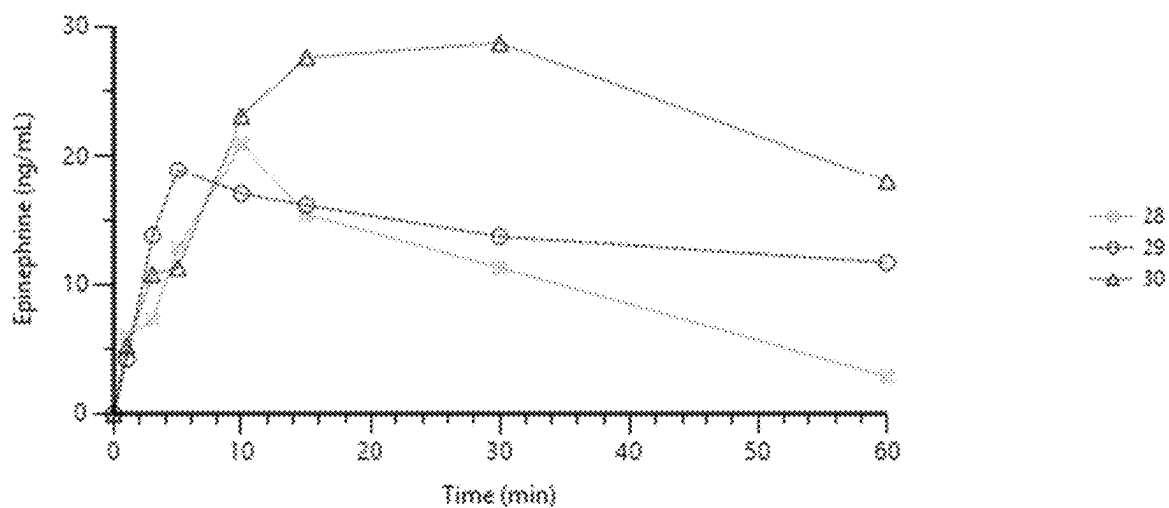


FIG. 62

Test_Article=Epinephrine + Niacin (2% w/w), Carrier=Sodium CMC

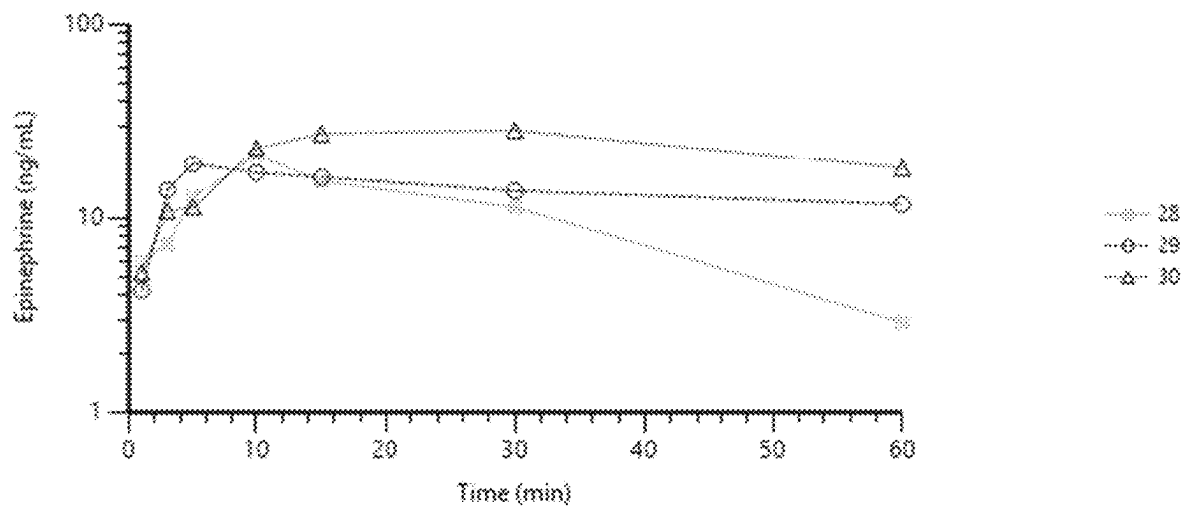


FIG. 63

Test_Article=Epinephrine + Polysorbate (0.5% w/w), Carrier=Lactose

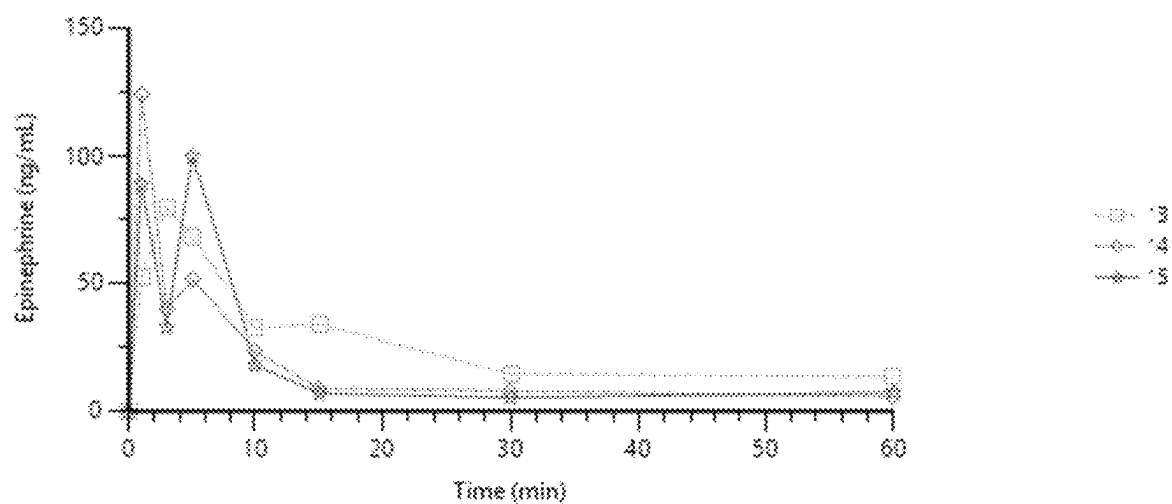


FIG. 64

Test_Article=Epinephrine + Polysorbate (0.5% w/w), Carrier=Lactose

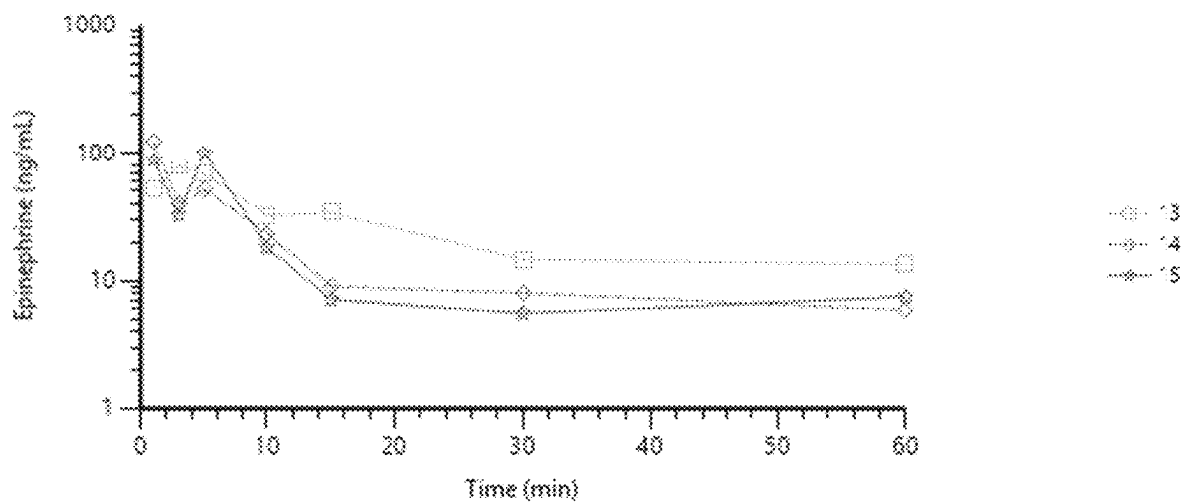


FIG. 65

Test_Article=Epinephrine + Polysorbate (0.5% w/w), Carrier=Sodium CMC

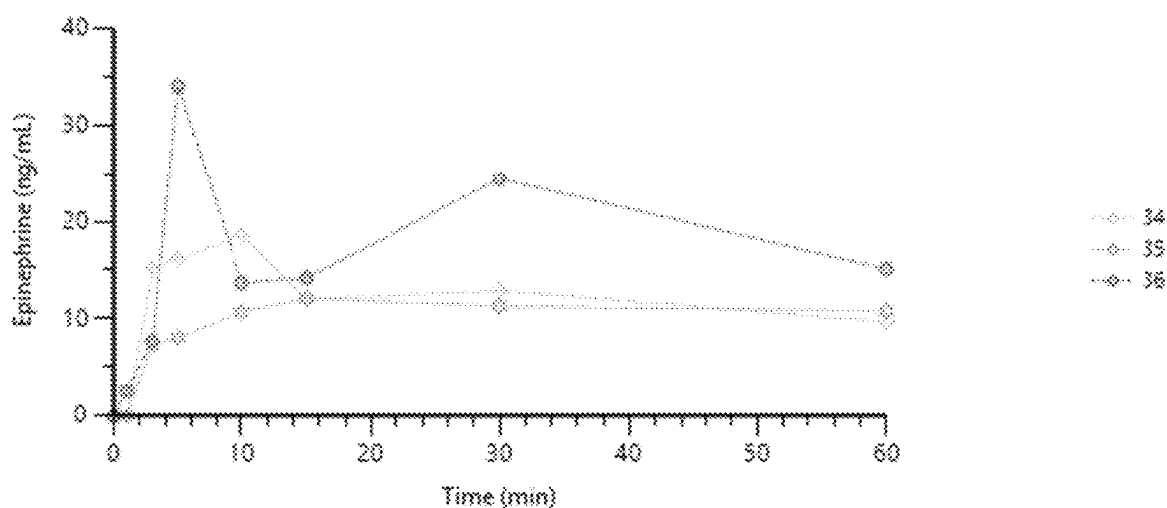


FIG. 66

Test_Article=Epinephrine + Polysorbate (0.5% w/w), Carrier=Sodium CMC

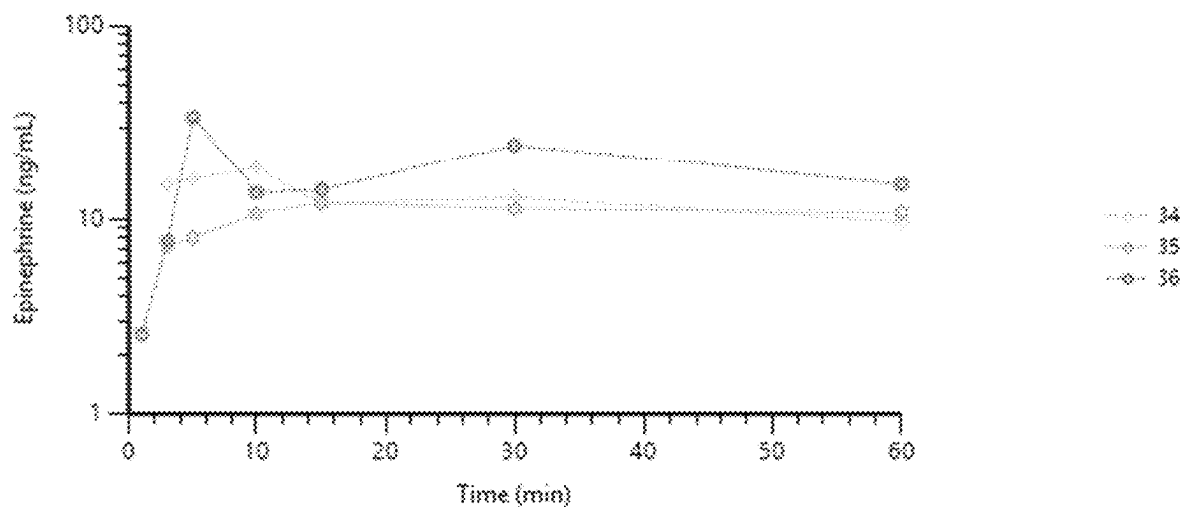


FIG. 67

Test_Article=Epinephrine intramuscular injection at 0.004 mg/kg, Carrier=Liquid

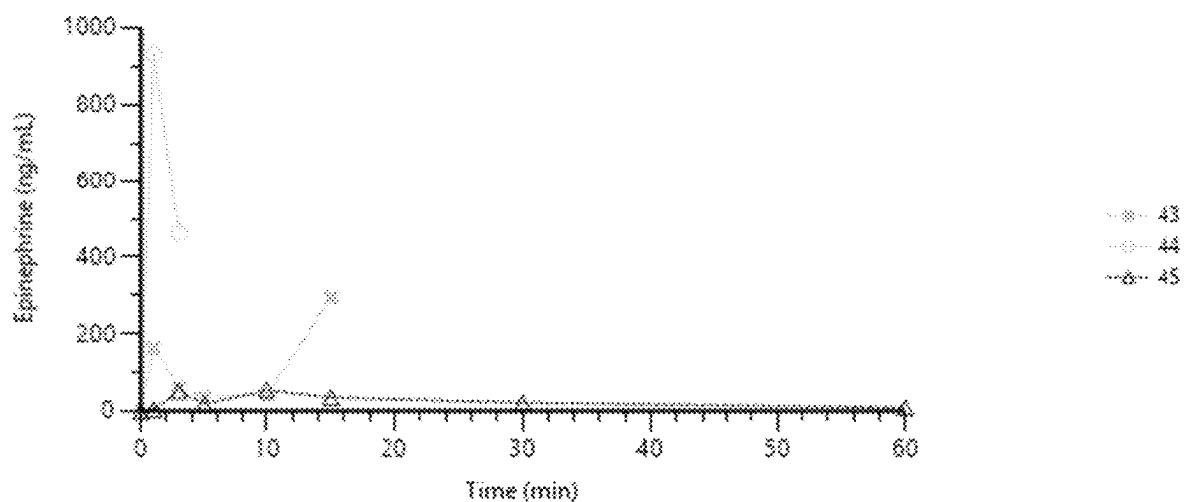


FIG. 68

Test_Article=Epinephrine intramuscular injection at 0.004 mg/kg, Carrier=Liquid

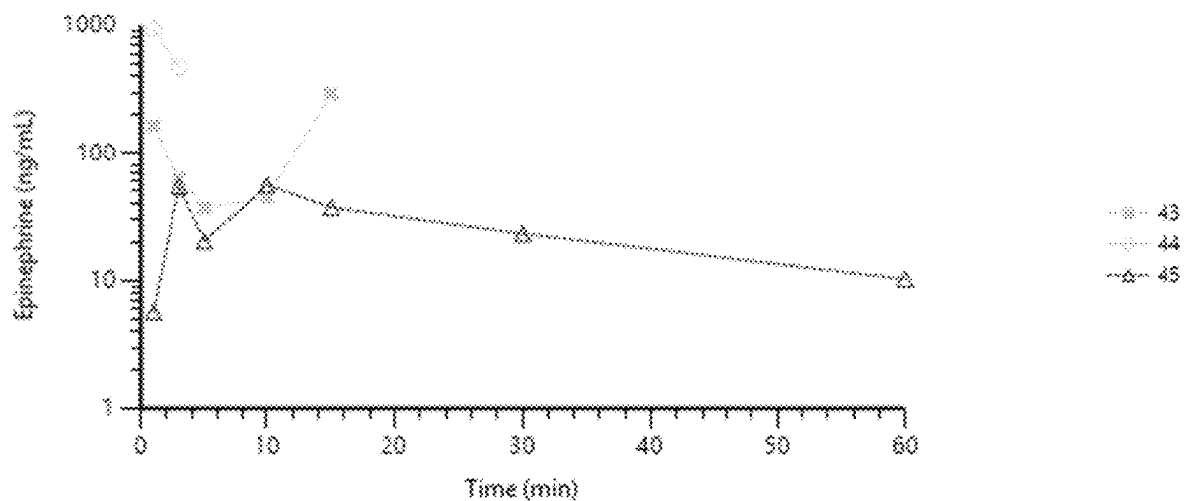


FIG. 69

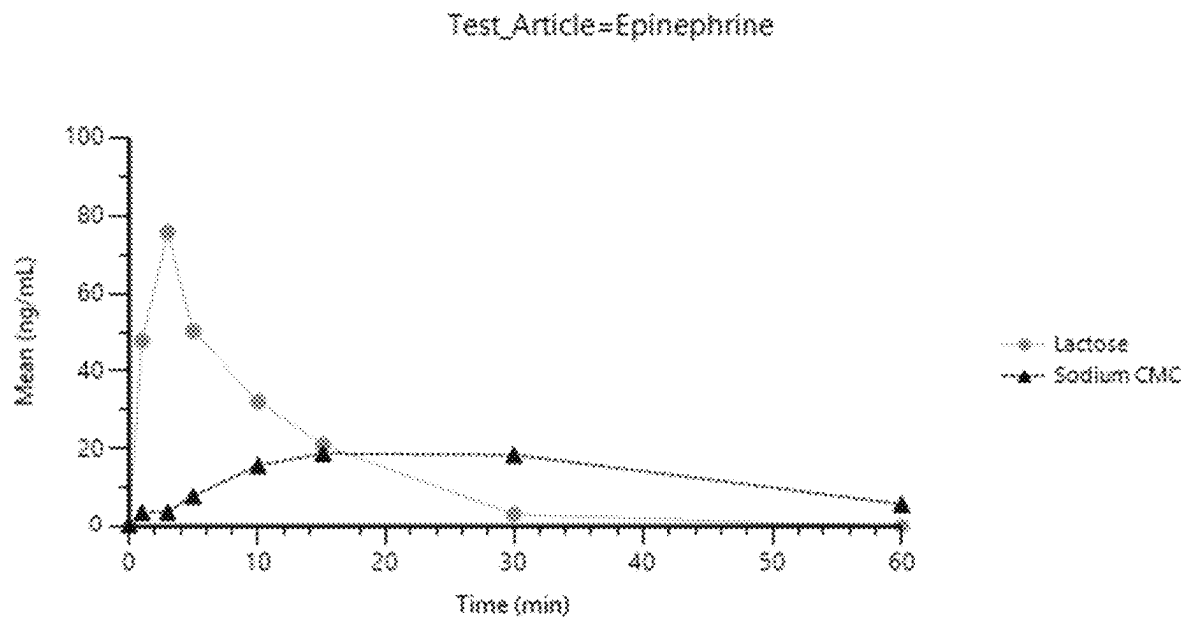


FIG. 70

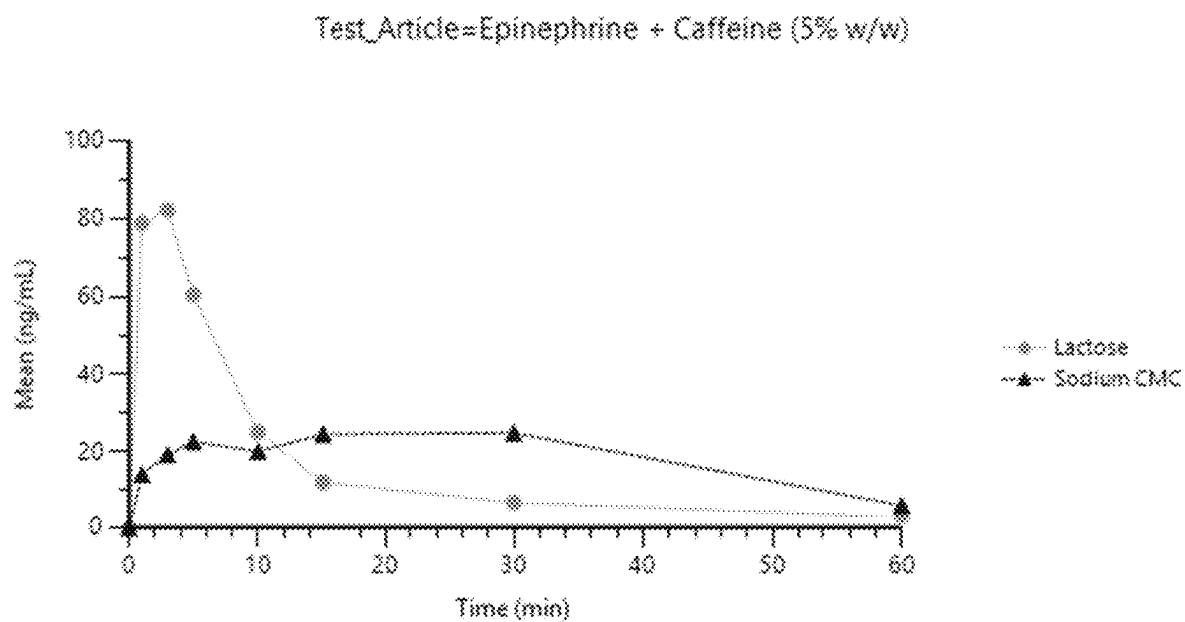


FIG. 71

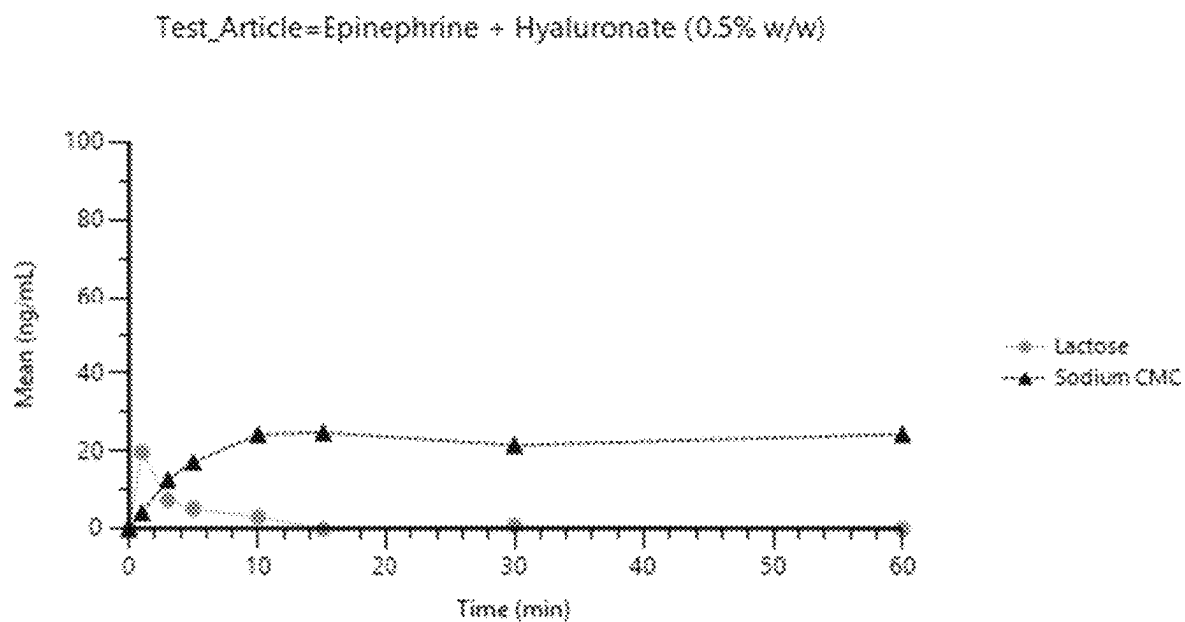


FIG. 72

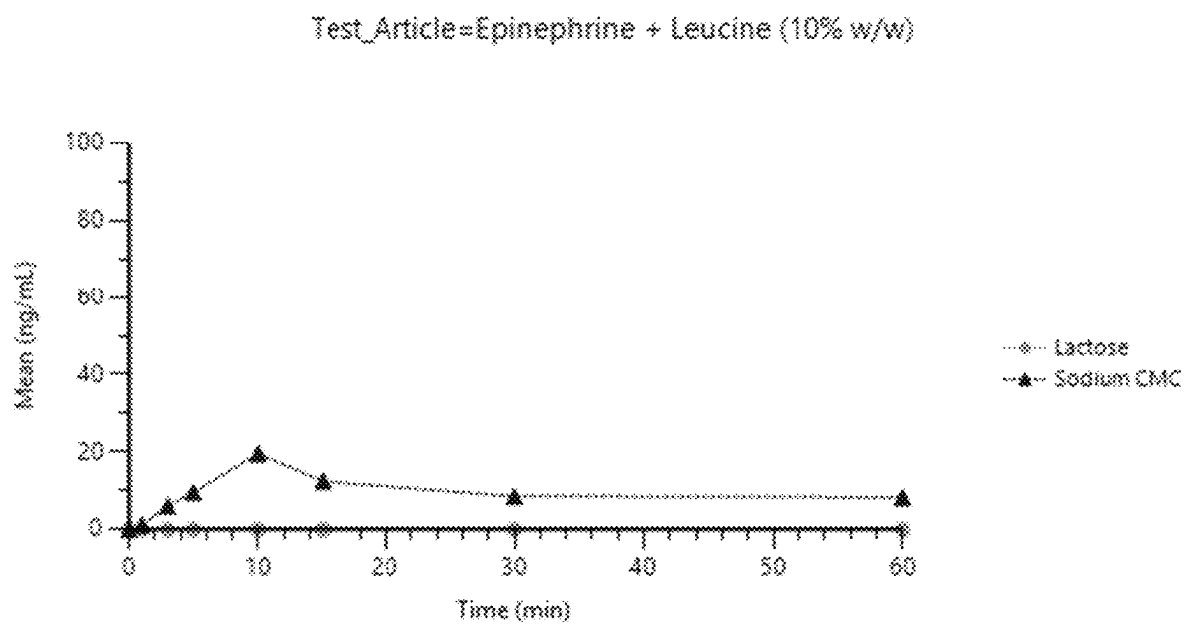


FIG. 73

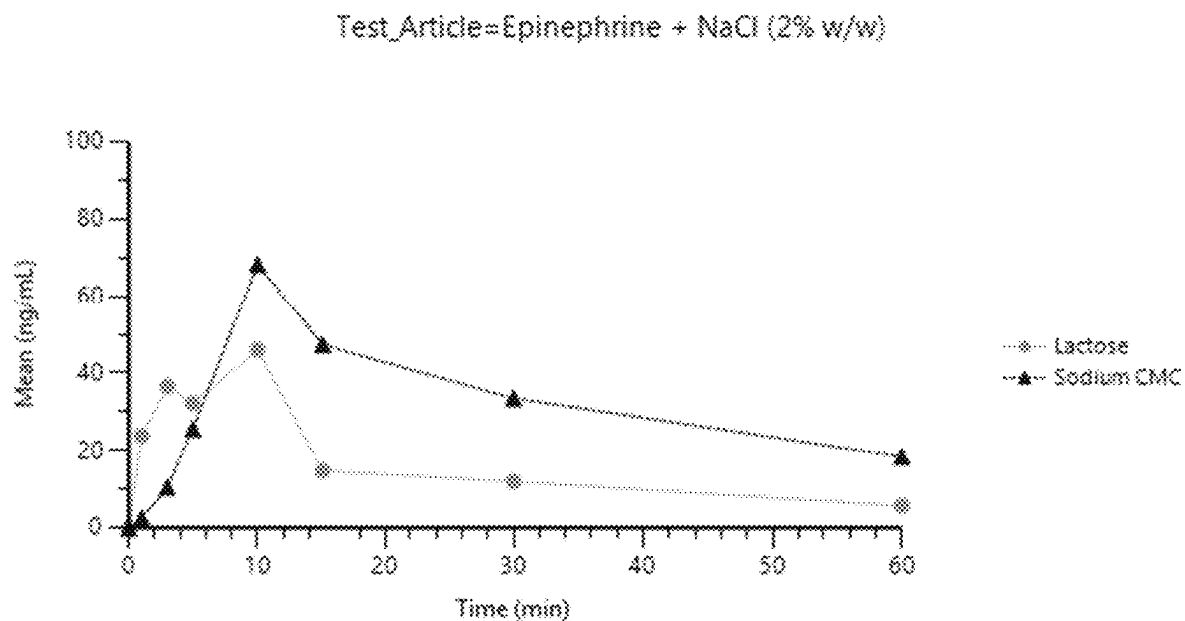


FIG. 74

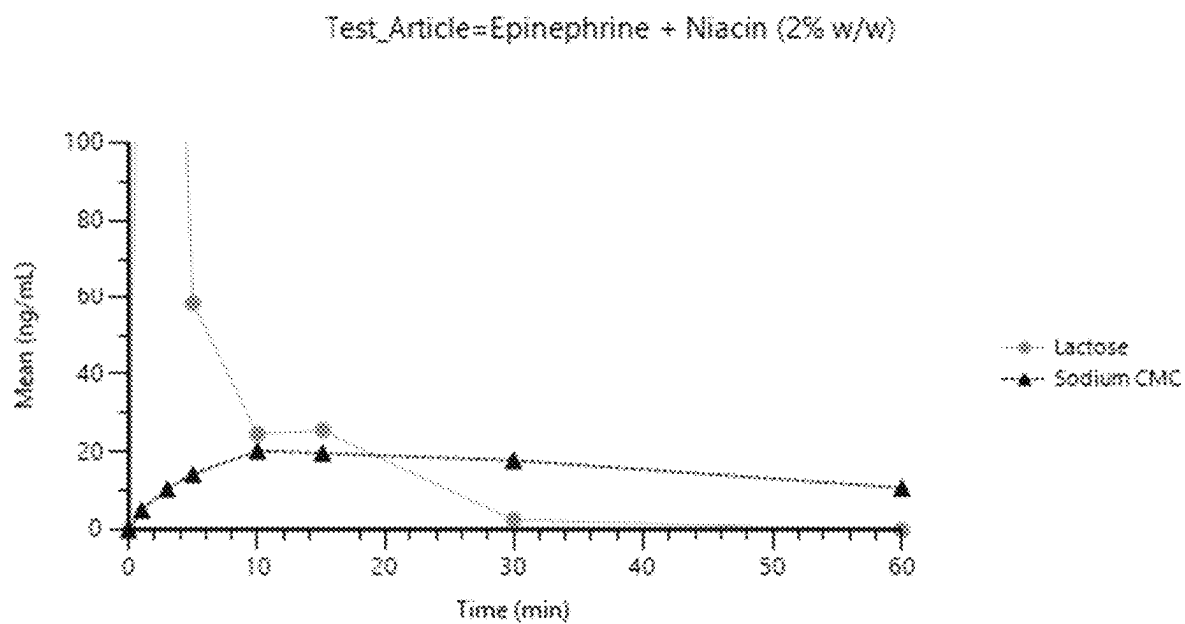


FIG. 75

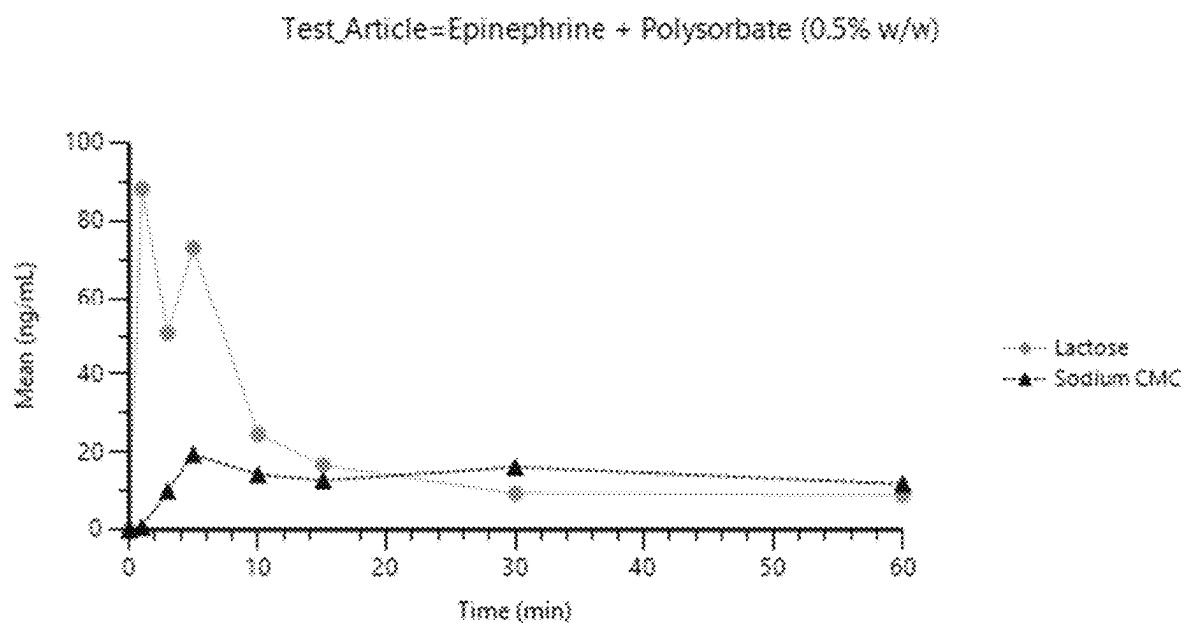


FIG. 76

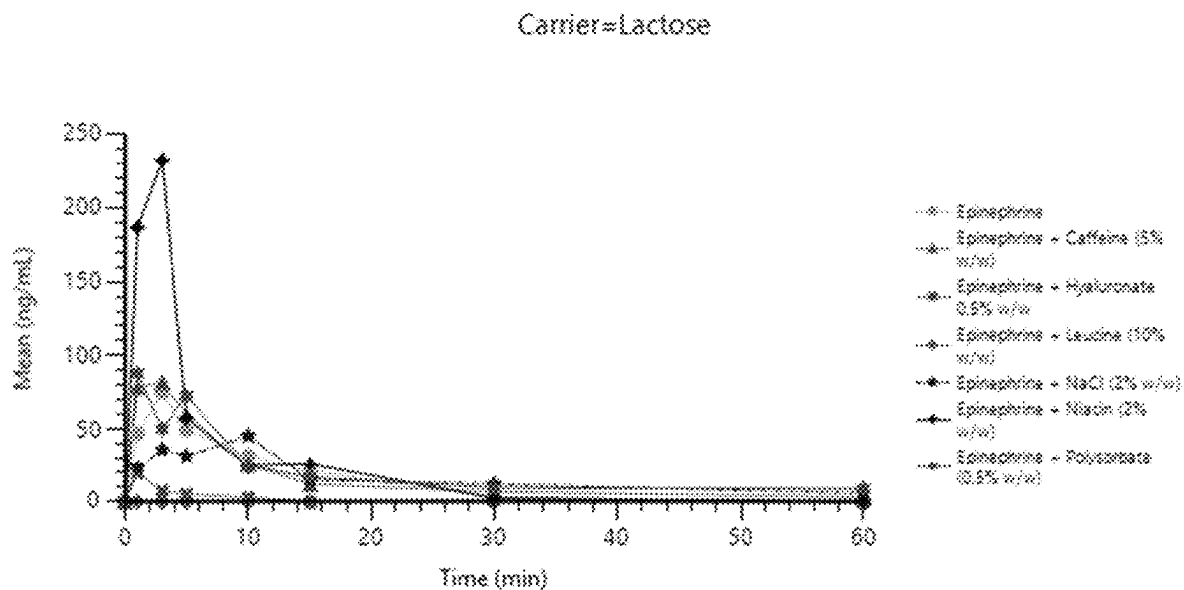


FIG. 77

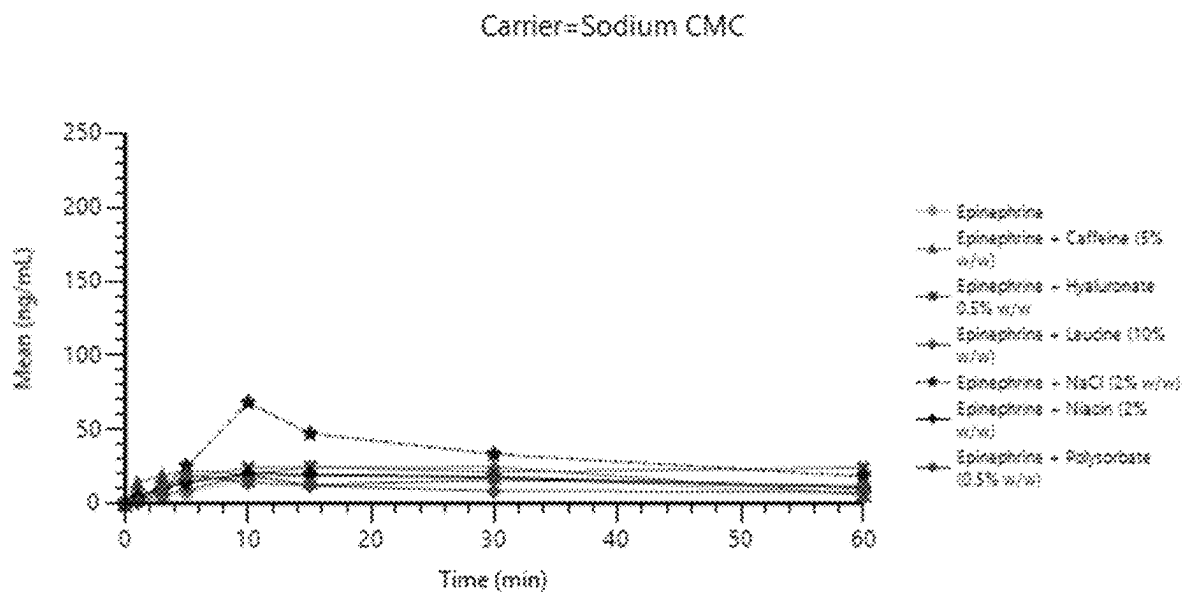


FIG. 78

Calibration Curve Data for Epinephrine in Rat K₂EDTA Plasma

Batch Number (Analysis Date)	Nominal Concentrations						Calibration Curve Parameters				
	5.00 (ng/mL)	10.0 (ng/mL)	25.0 (ng/mL)	100 (ng/mL)	300 (ng/mL)	900 (ng/mL)	1000 (ng/mL)	A ^a	B ^a	C ^a	RSQ ^b
1 (21 Oct 2021)	5.21 5.11	8.61 10.3	36.1 ^c 23.1	113 95.5	405 ^c 316	946 834	1040 961	4.36E-06	1.78E-02	6.29E-02	9.925E-01
2 (22 Oct 2021)	5.17 4.92	7.52 ^c 6.80 ^c	25.9 21.7	103 95.6	335 297	1010 628 ^c	857	5.77E-08	1.13E-03	2.06E-03	9.904E-01
3 (21 Oct 2021)	5.13 4.99	7.24 ^c 9.60	22.7 27.2	0.682 ^c 93.9	303 332	871 873	1040 992	1.83E-07	9.52E-03	-6.49E-03	9.955E-01
4 (22 Oct 2021)	3.22 ^d 5.19	9.24 10.5	25.4 21.4	89.3 106	320 337	930 920	910 999	1.74E-06	9.72E-03	4.42E-03	9.916E-01
5 (22 Oct 2021)	4.91 6.79 ^d	9.97 10.7	23.5 19.8 ^c	96.4 95.8	309 327	868 908	1030 967	6.32E-07	7.81E-03	-1.47E-02	9.971E-01
6 (23 Oct 2021)	5.45 4.70	9.42 7.52 ^c	17.1 ^c 18.9 ^c	100 94.8	317 316	881 841	1050 998	9.16E-07	8.00E-03	-6.91E-03	9.960E-01
7 (22 Oct 2021)	5.03 5.44	11.9 ^c 8.69	21.8 23.2	111 108	361 ^c 320	897 917	1050 903	7.70E-08	9.04E-04	-1.22E-03	9.900E-01
Mean	5.10	9.67	23.6	100	319	900	986	1.14E-06	7.84E-03	5.72E-03	9.933E-01
S.D.	0.215	0.756	1.96	7.35	12.3	47.1	60.2				
%CV	4.2	7.8	8.3	7.4	3.9	5.2	6.1				
%Bias	2.0	-3.3	-5.6	0.0	6.3	0.0	-1.4				
n	12	9	10	13	12	13	14	7	7	7	7

^a = Quadratic Regression: $y = Ax^2 + Bx + C$ where y is the peak area ratio of Epinephrine to Int. Std., x is the concentration of Epinephrine, and A, B, and C are regression constants. Regression weighted 1/x².

^a = Quadratic Regression: $y = Ax^2 + Bx + C$ where y is the peak area ratio of Epinephrine to Int. Std., x is the concentration of Epinephrine, and A, B, and C are regression constants. Regression weighted $1/x^2$.

^b = RSQ = R-Squared

^c = Value outside of acceptance criteria ($\pm 15\%$ theoretical). Value excluded from regression and summary statistics.

^d = Value outside of acceptance criteria ($\pm 20\%$ theoretical). Value excluded from regression and summary statistics.

FIG. 79

Quality Control Evaluation Data for Epinephrine in Rat K₂EDTA Plasma

Batch Number (Analysis Date)	Nominal Concentration			
	LMQC (15.0 ng/mL)	MQC (50.0 ng/mL)	HQC (800 ng/mL)	DQC ^a (800 ng/mL)
1 (21 Oct 2021)	14.1 15.1	43.3 45.8	886 728	
2 (22 Oct 2021)	14.8 9.83 ^b	37.7 ^b 42.9	884 751	
3 (21 Oct 2021)	12.0 ^b 13.4	44.5 102 ^b	815 798	
4 (22 Oct 2021)	13.6 11.6 ^b	46.8 103 ^b	793 793	
5 (22 Oct 2021)	14.7 17.2	48.8 44.1	791 817	
6 (23 Oct 2021)	16.5 12.5 ^b	50.6 53.7	764 680	
7 (22 Oct 2021)	14.1 8.88 ^b	49.9 49.0	695 763	799 649 ^b 838
Mean	13.5	54.4	783	762
S.D.	2.33	20.7	59.9	99.8
%CV	17.3	38.1	7.7	13.1
%Bias	-10.0	8.8	-2.1	-4.8
n	14	14	14	3

^a = Dilution QCs undiluted concentration 800 ng/mL; a 5-fold dilution with blank matrix was performed prior to extraction and analysis.

^b = Value outside of acceptance criteria ($\pm 15\%$ theoretical) but included in summary statistics.

FIG. 80

FIG. 81

		Time (min)											
		0.0	2.0	5.0	7.0	10.0	15.0	20.0	30.0	60.0	90.0	120.0	
Treatment	Animal	Epinephrine (pg/mL)											
	10%/40 mg w Tween	4428732	0.00	1090	642	550	541	499	490	673	725	861	1150
	TPO9		0.00	2800	2070	1270	971	879	1030	956	366	606	623
	TVO9		0.00	8680	4230	2680	2140	1810	1660	872	1100	1050	2040
	TWO9		0.00	6840	4690	3940	3550	2360	2490	1420	1400	1700	1590
	N		4	4	4	4	4	4	4	4	4	4	4
	Mean		0.00	4850	2910	2110	1800	1390	1420	980	898	1050	1350
	SD		0.00	3510	1890	1510	1350	851	860	316	449	467	606
	CV%			72.3	65.1	71.4	74.9	61.4	60.7	32.3	50.1	44.3	44.9
	Min		0.00	1090	642	550	541	499	490	673	366	606	623
	Median		0.00	4820	3150	1980	1560	1340	1350	914	913	956	1370
	Max		0.00	8680	4690	3940	3550	2360	2490	1420	1400	1700	2040

FIG. 82

Treatment		Animal	Time											
			0.0	3.0	5.0	7.0	10.0	15.0	20.0	30.0	60.0	90.0	120.0	
			Epinephrine											
10%/60mg		IQK9	0.00	4460	6340	6340	5630	5830	3850	2140	1600	1470	1600	
		TPO9	0.00	3850	3450	3450	3110	1550	1020	1110	439	606	794	
		TVO9	0.00	2320	1500	1500	1400	835	972	1130	1960	1020	1060	
N			3	3	3	3	3	3	3	3	3	3	3	
Mean			0.00	3540	3760	3760	3380	2740	1950	1460	1330	1030	1150	
SD			0.00	1100	2440	2440	2130	2700	1650	589	795	432	411	
CV%				31.1	64.7	64.7	63.0	98.6	84.6	40.3	59.6	41.9	35.7	
Min			0.00	2320	1500	1500	1400	835	972	1110	439	606	794	
Median			0.00	3850	3450	3450	3110	1550	1020	1130	1600	1020	1060	
Max			0.00	4460	6340	6340	5630	5830	3850	2140	1960	1470	1600	

FIG. 83

Treatment		Time											
		Epinephrine											
		0.0	2.0	5.0	7.0	10.0	15.0	20.0	30.0	60.0	90.0	120.0	
10%/60mg w Tween		Animal											
		TV09	0.00	7480	7680	8550	7180	4250	3220	3490	3860	5100	6550
		TWO9	0.00	5160	4650	3160	1760	1800	1940	1160	1110	1060	932
N		2	2	2	2	2	2	2	2	2	2	2	2
Mean		0.00	6320	6170	5860	4470	3030	2580	2330	2490	3080	3740	3740
SD		0.00	1640	2140	3810	3830	1730	905	1650	1940	2860	3970	3970
CV%			26.0	34.8	65.1	85.7	57.3	35.1	70.9	78.3	92.8	106.2	106.2
Min		0.00	5160	4650	3160	1760	1800	1940	1160	1110	1060	932	932
Median		0.00	6320	6170	5860	4470	3030	2580	2330	2490	3080	3740	3740
Max		0.00	7480	7680	8550	7180	4250	3220	3490	3860	5100	6550	6550

FIG. 84

Treatment		Time (min)											
		Epinephrine (pg/mL)											
		0.0	2.0	3.0	5.0	7.0	10.0	15.0	20.0	30.0	60.0	90.0	120.0
20%/30mg		Animal											
		4428732	0.00	5500	4640	3470	2240	1530	1390	1050	1010	743	1060
TPO9		0.00	9830		5730	5280	4240	2690	1920	1130	2110	2940	1160
TVO9		0.00	12200		6900	7400	4260	3930	2640	2050	1410	2150	1920
TWO9		0.00		5550	6850	5640	5290	3120	2770	1390	1330	1730	1690
N		4	2	2	4	4	4	4	4	4	4	4	4
Mean		0.00	11000	6030	6030	5450	4010	2820	2180	1410	1470	1890	1460
SD		0.00	1680	672	1070	1610	1280	1000	646	454	463	915	414
CV%			15.2	11.1	17.8	29.6	31.8	35.5	29.6	32.3	31.6	48.4	28.4
Min		0.00	9830	5550	4640	3470	2240	1530	1390	1050	1010	743	1060
Median		0.00	11000	6030	6290	5460	4250	2910	2280	1260	1370	1940	1430
Max		0.00	12200	6500	6900	7400	5290	3930	2770	2050	2110	2940	1920

FIG. 85

		Time (min)										
		0.0	2.0	5.0	7.0	10.0	15.0	20.0	30.0	60.0	90.0	120.0
Treatment	Animal	Epinephrine (pg/mL)										
20%/30mg w Tween	IQK9	0.00	2060	2310	2450	2190	1750	1670	2420	4080	2890	2840
	TPO9	0.00	535	900	1020	927	772	853	809	1260	1020	1160
	TWO9	0.00	3170	1950	1160	871	744	686	751	779	716	717
	N	3	3	3	3	3	3	3	3	3	3	3
	Mean	0.00	1920	1720	1540	1330	1090	1070	1330	2040	1540	1570
	SD	0.00	1320	733	788	746	573	527	947	1780	1180	1120
	CV%		68.8	42.6	51.1	56.1	52.6	49.2	71.4	87.4	76.3	71.2
	Min	0.00	535	900	1020	871	744	686	751	779	716	717
	Median	0.00	2060	1950	1160	927	772	853	809	1260	1020	1160
	Max	0.00	3170	2310	2450	2190	1750	1670	2420	4080	2890	2840

FIG. 86

		Time (min)										
		0.0	3.0	5.0	7.0	10.0	15.0	20.0	30.0	60.0	90.0	120.0
Treatment	Animal	Epinephrine (pg/mL)										
Epi-Pen IM Injection	4428732	0.00	272	401	355	305	319	596	316	372	840	862
	IQK9	0.00	2590	1810	1200	762	473	742	739	1530	1800	1080
	TPO9	0.00	417	724	541	564	1080	876	1200	1320	NS	1090
	TV09	0.00	838	816	834	1290	995	833	1010	2050	1700	1360
	N	4	4	4	4	4	4	4	4	4	3	4
	Mean	0.00	1030	938	733	730	717	762	816	1320	1450	1100
	SD	0.00	1070	608	369	417	377	124	383	701	528	204
	CV%		103.7	64.8	50.3	57.2	52.6	16.3	47.0	53.2	36.5	18.6
	Min	0.00	272	401	355	305	319	596	316	372	840	862
	Median	0.00	628	770	688	663	734	788	875	1430	1700	1090
	Max	0.00	2590	1810	1200	1290	1080	876	1200	2050	1800	1360

FIG. 87

Treatment	Dose (mg)	Animal	Cmax (pg/mL)	Tmax (min)	AUClast (min*pg/mL)
10%/40 mg w Tween	4.0	4428732	1150	120.00	92300
		TPO9	2800	2.00	87400
		TV09	8680	2.00	180000
		TWO9	6840	2.00	228000
		N	4	4	4
		Mean	4870	31.5	147000
		CV%	71.7	187.3	46.8
		Min	1150	2.00	87400
		Median	4820	2.00	136000
		Max	8680	120	228000
		Geometric Mean	3720	5.57	135000
		Geometric CV%	115.67	806.74	50.85

FIG. 88

Treatment	Dose (mg)	Animal	Cmax (pg/mL)	Tmax (min)	AUClast (min*pg/mL)
10%/60mg	6.0	IQK9	6340	5.00	278000
		TPO9	3850	3.00	116000
		TV09	2320	3.00	156000
		N	3	3	3
		Mean	4170	3.67	183000
		CV%	48.7	31.5	45.8
		Min	2320	3.00	116000
		Median	3850	3.00	156000
		Max	6340	5.00	278000
		Geometric Mean	3840	3.56	171000
		Geometric CV%	53.61	30.15	46.49

FIG. 89

Treatment	Dose (mg)	Animal	Cmax (pg/mL)	Tmax (min)	AUClast (min*pg/mL)
10%/60mg w Tween	6.0	TV09	8550	7.00	569000
		TWO9	5160	2.00	165000
N					
		Mean	2	2	2
		CV%	6860	4.50	367000
		Min	35.0	78.6	78.0
		Median	5160	2.00	165000
		Max	6860	4.50	367000
		Geometric Mean	8550	7.00	569000
		Geometric CV%	6640	3.74	306000
			36.88	109.17	107.71

FIG. 90

Treatment	Dose (mg)	Animal	Cmax (pg/mL)	Tmax (min)	AUClast (min*pg/mL)	
20%/30mg	6.0	4428732	6500	3.00	150000	
		TPO9	9830	2.00	283000	
		TVO9	12200	2.00	297000	
		TWO9	6850	5.00	243000	
		N				
		4	4	4		
		Mean	8850	3.00	243000	
		CV%	30.4	47.1	27.2	
		Min	6500	2.00	150000	
		Median	8340	2.50	263000	
		Max	12200	5.00	297000	
		Geometric Mean	8550	2.78	235000	
		Geometric CV%	30.70	45.62	31.97	

FIG. 91

Treatment	Dose (mg)	Animal	Cmax (pg/mL)	Tmax (min)	AUClast (min*pg/mL)
20%/30mg w Tween	6.0	IQK9	4080	60.00	346000
		TPO9	1260	60.00	122000
		TWO9	3170	2.00	98400
		N	3	3	3
		Mean	2840	40.7	189000
		CV%	50.7	82.3	72.4
		Min	1260	2.00	98400
		Median	3170	60.0	122000
		Max	4080	60.0	346000
		Geometric Mean	2540	19.3	161000
		Geometric CV%	68.27	680.28	75.66

FIG. 92

Treatment	Dose (mg)	Animal	Cmax (pg/mL)	Tmax (min)	AUClast (min*pg/mL)
Epi-Pen IM Injection	0.3	4428732	862	120.00	65100
		IQK9	2590	3.00	154000
		TPO9	1320	60.00	134000
		TV09	2050	60.00	175000
		N	4	4	4
		Mean	1710	60.8	132000
		CV%	44.9	78.6	36.1
		Min	862	3.00	65100
		Median	1690	60.0	144000
		Max	2590	120	175000
		Geometric Mean	1570	33.7	124000
		Geometric CV%	51.73	374.52	46.43

FIG. 93

		Animal					
		4428732	IQK9	TPO9	TV09	TV09	TWO9
Treatment	Dose (mg)	AUClast (min*pg/mL)					
10%/40 mg w Tween	4	92300	ND	87400	ND	180000	228000
10%/60mg	6	ND	278000	116000	156000	ND	ND
10%/60mg w Tween	6	ND	ND	ND	569000	ND	165000
20%/30mg	6	150000	ND	283000	ND	297000	243000
20%/30mg w Tween	6	ND	346000	122000	ND	ND	98400
Epi-Pen IM Injection	0.3	65100	154000	134000	175000	ND	ND

ND =Not Done

Treatment	Dose (mg)	Animal				
		4428732	IQK9	TPO9	TV09	TWO9
Relative Bioavailability (AUClast) (Treatment/EpiPen)*(DoseEpiPen/DoseTreatment)						
10%/40 mg w Tween	4	10.6%	ND	4.9%	ND	ND
10%/60mg	6	ND	9.0%	4.3%	4.5%	ND
10%/60mg w Tween	6	ND	ND	ND	16.3%	ND
20%/30mg	6	11.5%	ND	10.6%	ND	ND
20%/30mg w Tween	6	ND	11.2%	4.6%	ND	ND
Epi-Pen IM Injection	0.3	Reference	Reference	Reference	Reference	ND

ND =Not Done

FIG. 94

Treatment	Dose (mg)	Animal					
		4428732	IQK9	TPO9	TV09	TV09	TWO9
		Comparative Bioavailability (AUClast) (Treatment/EpiPen)					
10%/40 mg w Tween	4	142%	ND	65.2%	ND	ND	ND
10%/60mg	6	ND	181%	86.6%	89.1%	ND	ND
10%/60mg w Tween	6	ND	ND	ND	325%	ND	ND
20%/30mg	6	230%	ND	211%	ND	ND	ND
20%/30mg w Tween	6	ND	225%	91.0%	ND	ND	ND
Epi-Pen IM Injection	0.3	Reference	Reference	Reference	Reference	ND	ND

ND =Not Done

FIG. 95

Treatment = 10% epinephrine/40 mg with Tween, Dose Level = 4 mg

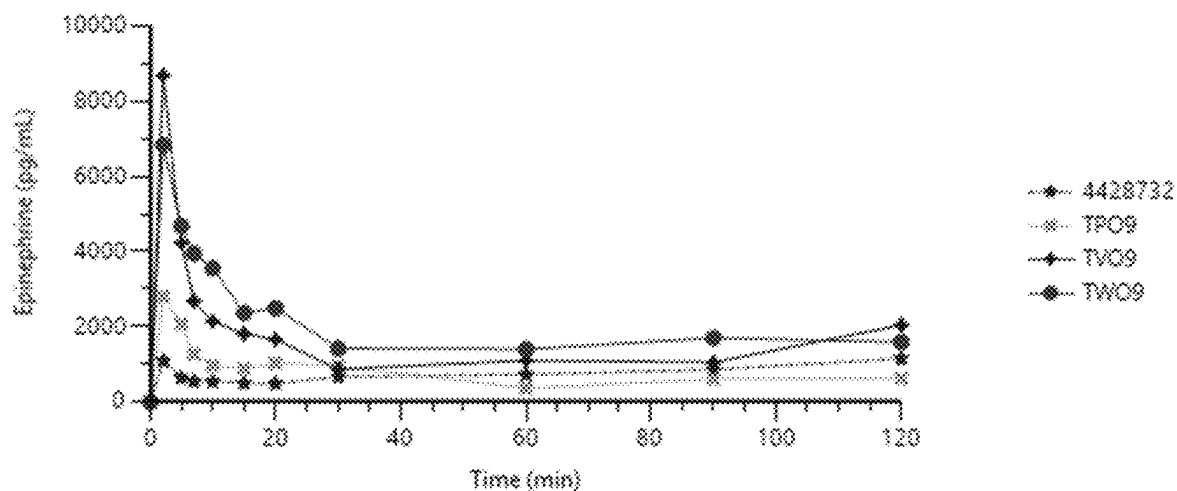


FIG. 96

Treatment = 10% epinephrine/60 mg, Dose Level = 6 mg

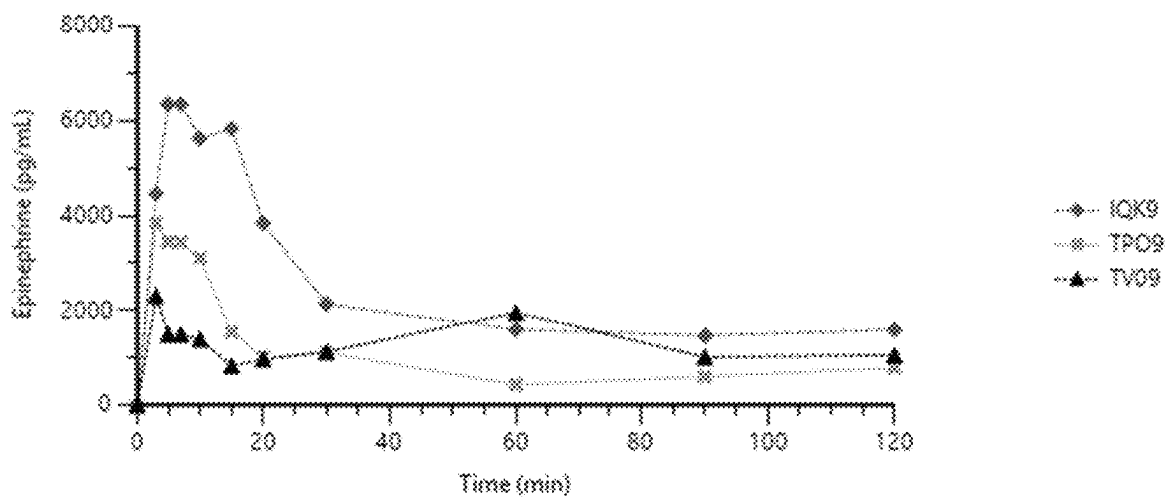


FIG. 97

Treatment = 10% epinephrine/60 mg with Tween, Dose Level = 6 mg

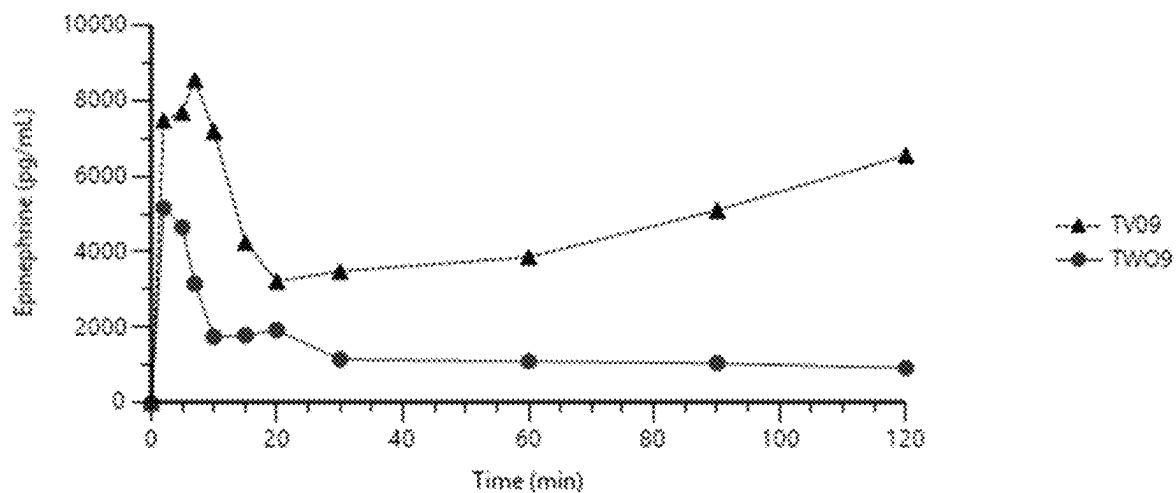


FIG. 98

Treatment = 20% epinephrine/30 mg, Dose Level = 6 mg

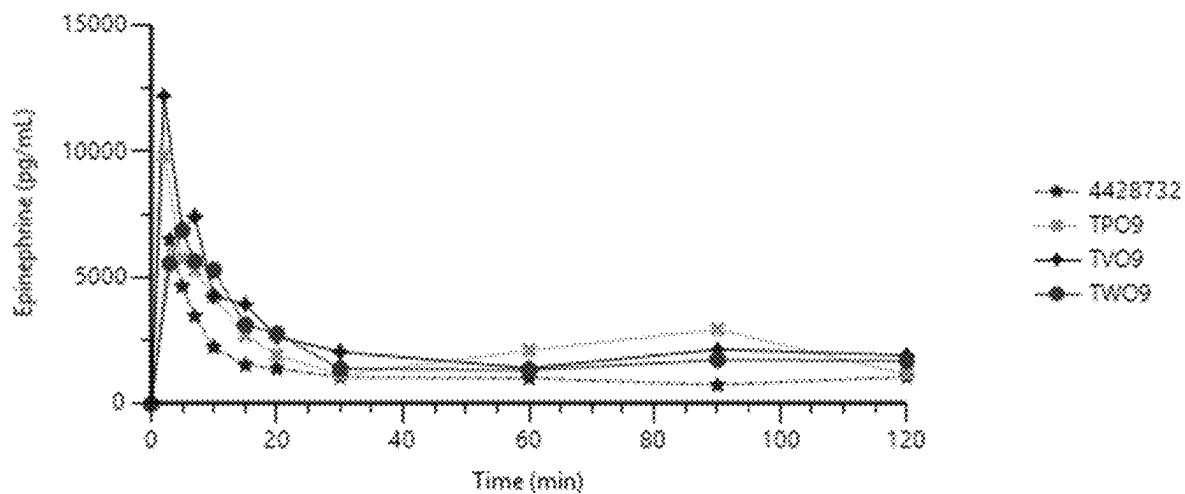


FIG. 99

Treatment = 20% epinephrine/30 mg with Tween, Dose Level = 6 mg

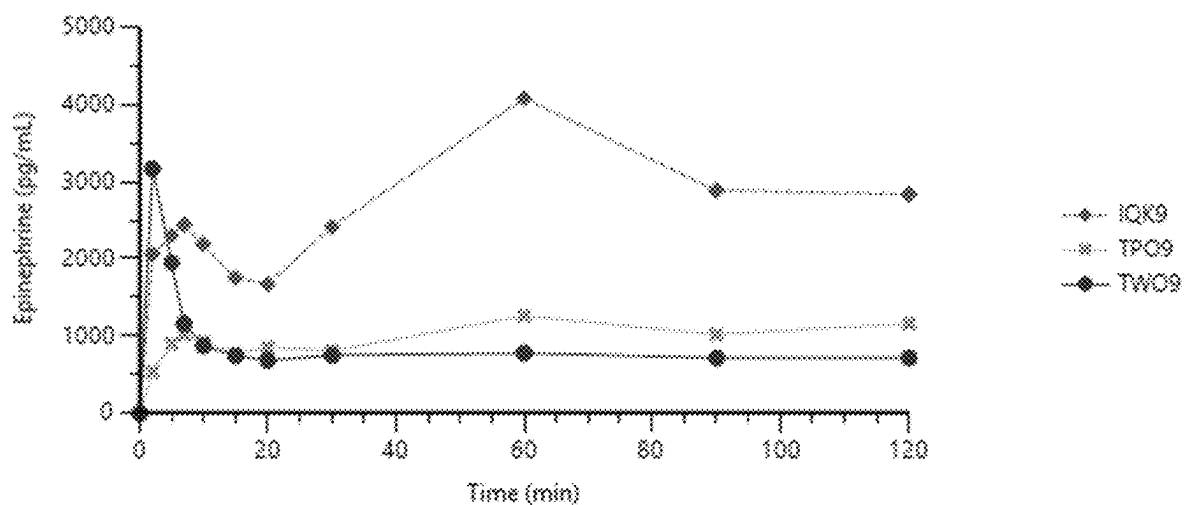


FIG. 100

Treatment = EPIPEN IM Injection, Dose Level = 0.3 mg

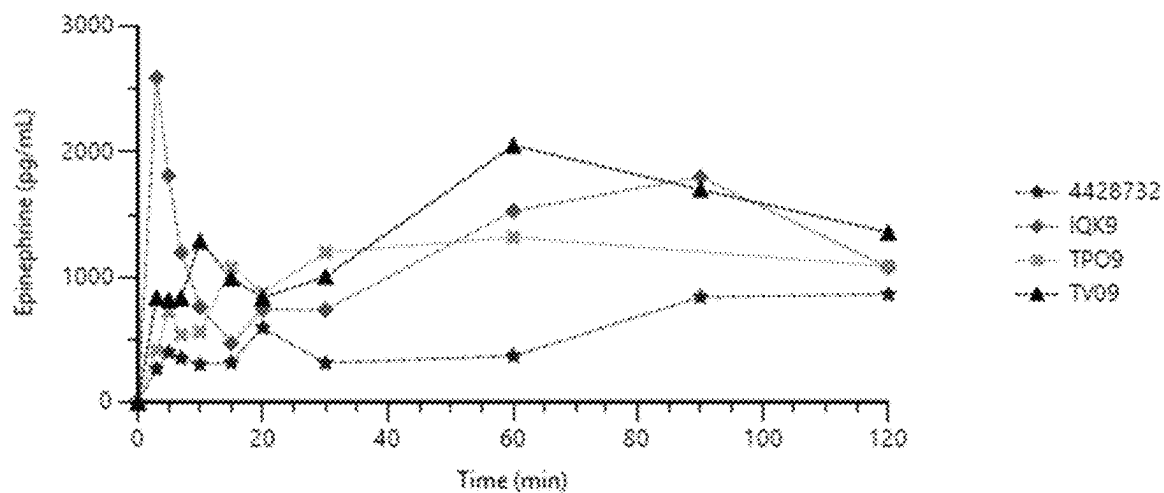


FIG. 101

Treatment = 10% epinephrine/40 mg with Tween, Dose Level = 4 mg

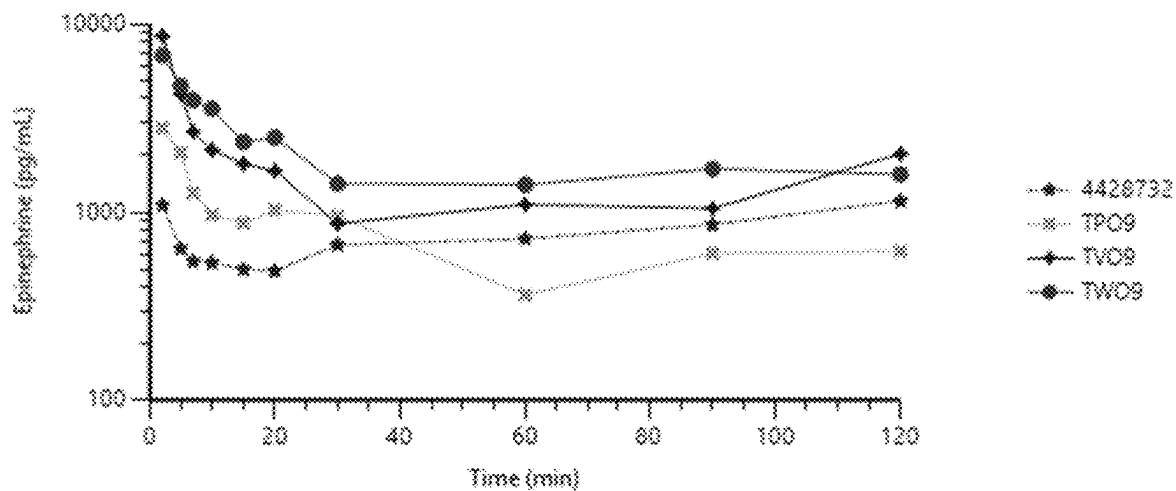


FIG. 102

Treatment = 10% epinephrine/60 mg, Dose Level = 6 mg

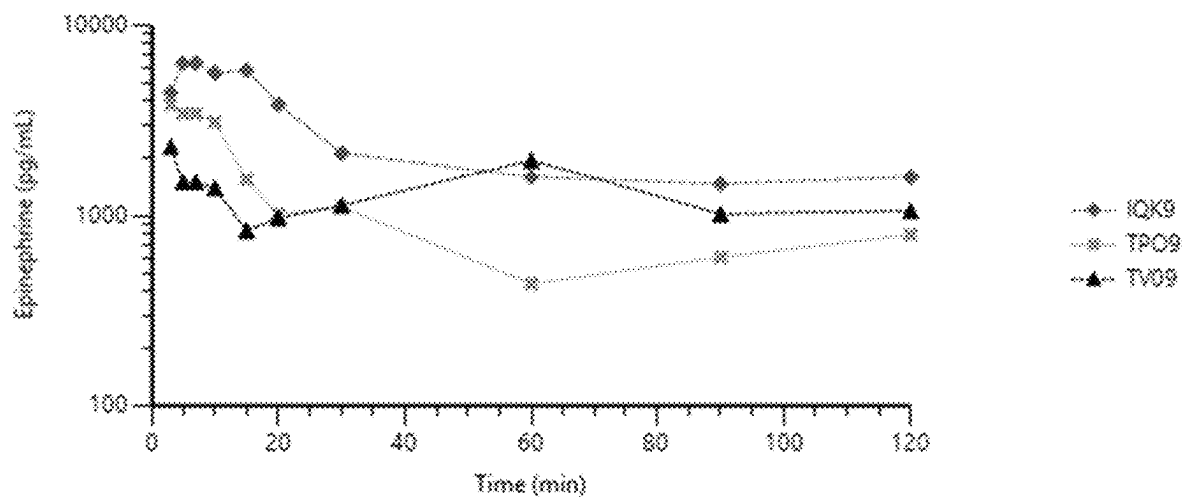


FIG. 103

Treatment = 10% epinephrine/60 mg with Tween, Dose Level = 6 mg

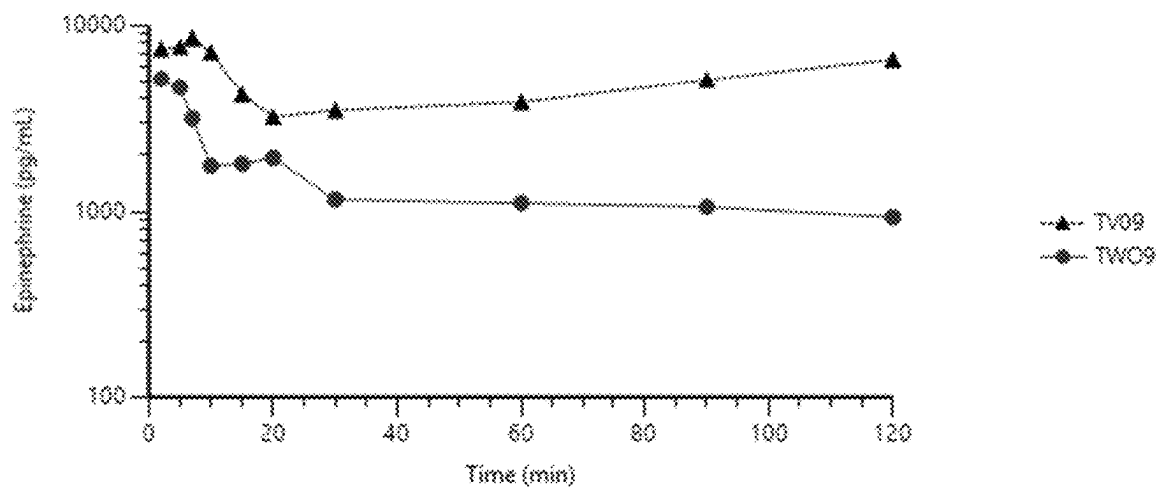


FIG. 104

Treatment = 20% epinephrine/30 mg, Dose Level = 6 mg

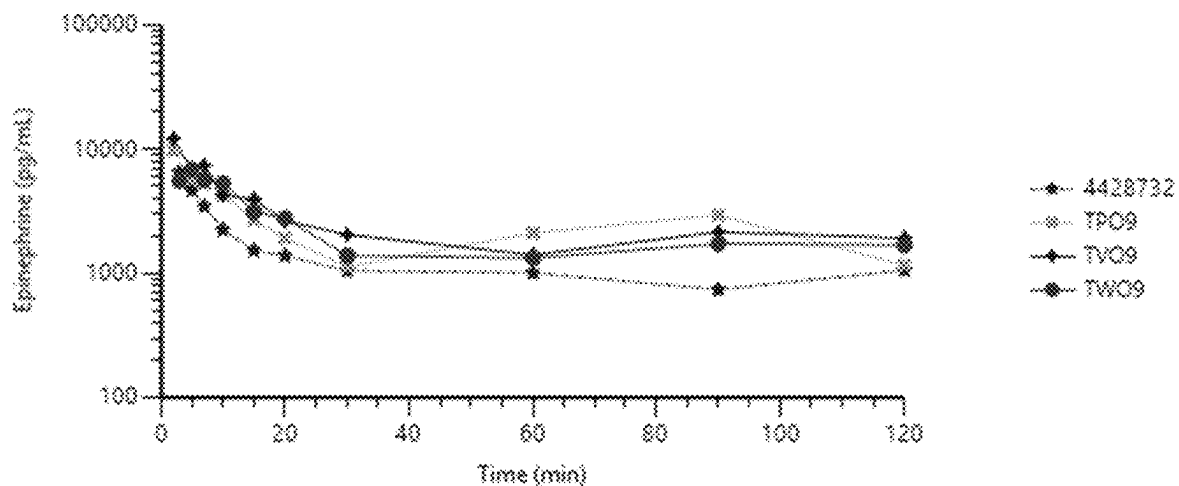


FIG. 105

Treatment = 20% epinephrine/30 mg with Tween, Dose Level = 6 mg

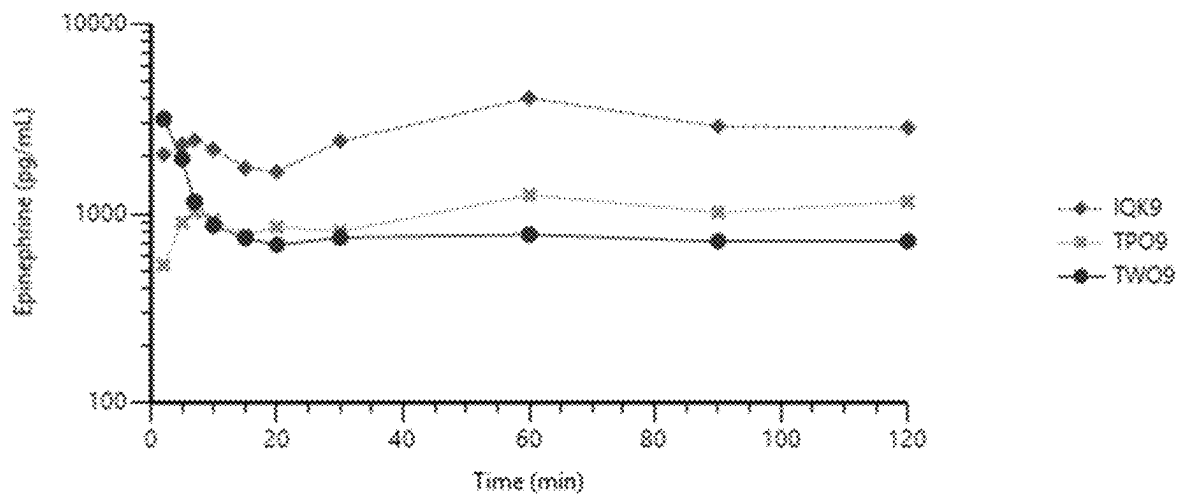


FIG. 106

Treatment = EPIPEN IM Injection, Dose Level = 0.3 mg

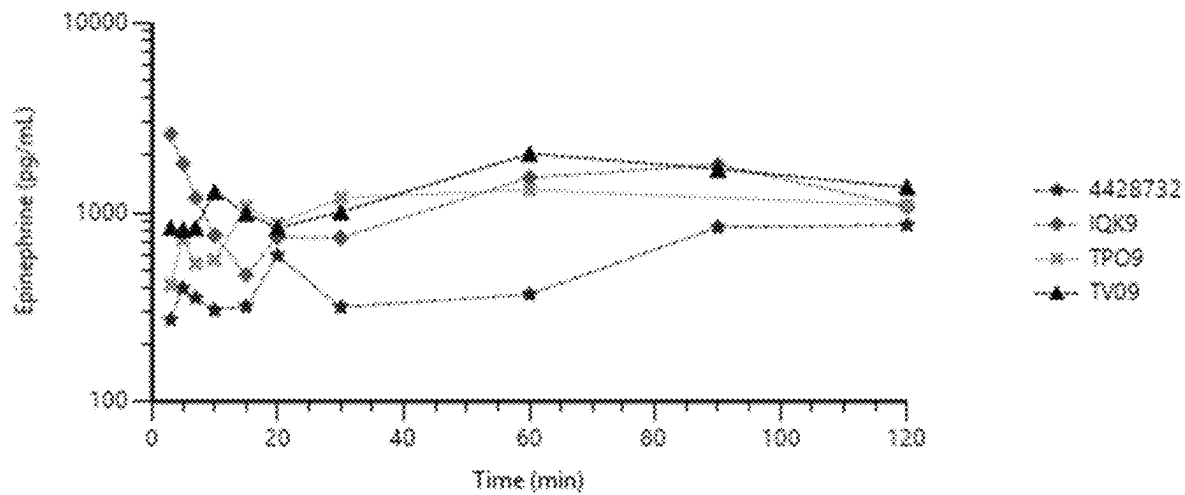


FIG. 107

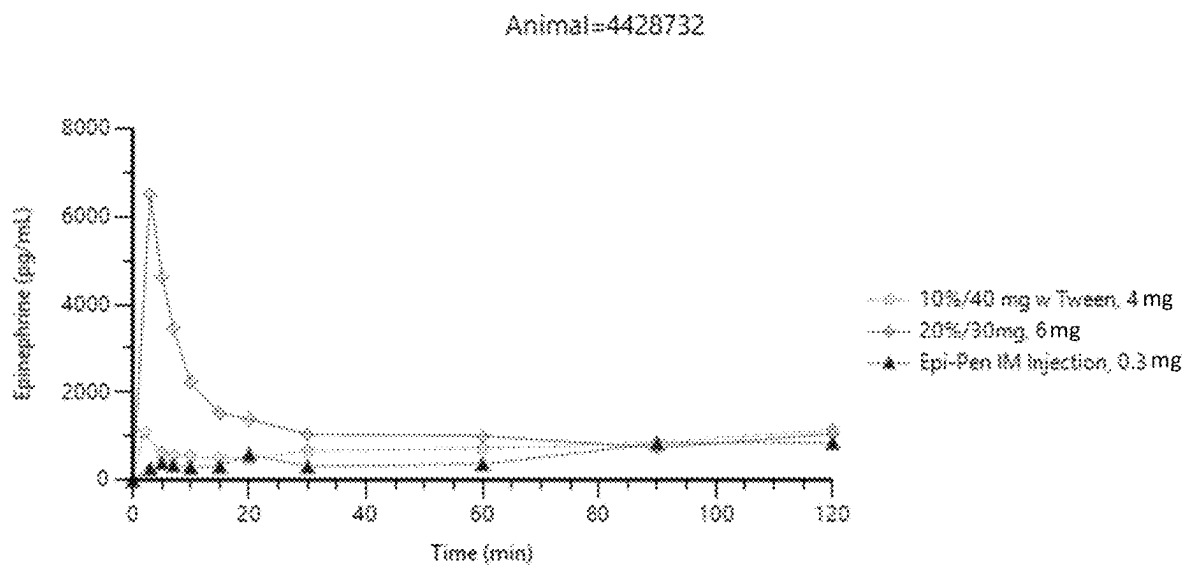


FIG. 108

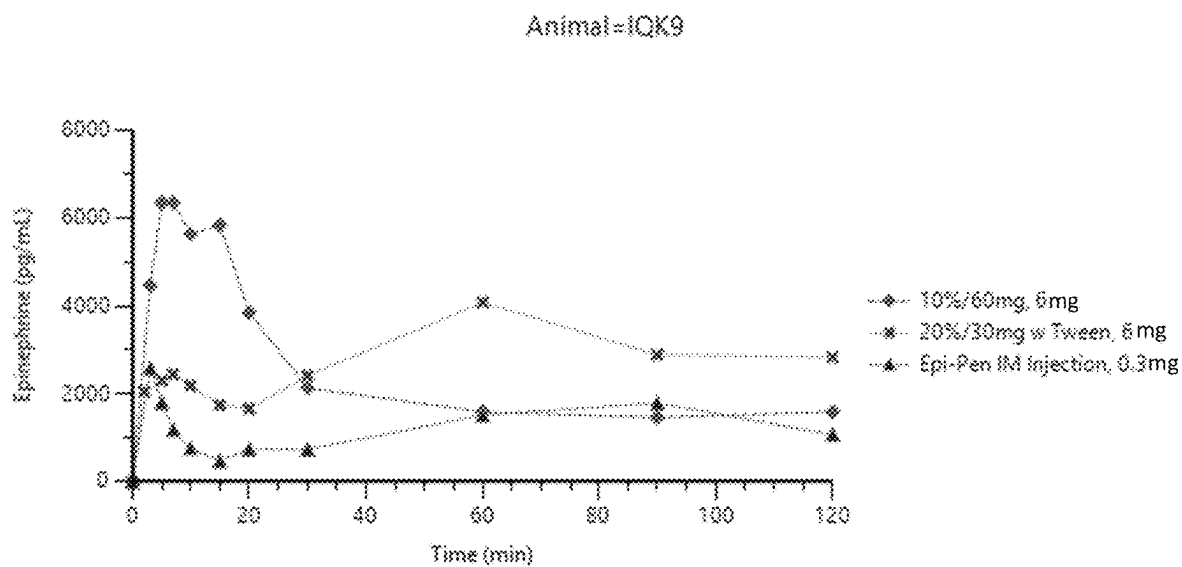


FIG. 109

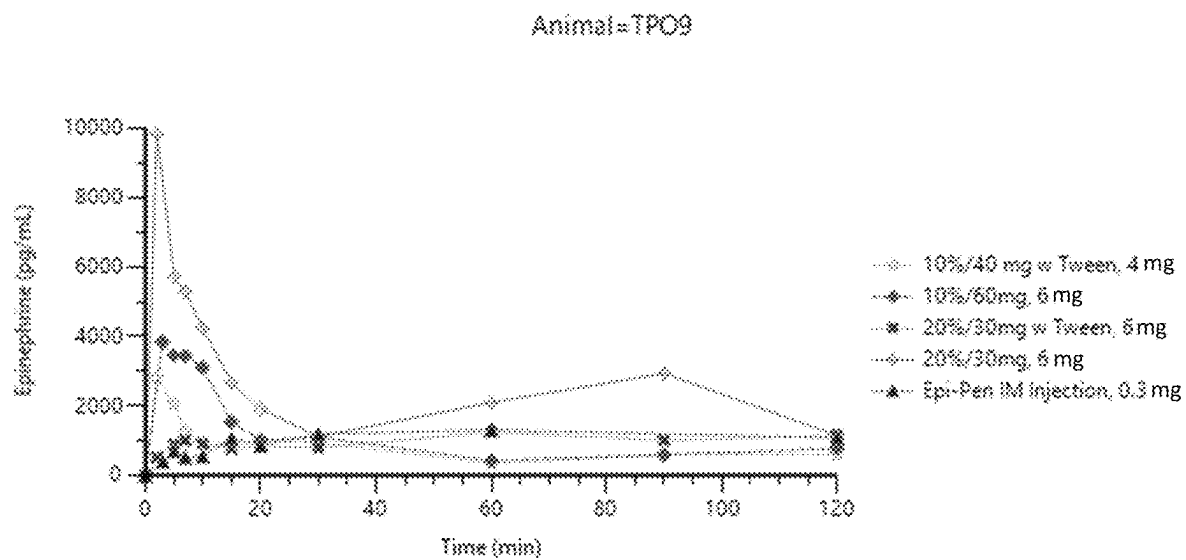


FIG. 110

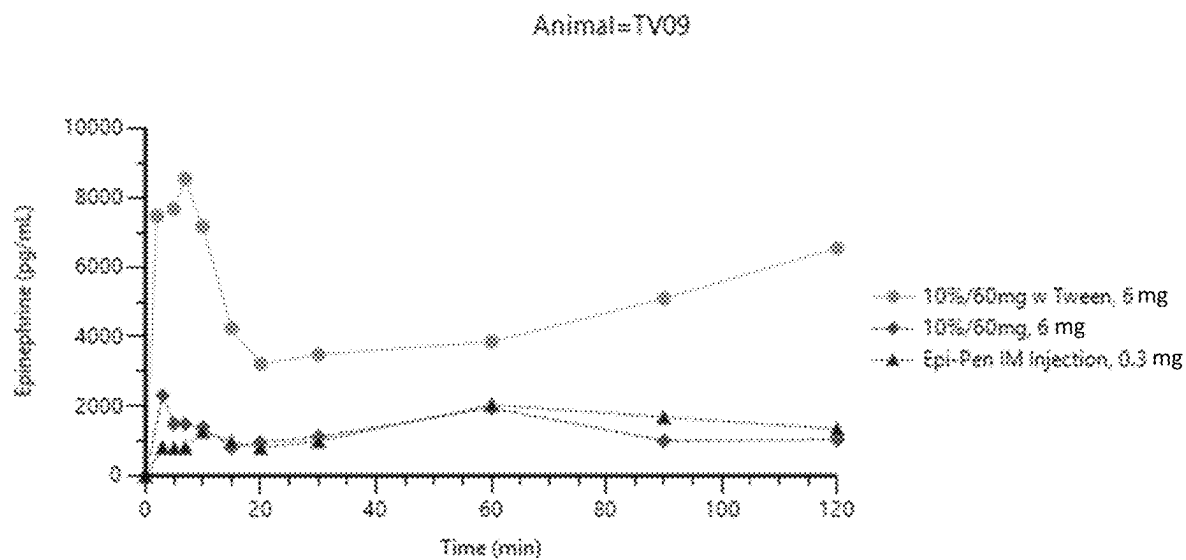


FIG. 111

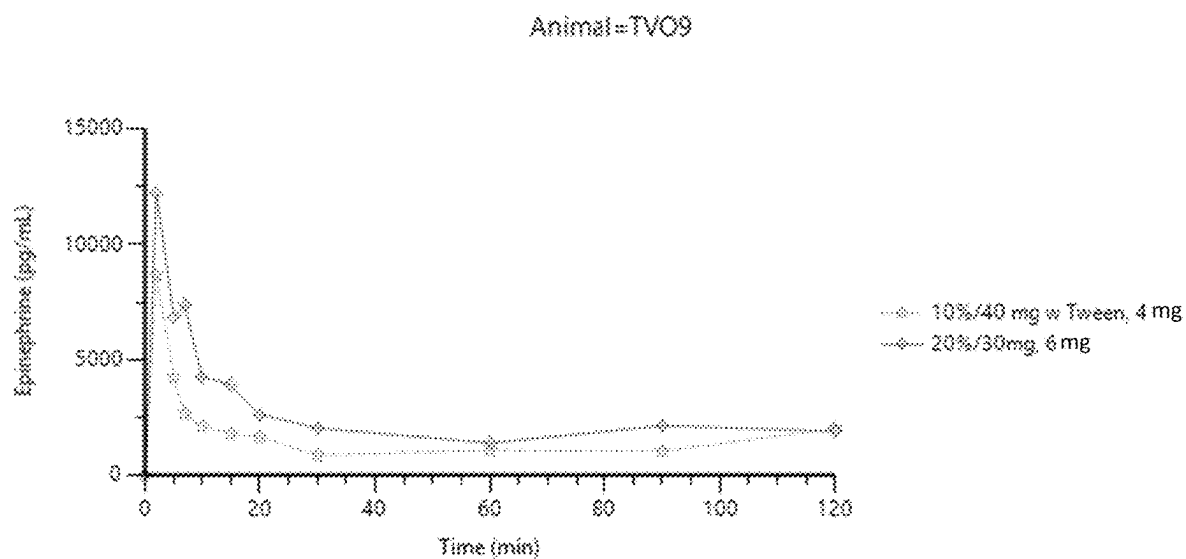


FIG. 112

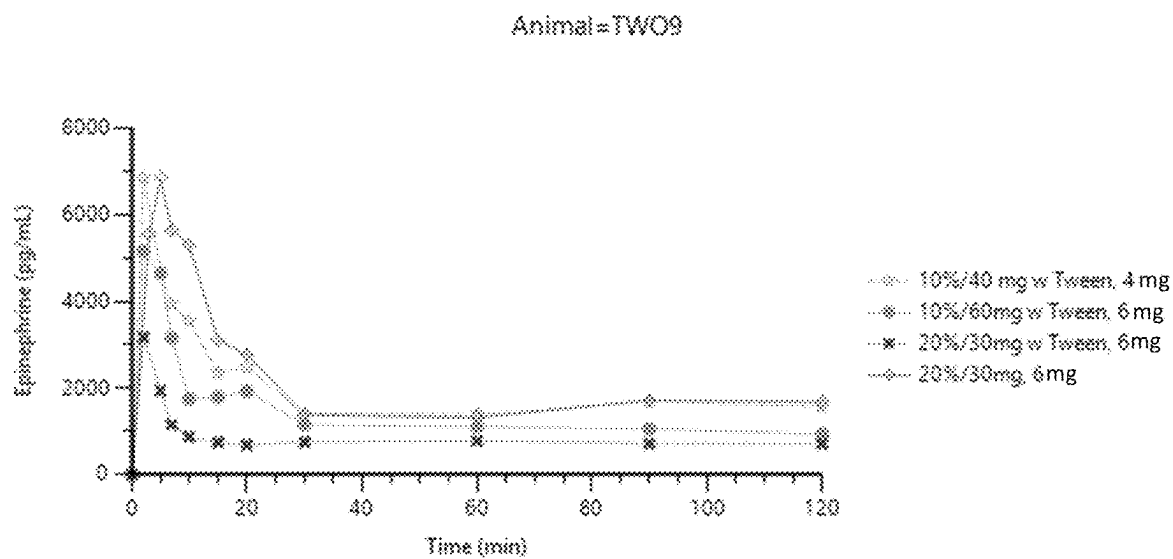


FIG. 113

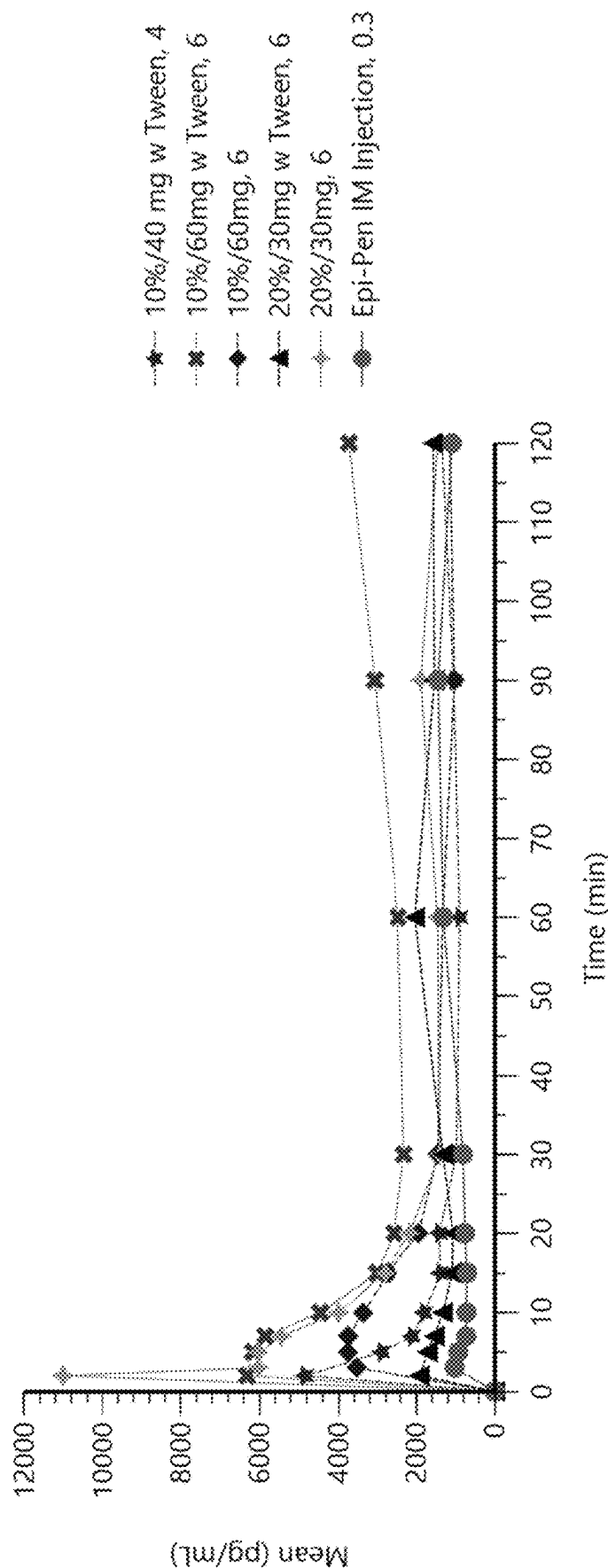


FIG. 114

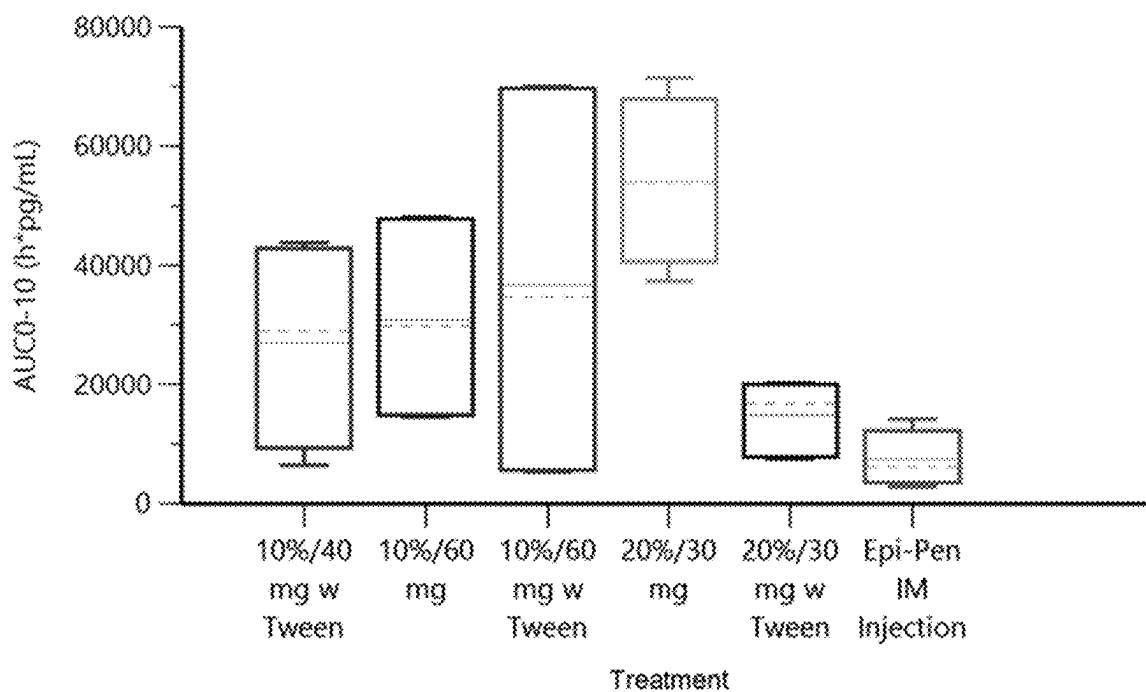


FIG. 115

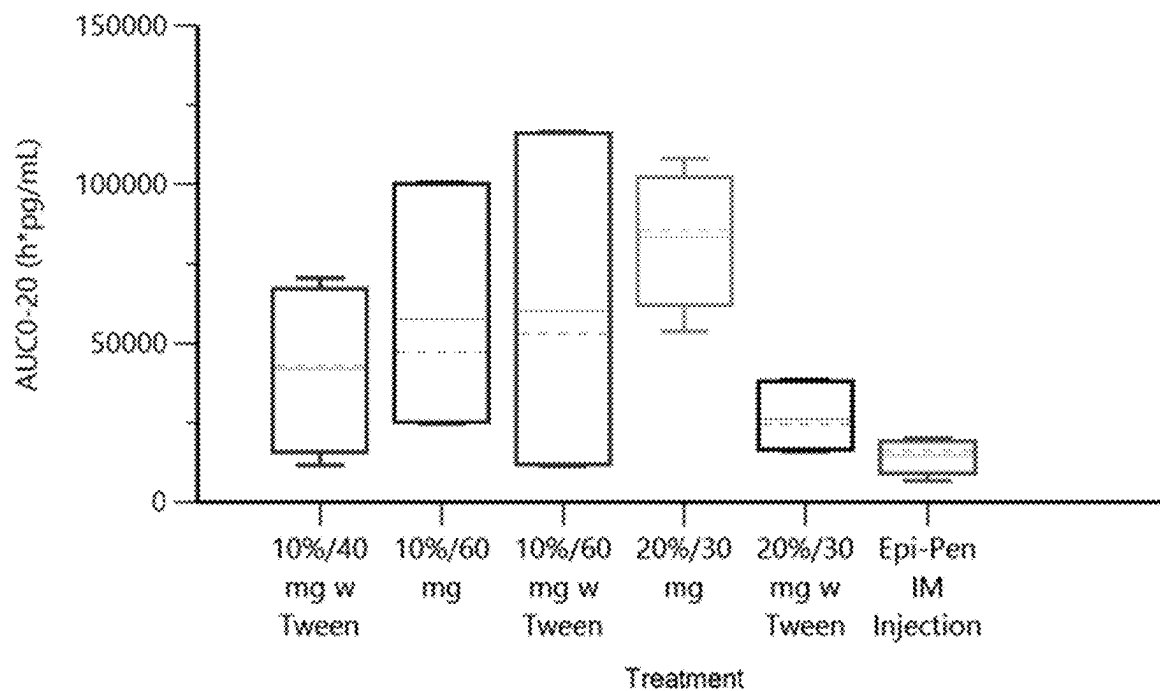


FIG. 116

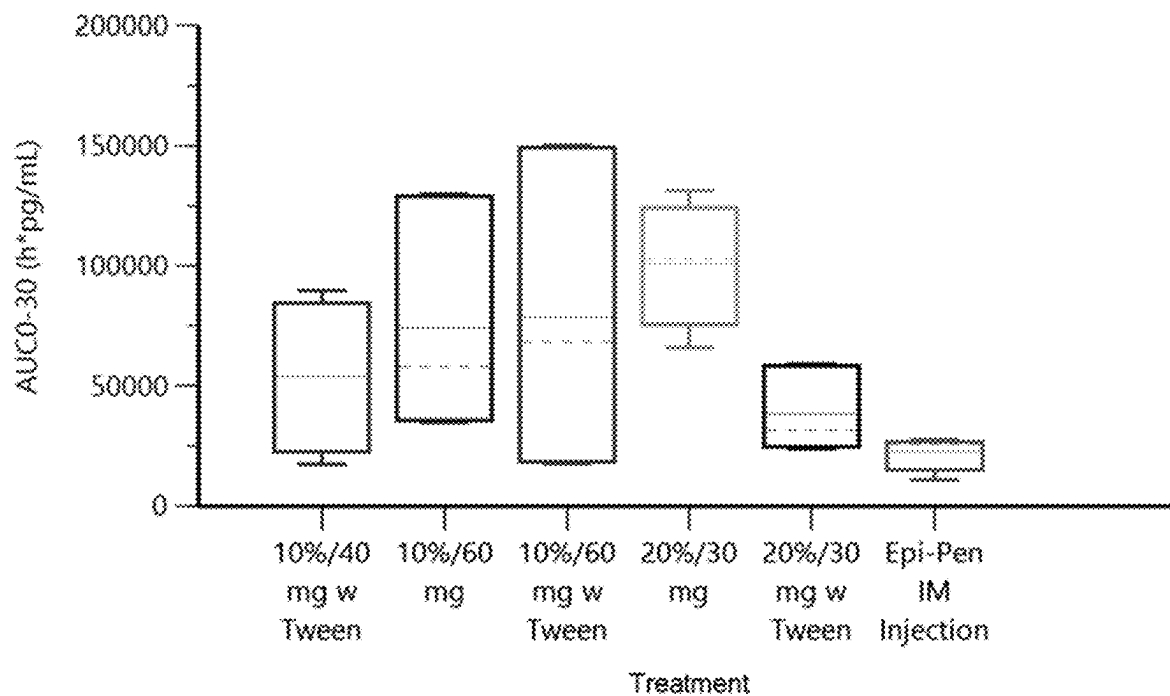


FIG. 117

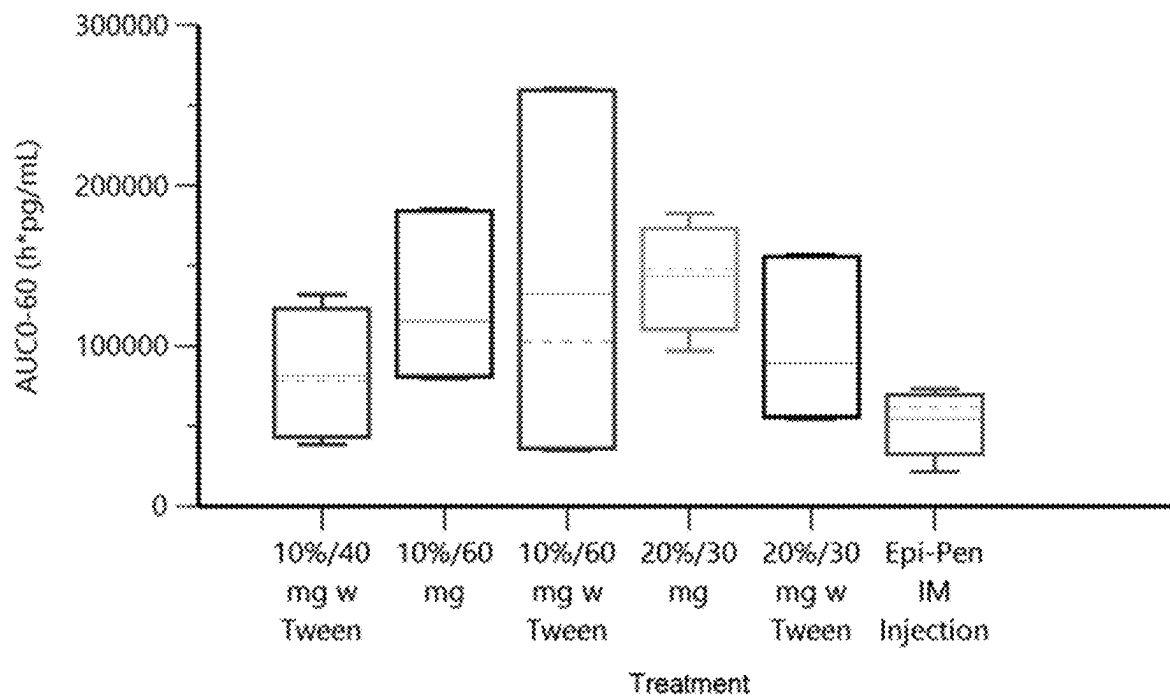


FIG. 118

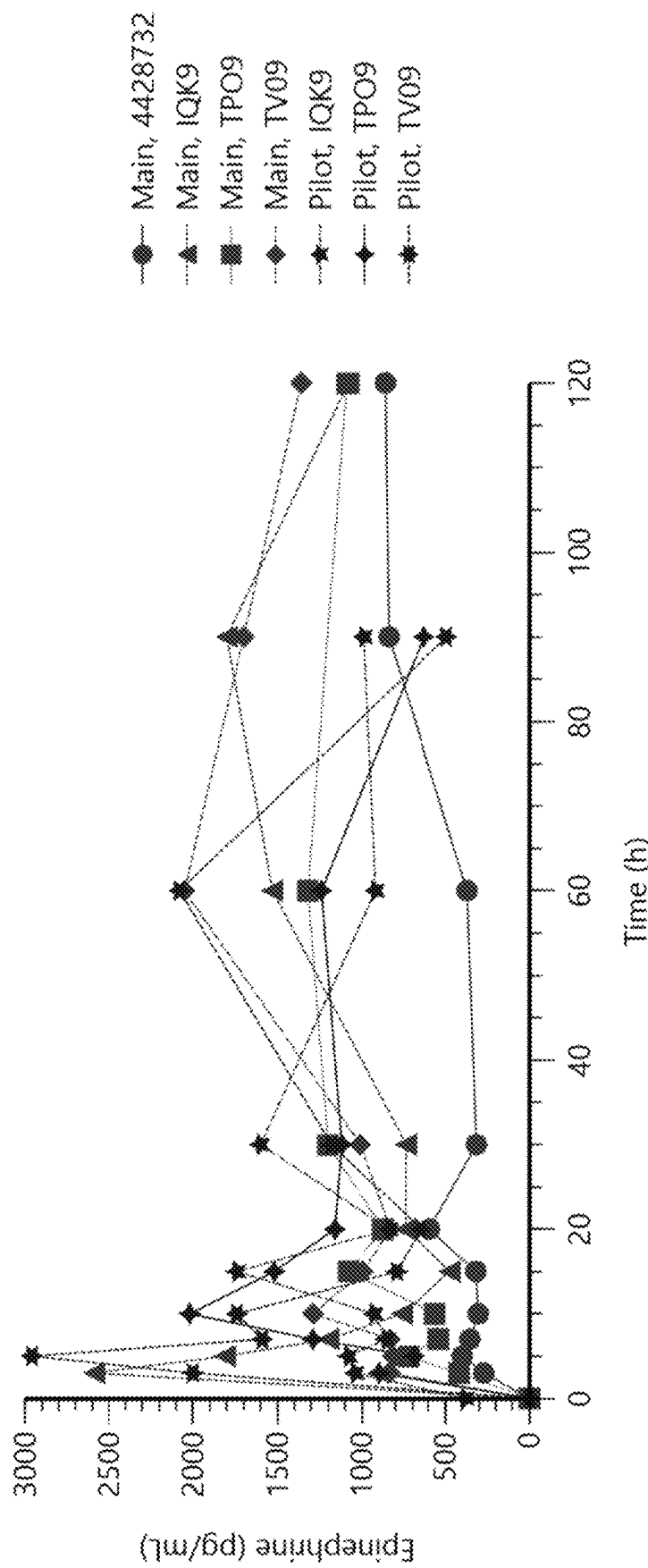


FIG. 119

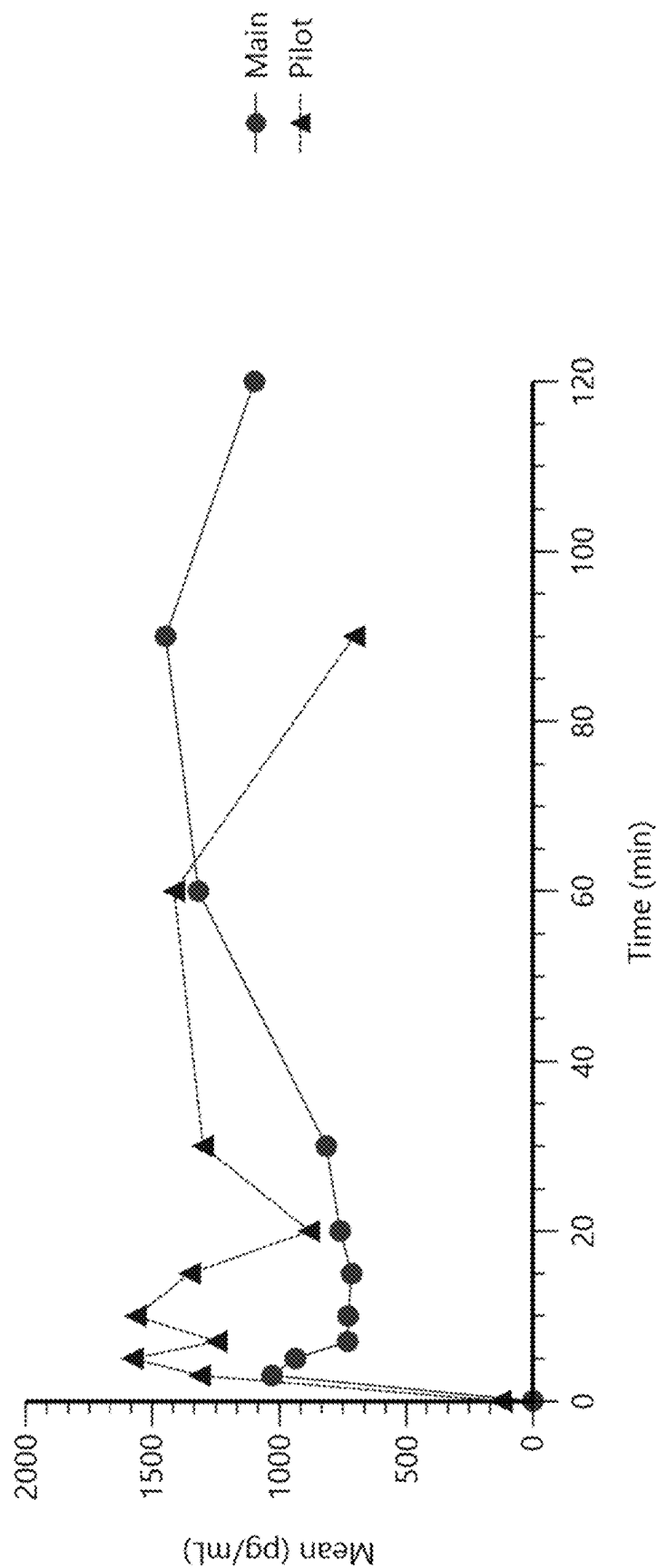


FIG. 120

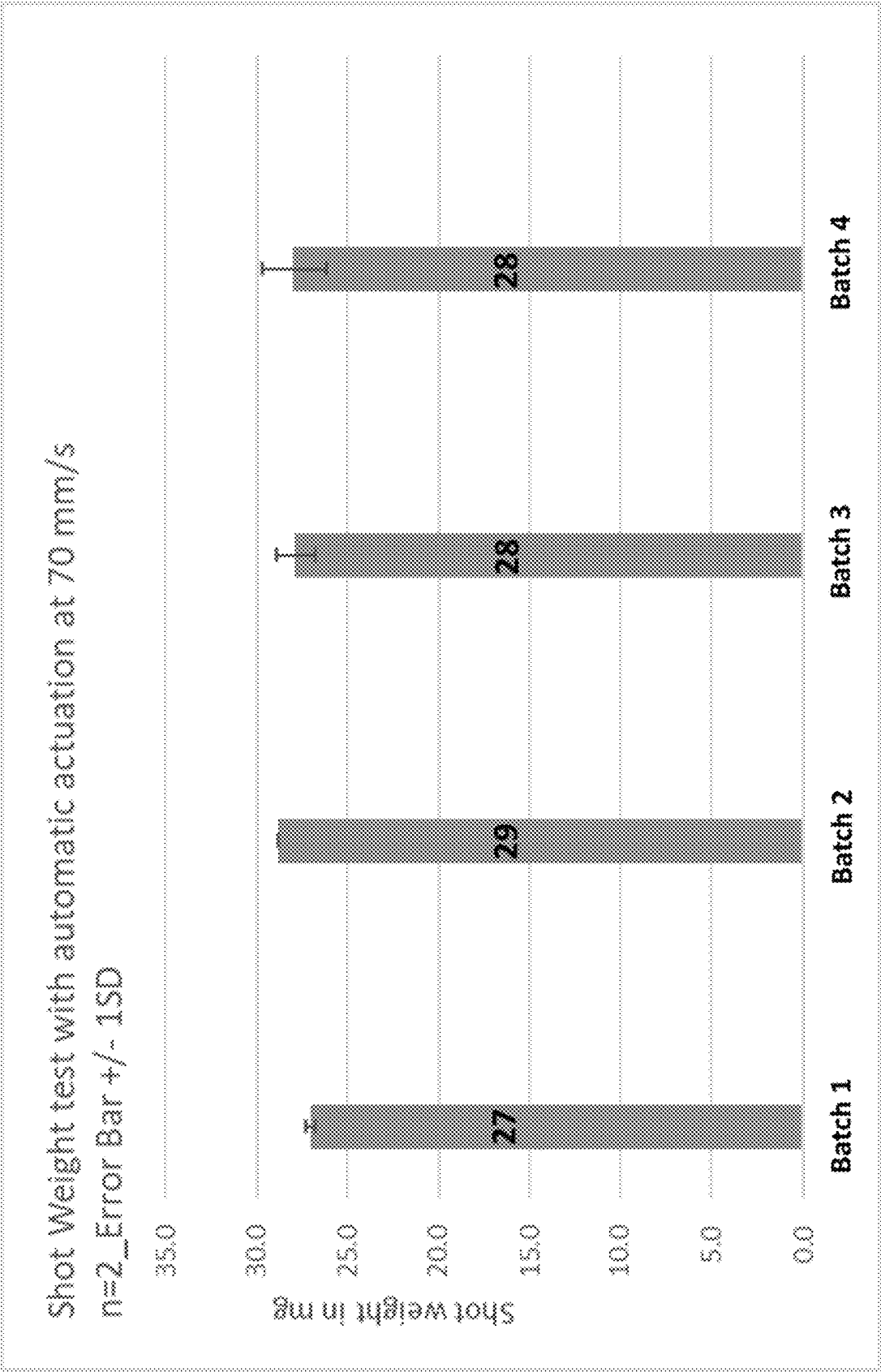


FIG. 121

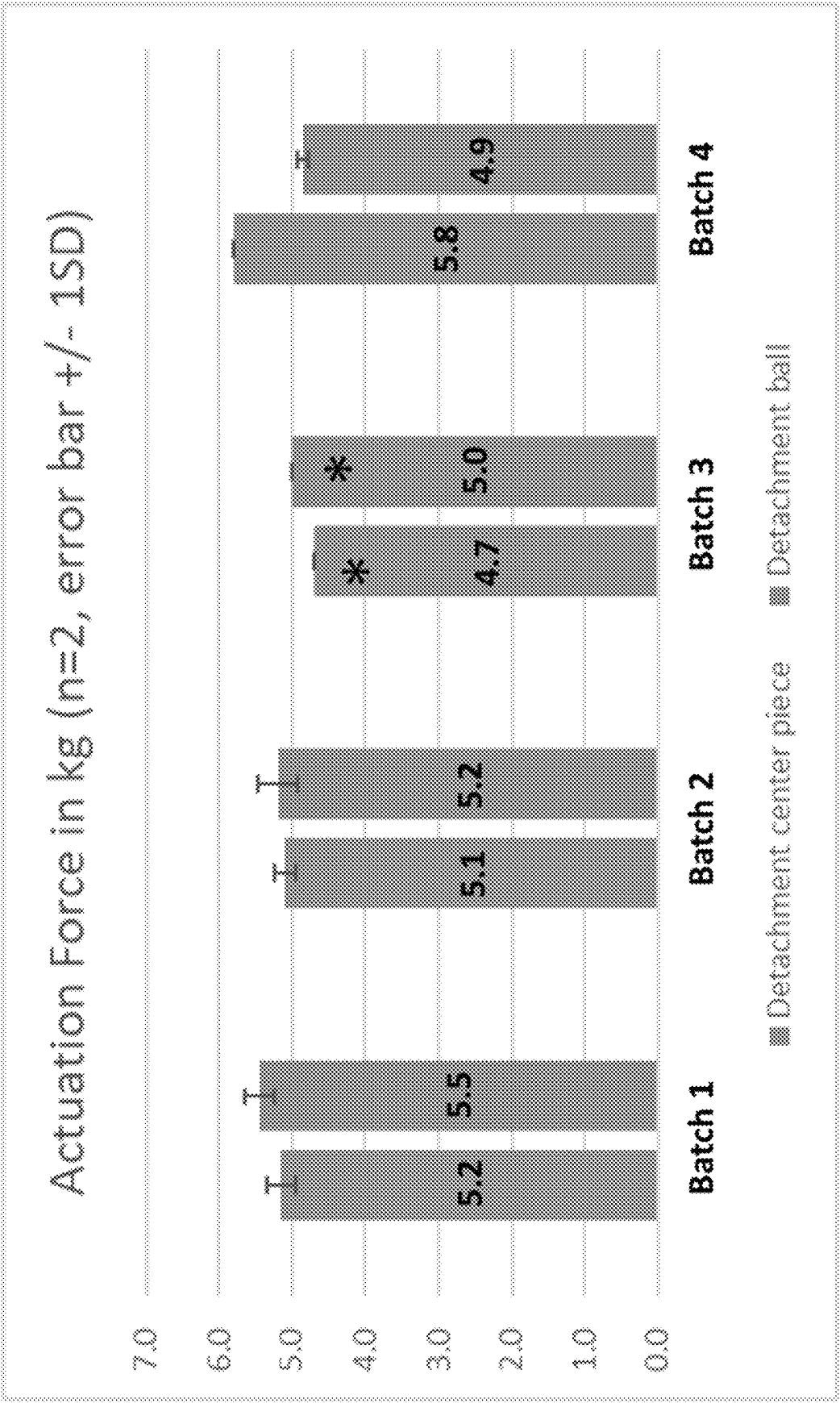


FIG. 122

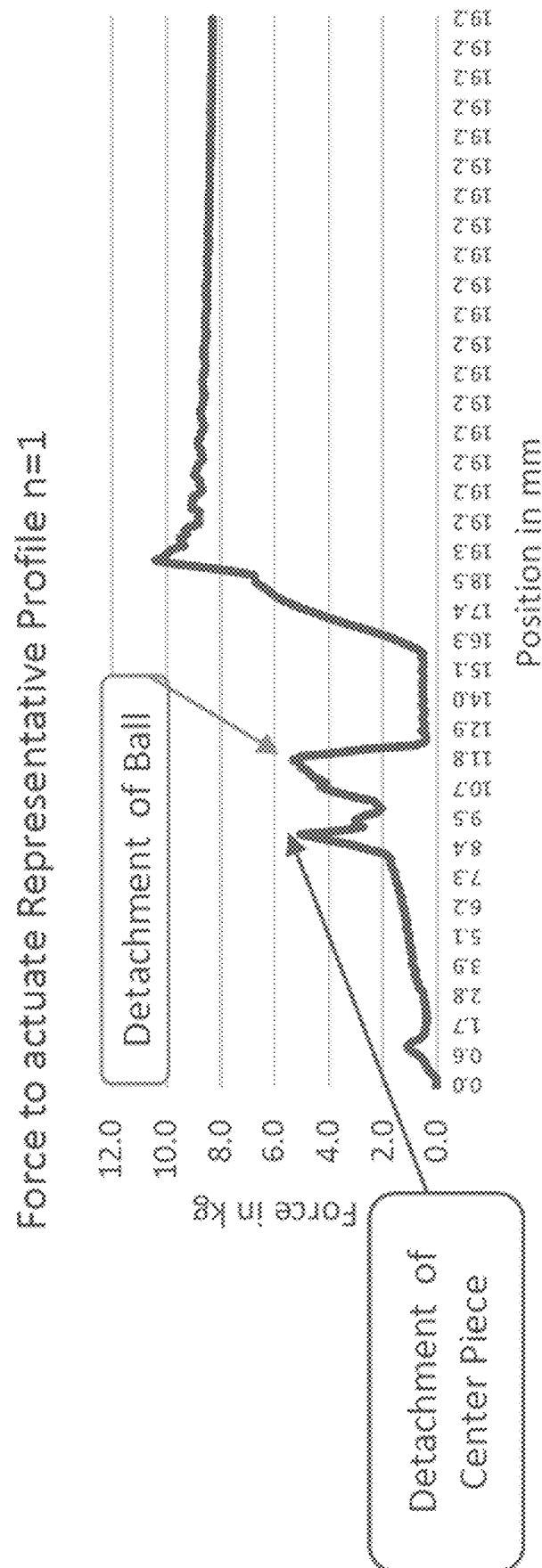


FIG. 123

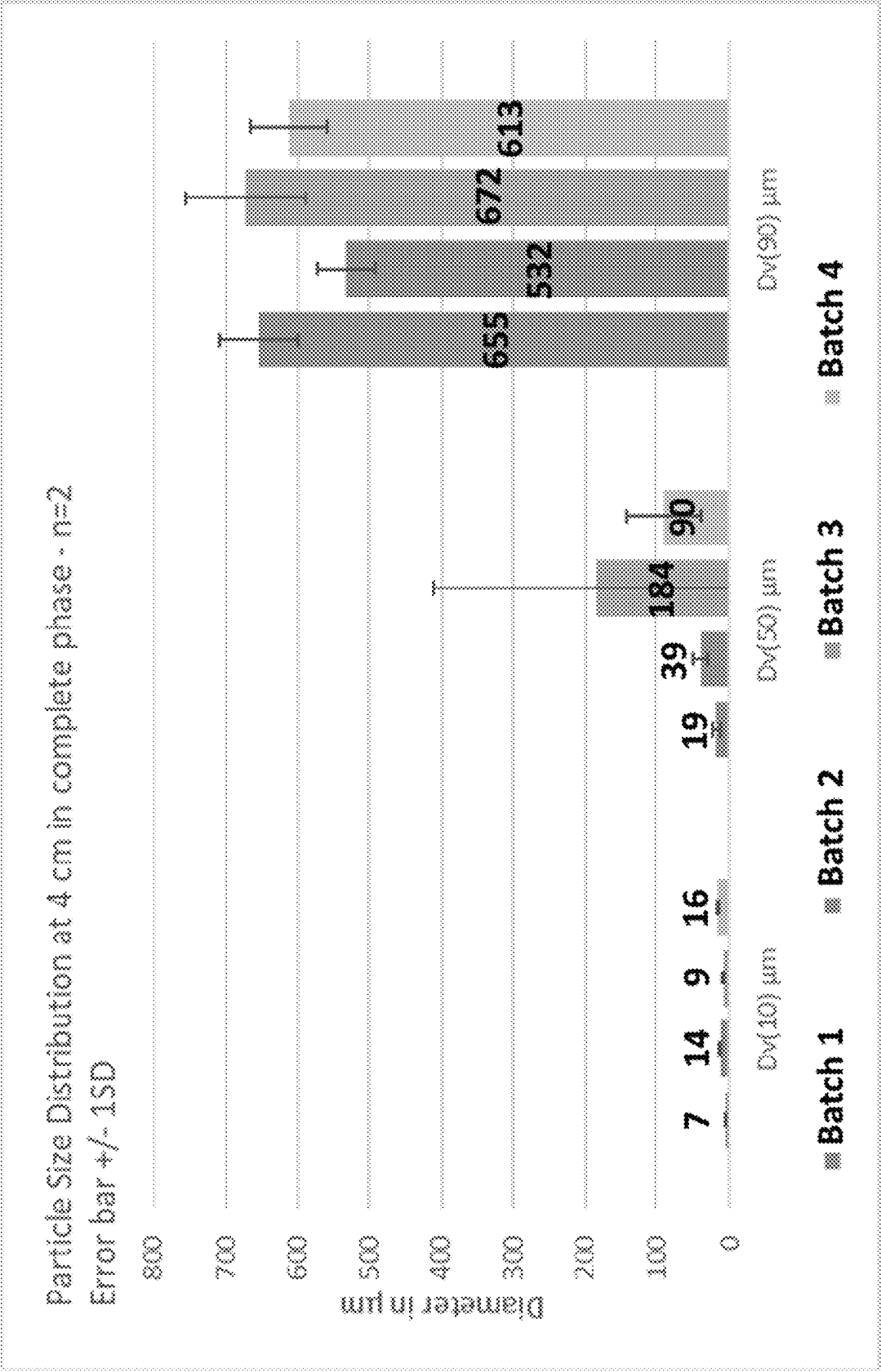


FIG. 124

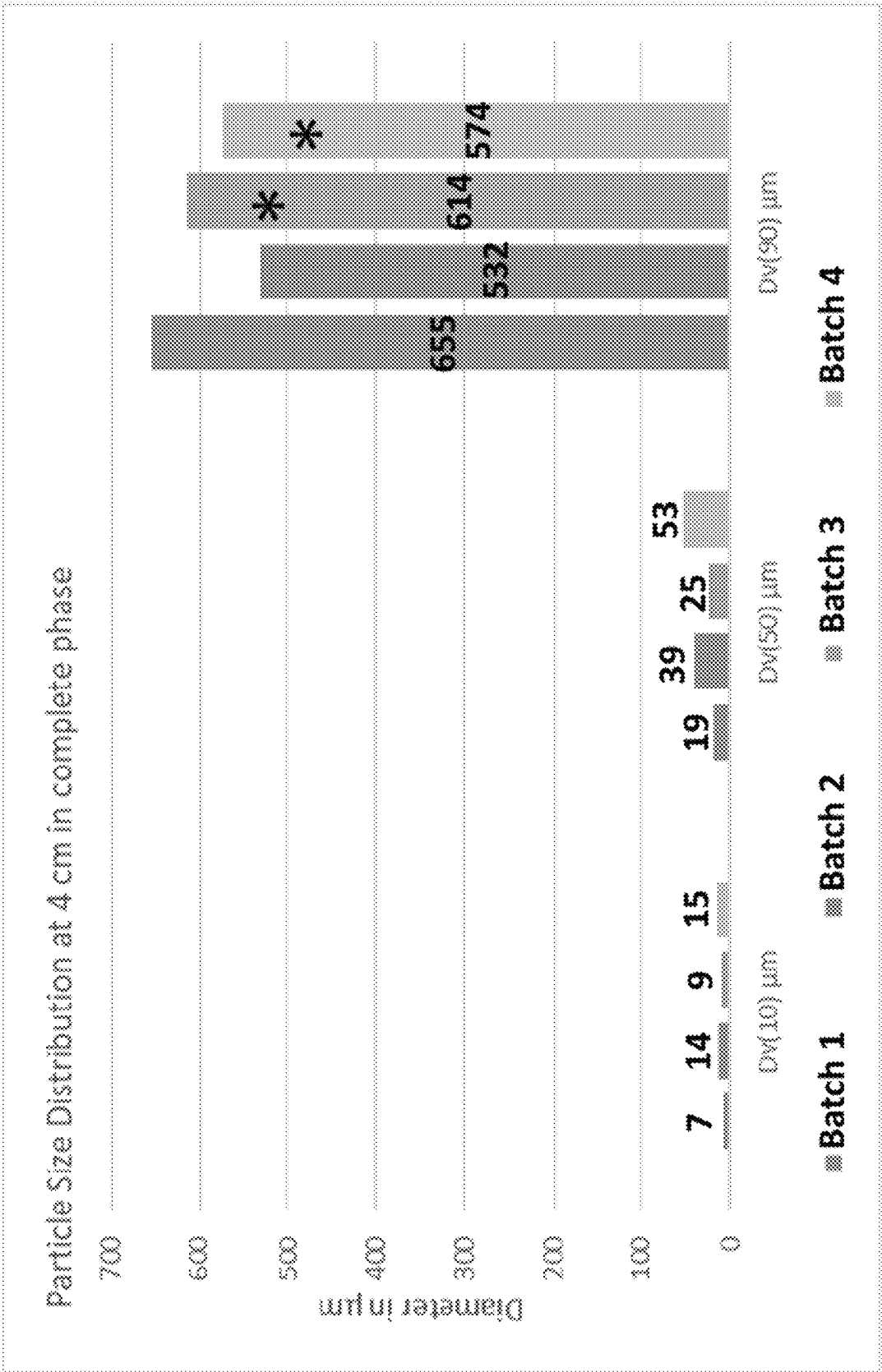


FIG. 125

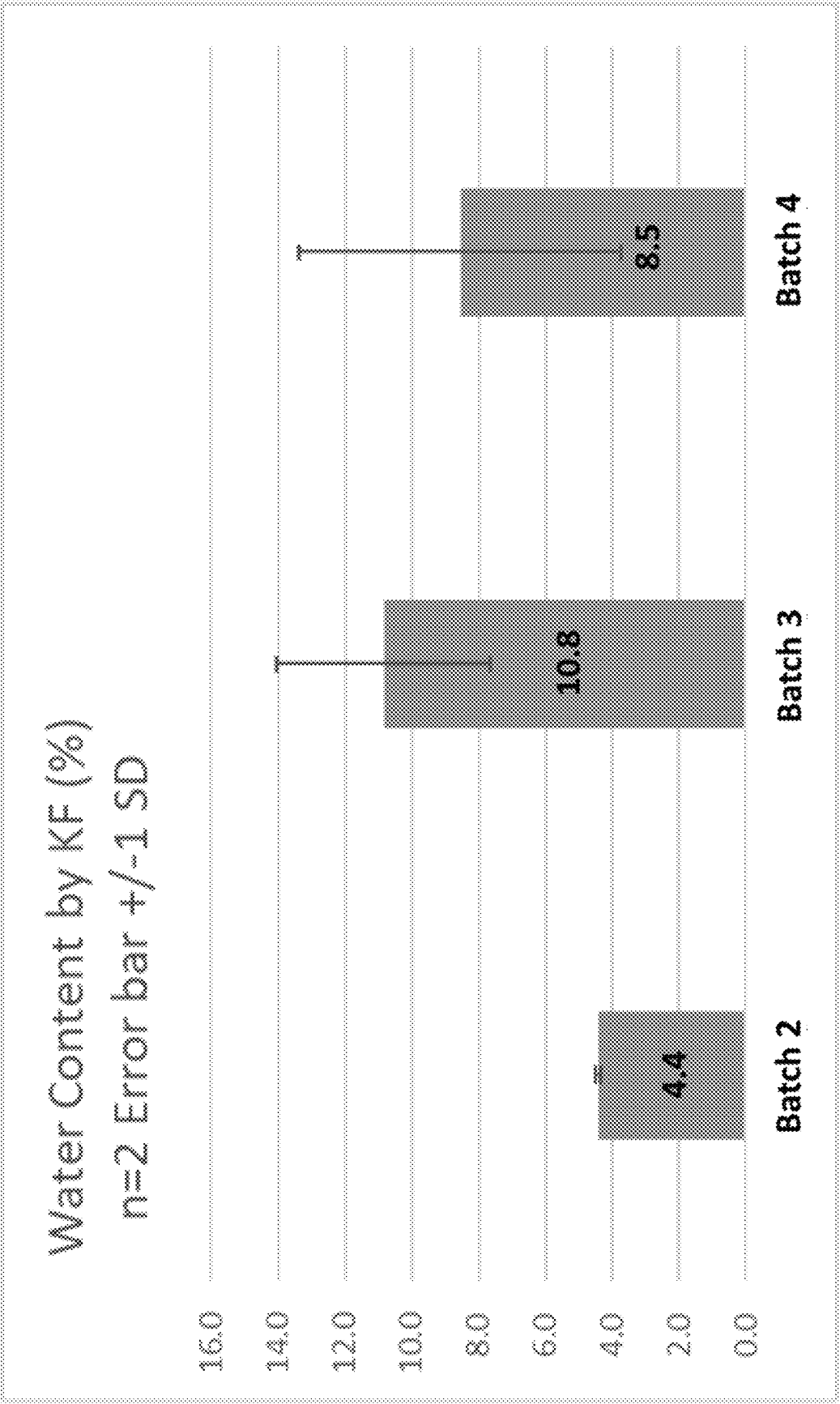
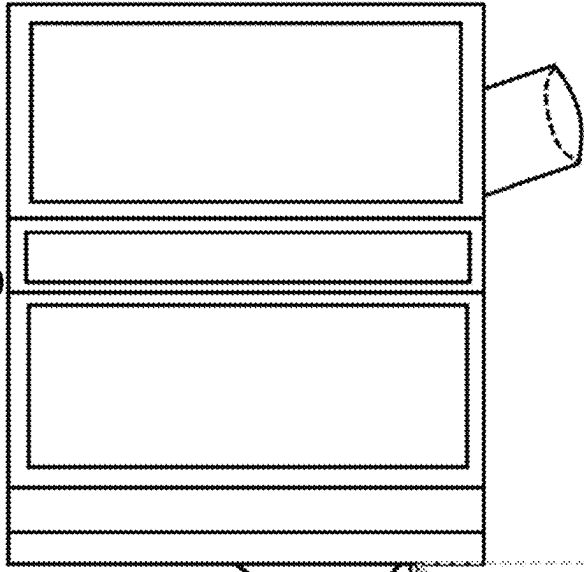


FIG. 126

Angle Y – delivery

angle



0°

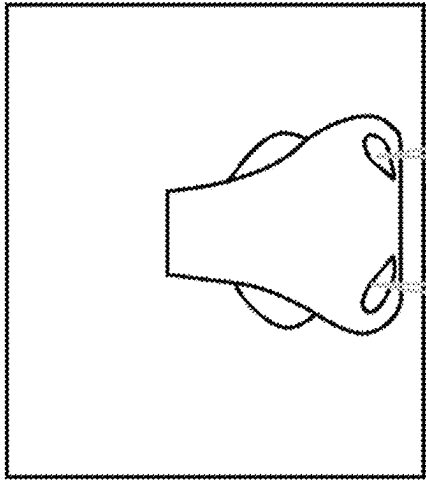
30°

45°

90°

FIG. 127

Angle X – angle from
the center wall



5°

5°

Center
wall

FIG. 128

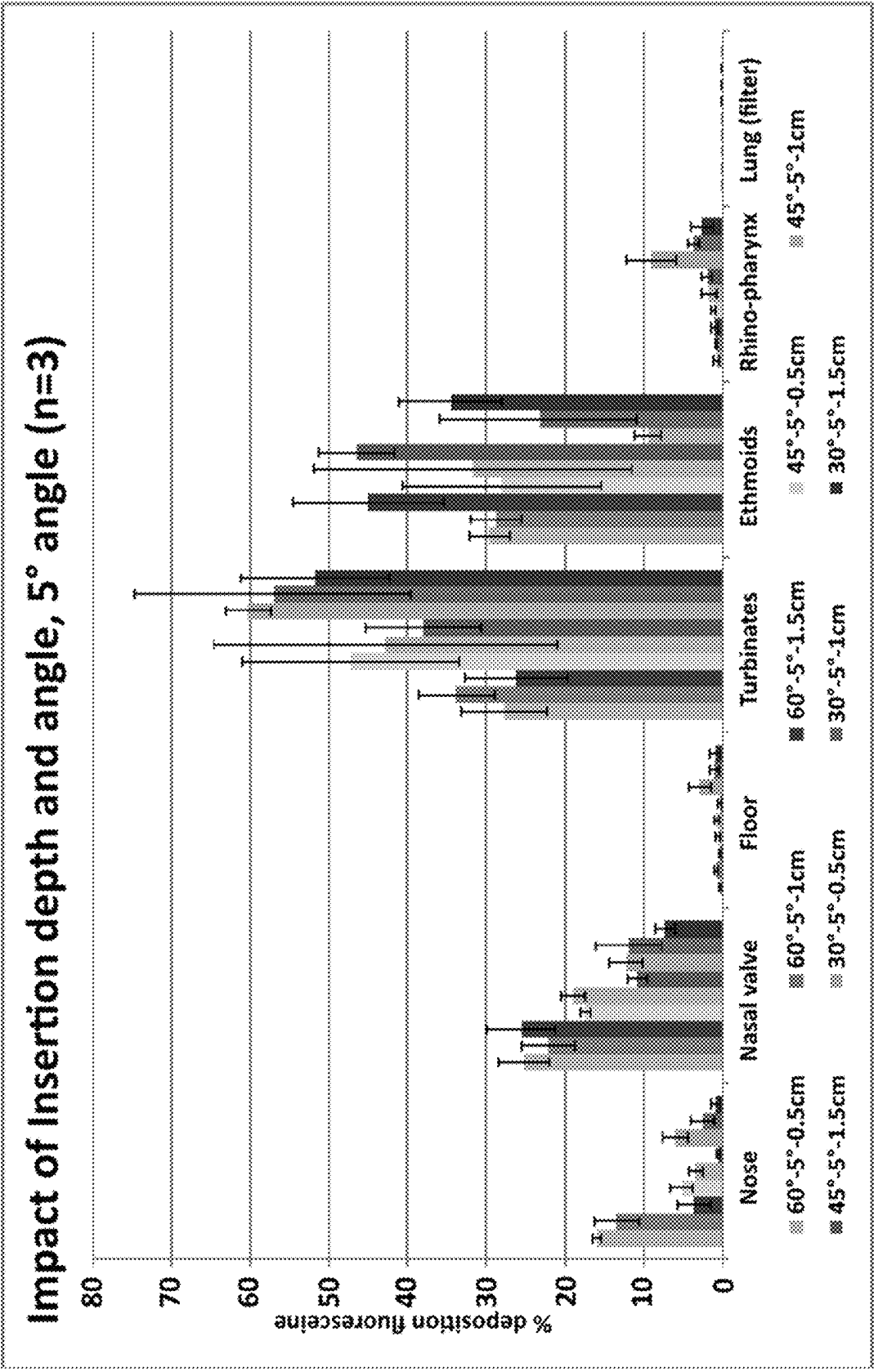


FIG. 129

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DRY POWDER FORMULATIONS OF EPINEPHRINE AND ASSOCIATED METHODS

CROSS REFERENCES TO RELATED APPLICATIONS

This application is related to and claims priority from the following U.S. patents and patent applications. This application is a continuation-in-part of U.S. patent application Ser. No. 17/836,588, filed Jun. 9, 2022, which claims priority to and the benefit of U.S. Provisional Patent Application No. 63/209,221, filed Jun. 10, 2021. Each of the above-listed applications is incorporated herein by reference in its entirety.

BACKGROUND OF THE INVENTION

1. Field of the Invention

The present invention relates to epinephrine, and more specifically to dry powder formulations of epinephrine and methods of treatment of various conditions using the dry power formulations.

2. Description of the Prior Art

It is generally known in the prior art to provide epinephrine for the treatment of anaphylactic reactions, which are commonly triggered by food allergies, insect bites or stings, vaccine injections, allergies to latex and other substances, allergic reactions to drugs, and other exposures. It is also generally known in the prior art to administer epinephrine for the treatment of other conditions, including anaphylactoid reaction, respiratory conditions including bronchospasm, hemodynamic collapse including hypotension, and for administration during cardiopulmonary arrest and other life-threatening conditions. Epinephrine is delivered intravenously and by needle and syringe injection, which may be subcutaneous or intra-muscular. Increasingly, epinephrine is delivered by auto-injectors, which may be used within medical facilities or in ambulatory settings by medical and non-medical personnel, caregivers, and patients themselves. Self-administration via auto-injector has dramatically improved the availability of epinephrine treatment in emergency situations. Alternatives that do not require a needle or syringe have been explored in the prior art.

Prior art patent documents include the following:

U.S. Pat. No. 9,789,071 for Intranasal formulation of epinephrine for the treatment of anaphylaxis by inventor Fleming, filed Jun. 27, 2013 and issued Oct. 17, 2017, is directed to pharmaceutical compositions of epinephrine for delivery to the nasal mucosa and methods of treating a subject in acute severe anaphylaxis, bronchospasm or during cardiopulmonary resuscitation (CPR). The composition further comprising agents, that either prevent localized degradation of epinephrine or enhance its absorption in the nasal mucosa to counter anaphylactic effects, symptoms or complications in a subject.

U.S. Pat. No. 8,747,813 for Inhalable epinephrine by inventors Batycky, et al., filed Mar. 6, 2013 and issued Jun. 10, 2014, is directed to particles for delivery of epinephrine to the respiratory system and methods for treating a patient in need of epinephrine. The particles and respirable compositions comprising the particles comprise the bioactive agent epinephrine, or a salt thereof, as a therapeutic agent. The particles are preferably formed by spray drying. Pref-

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erably, the particles and the respirable compositions are substantially dry and are substantially free of propellants. The patent discloses that the particles have aerodynamic characteristics that permit targeted delivery of epinephrine to the site(s) of action.

U.S. Pat. No. 8,415,397 for Inhalable epinephrine by inventors Batycky, et al., filed Apr. 8, 2011 and issued Apr. 9, 2013, is directed to particles for delivery of epinephrine to the respiratory system and methods for treating a patient in need of epinephrine. The particles and respirable compositions comprising the particles comprise the bioactive agent epinephrine, or a salt thereof, as a therapeutic agent. The particles are preferably formed by spray drying. Preferably, the particles and the respirable compositions are substantially dry and are substantially free of propellants. The patent discloses that the particles have aerodynamic characteristics that permit targeted delivery of epinephrine to the site(s) of action.

U.S. Pat. No. 7,954,491 for Low dose pharmaceutical powders for inhalations by inventor Hrkach, filed Jun. 14, 2004 and issued Jun. 7, 2011, is directed to a method of delivering an agent to the pulmonary system of a compromised patient, in a single breath-activated step, comprising administering a particle mass comprising an agent from an inhaler containing less than 5 milligrams of the mass, wherein at least about 50% of the mass in the receptacle is delivered to the pulmonary system of a patient. The patent also discloses receptacles containing the particle mass and the inhaler for use therein.

U.S. Pat. No. 7,947,742 for Inhalable epinephrine by inventors Batycky, et al., filed Jun. 26, 2003 and issued May 24, 2011, is directed to particles for delivery of epinephrine to the respiratory system and methods for treating a patient in need of epinephrine. The particles and respirable compositions comprising the particles comprise the bioactive agent epinephrine, or a salt thereof, as a therapeutic agent. The particles are preferably formed by spray drying. Preferably, the particles and the respirable compositions are substantially dry and are substantially free of propellants. The patent discloses that the particles have aerodynamic characteristics that permit targeted delivery of epinephrine to the site(s) of action.

SUMMARY OF THE INVENTION

The present invention relates to epinephrine, and more specifically to dry powder formulations of epinephrine and methods of treatment of various conditions using the dry power formulations.

It is an object of this invention to provide therapeutic formulations for enabling the absorption and bioavailability of epinephrine when sprayed into the human nasal passages. In all aspects of the present invention, epinephrine is the primary active ingredient to counteract the physiological changes that occur during anaphylaxis that leads to morbidity and mortality.

In one embodiment, the present invention provides an intranasal device for administration of a pharmaceutical composition including a reservoir and a means for discharging one or more doses of the pharmaceutical composition, wherein the reservoir contains a quantity of the pharmaceutical composition, wherein the pharmaceutical composition is a dry powder including epinephrine or a pharmaceutically acceptable salt thereof, wherein the pharmaceutical composition provides a dose of about 0.01 mg to about 10 mg of the epinephrine or the pharmaceutically acceptable salt thereof, a polysorbate, wherein the pharmaceutical compo-

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sition includes about 0.01% w/v to about 5% w/v of the polysorbate, and a carrier, wherein the pharmaceutical composition does not include an additional vasodilator, and wherein the intranasal device further includes a nasal probe and an actuator.

In another embodiment, the present invention provides a kit for intranasal administration of a pharmaceutical composition including at least one intranasal device, wherein each of the at least one intranasal device includes a reservoir and a means for discharging one or more doses of the pharmaceutical composition, wherein the reservoir contains a quantity of the pharmaceutical composition, and a pouch and/or a hard case, wherein the at least one intranasal device is enclosed in the pouch and/or the hard case, wherein the pharmaceutical composition is a dry powder including epinephrine or a pharmaceutically acceptable salt thereof, wherein the pharmaceutical composition provides a dose of about 0.01 mg to about 10 mg of the epinephrine or the pharmaceutically acceptable salt thereof, a polysorbate, wherein the pharmaceutical composition includes about 0.01% w/v to about 5% w/v of the polysorbate, and a carrier, wherein the pharmaceutical composition does not include an additional vasodilator, and wherein the at least one intranasal device further includes a nasal probe and an actuator.

In yet another embodiment, the present invention provides an intranasal device for administration of a pharmaceutical composition including a reservoir and a means for discharging one or more doses of the pharmaceutical composition, wherein the reservoir contains a quantity of the pharmaceutical composition, wherein the pharmaceutical composition is a spray-dried powder including epinephrine or a pharmaceutically acceptable salt thereof, wherein the pharmaceutical composition provides a dose of about 0.01 mg to about 10 mg of the epinephrine or the pharmaceutically acceptable salt thereof, polysorbate 80, wherein the pharmaceutical composition includes about 0.01% w/v to about 5% w/v of the polysorbate 80, and a carrier, wherein the pharmaceutical composition does not include an additional vasodilator, and wherein the intranasal device further includes a nasal probe and an actuator.

These and other aspects of the present invention will become apparent to those skilled in the art after a reading of the following description of the preferred embodiment when considered with the drawings, as they support the claimed invention.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a summary table for test formulations prepared according to the present invention.

FIG. 2 is another summary table for test formulations prepared according to the present invention.

FIG. 3 is yet another summary table for test formulations prepared according to the present invention.

FIG. 4 illustrates one embodiment of a nasal delivery device according to the present invention.

FIG. 5A illustrates a nasal delivery device at rest according to one embodiment of the present invention.

FIG. 5B illustrates positioning of fingers and a thumb on the nasal delivery device according to one embodiment of the present invention.

FIG. 5C illustrates discharge of the nasal delivery device according to one embodiment of the present invention.

FIG. 5D illustrates disposal of the nasal delivery device following use according to one embodiment of the present invention.

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FIG. 6 illustrates the nasal delivery device in an unprimed state, with a sealed chamber, pressure charged detachment of a centerpiece, a start of a discharge, and an end of a stroke according to one embodiment of the present invention.

FIG. 7 illustrates a delivery device and a secondary packaging in the form of a container according to one embodiment of the present invention.

FIG. 8 illustrates a secondary packaging in the form of a container according to one embodiment of the present invention.

FIG. 9 is a schematic diagram of a system of the present invention.

FIG. 10 is a table of epinephrine concentration-time data in rats for epinephrine with a lactose carrier.

FIG. 11 is a table of epinephrine concentration-time data in rats for epinephrine with a sodium carboxymethylcellulose (CMC) carrier.

FIG. 12 is a table of epinephrine concentration-time data in rats for epinephrine and caffeine (5% w/w) with a lactose carrier.

FIG. 13 is a table of epinephrine concentration-time data in rats for epinephrine and caffeine (5% w/w) with a sodium CMC carrier.

FIG. 14 is a table of epinephrine concentration-time data in rats for epinephrine and hyaluronate (0.5% w/w) with a lactose carrier.

FIG. 15 is a table of epinephrine concentration-time data in rats for epinephrine and hyaluronate (0.5% w/w) with a sodium CMC carrier.

FIG. 16 is a table of epinephrine concentration-time data in rats for epinephrine and leucine (10% w/w) with a lactose carrier.

FIG. 17 is a table of epinephrine concentration-time data in rats for epinephrine and leucine (10% w/w) with a sodium CMC carrier.

FIG. 18 is a table of epinephrine concentration-time data in rats for epinephrine and sodium chloride (2% w/w) with a lactose carrier.

FIG. 19 is a table of epinephrine concentration-time data in rats for epinephrine and sodium chloride (2% w/w) with a sodium CMC carrier.

FIG. 20 is a table of epinephrine concentration-time data in rats for epinephrine and niacin (2% w/w) with a lactose carrier.

FIG. 21 is a table of epinephrine concentration-time data in rats for epinephrine and niacin (2% w/w) with a sodium CMC carrier.

FIG. 22 is a table of epinephrine concentration-time data in rats for epinephrine and polysorbate (0.5% w/w) with a lactose carrier.

FIG. 23 is a table of epinephrine concentration-time data in rats for epinephrine and polysorbate (0.5% w/w) with a sodium CMC carrier.

FIG. 24 is a table of epinephrine concentration-time data in rats for epinephrine intramuscular injection at 0.7 mg/kg.

FIG. 25 is a table of epinephrine pharmacokinetic parameters in rats for epinephrine with a lactose carrier.

FIG. 26 is a table of epinephrine pharmacokinetic parameters in rats for epinephrine with a sodium CMC carrier.

FIG. 27 is a table of epinephrine pharmacokinetic parameters in rats for epinephrine and caffeine (5% w/w) with a lactose carrier.

FIG. 28 is a table of epinephrine pharmacokinetic parameters in rats for epinephrine and caffeine (5% w/w) with a sodium CMC carrier.

FIG. 50 is a linear graph of epinephrine concentration-time profiles in rats for epinephrine and hyaluronate (0.5% w/w) with a sodium CMC carrier.

FIG. 71 is a graph of mean epinephrine concentration-time profiles following IN administration in rats for epinephrine and caffeine (5% w/w) with a lactose carrier and epinephrine and caffeine (5% w/w) with a sodium CMC carrier.

FIG. 72 is a graph of mean epinephrine concentration-time profiles following IN administration in rats for epinephrine and hyaluronate (0.5% w/w) with a lactose carrier and epinephrine and hyaluronate (0.5% w/w) with a sodium CMC carrier.

FIG. 73 is a graph of mean epinephrine concentration-time profiles following IN administration in rats for epinephrine and leucine (10% w/w) with a lactose carrier and epinephrine and leucine (10% w/w) with a sodium CMC carrier.

FIG. 74 is a graph of mean epinephrine concentration-time profiles following IN administration in rats for epinephrine and sodium chloride (2% w/w) with a lactose carrier and epinephrine and sodium chloride (2% w/w) with a sodium CMC carrier.

FIG. 75 is a graph of mean epinephrine concentration-time profiles following IN administration in rats for epinephrine and niacin (2% w/w) with a lactose carrier and epinephrine and niacin (2% w/w) with a sodium CMC carrier.

FIG. 76 is a graph of mean epinephrine concentration-time profiles following IN administration in rats for epinephrine and polysorbate (0.5% w/w) with a lactose carrier and epinephrine and polysorbate (0.5% w/w) with a sodium CMC carrier.

FIG. 77 is a graph of mean epinephrine concentration-time profiles following IN administration in rats grouped by lactose carrier including epinephrine, epinephrine and caffeine (5% w/w), epinephrine and hyaluronate (0.5% w/w), epinephrine and leucine (10% w/w), epinephrine and sodium chloride (2% w/w), epinephrine and niacin (2% w/w), and epinephrine and polysorbate (0.5% w/w).

FIG. 78 is a graph of mean epinephrine concentration-time profiles following IN administration in rats grouped by sodium CMC carrier including epinephrine, epinephrine and caffeine (5% w/w), epinephrine and hyaluronate (0.5% w/w), epinephrine and leucine (10% w/w), epinephrine and sodium chloride (2% w/w), epinephrine and niacin (2% w/w), and epinephrine and polysorbate (0.5% w/w).

FIG. 79 is a table of calibration curve data for epinephrine in rat potassium EDTA plasma.

FIG. 80 is a table of quality control evaluation data for epinephrine in rat potassium EDTA plasma.

FIG. 81 illustrates a table of plasma epinephrine concentration-time data in dogs following a 40 mg dosage of 10% w/w epinephrine, 2% NaCl, 0.1% w/v polysorbate (TWEEN 80), and lactose.

FIG. 82 illustrates a table of plasma epinephrine concentration-time data in dogs following a 60 mg dosage of 10% w/w epinephrine, 2% NaCl, and lactose.

FIG. 83 illustrates a table of plasma epinephrine concentration-time data in dogs following a 60 mg dosage of 10% w/w epinephrine, 2% NaCl, 0.1% w/v polysorbate (TWEEN 80), and lactose.

FIG. 84 illustrates a table of plasma epinephrine concentration-time data in dogs following a 30 mg dosage of 20% w/w epinephrine, 2% NaCl, and lactose.

FIG. 85 illustrates a table of plasma epinephrine concentration-time data in dogs following a 30 mg dosage of 20% w/w epinephrine, 2% NaCl, 0.1% w/v polysorbate (TWEEN 80), and lactose.

FIG. 86 illustrates a table of plasma epinephrine concentration-time data in dogs following a 0.3 mg IM dosage of epinephrine using an EPIPEN.

FIG. 87 illustrates a table of epinephrine pharmacokinetic parameters in dogs following a 40 mg dosage of 10% w/w epinephrine, 2% NaCl, 0.1% w/v polysorbate (TWEEN 80), and lactose.

FIG. 88 illustrates a table of epinephrine pharmacokinetic parameters in dogs following a 60 mg dosage of 10% w/w epinephrine, 2% NaCl, and lactose.

FIG. 89 illustrates a table of epinephrine pharmacokinetic parameters in dogs following a 60 mg dosage of 10% w/w epinephrine, 2% NaCl, 0.1% w/v polysorbate (TWEEN 80), and lactose.

FIG. 90 illustrates a table of epinephrine pharmacokinetic parameters in dogs following a 30 mg dosage of 20% w/w epinephrine, 2% NaCl, and lactose.

FIG. 91 illustrates a table of epinephrine pharmacokinetic parameters in dogs following a 30 mg dosage of 20% w/w epinephrine, 2% NaCl, 0.1% w/v polysorbate (TWEEN 80), and lactose.

FIG. 92 illustrates a table of epinephrine pharmacokinetic parameters in dogs following a 0.3 mg IM dosage of epinephrine using an EPIPEN.

FIG. 93 illustrates a summary of epinephrine AUClast results.

FIG. 94 illustrates relative bioavailability calculations (AUClast).

FIG. 95 illustrates comparative bioavailability calculations (AUClast).

FIG. 96 illustrates a linear graph of individual epinephrine concentration-time profiles for a 40 mg dosage of 10% w/w epinephrine, 2% NaCl, 0.1% w/v polysorbate (TWEEN 80), and lactose.

FIG. 97 illustrates a linear graph of individual epinephrine concentration-time profiles for a 60 mg dosage of 10% w/w epinephrine, 2% NaCl, and lactose.

FIG. 98 illustrates a linear graph of individual epinephrine concentration-time profiles for a 60 mg dosage of 10% w/w epinephrine, 2% NaCl, 0.1% w/v polysorbate (TWEEN 80), and lactose.

FIG. 99 illustrates a linear graph of individual epinephrine concentration-time profiles for a 30 mg dosage of 20% w/w epinephrine, 2% NaCl, and lactose.

FIG. 100 illustrates a linear graph of individual epinephrine concentration-time profiles for a 30 mg dosage of 20% w/w epinephrine, 2% NaCl, 0.1% w/v polysorbate (TWEEN 80), and lactose.

FIG. 101 illustrates a linear graph of individual epinephrine concentration-time profiles for a 0.3 mg IM dosage of epinephrine using an EPIPEN.

FIG. 102 illustrates a log-linear graph of individual epinephrine concentration-time profiles for a 40 mg dosage of 10% w/w epinephrine, 2% NaCl, 0.1% w/v polysorbate (TWEEN 80), and lactose.

FIG. 103 illustrates a log-linear graph of individual epinephrine concentration-time profiles for a 60 mg dosage of 10% w/w epinephrine, 2% NaCl, and lactose.

FIG. 104 illustrates a log-linear graph of individual epinephrine concentration-time profiles for a 60 mg dosage of 10% w/w epinephrine, 2% NaCl, 0.1% w/v polysorbate (TWEEN 80), and lactose.

FIG. 105 illustrates a log-linear graph of individual epinephrine concentration-time profiles for a 30 mg dosage of 20% w/w epinephrine, 2% NaCl, and lactose.

FIG. 106 illustrates a log-linear graph of individual epinephrine concentration-time profiles for a 30 mg dosage of 20% w/w epinephrine, 2% NaCl, 0.1% w/v polysorbate (TWEEN 80), and lactose.

FIG. 107 illustrates a log-linear graph of individual epinephrine concentration-time profiles for a 0.3 mg IM dosage of epinephrine using an EPIPEN.

FIG. 108 illustrates a graph of epinephrine concentration-time profiles for a first dog.

FIG. 109 illustrates a graph of epinephrine concentration-time profiles for a second dog.

FIG. 110 illustrates a graph of epinephrine concentration-time profiles for a third dog.

FIG. 111 illustrates a graph of epinephrine concentration-time profiles for a fourth dog.

FIG. 112 illustrates a graph of epinephrine concentration-time profiles for a fifth dog.

FIG. 113 illustrates a graph of epinephrine concentration-time profiles for a sixth dog.

FIG. 114 illustrates a graph of mean plasma epinephrine concentration-time profiles by treatment.

FIG. 115 illustrates a graph of partial AUC for 0-10 minutes by treatment.

FIG. 116 illustrates a graph of partial AUC for 0-20 minutes by treatment.

FIG. 117 illustrates a graph of partial AUC for 0-30 minutes by treatment.

FIG. 118 illustrates a graph of partial AUC for 0-60 minutes by treatment.

FIG. 119 illustrates a graph of individual animal plasma epinephrine concentration-time profiles for 0.3 mg IM administration for a pilot study and a main study.

FIG. 120 illustrates a graph of mean plasma epinephrine concentration-time profiles for 0.3 mg IM administration for a pilot study and a main study.

FIG. 121 illustrates a graph of shot weight data.

FIG. 122 illustrates a graph of actuation force.

FIG. 123 illustrates a force to actuate representative profile.

FIG. 124 illustrates a graph of particle size distribution.

FIG. 125 illustrates a graph of particle size distribution with outliers removed.

FIG. 126 illustrates a graph of water content.

FIG. 127 illustrates delivery angle for a nasal device.

FIG. 128 illustrates angle from the center wall for a nasal device.

FIG. 129 illustrates impact of insertion depth and angle for a nasal device.

DETAILED DESCRIPTION

The present invention is generally directed to epinephrine, and more specifically to dry powder formulations of epinephrine and methods of treatment of various conditions using the dry power formulations.

In one embodiment, the present invention provides an intranasal device for administration of a pharmaceutical composition including a reservoir and a means for discharging one or more doses of the pharmaceutical composition, wherein the reservoir contains a quantity of the pharmaceutical composition, wherein the pharmaceutical composition is a dry powder including epinephrine or a pharmaceutically acceptable salt thereof, wherein the pharmaceutical composition provides a dose of about 0.01 mg to about 10 mg of the epinephrine or the pharmaceutically acceptable salt thereof, a polysorbate, wherein the pharmaceutical composition includes about 0.01% w/v to about 5% w/v of the polysorbate, and a carrier, wherein the pharmaceutical composition does not include an additional vasodilator, and wherein the intranasal device further includes a nasal probe and an actuator. In one embodiment, a median particle

diameter of the epinephrine or the pharmaceutically acceptable salt thereof is about 20 μm to about 75 μm . In one embodiment, the pharmaceutical composition further includes an anticaking agent. In one embodiment, the pharmaceutical composition further includes at least one antihistamine. In one embodiment, the pharmaceutical composition further includes at least one steroid. In one embodiment, the pharmaceutical composition further includes a catechol-o-methyl transferase (COMT) inhibitor. In one embodiment, the pharmaceutical composition further includes one or more agents selected from a group consisting of a mucosal permeation or penetration enhancer, a mucoadhesive, a mucosal transit slowing agent, a mucosal transport enhancer, or any combination thereof. In one embodiment, the intranasal device has 360° functionality and is constructed and configured to dispense a dose from any position. In one embodiment, the intranasal device further includes an intranasal device global positioning system (GPS) device or is coupled to a GPS device, wherein the intranasal device GPS or the GPS device is constructed and configured to provide location data. In one embodiment, the intranasal device further includes a communications interface, wherein the communications interface is constructed and configured to transmit data wirelessly to at least one remote device. In one embodiment, the carrier includes mannitol, a cyclodextrin, citric acid, lactose, and/or sodium carboxymethylcellulose. In one embodiment, the pharmaceutical composition does not include a preservative. In one embodiment, the quantity of the pharmaceutical composition is between about 5 mg and about 75 mg. In one embodiment, the intranasal device further includes a sensor that is adapted to detect a displacement or a deformation of a portion of the intranasal device when the dry powder is dispensed from the reservoir. In one embodiment, the intranasal device further includes a display and a power supply, a timer, a clock, and/or a printed circuit board, wherein the display is constructed and configured to display a time of dose dispensation. In one embodiment, the pharmaceutical composition further includes sodium chloride, wherein the pharmaceutical composition includes about 0.01 mg to about 5 mg of the sodium chloride.

In another embodiment, the present invention provides a kit for intranasal administration of a pharmaceutical composition including at least one intranasal device, wherein each of the at least one intranasal device includes a reservoir and a means for discharging one or more doses of the pharmaceutical composition, wherein the reservoir contains a quantity of the pharmaceutical composition, and a pouch and/or a hard case, wherein the at least one intranasal device is enclosed in the pouch and/or the hard case, wherein the pharmaceutical composition is a dry powder including epinephrine or a pharmaceutically acceptable salt thereof, wherein the pharmaceutical composition provides a dose of about 0.01 mg to about 10 mg of the epinephrine or the pharmaceutically acceptable salt thereof, a polysorbate, wherein the pharmaceutical composition includes about 0.01% w/v to about 5% w/v of the polysorbate, and a carrier, wherein the pharmaceutical composition does not include an additional vasodilator, and wherein the at least one intranasal device further includes a nasal probe and an actuator. In one embodiment, the hard case includes a desiccant plastic. In one embodiment, the pouch and/or the hard case is water resistant or waterproof. In one embodiment, the pouch and/or the hard case is Modular Lightweight Load-carrying Equipment (MOLLE)-compatible. In one embodiment, the pharmaceutical composition further includes sodium chlo-

ride, wherein the pharmaceutical composition includes about 0.01 mg to about 5 mg of the sodium chloride.

In yet another embodiment, the present invention provides an intranasal device for administration of a pharmaceutical composition including a reservoir and a means for discharging one or more doses of the pharmaceutical composition, wherein the reservoir contains a quantity of the pharmaceutical composition, wherein the pharmaceutical composition is a spray-dried powder including epinephrine or a pharmaceutically acceptable salt thereof, polysorbate 80, wherein the pharmaceutical composition provides a dose of about 0.01 mg to about 10 mg of the epinephrine or the pharmaceutically acceptable salt thereof, polysorbate 80, wherein the pharmaceutical composition includes about 0.01% w/v to about 5% w/v of the polysorbate 80, and a carrier, wherein the pharmaceutical composition does not include an additional vasodilator, and wherein the intranasal device further includes a nasal probe and an actuator. In one embodiment, the pharmaceutical composition further includes sodium chloride, wherein the pharmaceutical composition includes about 0.01 mg to about 5 mg of the sodium chloride.

Epinephrine is the gold standard for treatment of anaphylactic reactions, which are commonly triggered by food allergies, insect bites or stings, vaccine injections, allergies to latex and other substances, allergic reactions to drugs, and other exposures. Epinephrine is delivered intravenously and by needle and syringe injection, which may be subcutaneous or intra-muscular. Increasingly, epinephrine is delivered by auto-injectors, which may be used within medical facilities or in ambulatory settings by medical and non-medical personnel, caregivers, and patients themselves. Self-administration via auto-injector has dramatically improved the availability of epinephrine treatment in emergency situations.

However, auto-injectors have a number of known disadvantages. These include widespread fear of needles and the size of the units themselves, which may be cumbersome to carry. Many patients and caregivers avoid carrying epinephrine auto-injectors during hot weather or leaving them in automobiles or other places where the aqueous epinephrine formulation is liable to degrade and lose potency. These factors lead many patients or caregivers to not have epinephrine available during emergency anaphylactic events, or if it is available, they may be reluctant to use the device. Delayed access and delayed use of epinephrine auto-injectors has been associated with increased morbidity and mortality. See, e.g., Fleming, J. T., et al., "Early Treatment of Food-Induced Anaphylaxis with Epinephrine Is Associated with a Lower Risk of Hospitalization," *J. Allergy Clin Immunol Pract.*, January-February 2015; 3(1):57-62; doi: 10.1016/j.jaip.2014.07.004 (published online Sep. 8, 2014) and Gabrielli, S., et al., "Teenagers and those with severe reactions are more likely to use their epinephrine autoinjectors in cases of anaphylaxis in Canada," *J. Allergy & Clin. Immunology: In Practice, Clinical Communications*, Mar. 1, 2019, 7(3): P1073-1075.E3, doi.org/10.1016/j.jaip.2018.07.044, each of which is incorporated herein by reference in its entirety.

In addition, studies have shown that many patients do not use auto-injectors correctly. In one recent study of epinephrine auto-injector carriage and use practices among children, adolescents, and adults in the United States, the investigators reported that "Of the 242 patients, 54% were not able to recall all the steps for correct EAI use or completely failed to activate the device." Wasserman, S., et. al. "Epinephrine Autoinjectors: New Data, New Problems," *J Allergy Clin Immunol Pract.*, September-October 2017; 5(5):1180-1191,

doi: 10.1016/j.jaip.2017.06.027, which is incorporated herein by reference in its entirety.

Further, studies have shown that due to obesity many patients using auto-injectors do not receive the recommended dose via an intramuscular (IM) injection. One US study estimated that some thirty percent (30%) of both adults and children would not receive an IM injection. When injected, subcutaneously (SQ), a substantially lower plasma drug level and delayed delivery has been found. See Stetcher, D., et.al., "Epinephrine Auto-injectors: Is Needle Length Adequate for Delivery of Epinephrine Intramuscularly?" *Pediatrics*, July 2009, 124(1): 65-70, doi:10.1542/peds.2008-3388 and Simons F E, Gu X, Simons K J. Epinephrine absorption in adults: intramuscular versus subcutaneous injection. *J Allergy Clin Immunol.* 2001 November; 108(5):871-3. doi: 10.1067/mai.2001.119409. PMID: 11692118, each of which is incorporated herein by reference in its entirety.

Similarly, needle length may be inappropriate in devices used to treat infants and toddlers. Infants and young children may be uncooperative and unable to hold still while an injection is given, which may cause injuries (e.g., lacerations, bent needles, embedded needles). Further, the needle may strike bone in smaller patients. See, e.g., Greenberger P A, Wallace D V, Lieberman P L, Gregory S M. Contemporary issues in anaphylaxis and the evolution of epinephrine autoinjectors: What will the future bring? *Ann Allergy Asthma Immunol.* 2017 October; 119(4):333-338. doi: 10.1016/j.anai.2017.07.030. PMID: 28958374, which is incorporated herein by reference in its entirety. For additional information about injuries caused by auto-injectors, see Brown J C, Tuuri R E, Akhter S, Guerra L D, Goodman I S, Myers S R, Nozicka C, Manzi S, Long K, Turner T, Connors G P, Thompson R W, Park E. Lacerations and Embedded Needles Caused by Epinephrine Autoinjector Use in Children. *Ann Emerg Med.* 2016 March; 67(3):307-315.e8. doi: 10.1016/j.annemergmed.2015.07.011. Epub 2015 Oct. 9. PMID: 26452720, which is incorporated herein by reference in its entirety. Further, there are issues with unintentional epinephrine injections from auto-injectors, for example, while self-injecting epinephrine, injecting epinephrine into someone with anaphylaxis, or demonstrating use of the auto-injector. See, e.g., Simons F E, Lieberman P L, Read E J Jr, Edwards E S. Hazards of unintentional injection of epinephrine from autoinjectors: a systematic review. *Ann Allergy Asthma Immunol.* 2009 April; 102(4): 282-7. doi: 10.1016/S1081-1206(10)60332-8. PMID: 19441598, which is incorporated herein by reference in its entirety.

Delay or lack of usage of epinephrine in an anaphylactic event leads to increased morbidity and mortality. See, e.g., Chooniedass R, Temple B, Becker A. Epinephrine use for anaphylaxis: Too seldom, too late: Current practices and guidelines in health care. *Ann Allergy Asthma Immunol.* 2017 August; 119(2):108-110. doi: 10.1016/j.anai.2017.06.004. Epub 2017 Jul. 1. PMID: 28676208, which is incorporated herein by reference in its entirety. Delay in usage of epinephrine also leads to an increased risk for biphasic anaphylactic reactions. See, e.g., Miles L M, Ratnarajah K, Gabrielli S, Abrams E M, Protudjer J L P, Begin P, Chan E S, Upton J, Wasserman S, Watson W, Gerdtz J, Ben-Shoshan M. Community Use of Epinephrine for the Treatment of Anaphylaxis: A Review and Meta-Analysis. *J Allergy Clin Immunol Pract.* 2021 June; 9(6):2321-2333. doi: 10.1016/j.jaip.2021.01.038. Epub 2021 Feb. 4. PMID: 33549844, which is incorporated herein by reference in its entirety.

The aforementioned drawbacks of epinephrine auto-injectors have led to a search for new routes of delivering epinephrine including pulmonary, sublingual, and nasal delivery. Nasal delivery appears promising as it removes the needle-related concerns, provides a more portable device, and provides an easy-to-use delivery mechanism.

Nasal delivery of epinephrine, however, retains some of the drawbacks of injection if an aqueous formulation is employed. Aqueous formulations typically have shorter shelf lives than powders. Aqueous epinephrine formulations have preservatives, such as sodium bisulfate, which itself can cause an allergic reaction in many individuals. Additionally, when aqueous epinephrine formulations are delivered to the nasal passages, a significant portion of drug volume may be lost due to running down the back of the throat or back out the front of the nose when the device is removed. This may be exacerbated when a patient suffering an anaphylactic event is in a prone position. All of these factors lead to dose uncertainty for liquid nasal approaches. A particularly concerning drawback for aqueous formulations of epinephrine is that they are subject to degradation when exposed to high temperatures. When exposed to high temperature, even for short durations, as in outdoor summertime activity or when left in an automobile, epinephrine can degrade and lose potency. See Lacwik, P., et. al., "Single, short-time exposure to heat in a car during sunny day can decrease epinephrine concentration in autoinjectors: a real-life pilot study," *J. Allergy & Clin. Immunology*, Apr. 1, 2019, 7(4): P1362-1364, published Nov. 28, 2018, doi.org/10.1016/j.jaip.2018.10.027, which is incorporated herein by reference in its entirety. Degradation of epinephrine and loss of potency is also a serious issue when traveling or living in areas where temperatures are elevated most of the year (e.g., the tropics). Also see, e.g., (1) Patil Armenian, Danielle Campagne, Geoff Stroh, Crystal Ives Tallman, William Z. D. Zeng, Thomas Lin & Roy R. Gerona (2017) Hot and Cold Drugs: National Park Service Medication Stability at the Extremes of Temperature, *Prehospital Emergency Care*, 21:3, 378-385, DOI: 10.1080/10903127.2016.1258098; (2) Rachid O, Simons FE, Rawas-Qalaji M, Lewis S, Simons K J. Epinephrine doses delivered from auto-injectors stored at excessively high temperatures. *Drug Dev Ind Pharm*. 2016 January; 42(1):131-135. doi: 10.3109/03639045.2015.1035283. Epub 2015 May 22. PMID: 25997362; (3) Parish H G, Bowser C S, Morton J R, Brown J C. A systematic review of epinephrine degradation with exposure to excessive heat or cold. *Ann Allergy Asthma Immunol*. 2016 July; 117(1):79-87. doi: 10.1016/j.anai.2016.04.006. Epub 2016 May 21. PMID: 27221065; (4) Kassel L, Jones C, Mengesha A. Epinephrine drug degradation in autoinjector products. *J Allergy Clin Immunol Pract*. 2019 September-October; 7(7):2491-2493. doi: 10.1016/j.jaip.2019.04.028. Epub 2019 May 28. PMID: 31151888; (5) Rawas-Qalaji M, Simons F E, Collins D, Simons K J. Long-term stability of epinephrine dispensed in unsealed syringes for the first-aid treatment of anaphylaxis. *Ann Allergy Asthma Immunol*. 2009 June; 102(6):500-3. doi: 10.1016/S1081-1206(10)60124-X. PMID: 19558009; and (6) Gill M A, Kislik A Z, Gore L, Chandna A. Stability of advanced life support drugs in the field. *Am J Health Syst Pharm*. 2004 Mar. 15; 61(6):597-602. doi: 10.1093/ajhp/61.6.597. PMID: 15061431, each of which is incorporated herein by reference in its entirety.

Patients are aware of this issue and many will not carry epinephrine at times or will not store epinephrine in locations where high temperatures are likely. This behavior is due to both the concern over loss of potency and the concern

about the cost of replacing the device that has gone bad. As a result, epinephrine is often not available during emergencies, leading to greater morbidity.

Additionally, dry powder nasal sprays appear to be more readily absorbed during the initial minutes after delivery than aqueous nasal sprays. This may reflect the fact that a greater portion of the volume stays on the nasal mucosa. Further, aqueous nasal sprays frequently shown a biphasic peak blood plasma level. This is thought to reflect the gastrointestinal absorption of a portion of the spray that is swallowed (runs down the throat).

Nasal delivery of epinephrine in a dry powder eliminates the significant disadvantages of intramuscular or subcutaneous injection as well as the major drawbacks of aqueous formulations of the drug. The present invention describes dry powder formulations for delivery to the nasal passages by means of a nasal delivery device that are handheld.

Further, patients experience barriers to epinephrine usage, including, but not limited to, high cost of auto-injectors, lack of availability, lack of use even though a device is available (e.g., due to fear of using the device, due to an expired device), and incorrect technique. See, e.g., Prince B T, Mikhail I, Stukus D R. Underuse of epinephrine for the treatment of anaphylaxis: missed opportunities. *J Asthma Allergy*. 2018 Jun. 20; 11:143-151. doi: 10.2147/JAA.S159400. PMID: 29950873; PMCID: PMC6016581, which is incorporated herein by reference in its entirety. Both cost and supply are also factors that prevent epinephrine usage. See, e.g., Ponda P, Russell A F, Yu J E, Land M H, Crain M G, Patel K, Shroba J A, Sriaroon P. Access barriers to epinephrine autoinjectors for the treatment of anaphylaxis: A survey of practitioners. *J Allergy Clin Immunol Pract*. 2021 October; 9(10):3814-3815.e4. doi: 10.1016/j.jaip.2021.05.028. Epub 2021 Jun. 11. PMID: 34126272, which is incorporated herein by reference in its entirety. Further, low prescription rates of autoinjectors and lack of stock epinephrine in schools are barriers to use in pre-hospital settings. See, e.g., Ponda P, Russell A F, Yu J E, Land MI-1, Crain M G, Patel K, Shroba J A, Sriaroon P. Access barriers to epinephrine autoinjectors for the treatment of anaphylaxis: A survey of practitioners. *J Allergy Clin Immunol Pract*. 2021 October; 9(10):3814-3815.e4. doi: 10.1016/j.jaip.2021.05.028. Epub 2021 Jun. 11. PMID: 34126272, which is incorporated herein by reference in its entirety.

The present invention overcomes the drawbacks of earlier attempts to formulate epinephrine as a dry powder for nasal delivery. Such non-aqueous formulations reduce the susceptibility of the epinephrine product to heat-related degradation and extend product shelf life as previously discussed.

There are a number of physiological changes that occur during anaphylaxis with the most important leading to significant drop in blood pressure (hypotension, cardiovascular collapse) and bronchoconstriction (respiratory collapse). According to a US FDA product review, "the mechanism of action of epinephrine for anaphylaxis is based on its mixed pharmacology including its ability to activate α_1 , α_2 , β_1 , and β_2 receptors." See U.S. Food & Drug Administration. Multidisciplinary Review. NDA 201-739, S-008 and S-009. Auvi-Q (epinephrine injection, USP) Auto-Injector. September 2017, available at www.fda.gov/media/109468/download, which is incorporated herein by reference in its entirety. Epinephrine's ability to activate α -receptors may decrease mucosal edema and membrane leakage, and most importantly increase vasoconstriction and vascular resistance, which may counteract the significant drop in blood pressure seen in anaphylaxis. Epinephrine's β_2 effects

importantly leads to bronchodilation and can help reverse the severe bronchoconstriction that can occur during anaphylaxis.

One formulation (Fleming—U.S. Pat. No. 9,789,071) described the potential for delivery of epinephrine in a powder form, but this formulation depended on the use of an alpha-adrenergic blocker as a vasodilator. One of the primary therapeutic goals of administering epinephrine during anaphylactic events is to counteract the drop in peripheral circulatory blood pressure, so the use of alpha-adrenergic antagonists as disclosed by Fleming is problematic because these agents act to widen blood vessels in smooth muscle and counteract the intended therapeutic action of epinephrine itself. Furthermore, unlike the currently available injectable epinephrine devices, this approach promotes the administration of two active adrenergic receptor agents (i.e., drugs) versus just one drug. Administering more than one drug at a time may pose additional risks. Advantageously, the present invention does not require administration of an additional vasodilator.

There is a long-standing, unmet need for a formulation of epinephrine that is operable to be used without a needle or syringe. Additionally, there is a long-standing, unmet need for a dry powder formulation of epinephrine that is operable to be successfully delivered by the nasal route and which does not contain vasodilating ingredients in the alpha blocker drug class. Further, there is a long-standing, unmet need for the formulation to maintain its integrity when exposed to high temperature conditions. There is also a long-standing, unmet need for a dry formulation of epinephrine with a consistent particle size that does not enter the lungs. Advantageously, the dry powder formulation advances the ability to make epinephrine readily available in emergency situations for treatment of anaphylactic reactions and other indications.

Referring now to the drawings in general, the illustrations are for the purpose of describing one or more preferred embodiments of the invention and are not intended to limit the invention thereto.

Provided herein are pharmaceutical formulations comprising dry powder epinephrine and at least one dry powder enabling agent that have optimized systemic delivery of epinephrine through the nasal passages. The formulations preferably include at least one enabling agent, at least one carrier, and/or at least one flow agent that are combined in various ratios and employed to optimize the systemic delivery of epinephrine through the nasal passages. In some instances, the formulations include 0.1% to 75% w/w epinephrine. In one embodiment, the formulations include a salt of epinephrine. In one embodiment, the salt is epinephrine hydrochloride. Alternatively, the salt is epinephrine acetate, epinephrine tartrate, epinephrine bitartrate, epinephrine hydrogen tartrate, or epinephrine borate. In some instances, the formulations include 0.1% to 20% w/w of the at least one enabling agent. In some instances, the at least one enabling agent includes, but is not limited to, at least one mucoadhesive, at least one absorption enhancer, at least one permeability enhancer, at least one surfactant, at least one surface modifier, at least one sustained release agent, at least one vasodilator (e.g., a natural vasodilator, such as niacin), at least one anticaking agent, and at least one disintegrant. In some embodiments, the formulation includes at least one excipient that impacts the characteristics of the dry powder formulation that directly or indirectly affect the transport of epinephrine across the nasal mucosa. In some embodiments, the formulation includes at least one excipient that facilitates solubilization or processing during the manufacturing pro-

cess to result in a dry powder formulation having the desired characteristics. In some embodiments, the at least one enabling agent includes chitosan, hyaluronic acid (e.g., sodium hyaluronate), sodium carboxymethyl cellulose (NaCMC), cyclodextrin, niacin (e.g., nicotinic acid), caffeine, sodium taurocholate, carnitine hydrochloride, dimethyl-beta-cyclodextrin, polysorbate (e.g., Polysorbate 80 (e.g., TWEEN® 80), sodium chloride (NaCl), potassium chloride (KCl), Poloxamer 188, L-leucine, histidine, glycine, arginine, crospovidone, a polyacrylic acid polymer (e.g., CARBOPOL® 934), magnesium stearate, ethylenediaminetetraacetic acid (EDTA), sodium starch glycolate, lactose (e.g., D-lactose), microcrystalline cellulose, mannitol, and/or a mixture of mannitol and hydroxypropyl methylcellulose (HPMC). In one embodiment, the intranasal dry powder formulations are spray-dried powder formulations.

In some embodiments, a single dose of epinephrine is about 0.01 mg to about 10 mg. In some embodiments, the amount of epinephrine is at least about: 0.01 mg, 0.05 mg, 1.0 mg, 2.0 mg, 5.0 mg, or 10 mg in the formulation. In some embodiments, the epinephrine present in the formulation is about: 0.01 mg to 0.05 mg, 0.05 mg to 0.75 mg, 0.75 mg to 1.5 mg, 1.5 mg to 3.0 mg, 3.0 to 4.5 mg, 4.5 to 6.0 mg, 6.0 to 7.5 mg, 7.5 to 9.0 mg, or 9.0 to 10.0 mg. In some embodiments, the amount of epinephrine is operable to be about: 0.15, 0.3, 0.5, 0.75, 1, 1.5, 2, 3, 4, 5, 6, 7, 8, 9, or 10 mg in the formulation. In one instance, a single dose of epinephrine is about 0.75 mg. In one instance, a single dose of epinephrine is about 1.5 mg. In another instance, a single dose of epinephrine is about 3.0 mg. In some embodiments, the formulation, in the form of a single dose, contains about 0.01 mg to about 10 mg of epinephrine. In some embodiments, the formulation, in the form of a single dose, contains about 0.75 mg, 1.5 mg, or 3.0 mg of epinephrine. In some embodiments, based on a "standard" patient weight of 70 kg, the dose of epinephrine is operable to be adjusted according to an increased or decreased weight of the patient relative to the "standard" patient weight. In one embodiment, the dose of epinephrine is operable to be adjusted according to the increased or decreased weight of the patient at an increased or decreased increment of at least 0.01 mg/kg, respectively.

In one embodiment, a unit dosage of epinephrine herein ranges from about 0.01 mg to about 1 mg, for example about: 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, or 1 mg in the formulation. In another embodiment, the unit dosage of epinephrine is also at least about: 0.01, 0.1, 0.5, or 1 mg in the formulation.

In some embodiments, the epinephrine is about 0.25% to about 50% w/w of the weight of the formulation, for example about: 0.25%, 0.5%, 1%, 2%, 3%, 4%, 5%, 6%, 7%, 7.5%, 8%, 9%, 10%, 11%, 12%, 13%, 14%, 15%, 16%, 17%, 18%, 19%, 20%, 25%, 30%, 35%, 40%, 45%, or 50% w/w, based on the weight of the formulations and/or dosage units. For example, the epinephrine is about 4%, about 7.5%, or about 15% w/w of the weight of the formulation. In some embodiments, epinephrine is present in an amount of at least about: 0.25% w/w, 1% w/w, 5% w/w, 10% w/w, 20% w/w, 30% w/w, 40% w/w, or 50% w/w based on the weight of the formulations and/or dosage units. In some embodiments, epinephrine is present in an amount of about: 0.25% to 1% w/w, 1% to 5% w/w, 5% to 10% w/w, 10% to 20% w/w, 20% to 30% w/w, 30% to 40% w/w, or 40% to 50% w/w based on the total weight of the formulations and/or dosage units. In one embodiment, epinephrine is present in an amount of about 1% w/w to about 10% w/w based on the total weight of the formulations and/or dosage units.

In some embodiments, the dry powder formulations provided herein, when administered to a patient, produce a maximal blood concentration (C_{max}) of epinephrine that is at least about: 2- to 3-fold, 3- to 5-fold, 5- to 7-fold, or 7- to 10-fold more than the baseline level of epinephrine in the patient. In some embodiments, the dry powder formulations provided herein, when administered to a patient, produce a maximal blood concentration (C_{max}) of epinephrine at least 2-, 3-, 4-, 5-, 6-, 7-, 8-, 9-, or 10-fold more than the baseline level of epinephrine in the patient. In one embodiment, the dry powder formulations provided herein, when administered to a patient, produce a maximal blood concentration (C_{max}) of epinephrine at least 2-fold more than the baseline level of the epinephrine in the patient. In one embodiment, the formulations provided herein, when administered to a patient, increase the blood concentration of epinephrine by about 0.01 to 0.1 µg/mL. In one embodiment, the formulations provided herein, when administered to a patient, increase the blood concentration of epinephrine by about 0.01, 0.02, 0.03, 0.04, 0.05, 0.06, 0.07, 0.08, 0.09, or 0.1 µg/mL.

In some embodiments, the dry powder formulations provided herein, when administered to a patient, reach a maximal blood concentration of epinephrine in less than about 60 minutes (T_{max}) after administration. In some embodiments, the dry powder formulations provided herein, when administered to a patient, reach a maximal blood concentration (T_{max}) of epinephrine in less than about 60, 50, 40, 30, 20, 15, 10, 5, 3, or 1 minute(s) (T_{max}) after administration. In some embodiments, the dry powder formulations provided herein, when administered to a patient, reach a mean area under the curve (AUC) over a time period (e.g., 0-180 minutes) of epinephrine that is at least 20%, 30%, 40% 50%, 60%, 70%, 80%, 90%, 100%, 110%, 120%, 130%, 140%, or 150% of the mean AUC over the time period of an IV, IM, or SQ injected epinephrine. In some embodiments, the dry powder formulations provided herein, when administered to a patient, reach a mean AUC over a time period (e.g., 0-∞) of epinephrine that is at least 20%, 30%, 40% 50%, 60%, 70%, 80%, 90%, 100%, 110%, 120%, 130%, 140% or 150% of the mean AUC over the time period of an IV, IM, or SQ injected epinephrine. In some embodiments, the IV, IM, or SQ injected epinephrine contains 0.10 mg, 0.15 mg, 0.20 mg, 0.25 mg, 0.30 mg, 0.35 mg, 0.40 mg, 0.45 mg, 0.50 mg, 0.55 mg, 0.60 mg, 0.65 mg, 0.70 mg, 0.75 mg, 0.80 mg, 0.85 mg, 0.90 mg, 0.95 mg, or 1.0 mg of epinephrine. For example, the dry powder formulations provided herein, when administered to a patient, reach a mean AUC over a time period (e.g., 0-180 minutes) of epinephrine that is at least 80% of the mean AUC over the time period of a 0.15 mg IV injected epinephrine. In another instance, the dry powder formulations provided herein, when administered to a patient, reach a mean AUC over a time period (e.g., 0-∞) of epinephrine that is at least 80% of the mean AUC over the time period of a 0.15-1 mg IV injected epinephrine. In some embodiments, the IV, IM, or SQ injected epinephrine is injected by EPIPEN® auto-injector. In some embodiments, the dry powder formulations provided herein, when administered to a patient, reach a mean AUC over a time period (e.g., 0-180 minutes) of epinephrine that is at least 50,000 pg·min/mL, 100,000 pg·min/mL, 200,000 pg·min/mL, 300,000 pg·min/mL, 400,000 pg·min/mL, 500,000 pg·min/mL, 600,000 pg·min/mL, 700,000 pg·min/mL, 800,000 pg·min/mL, 900,000 pg·min/mL, or 1,000,000 pg·min/mL. In some embodiments, the dry powder formulations provided herein, when administered to a patient, reach a mean AUC over a time period (e.g., 0-∞) of epinephrine that is at least 50,000

pg·min/mL, 100,000 pg·min/mL, 200,000 pg·min/mL, 300,000 pg·min/mL, 400,000 pg·min/mL, 500,000 pg·min/mL, 600,000 pg·min/mL, 700,000 pg·min/mL, 800,000 pg·min/mL, 900,000 pg·min/mL, 1,000,000 pg·min/mL, 1,200,000 pg·min/mL, 1,400,000 pg·min/mL, 1,600,000 pg·min/mL, 1,800,000 pg·min/mL, or 2,000,000 pg·min/mL.

Pharmacokinetics of auto-injectors are discussed in Dreborg S, Kim H. The pharmacokinetics of epinephrine/adrenaline autoinjectors. *Allergy Asthma Clin Immunol*. 2021 Mar. 8; 17(1):25. doi: 10.1186/513223-021-00511-y. PMID: 33685510; PMCID: PMC7938517, which is incorporated herein by reference in its entirety.

In certain embodiments, the dry powder formulations and/or unit doses provided herein is operable to raise the blood concentration of epinephrine in a subject to about 0.02 µg/mL within about less than 1 minute to about 60 minutes (e.g., about: 60, 50, 40, 30, 20, 15, 10, 5, 3, <1 minutes), or about 10 µg/mL within about less than 1 minute to 15 minutes (e.g., about: <1, 3, 5, 10, 11, 12, 13, 14, or 15 minutes) of intranasal administration. In one embodiment, the formulation provided herein increase the blood concentration of epinephrine by about 0.01 to 0.04 µg/mL, for example 0.02 or 0.03 µg/mL, in about less than 1 minute to about 15 minutes (e.g., about: <1, 10, 11, 12, 13, 14, or 15 minutes), or about 3 µg/mL within about less than 1 minute to about 60 minutes (e.g., about: 60, 50, 40, 30, 20, 15, 10, 5, 3, <1 minutes).

In another embodiment, a single dose of epinephrine in the dry powder formulations and/or dosage units given intranasally is preferably bioequivalent (for example, in terms of peripheral blood levels, systemic exposure of epinephrine) to intravenously (IV), intramuscularly (IM), or subcutaneously (SQ) injected epinephrine (e.g., using EPIPEN® auto-injector of 0.3 mg for adult patients (e.g., anyone over 30 kg) or 1 mg IV epinephrine, using EPIPEN JR® auto-injector of 0.15 mg for pediatric patients or 0.01 mg/kg with a dilution of 0.1 mg/mL for IV epinephrine). In one example, bioequivalence includes a 90% confidence interval of a mean T_{max} (e.g., the time to reach maximal blood concentration), a mean C_{max} (e.g., maximal blood concentration), a mean AUC over a period of time (e.g., area under the plasma/serum/blood concentration-time curve from time zero to time t), and/or a mean AUC over a period of time from 0 to infinity (e.g., area under the plasma/serum/blood concentration-time curve from time zero to time infinity) are within 80.00% to 125.00% of a reference test. In one embodiment, bioequivalence is determined in a fasting state.

Vasodilators

In one embodiment, the dry powder formulation further includes a vasodilator; that is, a second vasodilator in addition to epinephrine. The vasodilator is preferably not an alpha-adrenergic blocker. The use of alpha-adrenergic antagonists is problematic because these agents act to widen blood vessels in smooth muscle and counteract the intended therapeutic action of epinephrine itself. Furthermore, unlike the currently available injectable epinephrine devices, this approach promotes the administration of two active adrenergic receptor agents (i.e., drugs) versus just one drug. In a preferred embodiment, the dry powder formulation does not include an additional vasodilator. Administering more than one drug at a time may pose additional risks.

In one embodiment, the vasodilator is caffeine or niacin. In some embodiments, a single dose of the formulation includes about 0.01 mg to about 10 mg of the vasodilator. In some embodiments, a vasodilator (e.g., niacin) herein is present in the formulation in an amount of about: 0.001 mg

to 0.01 mg, 0.01 mg to 0.05 mg, 0.05 to 0.1 mg, 0.1 to 0.5 mg, 0.5 to 0.75 mg, 0.75 mg to 1.5 mg, 1.5 mg to 3.0 mg, 3.0 to 4.5 mg, 4.5 to 6.0 mg, 6.0 to 7.5 mg, 7.5 to 9.0 mg, or 9.0 to 10.0 mg. In some embodiments, a vasodilator (e.g., niacin) herein is operable to be at least about: 0.001 mg, 0.01 mg, 0.1 mg, 0.5 mg, 1.0 mg, 2.0 mg, 5.0 mg, or 10.0 mg in the formulation and/or unit doses. In some embodiments, a single dose of the formulation comprises about 0.5 mg or about 1.0 mg. In some embodiments, the formulation, in the form of a single dose, contains about 0.01 mg to about 10 mg of the vasodilator. In some embodiments, the formulation, in the form of a single dose, contains about 0.5 mg or about 1.0 mg of the vasodilator.

In some embodiments, the amount of the vasodilator is about 0.005% to about 50% w/w of the weight of the formulation, for example about: 0.005%, 0.01%, 0.05%, 0.1%, 0.5%, 1%, 2%, 2.5%, 3%, 4%, 5%, 6%, 7%, 8%, 9%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, or 50% w/w of the weight of the formulation. In some embodiments, the amount of the vasodilator is about 2.5% w/w of the weight of the formulation.

Other vasodilators that are less preferred, but may be incorporated into the present invention, include ethanol. In one example, a powder form of ethanol is used as a vasodilator. However, some of the powder form of ethanol might be lost during the manufacturing process, so this embodiment is less preferred than other embodiments.

Enabling Agents

In one embodiment, the intranasal dry powder compositions and/or unit doses include at least one enabling agent. The at least one enabling agent includes, but is not limited to, at least one catechol-o-methyl transferase (COMT) inhibitor, at least one epinephrine potentiator, at least one atropine potentiator, at least one mucoadhesive, at least one absorption enhancer, at least one permeability enhancer, at least one surfactant, at least one surface modifier, at least one sustained release agent, at least one anticaking agent, at least one systemic vasodilator, at least one nasal mucosal vasodilator, at least one mucosal permeation enhancer, at least one agent that reduces mucosal transit time, at least one agent that increases mucosal absorption or adhesion or transport, at least one chelator, at least one steroid, at least one non-sulfite stabilizer, at least one preservative, at least one thickening agent, at least one humectant, at least one antihistamine, at least one solubilizing agent, at least one masking agent (e.g., taste, smell), at least one antioxidant, at least one viscosity enhancing agent, at least one dispersing agent, at least one pharmaceutically acceptable excipient, and/or at least one colorant. In some instances, the enabling agent includes at least one agent that reduces mucosal transit time, at least one agent that increases mucosal absorption and/or adhesion, at least one agent that enhances mucosal transport, or the enantiomers, diastereoisomers, racemates, or salts of such compounds with pharmaceutically acceptable counterions.

Additional details regarding nasal delivery of drugs, including information regarding enabling agents, are disclosed in (1) Bourganis V, Kammona O, Alexopoulos A, Kiparissides C. Recent advances in carrier mediated nose-to-brain delivery of pharmaceuticals. *Eur J Pharm Biopharm.* 2018 July; 128:337-362. doi: 10.1016/j.ejpb.2018.05.009. Epub 2018 May 4. PMID: 29733950; (2) Davis S S, Illum L. Absorption enhancers for nasal drug delivery. *Clin Pharmacokinet.* 2003; 42(13):1107-28. doi: 10.2165/00003088-200342130-00003. PMID: 14531723; (3) Gänger S, Schindowski K. Tailoring Compositions for Intranasal Nose-to-Brain Delivery: A Review on Architecture, Physico-

Chemical Characteristics and Mucociliary Clearance of the Nasal Olfactory Mucosa. *Pharmaceutics.* 2018 Aug. 3; 10(3):116. doi: 10.3390/pharmaceutics10030116. PMID: 30081536; PMCID: PMC6161189; and (4) Tiozzo Fasiolo L, Manniello M D, Tratta E, Buttini F, Rossi A, Sonvico F, Bortolotti F, Russo P, Colombo G. Opportunity and challenges of nasal powders: Drug composition and delivery. *Eur J Pharm Sci.* 2018 Feb. 15; 113:2-17. doi: 10.1016/j.ejps.2017.09.027. Epub 2017 Sep. 20. PMID: 28942007, each of which is incorporated herein by reference in its entirety.

In one embodiment, the intranasal dry powder compositions and/or unit doses include at least one COMT inhibitor. In some embodiments, the COMT inhibitor is a reversible COMT inhibitor. In one embodiment, the COMT inhibitor is entacapone or a pharmaceutically acceptable salt thereof. In some embodiments, a single dose of the COMT inhibitor is about 5 mg to about 800 mg. In one embodiment, a single dose of the COMT inhibitor is about 5 mg to about 200 mg. In some embodiments, the COMT inhibitor herein is present in the compositions in about: 0.05 mg to 0.1 mg, 0.1 mg to 0.5 mg, 0.5 mg to 0.75 mg, 0.75 mg to 1.5 mg, 1.5 mg to 3.0 mg, 3.0 mg to 4.5 mg, 4.5 mg to 6.0 mg, 6.0 mg to 7.5 mg, 7.5 mg to 9.0 mg, 9.0 mg to 10.0 mg, 10 mg to 25 mg, 25 mg to 50 mg, 50 mg to 100 mg, 100 mg to 150 mg, 150 mg to 200 mg, or 200 mg to 400 mg. In some embodiments, the COMT inhibitor herein is at least about: 0.001 mg, 0.01 mg, 0.1 mg, 0.5 mg, 1.0 mg, 2.0 mg, 5.0 mg, or 10.0 mg in the compositions and/or unit doses. In some embodiments, a single dose of the COMT inhibitor is about 0.5 mg, about 1.0 mg, about 5.0 mg, about 10.0 mg, about 25 mg, about 50 mg, about 75 mg, about 100 mg, about 150 mg, or about 200 mg. In some embodiments, the composition, in the form of a single dose, includes about 0.01 mg to about 25 mg of the COMT inhibitor. In some embodiments, the composition, in the form of a single dose, includes about 0.5 mg, about 1.0 mg, about 5.0 mg, about 10.0 mg, about 25 mg, about 50 mg, about 75 mg, about 100 mg, about 150 mg, or about 200 mg of the COMT inhibitor.

In one embodiment, the intranasal dry powder compositions and/or unit doses include at least one epinephrine potentiator. In one embodiment, the at least one epinephrine potentiator includes, but is not limited to, histidine, a flavonoid, a local anesthetic (e.g., benzocaine, lidocaine), a COMT inhibitor, levothyroxine sodium, at least one antihistamine, at least one tricyclic antidepressant, and/or a monoamine oxidase inhibitor (MAO) inhibitor. In one embodiment, the at least one tricyclic antidepressant includes amitriptyline, amoxapine, desipramine, doxepin, imipramine, nortriptyline, protriptyline, and/or trimipramine.

In one embodiment, the intranasal dry powder compositions and/or unit doses include at least one atropine potentiator. In one embodiment, the at least one atropine potentiator includes, but is not limited to, amantadine, at least one antihistamine, at least one tricyclic antidepressant, quinine, and/or disopyramide. In one embodiment, the at least one tricyclic antidepressant includes amitriptyline, amoxapine, desipramine, doxepin, imipramine, nortriptyline, protriptyline, and/or trimipramine.

In one embodiment, the intranasal dry powder compositions and/or unit doses include at least one mucoadhesive. In one embodiment, the at least one mucoadhesive includes a starch, crystalline cellulose, a cellulose derivative, a polymer (e.g., chitosan, a carbopol (e.g., carbopol 943), carboxiphil, carbomer), a polyacrylic acid or polyacrylic acid derivative, a protein (e.g., mucin, lactoferrin, transferrin),

and/or lecithin. See, e.g., (1) Takeuchi H, Thongborisute J, Matsui Y, Sugihara H, Yamamoto H, Kawashima Y. Novel mucoadhesion tests for polymers and polymer-coated particles to design optimal mucoadhesive drug delivery systems. *Adv Drug Deliv Rev.* 2005 Nov. 3; 57(11):1583-94. doi: 10.1016/j.addr.2005.07.008. Epub 2005 Sep. 16. PMID: 16169120; (2) D Tabor, Surface forces and surface interactions, *J. Colloid Interface Sci.*, Volume 58, Issue 1, 1977, Pages 2-13, [https://doi.org/10.1016/0021-9797\(77\)90366-6](https://doi.org/10.1016/0021-9797(77)90366-6); and (3) Robert J Good, Surface free energy of solids and liquids: Thermodynamics, molecular forces, and structure, *J. Colloid Interface Sci.*, Volume 59, Issue 3, 1977, Pages 398-419, [https://doi.org/10.1016/0021-9797\(77\)90034-0](https://doi.org/10.1016/0021-9797(77)90034-0).

In one embodiment, the intranasal dry powder compositions and/or unit doses include at least one absorption enhancer. In one embodiment, the at least one absorption enhancer includes a flavonoid (e.g., Vitamin P-like compound), vasopressin, methylphenidate, tropolone, desmethyl papaverine, pyrogallol, an amino acid (e.g., histidine), an antihistamine, an amphetamine, a local anesthetic, norepinephrine, isoproterenol, hydrocortisone, tripeleminamine, bufotenine, harmine, methergine, a ganglionic blocker, guanethidine, mescaline, cocaine, lysergic acid diethylamide (LSD), or an enantiomer, diastereoisomer, racemate, prodrug, or salt of such compounds.

In one embodiment, the intranasal dry powder compositions and/or unit doses include at least one permeability enhancer and/or at least one mucosal permeation enhancer. In one embodiment, the at least one permeability enhancer and/or at least one mucosal permeation enhancer increases a fraction of the at least one active pharmaceutical ingredient that reaches circulation by at least about 10%, at least about 25%, preferably at least 50%, and most preferably at least 100%. In one embodiment, the at least one permeability enhancer includes a bile salt, an alkyl glycoside, a polymer, a tight junction modulation peptide, a lipid, a surfactant, a cyclodextrin, a chelator (e.g., EDTA), a Hsieh enhancer, a cyclic lactone, a cyclic diester, a cyclic ketone, a fatty acid, a salicylate, and/or an amphiphilic steroid (e.g., a fusidic acid derivative). In one embodiment, the alkyl glycoside includes an alkyl chain having between 8 and 20 carbons. In one embodiment, the alkyl glycoside is undecyl maltoside, dodecyl maltoside, tridecyl maltoside, tetradecyl maltoside, sucrose mono-dodecanoate, sucrose mono-tridecanoate, sucrose mono-tetradecanoate, and/or dodecyl-beta-D-maltoside. Tight junction modulating peptides are described in U.S. Patent Publication No. 20090220435, which is incorporated herein by reference in its entirety. In one embodiment, the cyclodextrin includes alpha-cyclodextrin, beta-cyclodextrin, gamma-cyclodextrin, hydroxypropyl-beta-cyclodextrin, and/or sulfobutylether beta-cyclodextrin. In one embodiment, the lipid includes 1,2-Dioleoyl-sn-Glycero-3 Ethylphosphocholine, 1,2-di-O-phytanyl-glycero-3-phosphocholine, 1-O-hexadecyl-2-acetyl-sn-glycerol, 1-O-octadecyl-2-O-methyl-glycerol-3-phosphocholine, 16:0-09:0(ALDO)PC, 16:0-09:0(COOH)PC, 3-beta-hydroxy-5alpha-cholest-8(14)-en-15-one, C10 sucrose, C12 maltose, C12 sucrose, C14 maltose, C16-09:0, C6 glucose, C6 maltose, C7 glucose, C8 glucose, Cardiolipin (sodium salt), Ceramide (brain porcine), Ceramide C10:0, Ceramide C12:0, Ceramide C14:0, Ceramide C16:0, Ceramide C17:0, Ceramide C18:0, Ceramide C18:1, Ceramide C20:0, Ceramide C24:0, Ceramide C24:1, Ceramide C2:0, Ceramide C4:0, Ceramide C6:0, Ceramide C8:0, Cerebroside (brain porcine), Cerebroside Sulfatide (porcine), Dimethyl sphingosine, Egg Ceramide, Galactosyl sphingosine, Glucosyl-sphingosine, Lactosyl(B) Sphingosine, Lyso-PAF, N-acetyl

ceramide-1-phosphate, N-octanoyl ceramide-1-phosphate, PGPC1, POVPC, Phosphatidylinositol (Soy), Phosphatidylinositol (bovine), Platelet-Activation Factor, Porcine brain ganglioside, Sphingomyelin (brain porcine), Sphingosine-1-phosphate, and trimethylsphingosine. The lipid is preferably a glycosylated sphingosine, an alkylglucoside, an oxidized lipid, and/or an ether lipid (PAF). In one embodiment, the fatty acid is sodium caprate, sodium laurate, sodium caprylate, capric acid, lauric acid, caprylic acid, and/or an acyl carnitine (e.g., palmitoyl carnitine, stearoyl carnitine, myristoyl carnitine, lauroyl carnitine). In one embodiment, the salicylate is sodium salicylate, 5-methoxy salicylate, and methyl salicylate. Hsieh enhancers are described in U.S. Pat. Nos. 5,023,252 and 5,731,303, each of which is incorporated herein by reference in its entirety. Cyclic lactones, cyclic diesters, and cyclic ketones are described in U.S. Pat. No. 8,481,043, which is incorporated herein by reference in its entirety. Amphiphilic steroids are discussed in U.S. Pat. Nos. 4,548,922 and 4,746,508, each of which is incorporated herein by reference in its entirety. In one preferred embodiment, the at least one permeability enhancer and/or the at least one mucosal permeation enhancer is a generally accepted as safe (GRAS) pharmaceutical excipient. Alternatively, the at least one permeability enhancer and/or the at least one mucosal permeation enhancer is a near-GRAS excipient and/or a non-GRAS excipient. In one embodiment, the at least one permeation enhancer and/or the at least one mucosal permeation enhancer is about 1% to about 30% w/w of the weight of the composition.

In one embodiment, the intranasal dry powder compositions and/or unit doses include at least one surfactant. The at least one surfactant is a non-ionic surfactant, an ionic surfactant, a cationic surfactant, an anionic surfactant, and/or a zwitterionic surfactant. Examples of the at least one surfactant compatible with the present invention include, but are not limited to, sodium glycocholate, sodium taurocholate, polyoxyethylene lauryl ether, polyacrylic acid gel, sodium lauryl sulfate, polysorbate, and/or sodium deoxycholate.

In one embodiment, the intranasal dry powder compositions and/or unit doses do not include a surfactant. Some liquid compositions of drugs require a surfactant to prevent aggregation of the active ingredient. Advantageously, dry powder compositions do not require a surfactant.

In one embodiment, the intranasal dry powder compositions and/or unit doses include at least one surface modifier. In one embodiment, the at least one surface modifier includes a lubricant (e.g., magnesium stearate), a fluidizing agent (e.g., talc, silicon dioxide), a nitric oxide (NO) stimulator, chitosan, and/or a chitosan derivative.

In one embodiment, the intranasal dry powder compositions and/or unit doses include at least one sustained release agent. In one embodiment, the at least one sustained release agent is achieved by manipulating one or more of the at least one active pharmaceutical ingredient to control its dissolution and/or the composition in which the at least one active pharmaceutical ingredient is suspended. In one embodiment, excipients with mucoadhesive and/or viscosity enhancing characteristics are incorporated. Additionally or alternatively, the composition is operable to reversibly diminish mucociliary clearance.

In one embodiment, the intranasal dry powder compositions and/or unit doses include at least one anticaking agent. The at least one anticaking agent includes, but is not limited to, tribasic calcium phosphate. In one embodiment, the at least one anticaking agent is about 0.5% to about 5% w/w of

the weight of the composition. In some embodiments, the at least one anticaking agent has an average particle diameter of about 100 μm or less, for example about: 90 to 100 μm , 80 to 90 μm , 70 to 80 μm , 60 to 70 μm , 50 to 60 μm , 40 to 50 μm , 30 to 40 μm , 20 to 30 μm , or 10 to 20 μm . In some

embodiments, the at least one anticaking agent has an average particle diameter of about 30 μm to 100 μm . In one embodiment, the intranasal dry powder compositions and/or unit doses include at least one systemic vasodilator and/or at least one nasal mucosal vasodilator. In one embodiment, the at least one systemic vasodilator and/or the at least one nasal mucosal vasodilator includes an angiotensin-converting enzyme (ACE) inhibitor (e.g., Benazepril (Lotensin), Captopril (Capoten), Enalapril (Vasotec), Fosinopril (Monopril), Lisinopril (Prinivil, Zestril), Minoxidil (Loniten), Meoexipril (Univasc), Perindopril (Aceon), Quinapril (Accupril), Ramipril (Altace), Trandolapril (Mavik)), an angiotensin II receptor antagonist (e.g., Losartan, Candesartan, Valsartan, Irbesartan, Telmisartan, Eprosartan, Olmesartan, Azilsartan), phenolamine, nitroglycerine, hydralazine, isosorbide mononitrate, isosorbide dinitrate, papaverine hydrochloride or mesylate, cocaine, ethyl nitrate, diltiazem, urapidil, nicorandil, sodium nitroprusside, glyceryl trinitrate-verapamil, phenoxybenzamine, dopexamine, chlorpromazine, propiverine hydrochloride, or an enantiomer, diastereoisomer, racemate, prodrug, or salt of such compounds. In a preferred embodiment, the at least one systemic vasodilator and/or the at least one nasal mucosal vasodilator is phenolamine.

In one embodiment, the intranasal dry powder compositions and/or unit doses include at least one agent that reduces mucosal transit time. In one embodiment, the at least one agent that reduces mucosal transit time includes a polyacrylate mucoadhesive agent and/or a peptide. See, e.g., WIPO Publication No. WO2003037355, which is incorporated herein by reference in its entirety.

In one embodiment, the intranasal dry powder compositions and/or unit doses include at least one agent that increases mucosal absorption or adhesion or transport. In one embodiment, the at least one agent that increases mucosal absorption or adhesion or transport includes a surfactant, gelling microsphere, chitosan, sodium lauryl sulfate, sodium salicylate, oleic acid, lecithin, dehydrated alcohol, Tween, polyoxyl 40 stearate, polyoxyl ethylene 40 stearate, propylene glycol, hydroxyl fatty acid ester of polyethylene glycol, glycerol monooleate, fusieates, a bile salt, octoxynol, polysorbate 20, polysorbate 80, DDPC, DPPC, a chelator (e.g., EDTA, EGTA, citrate), and/or a surfactant. See, e.g., (1) Illum L and Fisher A N (1997) Intranasal delivery of peptides and proteins, in *Inhalation Delivery of Therapeutic Peptides and Proteins* (Adjei A L and Gupta P K eds), Marcel Dekker, New York and (2) Costantino H R, Illum L, Brandt G, Johnson P H, Quay S C. Intranasal delivery: physicochemical and therapeutic aspects. *Int J Pharm.* 2007 Jun. 7; 337(1-2):1-24. doi: 10.1016/j.ijpharm.2007.03.025. Epub 2007 Mar. 25. PMID: 17475423, each of which is incorporated herein by reference in its entirety.

In one embodiment, the intranasal dry powder compositions and/or unit doses include at least one chelator. In one embodiment, the at least one chelator includes ethylenediaminetetraacetic acid (EDTA), ethylene glycol-bis(β -aminoethyl ether)-N,N,N',N'-tetraacetic acid (EGTA), and/or citrate.

In one embodiment, the intranasal dry powder compositions and/or unit doses include at least one steroid. In one embodiment, the at least one steroid includes at least one

corticosteroid. In one embodiment, the at least one steroid is hydrocortisone, beclomethasone, fluticasone, triamcinolone, flunisolide, mometasone, ciclesonide, and/or budesonide.

In one embodiment, the intranasal dry powder compositions and/or unit doses include at least one non-sulfite stabilizer. In one embodiment, the at least one non-sulfite stabilizer is ascorbic acid.

In one embodiment, the intranasal dry powder compositions and/or unit doses include at least one preservative. The at least one preservative includes, but is not limited to a paraben, benzalkonium chloride, phenyl ethyl alcohol, ethylenediaminetetraacetic acid (EDTA), benzoyl alcohol, a thiol, glutathione, glutathione reductase, glutathione peroxidase, hydroquinone, amikasin sulfate, apomorphine hydrochloride, metaraminol, levobunolol, levobunolol hydrochloride, acamprosate calcium, fenoldopam, hydrocortisone/neomycin sulfate/polymyxin B, dexamethasone sodium phosphate, hydromorphone, dobutamine, epinephrine, etidicaine/epinephrine bitartrate, gentamycin, tinzaparin, isoproterenol, ketoconazole, sodium sulfacetamide, norepinephrine, bupivacaine/epinephrine bitartrate, morphine, tobramycin, rotigotine, orphenadrine, procaine, nalbuphine, oxytetracycline, nortriptyline, perphenazine, promethazine hydrochloride, prednisolone acetate, propofol, mesalamine, trimethoprim/sulfamethoxazole, carisoprodol/aspirin/codeine, streptomycin, mafenide acetate, tetracycline hydrochloride, pentazocine lactate, chlorpromazine, triethylperazine maleate, fluorinolone acetone/hydroquinone/tretinoin, acetaminophen/codeine, doxycycline calcium, and/or lidocaine/epinephrine. In one embodiment, the at least one preservative is about 0.01% to about 5% w/w of the weight of the composition, for example about: 0.01%, 0.05%, 0.1%, 0.2%, 0.5%, 1%, 2%, 3%, 4%, or 5% w/w based on the weight of the compositions and/or unit doses. In one embodiment, the at least one preservative is about 0.01% to 5%, 0.02% to 4%, or 0.05% to 2.5% w/w based on the weight of the compositions and/or unit doses. In a preferred embodiment, the at least one preservative is sulfite-free.

In a preferred embodiment, the intranasal dry powder compositions and/or unit doses do not include a preservative. Preservatives (e.g., sodium bisulfite, which is present in EpiPen® and other autoinjectors) can cause an allergic reaction in some individuals. Because aqueous compositions of drugs are often sensitive to light and heat, they generally include a preservative to improve stability. The intranasal dry powder compositions and/or unit doses of the present invention advantageously do not require a preservative. Additionally, not including a preservative in the intranasal dry powder compositions and/or unit doses reduces the risk of further allergic reaction(s) and/or sensitivities. "Despite documentation of sensitivity, sulfites should not be withheld from patients experiencing a life-threatening emergency. Non-sulfited alternatives are often available, and should be used preferentially." See, e.g., Roth J V, Shields A. A dilemma: How does one treat anaphylaxis in the sulfite allergic patient since epinephrine contains sodium metabisulfite? *Anesth Analg.* 2004 May; 98(5):1499; author reply 1500. doi: 10.1213/01.ane.0000120092.39021.f2. PMID: 15105239, which is incorporated herein by reference in its entirety. Also see, e.g., Susan C. Smolinske (1992) Review of Parenteral Sulfite Reactions, *Journal of Toxicology: Clinical Toxicology*, 30:4, 597-606, DOI: 10.3109/15563659209017945, which is incorporated herein by reference in its entirety. Drugs without sulfites are often available in a medical setting (e.g., hospital, clinic) because environmental conditions can be controlled. Aqueous prepa-

rations and auto-injectors generally contain preservatives because they are intended for ambient use. There is a long-standing, unmet need for epinephrine that does not contain a preservative.

In one embodiment, the intranasal dry powder compositions and/or unit doses include at least one thickening agent. In one embodiment, the at least one thickening agent includes microcrystalline cellulose and/or carboxymethylcellulose sodium.

In one embodiment, the intranasal dry powder compositions and/or unit doses include at least one humectant. The at least one humectant includes, but is not limited to, glycerine, glycerol, sorbitol, mannitol, and/or vegetable oil.

In one embodiment, the intranasal dry powder compositions and/or unit doses include at least one antihistamine. In one embodiment, the at least one antihistamine includes an H_1 receptor antagonist and/or an H_2 receptor antagonist. In one embodiment, the H_1 receptor antagonist includes an ethylenediamine, an ethanolamine, an alkylamine, a piperazine, a tricyclic, and/or a tetracyclic. In one embodiment, the at least one antihistamine includes loratadine, desloratadine, diphenhydramine, doxylamine, fexofenadine, chlorpheniramine, hydroxyzine, cetirizine, levocetirizine, brompheniramine, clemastine, carbinoxamine, azelastine, emadastine, mepyramine, promethazine, cyproheptadine, doxepin, mirazapine, cimetidine, famotidine, nizatidine, roxatidine, lafutidine, and/or levocabastine. Additional information regarding antihistamines is disclosed in U.S. Patent Publication No. 20100055152 and U.S. Pat. No. 8,263,581, each of which is incorporated herein by reference in its entirety.

In one embodiment, the intranasal dry powder compositions and/or unit doses include at least one solubilizing agent. The at least one solubilizing agent includes, but is not limited to, a glycol, an alcohol, 2-(2-ethoxyethoxy)ethanol, a cyclodextrin, and/or a glyceride (e.g., a medium chain glyceride, LABRASOL®).

In one embodiment, the intranasal dry powder compositions and/or unit doses include at least one masking agent (e.g., taste, smell). In a preferred embodiment, the at least one masking agent includes, but is not limited to, at least one sweetener and/or at least one flavoring agent. The at least one sweetener includes, but is not limited to, saccharin (e.g., sodium salt, calcium salt), fructose, dextrose, aspartame, acesulfame potassium, glycerin, sucralose, maltodextrin, sucrose, glucose, maltose, xylitol, sorbitol, erythritol, and/or mannitol. In one embodiment, the at least one masking agent includes phenethyl alcohol, vanilla, cherry, cinnamon, lavender, lemon, menthol, orange, peppermint, spearmint, raspberry, strawberry, grape, ethyl vanillin, coriander, ginger, nutmeg, cardamom, butterscotch, cocoa, acacia syrup, anethole, anise oil, benzaldehyde, ethyl acetate, methyl salicylate, and/or tolu. In one embodiment, the at least one masking agent is about 0.001% to about 1% w/w of the weight of the composition, for example about: 0.001%, 0.005%, 0.01%, 0.02%, 0.05%, 0.1%, 0.2%, 0.3%, 0.4%, 0.5%, 0.6%, 0.7%, 0.8%, 0.9%, or 1% w/w based on the weight of the compositions and/or unit doses. In one embodiment, the at least one masking agent is about 0.01% to 0.5%, 0.02% to 0.2%, or 0.015% to 0.15% w/w based on the weight of the compositions and/or unit doses.

In one embodiment, the intranasal dry powder compositions and/or unit doses include at least one antioxidant. The at least one antioxidant includes, but is not limited to, sodium metabisulfite, sodium bisulfate, butylated hydroxytoluene, tocopherol, ascorbic acid (Vitamin C), glutathione, glutathione reductase, glutathione peroxidase, superoxide dismutase (CuZn-SOD), superoxide reductase, carnosine,

ergothione, ovothioli, lipoic acid, thioctic acid, thioredoxin peroxidase, and/or recombinant thermostable variants thereof. In one embodiment, the at least one antioxidant is about 0.0001% to about 10% w/w of the weight of the composition, for example about: 0.0001%, 0.0005%, 0.001%, 0.005%, 0.01%, 0.02%, 0.05%, 0.1%, 0.5%, 1%, 2%, 3%, 4%, 5%, 6%, 7%, 8%, 9%, or 10% w/w based on the weight of the compositions and/or unit doses. In one embodiment, the at least one antioxidant is about 0.001% to 5%, 0.05% to 2%, or 0.1% to 1% w/w based on the weight of the compositions and/or unit doses.

In one embodiment, the intranasal dry powder compositions and/or unit doses include at least one viscosity enhancing agent. The at least one viscosity enhancing agent includes, but is not limited to, a cellulose derivative (e.g., crystalline cellulose, amorphous cellulose, methylcellulose, carboxymethylcellulose, ethylcellulose, hypromellose, hydroxypropyl cellulose, or a salt thereof), carrageenan, guar gum, an alginate, a carbomer, a polyethylene glycol, propylene glycol, a polyvinyl alcohol, xanthan gum, a polyvinylpyrrolidone (PVP), chitosan, a polysaccharide, a starch, and/or carbopol. In one embodiment, the at least one viscosity enhancing agent is about 0.1% to about 10% w/w of the weight of the composition, for example about: 0.1%, 0.5%, 1%, 2%, 3%, 4%, 5%, 6%, 7%, 7.5%, 8%, 9%, or 10% w/w based on the weight of the compositions and/or unit doses.

In one embodiment, the intranasal dry powder compositions and/or unit doses include at least one dispersing agent. In one embodiment, the at least one dispersing agent includes citric acid.

In one embodiment, the intranasal dry powder compositions and/or unit doses include at least one colorant. In a preferred embodiment, the at least one colorant is non-allergenic.

In one embodiment, the intranasal dry powder compositions and/or unit doses include at least one buffering agent. The at least one buffering agent includes, but is not limited to, a phosphate, a citrate, a succinate, histidine, glycine, arginine, malic acid, tartaric acid, acetic acid, benzoic acid, lactic acid, ascorbic acid, ammonium chloride, sodium chloride, potassium chloride, zinc chloride, calcium chloride, sodium acetate trihydrate, and/or triethanolamine. In one embodiment, the at least one buffering agent is about 0.10% to about 3% w/w of the weight of the composition, for example about: 0.01%, 0.05%, 0.1%, 0.2%, 0.3%, 0.4%, 0.5%, 0.6%, 0.7%, 0.8%, 0.9%, 1%, 1.5%, 2%, 2.5%, or 3% w/w based on the weight of the compositions and/or unit doses. In one embodiment, the at least one buffering agent is about 0.05% to 2.5% w/w based on the weight of the compositions and/or unit doses.

In one embodiment, the compositions of the present invention do not include a pH buffer. For example, the most stable pH for liquid epinephrine is between 2.5 and 5. A pH buffer (e.g., hydrochloric acid) is often to liquid formulations to reduce the pH. However, a low pH (e.g., between 2.5 and 5) may cause some discomfort for patients. Advantageously, the compositions of the present invention do not require a pH buffer.

60 Carriers and Excipients

In some embodiments, the dry powder composition further includes at least one carrier and/or excipient (e.g., at least one pharmaceutically acceptable carrier and/or excipient). In one embodiment, the at least one carrier and/or excipient includes, but is not limited to, lactose (e.g., D-lactose, lactose monohydrate), sucrose, glucose, dextrose, trehalose, sodium carboxymethylcellulose (CMC), manni-

tol, sorbitol, malitol, xylitol, maltose, cellulose and derivatives, starch and derivatives, microcrystalline cellulose, hypromellose acetate succinate (HPMCAS), a cyclodextrin (e.g., dimethyl-beta-cyclodextrin), calcium carbonate, citric acid, tartaric acid, glycine, leucine, polyvinyl pyrrolidone (PVP), a polyethylene glycol, polysorbate (e.g., Polysorbate 80 (e.g., TWEEN® 80)), chitosan, hyaluronic acid (e.g., sodium hyaluronate), sodium carboxymethyl cellulose (NaCMC), magnesium stearate, calcium stearate, an alkyl saccharide (e.g., n-Dodecyl β-D-Maltoside (DDM)), niacin, ethanol (e.g., dried ethanol), caffeine, benzalkonium chloride, ubiquinone (i.e., coenzyme Q10), magnesium oxide, sodium chloride, dodecylphosphocholine (DPC), silicone, gelatin, a polyacrylic acid polymer (e.g., CARBOPOL® 934), sodium taurocholate, carnitine hydrochloride, Poloxamer 188, histidine, arginine, crospovidone, ethylenediaminetetraacetic acid (EDTA), sodium starch glycolate, and/or a mixture of mannitol and hydroxypropyl methylcellulose (HPMC). In one embodiment, the at least one carrier and/or excipient includes at least one carbohydrate. In one embodiment, the at least one carbohydrate includes at least one monosaccharide, at least one disaccharide, at least one cyclodextrin, at least one polysaccharide, at least one starch, and/or at least one cellulose. In one embodiment, the at least one carrier and/or excipient includes at least one alkyl saccharide. In one embodiment, the at least one alkyl saccharide includes a sucrose ester, a sugar ether, a lipid ester, and/or an alkyl glycoside. Alkyl saccharides, sucrose esters, sugar ethers, and/or alkyl glycosides are described in WIPO Publication Nos. WO2016016431, WO2015095389, and WO2018089709, U.S. Pat. Nos. 9,895,444 and 10,653,690, and U.S. Publication No. 20190269782, each of which is incorporated herein by reference in its entirety. In another embodiment, the at least one carrier and/or excipient does not include a sucrose ester, a sugar ether, and/or an alkyl glycoside. In one embodiment, the at least one carrier and/or excipient includes at least one salt. The at least one salt includes, but is not limited to, sodium chloride, potassium chloride, sodium phosphate, calcium phosphate, calcium sulfate, and/or magnesium sulfate.

In some embodiments, the at least one carrier and/or excipient includes a first cellulose and/or a second cellulose. In some embodiments, the first cellulose is a crystalline cellulose. In some embodiments, the first cellulose is a microcrystalline cellulose. In some embodiments, the first cellulose has an average particle diameter of about 100 μm or less, for example about: 90 to 100 μm, 80 to 90 μm, 70 to 80 μm, 60 to 70 μm, 50 to 60 μm, 40 to 50 μm, 30 to 40 μm, 20 to 30 μm, or 10 to 20 μm. In some embodiments, the first cellulose has an average particle diameter of less than about 100 μm, 90 μm, 80 μm, 70 μm, 60 μm, 50 μm, 40 μm, 30 μm, 20 μm, 10 μm, or 5 μm. In some embodiments, the first cellulose has an average particle diameter of about 30 μm or less. In some embodiments, the at least one carrier and/or excipient includes a second cellulose. In some embodiments, the second cellulose is a crystalline cellulose. In some embodiments, the second cellulose is a microcrystalline cellulose. In some embodiments, the at least one carrier and/or excipient further includes a starch. In some embodiments, the at least one carrier and/or excipient includes a second cellulose and starch. In some embodiments, the second cellulose and/or starch have an average particle diameter of about 30 to about 100 μm, for example about: 30-40 μm, 30-50 μm, 30-60 μm, 30-70 μm, 30-80 μm, or 30-90 μm. In some embodiments, the second cellulose and/or starch have an average particle diameter of less than about 100 μm, 90 μm, 80 μm, 70 μm, 60 μm, 50 μm, 40 μm,

30 μm, 20 μm, 10 μm, or 5 μm. In some embodiments, the second cellulose, the starch, or the second cellulose and starch each individually has an average particle diameter of about 30 to about 100 μm.

In one embodiment, the at least one carrier and/or excipient present in the intranasal dry powder compositions is a mixture of a first microcrystalline cellulose, a second microcrystalline cellulose, a starch, and/or tribasic calcium phosphate. In one embodiment, the at least one carrier and/or excipient includes: i) a first crystalline cellulose with an average particle diameter of about 30 μm or less, for example about: 30-10 μm, 30-15 μm, 30-20 μm, or 30-25 μm; ii) tribasic calcium phosphate; and iii) a second crystalline cellulose, or starch, with an average particle diameter of about 30 to about 100 μm, for example about: 30-40 μm, 30-50 μm, 30-60 μm, 30-70 μm, 30-80 μm, or 30-90 μm. See, e.g., U.S. Pat. No. 8,337,817, which is incorporated herein by reference in its entirety.

In one embodiment, the at least one carrier and/or excipient includes particles having an average diameter of 1 μm to 100 μm. This is applicable to unimodal or multimodal compositions. In a preferred embodiment, the at least one carrier and/or excipient includes particles having an average diameter of at least 15 μm. Advantageously, an average diameter greater than 15 μm prevents particles from entering the lungs. In a preferred embodiment, the at least one carrier and/or excipient includes particles having an average diameter of about 50 μm. In one embodiment, the at least one carrier and/or excipient includes particles having an average diameter of about 25 μm to about 75 μm.

In one embodiment, an average particle diameter of a dry powder composition is determined using a laser-diffraction particle size distribution analyzer. In some embodiments, an average particle diameter of a dry powder composition is determined using sieve sorting.

The compositions of the present invention preferably do not include any liquid carriers (e.g., water, alcohol, and/or propylene glycol). Liquid carriers often require additional preservatives to improve stability. Advantageously, dry powder compositions do not require a preservative, which reduces the risk for allergic reactions.

Further, the formulations of the present invention preferably do not include a surfactant. Some liquid formulations of epinephrine require a surfactant to prevent aggregation of the active ingredient. Advantageously, dry powder formulations do not require a surfactant.

Active Ingredient Particle Characteristics

The epinephrine and other enabling agents are operable to be individually substantially amorphous or crystalline. In some embodiments, the formulations and/or unit doses provided herein are in the form of particles, and the shapes of the particles are operable to be individually, e.g., cylindrical, discoidal, spherical, tabular, ellipsoidal, angular, and/or irregular.

In some embodiments, the average particle diameter of the epinephrine, enabling agent, and/or carrier are, individually, up to 100 μm, up to 50 μm, or up to 30 μm. In a preferred embodiment, the average particle diameter of the epinephrine, enabling agent, and/or carrier are, individually, less than or equal to 50 μm. In one embodiment, the average particle diameter of the epinephrine, enabling agent, and/or carrier are, individually, about: 10-20, 20-30, 30-40, 40-50, 50-60, 60-70, 70-80, 80-90, or 90-100 μm. In another embodiment, the average particle diameter of the epinephrine, enabling agent, and/or carrier are, individually, about: 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, or 100 μm.

In some embodiments, the median particle diameter of the epinephrine powder herein is about 30 μm (e.g., 28.7 μm). In some embodiments, the median particle diameter of the epinephrine powder herein is about: 10-50, 20-40, or 25-35 μm . In one embodiment, 90% of the epinephrine particles herein have a particle diameter under about 50 μm (e.g., about 45.5 μm). In another embodiment, 90% of the epinephrine particles herein have a particle diameter under about: 40, 45, 35, 30, 25, or 20 μm . In yet another embodiment, about 10% of the epinephrine particles herein have a particle diameter under about 20 μm (e.g., about 17.3 μm). In still another embodiment, about 10% of the epinephrine particles herein have a particle diameter under about: 19, 18, 17, 16, 15, 14, 13, 12, 11, or 10 μm .

In a preferred embodiment, the average particle size, and/or the mean particle size is greater than 15 μm . Advantageously, an average particle size and/or a mean particle size greater than 15 μm avoids any entry of the particles into the lungs. In a preferred embodiment, the average particle size and/or the mean particle size is about 50 μm . In one embodiment, the average particle size and/or the mean particle size is between about 25 μm and about 75 μm .

In one embodiment, the formulations or dosage units herein are not or do not include spray-dried particles. In some embodiments, the formulations herein do not possess a fine particle fraction of less than 5.6 microns of at least about 45 percent. In some embodiments, the formulations herein do not include particles including: (a) about 11 to about 21 weight percent epinephrine bitartrate; (b) about 62 to about 82 weight percent leucine; and/or (c) about 7 to about 17 weight percent sodium tartrate.

Spray Drying Formulation Preparation

Test formulations were prepared as stock solutions in a mixture of water and ethanol (EtOH). The ratio of water to ethanol for each test formulation is included in the figures. The spray-dried powder formulations were prepared using a Buchi B-290 Mini Spray Dryer. Each formulation was pumped into the heated inlet and then aspirated through a sonicated nozzle with compressed air into the heated spray cylinder where it dries, with the resultant powder collected through a cyclone and into a collection vessel. By controlling the concentration of the formulation within the feed solution, the spray dryer pump rate, inlet temperature, aspirator power and nozzle power, average particle size, and yield were manipulated.

Particle size distribution was determined for each formulation prepared with the following instrumentation and parameters to determine particle size. D-lactose (SIGMA-ALDRICH), sodium carboxymethylcellulose (Na-CMC) (SIGMA-ALDRICH), and trehalose (ACROS) were evaluated using a Malvern Mastersizer 2000 (MALVERN PANALYTICAL) with a propan-2-ol dispersant (refractive index 1.39) using a Mie scattering model. A measurement duration of 15 seconds and a measurement pause of 10 seconds was used. A total of six measurements were obtained. Three of the six measurements were performed with sonication and the other three were performed without sonication. Table 1 includes additional information for each additive.

TABLE 1

	Lactose	Na-CMC	Trehalose
Particle refractive index	1.347	1.515	1.652
	(absorption 0.1)	(absorption 0)	(absorption 0)
Pump Speed (rpm)	1800	1500	1800 (without sonication)
			1500 (with sonication)

TABLE 1-continued

	Lactose	Na-CMC	Trehalose
Sonication power (%)	10	10	25

FIGS. 1-3 are summary tables for test formulations prepared, including composition, preparation parameters, and particle size distribution results. Batches of up to approximately 1 g of spray-dried product were made, with a yield of approximately 25%. The concentration of epinephrine was kept constant at 10% w/w throughout the formulation preparation process. No correction was made for salt to base for the drug substance. For the initial formulations, only active pharmaceutical ingredient (API) and backbone (or carrier) combinations were prepared. After instrument parameters were optimized, the other excipients under test were added into the formulation blends functioning as potential surface modifiers, vasodilators, surfactants, mucoadhesives, and/or controlled release modifiers. In a preferred embodiment, the formulation includes lactose and/or Na-CMC.

Leucine (added as a surface modifier) was added to the formulation at 10% w/w. In the lactose formulations, the introduction of leucine had a significant impact on particle size, which remained consistently and constantly below target despite changes made in nozzle power. Changes in nozzle power typically lead to corresponding changes in particle size. Acceptable particle size was achieved in a formulation including epinephrine, Na-CMC, and leucine.

Additional additives examined include sodium chloride (NaCl), niacin, polysorbate, caffeine, hyaluronate, chitosan, and Carbomer 934. Sodium chloride was introduced at a concentration of 2% w/w as another surface modifier. Formulations in both lactose and Na-CMC were successfully prepared with acceptable particle size distribution. Niacin (nicotinic acid) was successfully introduced into the formulation at a concentration of 2% w/w to function as a vasodilator. Polysorbate (TWEEN 80) was successfully introduced into the formulation at a concentration of 0.5% w/w to function as a surfactant. Caffeine was successfully introduced into the formulation at a concentration of 2% w/w to function as a controlled release additive. Hyaluronate was successfully introduced into the formulation at a concentration of 0.5% w/w to function as a mucoadhesive. Chitosan, introduced into the formulation as a potential permeation enhancer at a concentration of 10%, was found only to be soluble at a low pH and also was too viscous to nebulize in the spray drier nozzle. Carbomer 934, introduced as another potential mucoadhesive, was assessed in the lactose carrier only.

The lactose-based feed solutions were clear and colorless in appearance and the spray-dried formulations are white powders. In contrast, the Na-CMC feed solutions had some color ranging from a clear pale straw color to a clear deep red solution. The Na-CMC spray-dried powders ranged in color from off-white to yellow powders.

Delivery Devices and Packaging

In one embodiment, provided herein is therapeutic product including: (a) a dose of an intranasal dry powder formulations disclosed herein; (b) a dry powder nasal delivery device; and/or (c) a secondary packaging for the device that provides protection against humidity, light, or oxygen or other gases that may reduce the stability or physical characteristics of the dry powder formulations disclosed herein.

An example of such a delivery device is the UDS-P nasal delivery device manufactured by Aptar Pharma.

Nasal devices are disclosed in U.S. Pat. Nos. 10,814,079; 10,806,870; 10,668,228; 9,808,818; 9,156,048; 8,734,392; 8,016,209; 7,950,391; 7,946,455; 7,878,352; 7,389,946; 7,387,265; 7,353,971; 7,216,781; 7,011,234; 6,877,672; 6,725,857; 6,708,846; 6,679,248; 6,626,379; 6,484,715; 6,461,322; 6,450,216; 6,427,680; 6,425,499; 6,398,074; 6,367,473; 6,264,065; 6,234,366; 6,209,760; 6,179,164; 6,029,663; 5,901,883; 5,568,884; 5,328,099; and 5,240,149, U.S. Publication Nos. 20190358417, 20160318051, 20150299846, 20140103064, 20140034663, 20140000588, 20130312740, 20130171334, 20130171330, 20130149459, 20130081953, 20130022750, 20120318677, 20110233232, 20110194110, 20100078447, 20070272764, and 20040084554, and U.S. application Ser. Nos. 16/814,997 and 17/687,167, each of which is incorporated herein by reference in its entirety.

In one embodiment, the delivery device includes at least one nasal probe that is operable to be replaced between discharges, so that the device is operable to be used to treat two or more individuals. The packaging of the delivery device includes with a plurality of replaceable nasal probes (e.g., corresponding to a number of doses in the device). In contrast to the existing injectors, wherein the needle comes in contact with tissue and blood of each individual, this embodiment of the delivery device advantageously enables the more rapid treatment of a plurality of patients. Additionally, this embodiment of the delivery device and the set of replaceable nasal probes reduces the physical space required for carriage or storage of the quantity of pharmaceutical agent needed to respond to a plurality of patients (e.g., on a battlefield, in an ambulance, etc.).

In one embodiment, the delivery device includes a reservoir that holds the dry powder composition. The delivery device preferably includes a reservoir and means for expelling a pharmaceutical dose in the form of a spray. The reservoir is operable to contain a plurality of pharmaceutical doses (e.g., at least 4 doses, at least 8 doses, at least 20 doses, at least 50 doses, etc.). In one embodiment, the reservoir has a fill weight of at least 10 mg. In another embodiment, the reservoir has a fill weight of between about 10 mg and about 80 mg (e.g., about 10 mg, 20 mg, 30 mg, 40 mg, 50 mg, 60 mg, 70 mg, or 80 mg). In yet another embodiment, the reservoir has a fill weight of between about 20 mg and about 60 mg. In one embodiment, the reservoir has a fill volume of at least 50 mm³. In another embodiment, the reservoir has a fill volume of between about 50 mm³ and about 300 mm³ (e.g., about 50 mm³, 80 mm³, 100 mm³, 130 mm³, 150 mm³, 175 mm³, 200 mm³, 225 mm³, 250 mm³, 275 mm³, or 300 mm³). In one embodiment, the delivery device includes a plurality of individual reservoirs, each containing a pharmaceutical dose (e.g., blisters). In some embodiments, the delivery device is disposable. In some embodiments, the delivery device is reusable. In some embodiments, the delivery device is recyclable. In some embodiments, the package further includes one intranasal delivery device.

In a preferred embodiment, the delivery device does not require priming or shaking. The delivery device is preferably operable to dispense a dose from any position (i.e., 360° functionality).

FIG. 4 illustrates one embodiment of a nasal delivery device according to the present invention. The nasal delivery device 100 includes a plurality of components. In a preferred embodiment, the plurality of components includes a ball 102, a center piece 104, a container or reservoir 106, an actuator 108, a bottom 110, and a piston 112. In one

embodiment, the plurality of components is formed of at least one plastic. The at least one plastic includes, but is not limited to, polypropylene (e.g., high density polypropylene (HDPE), linear low-density polyethylene (LLDPE)) and/or polyethylene. In one embodiment, one or more of the plurality of components further includes a dye or a colorant.

FIGS. 5A-5D illustrate one embodiment of a method of using a nasal delivery device according to the present invention. FIG. 5A illustrates one embodiment of the nasal delivery device 100 at rest. The nasal delivery device 100 includes a nasal probe 120 and a push button 130. FIG. 5B illustrates positioning of a first finger 140 and a second finger 142 on the nasal delivery device 130 according to one embodiment of the present invention. A thumb 150 is positioned on the push button 130 of the nasal delivery device 100. To discharge the nasal delivery device 100, the thumb 150 presses up on the push button 130 of the nasal delivery device 100 as shown in FIG. 5C. Discharge causes particles 160 to be expelled from the nasal delivery device 100 (e.g., into a nasal passage). The nasal delivery device 100 is preferably operable to be disposed of (e.g., in a recycling can 190) following use as shown in FIG. 5D.

FIG. 6 illustrates the nasal delivery device in an unprimed state, with a sealed chamber, pressure charged detachment of a centerpiece, a start of a discharge, and an end of a stroke according to one embodiment of the present invention.

In one embodiment, the delivery device includes a counter or indicator. In one embodiment, the counter or the indicator is mechanical. Alternatively, the counter or indicator is electronic. In one embodiment, the electronic counter or indicator includes a sensor that is adapted to detect a displacement or a deformation of a portion of the delivery device (e.g., when the dry powder formulation is dispensed). In one embodiment, the electronic counter or indicator includes a display (e.g., LCD screen), a power supply (e.g., battery, rechargeable battery), a timer, a clock, at least one processor, at least one memory, a communication interface, and/or a printed circuit board (PCB). The sensor preferably transmits a signal to the PCB that causes the display to change when the delivery device is actuated. The at least one memory is operable to store information generated by the delivery device and/or the sensor. In one embodiment, the communications interface is operable to transmit data wirelessly (e.g., via BLUETOOTH®). In one embodiment, the data is transmitted wirelessly to at least one remote device (e.g., smartphone, tablet, etc.). The at least one remote device preferably includes a mobile application with a graphical user interface (GUI). In one embodiment, the mobile application tracks an expiration date of a delivery device, tracks use of the delivery device (e.g., remaining doses), and/or prompts ordering of another delivery device after use of the delivery device. In one embodiment, the at least one remote device is operable to provide messaging and/or notifications between a user and a third party (e.g., healthcare provider, parent, caregiver, emergency services, pharmacy). For example, the at least one remote device provides a notification to the third party when the delivery device dispenses a dose and/or provides a notification on a screen of the at least one remote device. In one embodiment, the delivery device displays a time of dose dispensation on the display (e.g., from the timer or the clock). Advantageously, this alerts the third party to the emergency situation (e.g., allergic reaction, anaphylaxis, cardiac arrest) and/or use of the device (e.g., to order a new delivery device). In the case of an anaphylactic reaction, conveying to the third party the time of dose dispensation is significant, as in many anaphylactic events a second dose is required. Thus, the

knowledge of how much time has passed since the initial dosing is operable to guide the decision of whether to administer subsequent doses. In one embodiment, the at least one remote device is operable to transmit the data to at least one remote server. In another embodiment, the delivery device further includes an accelerometer and/or a gyroscope to detect movement of the delivery device. In one embodiment, the remote device is operable to transmit location data (e.g., to the third party) after the delivery device is discharged. In one embodiment, the location data is obtained from the remote device. Additionally, or alternatively, the device further includes a global positioning system (GPS) device or is coupled to a GPS device operable to provide location data. Advantageously, this alerts the third party to the emergency situation (e.g., allergic reaction, anaphylaxis, cardiac arrest). In one example, a third party (e.g., emergency services, a parent, and/or a healthcare provider) is alerted when the delivery device is discharged. In a preferred embodiment, the location of the delivery device is provided to the third party. Wireless communication in delivery devices is disclosed in U.S. Pat. Nos. 10,967,140; 7,861,943; and 6,886,556 and U.S. Patent Publication Nos. 20200164164, 20200246562, 20200155775, and 20190134322, each of which is incorporated herein by reference in its entirety.

The delivery device is preferably packaged in at least one secondary packaging. The at least one secondary packaging is operable to protect the delivery device from external elements (e.g., light, humidity, oxygen or other gases). The at least one secondary package includes, but is not limited to, a vial, a tube, a container, a bottle, a box, and/or a carton. In one embodiment, the at least one secondary package includes a desiccant or other agents that assist with stability of the epinephrine formulation (e.g., by preventing effects of temperature, light, humidity, oxygen or other gases). In one embodiment, the desiccant is included as a liner (e.g., a tube liner). In one embodiment, the at least one secondary package is formed of a plastic. In one embodiment, the plastic is a desiccant plastic. In one embodiment, the desiccant plastic includes a base polymer, a channeling agent, and a desiccant. Such materials are described in, for example U.S. Pat. Nos. 5,911,937; 6,080,350; 6,124,006; 6,130,263; 6,174,952; 6,194,079; 6,214,255; 6,221,446; 6,486,231; 7,005,459; and 9,902,788, each of which is incorporated herein by reference in its entirety. Advantageously, the desiccant removes moisture within the packaging and improves the stability of the API in the delivery device.

In one embodiment, the delivery device and/or one or more of the at least one secondary packaging includes a tamper resistant seal. In one embodiment, the one or more of the at least one secondary packaging and/or the delivery device includes a sensor to detect if the tamper resistant seal is removed. In one embodiment, a notification that the sensor detected the removal of the tamper resistant seal is transmitted to the mobile application and/or a third party. Advantageously, this provides notification that the delivery device is exposed to external elements (e.g., humidity). Additionally or alternatively, one or more of the at least one secondary packaging is child resistant.

In one embodiment, one or more of the at least one secondary packaging includes an authentication method to ensure that the delivery device enclosed in the at least one secondary packaging is from the manufacturer (e.g., and not counterfeit). In one embodiment, the authentication method includes, but is not limited to, at least one code (e.g., serial number, bar code), at least one image, at least one text,

and/or at least one tracker (e.g., RFID chip). In one embodiment, the at least one authentication method is verifiable via the mobile application.

FIG. 7 illustrates one embodiment of a delivery device **100** and a secondary packaging in the form of a container **200**. The delivery device **100** is operable to be stored in the container **200**. The container **200** includes a base **210** and a lid **220**. In one embodiment, the lid **220** is connected and/or attached to the base **210**. Alternatively, the lid **220** is not connected and/or attached to the base **210** (e.g., screw on lid). Examples of a container compatible with the present invention include, but are not limited to, those disclosed in U.S. Pat. Nos. 9,834,341; 10,472,136; and 10,974,887, each of which is incorporated herein by reference in its entirety.

FIG. 8 illustrates one embodiment of a secondary packaging in the form of a container **200**. The container **200** includes a base **210** and a lid **220**. The container **200** is operable to store the delivery device (not shown). The lid **220** further includes a hole **230**. The hole **230** is operable to attach the container **200** to a ring. The ring is further operable to attach the container **200** to a keychain or a set of keys, a backpack, a purse, or other personal item. Advantageously, this helps to ensure that the delivery device is conveniently located at all times.

In one embodiment, the at least one secondary package (e.g., carton) includes a first delivery device and a second delivery device. When providing epinephrine, the lowest effective dose is desired. If a patient does not adequately respond to delivery of a first dose from the first delivery device, the second delivery device is operable to provide a second dose. Advantageously, this also ensures that a second dose is available if the first delivery device is not used properly. However, unlike auto-injectors, the nasal delivery device is not subject to a syringe misfiring.

Training Device

In one embodiment, the present invention includes a training device. Advantageously, the training device educates a patient on proper use of the nasal delivery device, providing the patient with a greater level of confidence in the event of an emergency situation when the delivery device must be used. Patients with auto-injectors routinely receive training with a training device. Such as in a prescribing physician office or at a dispensing pharmacy. This creates familiarity with the device operation and is intended to reduce errors in usage and hesitancy to use the device in an emergency. Delay in administering epinephrine is associated with increased morbidity. However, many patients fear needles despite the training for auto-injectors. There is a long-standing, unmet need for a training device for epinephrine delivery that provides confidence in device operation while not invoking a fear of needles.

Methods of Treatment

Provided herein are methods of treating a patient by intranasally administering the dry powder formulation provided herein. In some embodiments, the patient being treated is experiencing the symptoms of at least one of an anaphylactic reaction, an anaphylactoid reaction, cardiac arrest, exposure to a toxic synthetic organophosphorus compound, or exposure to cyanide or hydrogen sulfide. In some embodiments, the patient is experiencing one or more of bronchoconstriction (bronchospasm), hypotension (low blood pressure), hypotensive shock, and/or cardiac arrest (e.g., including minimal or no cardiac activity). In some instances, the patient in need of treatment is experiencing cardiac arrest and/or bronchospasm. In some embodiments, the patient is experiencing bronchoconstriction. In some embodiments, the patient is experiencing hemodynamic

collapse. In some embodiments, the patient has hypotension. In some embodiments, the patient is experiencing hypotensive shock. In some embodiments, the patient is experiencing cardiac arrest. In some embodiments, the dry powder formulation is administered to a patient in a unit dose form as disclosed herein. In some embodiments, the dry powder formulation is administered to a patient by using a delivery device or product as disclosed herein. In some embodiments, as discussed above, the dry powder formulation provided herein provides a fast onset time.

In some embodiments, the patient is experiencing symptoms of an anaphylactic reaction. Anaphylaxis is a severe allergic reaction to venom, food, or medication. Most cases are caused by a bee sting or eating foods that are known to cause allergies (e.g., peanuts, tree nuts, shellfish). Anaphylaxis causes a series of symptoms known as anaphylactic shock, which include at least one of a skin reaction (e.g., hives, itching, flushed or pale skin), low blood pressure (hypotension), constriction of the airways, a swollen tongue or throat, a weak and rapid pulse, nausea, vomiting, diarrhea, dizziness, and/or fainting. In some embodiments, the patient is experiencing symptoms of an anaphylactoid reaction. Anaphylactoid reactions are defined as those reactions that produce the same clinical picture with anaphylaxis but are not IgE mediated, occur through a direct nonimmune-mediated release of mediators from mast cells and/or basophils, or result from direct complement activation. In some instances, the patient may be experiencing at least one of bronchoconstriction (bronchospasm), hypotension (low blood pressure), or hypotensive shock. In some instances, the patient may experience cardiac arrest if the patient's peripheral blood pressure drops sufficiently. The patient is provided with an effective dose of the intranasal epinephrine of the present invention. Emergency services often take up to 15 minutes (or longer) to arrive after a patient has an anaphylactic reaction. Advantageously, providing intranasal epinephrine before emergency services arrives reduces mortality.

In some embodiments, the patient is experiencing symptoms of cardiac arrest, including minimal or no cardiac activity. Such patients may also be experiencing hypotension and/or hypotensive shock. In some instances, the patient is in need of cardiopulmonary resuscitation (CPR) and/or basic or advanced cardiac life support (ACLS). The patient is provided with an effective dose of the intranasal epinephrine of the present invention. Emergency services often take up to 15 minutes (or longer) to arrive after a patient goes into cardiac arrest. Advantageously, providing intranasal epinephrine before emergency services arrives reduces mortality. See, e.g., Okubo M, Komukai S, Callaway C W, Izawa J. Association of Timing of Epinephrine Administration With Outcomes in Adults With Out-of-Hospital Cardiac Arrest. *JAMA Netw Open*. 2021 Aug. 2; 4(8):e2120176. Doi: 10.1001/jamanetworkopen.2021.20176. PMID: 34374770; PMCID: PMC8356068, which is incorporated herein by reference in its entirety. Further, it may be difficult to place an IV in a patient experiencing symptoms of cardiac arrest. The present invention advantageously is operable to be used without needing to place an IV. In one embodiment, at least one nasal delivery device of the present invention is included in a kit with a defibrillator (e.g., automated external defibrillator (AED)).

In some embodiments, the patient is experiencing symptoms of exposure to a toxic synthetic organophosphorus compound. In one embodiment, the toxic synthetic organophosphorus compound includes, but is not limited to, nerve

agents (e.g., sarin, soman, venomous agent X (VX)), a pesticide, and/or an insecticide. The patient is provided with an effective dose of the intranasal epinephrine of the present invention. Emergency services often take up to 15 minutes (or longer) to arrive after a patient is exposed to the toxic synthetic organophosphorus compound. Advantageously, providing intranasal epinephrine before emergency services arrives reduces mortality.

In some embodiments, the patient is experiencing symptoms of exposure to cyanide or hydrogen sulfide. The patient is provided with an effective dose of the intranasal epinephrine of the present invention. See, e.g., Judenherc-Haouzi A, Sonobe T, Bebart V S, Haouzi P. On the Efficacy of Cardio-Pulmonary Resuscitation and Epinephrine Following Cyanide- and H₂S Intoxication-Induced Cardiac Asystole. *Cardiovasc Toxicol*. 2018 October; 18(5):436-449. Doi: 10.1007/s12012-018-9454-2. PMID: 29644580; PMCID: PMC6126935, which is incorporated herein by reference in its entirety. See also, e.g., Bebart V S, Pitotti R L, Dixon P S, Valtier S, Esquivel L, Bush A, Little C M. Hydroxocobalamin and epinephrine both improve survival in a swine model of cyanide-induced cardiac arrest. *Ann Emerg Med*. 2012 October; 60(4):415-22. Doi: 10.1016/j.annemergmed.2012.02.002. Epub 2012 Mar. 15. PMID: 22424656, which is incorporated herein by reference in its entirety.

In some embodiments, a single dose of the dry powder epinephrine formulation provided herein is administered to the patient. In some embodiments, more than one dose of the dry powder epinephrine formulation provided herein is administered to the patient. In some embodiments, the dry powder epinephrine formulation is administered more than once to the patient if a first administered dose of epinephrine does not increase the patient's arterial pressure to a normal range. In some instances, administration of the formulation is repeated (e.g., every 5-20 minutes) as necessary.

In some embodiments, intranasally administering the dry powder formulations provided herein results in application of the dry powder formulations to at least one mucosal surface of a nasal cavity or cavities of the patient (e.g., the mucosal surfaces of the anterior regions of the nose, the frontal sinus, the maxillary sinuses, and/or on each of the mucosal surfaces that overlie the turbinates covering the conchas).

In some embodiments, administration of the dry powder formulation intranasally to the patient is sufficient to increase an arterial pressure in the patient within 60 minutes, 50 minutes, 40 minutes, 30 minutes, 20 minutes, 10 minutes, 5 minutes, 3 minutes, 2 minutes, or less than 1 minute after administration. In some embodiments, the intranasal dry powder formulation is sufficient to increase a mean arterial pressure in the patient within 60 minutes, 50 minutes, 40 minutes, 30 minutes, 20 minutes, 10 minutes, 5 minutes, 3 minutes, 2 minutes, or less than 1 minute after administration. In some embodiments, the intranasal dry powder formulation is sufficient to increase a coronary perfusion pressure in the patient within 60 minutes, 50 minutes, 40 minutes, 30 minutes, 20 minutes, 10 minutes, 5 minutes, 3 minutes, 2 minutes, or less than 1 minute after administration. In some embodiments, the intranasal dry powder formulation is sufficient to resume a spontaneous circulation in the patient within 60 minutes, 50 minutes, 40 minutes, 30 minutes, 20 minutes, 10 minutes, 5 minutes, 3 minutes, 2 minutes, or less than 1 minute after administration. In some embodiments, the intranasal dry powder formulation is sufficient to relieve a bronchoconstriction in the patient within 60 minutes, 50 minutes, 40 minutes, 30 minutes, 20

minutes, 10 minutes, 5 minutes, 3 minutes, 2 minutes, or less than 1 minute after administration.

In some instances, the formulation is administered to the patient in the form of a unit dose as described above. In some embodiments, a single dose of the dry powder formulation includes about 0.01 mg to about 10 mg epinephrine. In some embodiments, the amount of epinephrine in a single dose of the formulation is at least: 0.01 mg, 0.05 mg, 1.0 mg, 2.0 mg, 5.0 mg, or 10 mg. In some embodiments, a single dose of the formulation includes about: 0.01 mg to 0.05 mg, 0.05 mg to 0.75 mg, 0.75 mg to 1.5 mg, 1.5 mg to 3.0 mg, 3.0 to 4.5 mg, 4.5 to 6.0 mg, 6.0 to 7.5 mg, 7.5 to 9.0 mg, or 9.0 to 10.0 mg epinephrine. In some embodiments, a single dose of the formulation includes about: 0.15, 0.3, 0.5, 0.75, 1, 1.5, 2, 3, 4, 5, 6, 7, 8, 9, or 10 mg epinephrine. In one instance, a single dose includes about 0.75 mg epinephrine. In one instance, a single dose includes about 1.5 mg epinephrine. In another instance, a single dose includes about 3.0 mg epinephrine. In some embodiments, a single dose includes about 0.01 mg to about 10 mg of epinephrine. In some embodiments, a single dose includes about: 0.75 mg, 1.5 mg, or 3.0 mg of epinephrine. In some embodiments, based on a "standard" patient weight of 70 kg, the dose of epinephrine is adjusted according to an increased or decreased weight of the patient relative to the "standard" patient weight. In one embodiment, the adjustment according to the increased or decreased weight of the patient is at an increased or decreased increment of at least 0.01 mg/kg, respectively. In some embodiments, the delivered dose of the dry powder formulation (i.e., the amount of the formulation delivered to the patient's nasal passages) when administered to the patient using the devices described herein is substantially similar to the unit dose of the formulation. In some embodiments, the dose administered to the patient results in the systemic blood bioequivalent of intravenously (IV), intramuscularly (IM) or subcutaneously (SQ) administered epinephrine (e.g., the systemic blood level achieved with administration of 1 mg epinephrine by IV or 0.3 mg epinephrine by EPIPEN® device for adults, 0.15 mg epinephrine by EPIPEN JR® device for pediatric patients or 0.01 mg/kg with a dilution of 0.1 mg/mL for IV epinephrine), referred to herein as "a nasal loading dose".

In some embodiments, the dry powder formulations provided herein, when administered to a patient, produce a maximal blood concentration (Cmax) of epinephrine that is at least about: 2- to 3-fold, 3- to 5-fold, 5- to 7-fold, or 7- to 10-fold more than the baseline level of epinephrine in the patient. In some embodiments, the dry powder formulations provided herein, when administered to a patient, produce a maximal blood concentration (Cmax) of epinephrine at least 2-, 3-, 4-, 5-, 6-, 7-, 8-, 9-, or 10-fold more than the baseline level of epinephrine in the patient. In one embodiment, the dry powder formulations provided herein, when administered to a patient, produce a maximal blood concentration (Cmax) of epinephrine at least 2-fold more than the baseline level of the epinephrine in the patient. In one embodiment, the formulations provided herein, when administered to a patient, increase the blood concentration of epinephrine by about 0.01 to 0.1 µg/mL. In one embodiment, the formulations provided herein, when administered to a patient, increase the blood concentration of epinephrine by about: 0.01, 0.02, 0.03, 0.04, 0.05, 0.06, 0.07, 0.08, 0.09, or 0.1 µg/mL.

In some embodiments, the dry powder formulations provided herein, when administered to a patient, reach a maximal blood concentration of epinephrine in less than about 60 minutes (Tmax) after administration. In some embodiments,

the dry powder formulations provided herein, when administered to a patient, reach a maximal blood concentration (Tmax) of epinephrine in less than about: 60, 50, 40, 30, 20, 15, 10, 5, 3, or less than 1 minute (Tmax) after administration. In some embodiments, the dry powder formulations provided herein, when administered to a patient, reach a mean AUC in a time period (e.g., 0-180 minutes) of epinephrine that is at least 20%, 30%, 40% 50%, 60%, 70%, 80%, 90%, 100%, 110%, 120%, 130%, 140% or 150% of the mean AUC in the time period of an IV, IM, or SQ injected epinephrine. In some embodiments, the dry powder formulations provided herein, when administered to a patient, reaches a mean AUC in a time period (e.g., 0-∞) of epinephrine that is at least 20%, 30%, 40% 50%, 60%, 70%, 80%, 90%, 100%, 110%, 120%, 130%, 140% or 150% of the mean AUC in the time period of an IV, IM, or SQ injected epinephrine. In some embodiments, the IV, IM, or SQ injected epinephrine contains 0.10 mg, 0.15 mg, 0.20 mg, 0.25 mg, 0.30 mg, 0.35 mg, 0.40 mg, 0.45 mg, 0.50 mg, 0.55 mg, 0.60 mg, 0.65 mg, 0.70 mg, 0.75 mg, 0.80 mg, 0.85 mg, 0.90 mg, 0.95 mg, or 1.0 mg of epinephrine. For example, the dry powder formulations provided herein, when administered to a patient, reach a mean AUC in a time period (e.g., 0-180 minutes) of epinephrine that is at least 80% of the mean AUC in the time period of a 0.15 mg IV injected epinephrine. In another instance, the dry powder formulations provided herein, when administered to a patient, reach a mean AUC in a time period (e.g., 0-∞) of epinephrine that is at least 80% of the mean AUC in the time period of a 0.15-1 mg IV injected epinephrine. In some embodiments, the IV, IM, or SQ injected epinephrine is injected by EPIPEN® auto-injector (e.g., EPIPEN® 0.3 mg, EPIPEN JR® 0.15 mg). In some embodiments, the dry powder formulations provided herein, when administered to a patient, reach a mean AUC in a time period (e.g., 0-180 minutes) of epinephrine that is at least 50,000 pg-min/mL, 100,000 pg-min/mL, 200,000 pg-min/mL, 300,000 pg-min/mL, 400,000 pg-min/mL, 500,000 pg-min/mL, 600,000 pg-min/mL, 700,000 pg-min/mL, 800,000 pg-min/mL, 900,000 pg-min/mL, or 1,000,000 pg-min/mL. In some embodiments, the dry powder formulations provided herein, when administered to a patient, reach a mean AUC in a time period (e.g., 0-∞) of epinephrine that is at least 50,000 pg-min/mL, 100,000 pg-min/mL, 200,000 pg-min/mL, 300,000 pg-min/mL, 400,000 pg-min/mL, 500,000 pg-min/mL, 600,000 pg-min/mL, 700,000 pg-min/mL, 800,000 pg-min/mL, 900,000 pg-min/mL, 1,000,000 pg-min/mL, 1,200,000 pg-min/mL, 1,400,000 pg-min/mL, 1,600,000 pg-min/mL, 1,800,000 pg-min/mL, or 2,000,000 pg-min/mL.

In certain embodiments, the dry powder formulations and/or unit doses provided herein are operable to raise the blood concentration of epinephrine in a subject to about 0.02 µg/mL within about less than 1 minute to about 60 minutes (e.g., about: 60, 50, 40, 30, 20, 15, 10, 5, 3, <1 minute). In one embodiment, the dry powder formulations and/or unit doses provided herein are operable to raise the blood concentration of epinephrine in a subject about 10 µg/mL within about less than 1 minute to about 15 minutes (e.g., about: <1, 3, 5, 10, 11, 12, 13, 14, or 15 minutes) of intranasal administration. In one embodiment, the formulations provided herein increase the blood concentration of epinephrine by about 0.01 to about 0.04 µg/mL (e.g., 0.02 or 0.03 µg/mL) in about less than 1 minute to about 15 minutes (e.g., about: <1, 10, 11, 12, 13, 14, or 15 minutes). In one embodiment, the formulations provided herein increase the blood concen-

tration of epinephrine by about 3 µg/mL in about less than 1 minute to about 60 minutes (e.g., about: 60, 50, 40, 30, 20, 15, 10, 5, 3, <1 minute).

A single dose of epinephrine in the dry powder formulations and/or dosage units given intranasally is preferably bioequivalent (for example, in terms of peripheral blood levels, systemic exposure of epinephrine) to intravenously (IV), intramuscularly (IM), or subcutaneously (SQ) injected epinephrine (e.g., using EPIPEN® auto-injector of 0.3 mg for adult patients or 1 mg IV epinephrine, using EPIPEN® auto-injector of 0.15 mg for pediatric patients or 0.01 mg/kg with a dilution of 0.1 mg/mL for IV epinephrine). In one example, bioequivalence includes a 90% confidence interval of a mean Tmax (e.g., the time to reach maximal blood concentration), a mean Cmax (e.g., maximal blood concentration), a mean AUC over a period of time (e.g., area under the plasma/serum/blood concentration-time curve from time zero to time t), and/or a mean AUC over a period of time from 0 to infinity (e.g., area under the plasma/serum/blood concentration-time curve from time zero to time infinity) are within 80.00% to 125.00% of a reference test. In one embodiment, bioequivalence is determined in a fasting state.

In some instances, the patient in need of treatment is experiencing cardiac arrest and/or bronchospasm. In some embodiments, the method includes administering a nasal loading dose of a formulation as provided herein, wherein the nasal loading dose includes about 0.05 mg to about 10 mg of epinephrine (e.g., 0.5 to 5 mg, or 0.75, 1.5, or 3.0 mg), about 1 µg to about 10 mg of a second vasodilator (a vasodilator other than epinephrine; e.g., niacin) (e.g., 0.1 to 5 mg, 0.1 to 1 mg, or 0.5 mg); and at least one pharmaceutically acceptable carrier in an amount of about 1 mg to about 50 mg (e.g., 10 to 30 mg, 15 to 20 mg, or 18 mg). In some instances, the dry powder formulation includes at least one agent that reduces mucosal transit time, at least one agent that increases mucosal absorption and/or adhesion, and/or at least one agent that enhances mucosal transport, or the enantiomers, diastereoisomers, racemates, or salts of such compounds with pharmaceutically acceptable counterions. In one embodiment, an amount of the at least one agent that reduces mucosal transit time, the at least one agent that increases mucosal absorption and/or adhesion, and/or the at least one agent that enhances mucosal transport, or the enantiomers, diastereoisomers, racemates, or salts of such compounds with pharmaceutically acceptable counterions is synergistic for the treatment of bronchospasm and/or cardiac arrest. In some instances, the formulations provided herein include low doses that, when administered intranasally to a subject, result in a sufficiently high peak blood plasma concentration of epinephrine of at least: 2-fold, 3-5 fold, 5-7 fold, or 7-10 fold more than baseline levels rapidly after administration (e.g., within 60, 50, 40, 30, 20, 15, 10, 5, 3, or <1 minute) so as to be effective in the treatment or reducing the symptoms of bronchospasm, and/or cardiac arrest.

In one embodiment, the invention provides a pharmaceutical formulation including dry powder epinephrine and chitosan (e.g., dry powder chitosan) as an enabling agent to optimize the systemic delivery of epinephrine through the nasal passages. In some embodiments, chitosan and at least one excipient (e.g., at least one enabling agent, at least one carrier, and/or at least one flow agent) are combined in various ratios and employed to optimize the systemic delivery of epinephrine through the nasal passages. In some embodiments, the concentration of the at least one enabling agent ranges from 0% and 20% w/w. The at least one enabling agent includes, but is not limited to, at least one

mucoadhesive, at least one absorption enhancer, at least one permeability enhancer, at least one surfactant, at least one surface modifier, at least one sustained release agent, at least one vasodilator, at least one anticaking agent, and at least one disintegrant. In some embodiments, the formulation includes at least one excipient that impacts the characteristics of the dry powder formulation that directly or indirectly affect the transport of epinephrine across the nasal mucosa. In some embodiments, the formulation includes at least one excipient that facilitates solubilization or processing so as to provide a spray-dried powder having the desired characteristics.

In one embodiment, the invention provides a pharmaceutical formulation including dry powder epinephrine and hyaluronic acid (e.g., dry powder hyaluronic acid, dry powder sodium hyaluronate) as an enabling agent to optimize the systemic delivery of epinephrine through the nasal passages. In some embodiments, hyaluronic acid and at least one excipient (e.g., at least one enabling agent, at least one carrier, and/or at least one flow agent) are combined in various ratios and employed to optimize the systemic delivery of epinephrine through the nasal passages. In some embodiments, the concentration of the at least one enabling agent ranges from 0% and 20% w/w. The at least one enabling agent includes, but is not limited to, at least one mucoadhesive, at least one absorption enhancer, at least one permeability enhancer, at least one surfactant, at least one surface modifier, at least one sustained release agent, at least one vasodilator, at least one anticaking agent, and at least one disintegrant. In some embodiments, the formulation includes at least one excipient that impacts the characteristics of the dry powder formulation that directly or indirectly affect the transport of epinephrine across the nasal mucosa. In some embodiments, the formulation includes at least one excipient that facilitates solubilization or processing so as to provide a spray-dried powder having the desired characteristics.

In a preferred embodiment, the invention provides a pharmaceutical formulation including dry powder epinephrine and sodium carboxymethyl cellulose (NaCMC) (e.g., dry powder NaCMC) as an enabling agent to optimize the systemic delivery of epinephrine through the nasal passages. In some embodiments, NaCMC and at least one excipient (e.g., at least one enabling agent, at least one carrier, and/or at least one flow agent) are combined in various ratios and employed to optimize the systemic delivery of epinephrine through the nasal passages. In some embodiments, the concentration of the at least one enabling agent ranges from 0% and 20% w/w. The at least one enabling agent includes, but is not limited to, at least one mucoadhesive, at least one absorption enhancer, at least one permeability enhancer, at least one surfactant, at least one surface modifier, at least one sustained release agent, at least one vasodilator, at least one anticaking agent, and at least one disintegrant. In some embodiments, the formulation includes at least one excipient that impacts the characteristics of the dry powder formulation that directly or indirectly affect the transport of epinephrine across the nasal mucosa. In some embodiments, the formulation includes at least one excipient that facilitates solubilization or processing so as to provide a spray-dried powder having the desired characteristics.

In one embodiment, the invention provides a pharmaceutical formulation including dry powder epinephrine and cyclodextrin (e.g., dry powder cyclodextrin) as an enabling agent to optimize the systemic delivery of epinephrine through the nasal passages. In some embodiments, cyclodextrin and at least one excipient (e.g., at least one enabling

agent, at least one carrier, and/or at least one flow agent) are combined in various ratios and employed to optimize the systemic delivery of epinephrine through the nasal passages. In some embodiments, the concentration of the at least one enabling agent ranges from 0% and 20% w/w. The at least one enabling agent includes, but is not limited to, at least one mucoadhesive, at least one absorption enhancer, at least one permeability enhancer, at least one surfactant, at least one surface modifier, at least one sustained release agent, at least one vasodilator, at least one anticaking agent, and at least one disintegrant. In some embodiments, the formulation includes at least one excipient that impacts the characteristics of the dry powder formulation that directly or indirectly affect the transport of epinephrine across the nasal mucosa. In some embodiments, the formulation includes at least one excipient that facilitates solubilization or processing so as to provide a spray-dried powder having the desired characteristics.

In one embodiment, the invention provides a pharmaceutical formulation including dry powder epinephrine and niacin (e.g., dry powder niacin, dry powder nicotinic acid) as an enabling agent to optimize the systemic delivery of epinephrine through the nasal passages. In some embodiments, niacin and at least one excipient (e.g., at least one enabling agent, at least one carrier, and/or at least one flow agent) are combined in various ratios and employed to optimize the systemic delivery of epinephrine through the nasal passages. In some embodiments, the concentration of the at least one enabling agent ranges from 0% and 20% w/w. The at least one enabling agent includes, but is not limited to, at least one mucoadhesive, at least one absorption enhancer, at least one permeability enhancer, at least one surfactant, at least one surface modifier, at least one sustained release agent, at least one vasodilator, at least one anticaking agent, and at least one disintegrant. In some embodiments, the formulation includes at least one excipient that impacts the characteristics of the dry powder formulation that directly or indirectly affect the transport of epinephrine across the nasal mucosa. In some embodiments, the formulation includes at least one excipient that facilitates solubilization or processing so as to provide a spray-dried powder having the desired characteristics.

In one embodiment, the invention provides a pharmaceutical formulation including dry powder epinephrine and caffeine (e.g., dry powder caffeine) as an enabling agent to optimize the systemic delivery of epinephrine through the nasal passages. In some embodiments, caffeine and at least one excipient (e.g., at least one enabling agent, at least one carrier, and/or at least one flow agent) are combined in various ratios and employed to optimize the systemic delivery of epinephrine through the nasal passages. In some embodiments, the concentration of the at least one enabling agent ranges from 0% and 20% w/w. The at least one enabling agent includes, but is not limited to, at least one mucoadhesive, at least one absorption enhancer, at least one permeability enhancer, at least one surfactant, at least one surface modifier, at least one sustained release agent, at least one vasodilator, at least one anticaking agent, and at least one disintegrant. In some embodiments, the formulation includes at least one excipient that impacts the characteristics of the dry powder formulation that directly or indirectly affect the transport of epinephrine across the nasal mucosa. In some embodiments, the formulation includes at least one excipient that facilitates solubilization or processing so as to provide a spray-dried powder having the desired characteristics.

In one embodiment, the invention provides a pharmaceutical formulation including dry powder epinephrine and sodium taurocholate (e.g., dry powder sodium taurocholate) as an enabling agent to optimize the systemic delivery of epinephrine through the nasal passages. In some embodiments, sodium taurocholate and at least one excipient (e.g., at least one enabling agent, at least one carrier, and/or at least one flow agent) are combined in various ratios and employed to optimize the systemic delivery of epinephrine through the nasal passages. In some embodiments, the concentration of the at least one enabling agent ranges from 0% and 20% w/w. The at least one enabling agent includes, but is not limited to, at least one mucoadhesive, at least one absorption enhancer, at least one permeability enhancer, at least one surfactant, at least one surface modifier, at least one sustained release agent, at least one vasodilator, at least one anticaking agent, and at least one disintegrant. In some embodiments, the formulation includes at least one excipient that impacts the characteristics of the dry powder formulation that directly or indirectly affect the transport of epinephrine across the nasal mucosa. In some embodiments, the formulation includes at least one excipient that facilitates solubilization or processing so as to provide a spray-dried powder having the desired characteristics.

In one embodiment, the invention provides a pharmaceutical formulation including dry powder epinephrine and carnitine hydrochloride (e.g., dry powder carnitine hydrochloride) as an enabling agent to optimize the systemic delivery of epinephrine through the nasal passages. In some embodiments, carnitine hydrochloride and at least one excipient (e.g., at least one enabling agent, at least one carrier, and/or at least one flow agent) are combined in various ratios and employed to optimize the systemic delivery of epinephrine through the nasal passages. In some embodiments, the concentration of the at least one enabling agent ranges from 0% and 20% w/w. The at least one enabling agent includes, but is not limited to, at least one mucoadhesive, at least one absorption enhancer, at least one permeability enhancer, at least one surfactant, at least one surface modifier, at least one sustained release agent, at least one vasodilator, at least one anticaking agent, and at least one disintegrant. In some embodiments, the formulation includes at least one excipient that impacts the characteristics of the dry powder formulation that directly or indirectly affect the transport of epinephrine across the nasal mucosa. In some embodiments, the formulation includes at least one excipient that facilitates solubilization or processing so as to provide a spray-dried powder having the desired characteristics.

In one embodiment, the invention provides a pharmaceutical formulation including dry powder epinephrine and dimethyl-beta-cyclodextrin (e.g., dry powder dimethyl-beta-cyclodextrin) as an enabling agent to optimize the systemic delivery of epinephrine through the nasal passages. In some embodiments, dimethyl-beta-cyclodextrin and at least one excipient (e.g., at least one enabling agent, at least one carrier, and/or at least one flow agent) are combined in various ratios and employed to optimize the systemic delivery of epinephrine through the nasal passages. In some embodiments, the concentration of the at least one enabling agent ranges from 0% and 20% w/w. The at least one enabling agent includes, but is not limited to, at least one mucoadhesive, at least one absorption enhancer, at least one permeability enhancer, at least one surfactant, at least one surface modifier, at least one sustained release agent, at least one vasodilator, at least one anticaking agent, and at least one disintegrant. In some embodiments, the formulation

includes at least one excipient that impacts the characteristics of the dry powder formulation that directly or indirectly affect the transport of epinephrine across the nasal mucosa. In some embodiments, the formulation includes at least one excipient that facilitates solubilization or processing so as to provide a spray-dried powder having the desired characteristics.

In one embodiment, the invention provides a pharmaceutical formulation including dry powder epinephrine and polysorbate (e.g., dry powder polysorbate) as an enabling agent to optimize the systemic delivery of epinephrine through the nasal passages. In some embodiments, polysorbate and at least one excipient (e.g., at least one enabling agent, at least one carrier, and/or at least one flow agent) are combined in various ratios and employed to optimize the systemic delivery of epinephrine through the nasal passages. In some embodiments, the concentration of the at least one enabling agent ranges from 0% and 20% w/w. The at least one enabling agent includes, but is not limited to, at least one mucoadhesive, at least one absorption enhancer, at least one permeability enhancer, at least one surfactant, at least one surface modifier, at least one sustained release agent, at least one vasodilator, at least one anticaking agent, and at least one disintegrant. In some embodiments, the formulation includes at least one excipient that impacts the characteristics of the dry powder formulation that directly or indirectly affect the transport of epinephrine across the nasal mucosa. In some embodiments, the formulation includes at least one excipient that facilitates solubilization or processing so as to provide a spray-dried powder having the desired characteristics.

In one embodiment, the invention provides a pharmaceutical formulation including dry powder epinephrine and sodium chloride (NaCl) (e.g., dry powder NaCl) as an enabling agent to optimize the systemic delivery of epinephrine through the nasal passages. In some embodiments, NaCl and at least one excipient (e.g., at least one enabling agent, at least one carrier, and/or at least one flow agent) are combined in various ratios and employed to optimize the systemic delivery of epinephrine through the nasal passages. In some embodiments, the concentration of the at least one enabling agent ranges from 0% and 20% w/w. The at least one enabling agent includes, but is not limited to, at least one mucoadhesive, at least one absorption enhancer, at least one permeability enhancer, at least one surfactant, at least one surface modifier, at least one sustained release agent, at least one vasodilator, at least one anticaking agent, and at least one disintegrant. In some embodiments, the formulation includes at least one excipient that impacts the characteristics of the dry powder formulation that directly or indirectly affect the transport of epinephrine across the nasal mucosa. In some embodiments, the formulation includes at least one excipient that facilitates solubilization or processing so as to provide a spray-dried powder having the desired characteristics.

In one embodiment, the invention provides a pharmaceutical formulation comprising dry powder epinephrine and potassium chloride (KCl) (e.g., dry powder KCl) as an enabling agent to optimize the systemic delivery of epinephrine through the nasal passages. In some embodiments, KCl and at least one excipient (e.g., at least one enabling agent, at least one carrier, and/or at least one flow agent) are combined in various ratios and employed to optimize the systemic delivery of epinephrine through the nasal passages. In some embodiments, the concentration of the at least one enabling agent ranges from 0% and 20% w/w. The at least one enabling agent includes, but is not limited to, at least one

mucoadhesive, at least one absorption enhancer, at least one permeability enhancer, at least one surfactant, at least one surface modifier, at least one sustained release agent, at least one vasodilator, at least one anticaking agent, and at least one disintegrant. In some embodiments, the formulation includes at least one excipient that impacts the characteristics of the dry powder formulation that directly or indirectly affect the transport of epinephrine across the nasal mucosa. In some embodiments, the formulation includes at least one excipient that facilitates solubilization or processing so as to provide a spray-dried powder having the desired characteristics.

In one embodiment, the invention provides a pharmaceutical formulation including dry powder epinephrine and 2-(2-propoxypropoxy)ethanol (also known as Poloxamer 188) (e.g., dry powder Poloxamer 188) as an enabling agent to optimize the systemic delivery of epinephrine through the nasal passages. In some embodiments, Poloxamer 188 and at least one excipient (e.g., at least one enabling agent, at least one carrier, and/or at least one flow agent) are combined in various ratios and employed to optimize the systemic delivery of epinephrine through the nasal passages. In some embodiments, the concentration of the at least one enabling agent ranges from 0% and 20% w/w. The at least one enabling agent includes, but is not limited to, at least one mucoadhesive, at least one absorption enhancer, at least one permeability enhancer, at least one surfactant, at least one surface modifier, at least one sustained release agent, at least one vasodilator, at least one anticaking agent, and at least one disintegrant. In some embodiments, the formulation includes at least one excipient that impacts the characteristics of the dry powder formulation that directly or indirectly affect the transport of epinephrine across the nasal mucosa. In some embodiments, the formulation includes at least one excipient that facilitates solubilization or processing so as to provide a spray-dried powder having the desired characteristics.

In one embodiment, the invention provides a pharmaceutical formulation including dry powder epinephrine and leucine (e.g., dry powder leucine, dry powder L-leucine) as an enabling agent to optimize the systemic delivery of epinephrine through the nasal passages. In some embodiments, leucine and at least one excipient (e.g., at least one enabling agent, at least one carrier, and/or at least one flow agent) are combined in various ratios and employed to optimize the systemic delivery of epinephrine through the nasal passages. In some embodiments, the concentration of the at least one enabling agent ranges from 0% and 20% w/w. The at least one enabling agent includes, but is not limited to, at least one mucoadhesive, at least one absorption enhancer, at least one permeability enhancer, at least one surfactant, at least one surface modifier, at least one sustained release agent, at least one vasodilator, at least one anticaking agent, and at least one disintegrant. In some embodiments, the formulation includes at least one excipient that impacts the characteristics of the dry powder formulation that directly or indirectly affect the transport of epinephrine across the nasal mucosa. In some embodiments, the formulation includes at least one excipient that facilitates solubilization or processing so as to provide a spray-dried powder having the desired characteristics.

In one embodiment, the invention provides a pharmaceutical formulation including dry powder epinephrine and histidine (e.g., dry powder histidine) as an enabling agent to optimize the systemic delivery of epinephrine through the nasal passages. In some embodiments, histidine and at least one excipient (e.g., at least one enabling agent, at least one

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carrier, and/or at least one flow agent) are combined in various ratios and employed to optimize the systemic delivery of epinephrine through the nasal passages. In some embodiments, the concentration of the at least one enabling agent ranges from 0% and 20% w/w. The at least one enabling agent includes, but is not limited to, at least one mucoadhesive, at least one absorption enhancer, at least one permeability enhancer, at least one surfactant, at least one surface modifier, at least one sustained release agent, at least one vasodilator, at least one anticaking agent, and at least one disintegrant. In some embodiments, the formulation includes at least one excipient that impacts the characteristics of the dry powder formulation that directly or indirectly affect the transport of epinephrine across the nasal mucosa. In some embodiments, the formulation includes at least one excipient that facilitates solubilization or processing so as to provide a spray-dried powder having the desired characteristics.

In one embodiment, the invention provides a pharmaceutical formulation including dry powder epinephrine and glycine (e.g., dry powder glycine) as an enabling agent to optimize the systemic delivery of epinephrine through the nasal passages. In some embodiments, glycine and at least one excipient (e.g., at least one enabling agent, at least one carrier, and/or at least one flow agent) are combined in various ratios and employed to optimize the systemic delivery of epinephrine through the nasal passages. In some embodiments, the concentration of the at least one enabling agent ranges from 0% and 20% w/w. The at least one enabling agent includes, but is not limited to, at least one mucoadhesive, at least one absorption enhancer, at least one permeability enhancer, at least one surfactant, at least one surface modifier, at least one sustained release agent, at least one vasodilator, at least one anticaking agent, and at least one disintegrant. In some embodiments, the formulation includes at least one excipient that impacts the characteristics of the dry powder formulation that directly or indirectly affect the transport of epinephrine across the nasal mucosa. In some embodiments, the formulation includes at least one excipient that facilitates solubilization or processing so as to provide a spray-dried powder having the desired characteristics.

In one embodiment, the invention provides a pharmaceutical formulation including dry powder epinephrine and arginine (e.g., dry powder arginine) as an enabling agent to optimize the systemic delivery of epinephrine through the nasal passages. In some embodiments, arginine and at least one excipient (e.g., at least one enabling agent, at least one carrier, and/or at least one flow agent) are combined in various ratios and employed to optimize the systemic delivery of epinephrine through the nasal passages. In some embodiments, the concentration of the at least one enabling agent ranges from 0% and 20% w/w. The at least one enabling agent includes, but is not limited to, at least one mucoadhesive, at least one absorption enhancer, at least one permeability enhancer, at least one surfactant, at least one surface modifier, at least one sustained release agent, at least one vasodilator, at least one anticaking agent, and at least one disintegrant. In some embodiments, the formulation includes at least one excipient that impacts the characteristics of the dry powder formulation that directly or indirectly affect the transport of epinephrine across the nasal mucosa. In some embodiments, the formulation includes at least one excipient that facilitates solubilization or processing so as to provide a spray-dried powder having the desired characteristics.

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In one embodiment, the invention provides a pharmaceutical formulation including dry powder epinephrine and crospovidone (e.g., dry powder crospovidone) as an enabling agent to optimize the systemic delivery of epinephrine through the nasal passages. In some embodiments, crospovidone and at least one excipient (e.g., at least one enabling agent, at least one carrier, and/or at least one flow agent) are combined in various ratios and employed to optimize the systemic delivery of epinephrine through the nasal passages. In some embodiments, the concentration of the at least one enabling agent ranges from 0% and 20% w/w. The at least one enabling agent includes, but is not limited to, at least one mucoadhesive, at least one absorption enhancer, at least one permeability enhancer, at least one surfactant, at least one surface modifier, at least one sustained release agent, at least one vasodilator, at least one anticaking agent, and at least one disintegrant. In some embodiments, the formulation includes at least one excipient that impacts the characteristics of the dry powder formulation that directly or indirectly affect the transport of epinephrine across the nasal mucosa. In some embodiments, the formulation includes at least one excipient that facilitates solubilization or processing so as to provide a spray-dried powder having the desired characteristics.

In one embodiment, the invention provides a pharmaceutical formulation including dry powder epinephrine and a polyacrylic acid polymer (e.g., CARBOPOL® 934) (e.g., a dry powder polyacrylic acid polymer) as an enabling agent to optimize the systemic delivery of epinephrine through the nasal passages. In some embodiments, the polyacrylic acid polymer and at least one excipient (e.g., at least one enabling agent, at least one carrier, and/or at least one flow agent) are combined in various ratios and employed to optimize the systemic delivery of epinephrine through the nasal passages. In some embodiments, the concentration of the at least one enabling agent ranges from 0% and 20% w/w. The at least one enabling agent includes, but is not limited to, at least one mucoadhesive, at least one absorption enhancer, at least one permeability enhancer, at least one surfactant, at least one surface modifier, at least one sustained release agent, at least one vasodilator, at least one anticaking agent, and at least one disintegrant. In some embodiments, the formulation includes at least one excipient that impacts the characteristics of the dry powder formulation that directly or indirectly affect the transport of epinephrine across the nasal mucosa. In some embodiments, the formulation includes at least one excipient that facilitates solubilization or processing so as to provide a spray-dried powder having the desired characteristics.

In one embodiment, the invention provides a pharmaceutical formulation including dry powder epinephrine and magnesium stearate (e.g., dry powder magnesium stearate) as an enabling agent to optimize the systemic delivery of epinephrine through the nasal passages. In some embodiments, magnesium stearate and at least one excipient (e.g., at least one enabling agent, at least one carrier, and/or at least one flow agent) are combined in various ratios and employed to optimize the systemic delivery of epinephrine through the nasal passages. In some embodiments, the concentration of the at least one enabling agent ranges from 0% and 20% w/w. The at least one enabling agent includes, but is not limited to, at least one mucoadhesive, at least one absorption enhancer, at least one permeability enhancer, at least one surfactant, at least one surface modifier, at least one sustained release agent, at least one vasodilator, at least one anticaking agent, and at least one disintegrant. In some embodiments, the formulation includes at least one excipient

that impacts the characteristics of the dry powder formulation that directly or indirectly affect the transport of epinephrine across the nasal mucosa. In some embodiments, the formulation includes at least one excipient that facilitates solubilization or processing so as to provide a spray-dried powder having the desired characteristics.

Alternatively, in a preferred embodiment, the present invention does not include magnesium stearate. Magnesium stearate may cause allergic reactions in some individuals. See, e.g., Tammaro A, Abruzeze C, Narcisi A, Cortesi G, Persechino F, Parisella F R, Persechino S. Magnesium stearate: an underestimated allergen. *J Biol Regul Homeost Agents*. 2012 October-December; 26(4):783-4. PMID: 23241129, which is incorporated herein by reference in its entirety. Advantageously, sodium chloride is safer to use than magnesium stearate.

In one embodiment, the invention provides a pharmaceutical formulation including dry powder epinephrine and ethylenediaminetetraacetic acid (EDTA) (e.g., dry powder EDTA) as an enabling agent to optimize the systemic delivery of epinephrine through the nasal passages. In some embodiments, EDTA and at least one excipient (e.g., at least one enabling agent, at least one carrier, and/or at least one flow agent) are combined in various ratios and employed to optimize the systemic delivery of epinephrine through the nasal passages. In some embodiments, the concentration of the at least one enabling agent ranges from 0% and 20% w/w. The at least one enabling agent includes, but is not limited to, at least one mucoadhesive, at least one absorption enhancer, at least one permeability enhancer, at least one surfactant, at least one surface modifier, at least one sustained release agent, at least one vasodilator, at least one anticaking agent, and at least one disintegrant. In some embodiments, the formulation includes at least one excipient that impacts the characteristics of the dry powder formulation that directly or indirectly affect the transport of epinephrine across the nasal mucosa. In some embodiments, the formulation includes at least one excipient that facilitates solubilization or processing so as to provide a spray-dried powder having the desired characteristics.

In one embodiment, the invention provides a pharmaceutical formulation including dry powder epinephrine and sodium starch glycolate (e.g., dry powder sodium starch glycolate) as an enabling agent to optimize the systemic delivery of epinephrine through the nasal passages. In some embodiments, sodium starch glycolate and at least one excipient (e.g., at least one enabling agent, at least one carrier, and/or at least one flow agent) are combined in various ratios and employed to optimize the systemic delivery of epinephrine through the nasal passages. In some embodiments, the concentration of the at least one enabling agent ranges from 0% and 20% w/w. The at least one enabling agent includes, but is not limited to, at least one mucoadhesive, at least one absorption enhancer, at least one permeability enhancer, at least one surfactant, at least one surface modifier, at least one sustained release agent, at least one vasodilator, at least one anticaking agent, and at least one disintegrant. In some embodiments, the formulation includes at least one excipient that impacts the characteristics of the dry powder formulation that directly or indirectly affect the transport of epinephrine across the nasal mucosa. In some embodiments, the formulation includes at least one excipient that facilitates solubilization or processing so as to provide a spray-dried powder having the desired characteristics.

In a preferred embodiment, the invention provides a pharmaceutical formulation including dry powder epineph-

rine and lactose (e.g., dry powder lactose, dry powder D-lactose) as an enabling agent to optimize the systemic delivery of epinephrine through the nasal passages. In some embodiments, lactose and at least one excipient (e.g., at least one enabling agent, at least one carrier, and/or at least one flow agent) are combined in various ratios and employed to optimize the systemic delivery of epinephrine through the nasal passages. In some embodiments, the concentration of the at least one enabling agent ranges from 0% and 20% w/w. The at least one enabling agent includes, but is not limited to, at least one mucoadhesive, at least one absorption enhancer, at least one permeability enhancer, at least one surfactant, at least one surface modifier, at least one sustained release agent, at least one vasodilator, at least one anticaking agent, and at least one disintegrant. In some embodiments, the formulation includes at least one excipient that impacts the characteristics of the dry powder formulation that directly or indirectly affect the transport of epinephrine across the nasal mucosa. In some embodiments, the formulation includes at least one excipient that facilitates solubilization or processing so as to provide a spray-dried powder having the desired characteristics.

In one embodiment, the invention provides a pharmaceutical formulation including dry powder epinephrine and microcrystalline cellulose (e.g., dry powder microcrystalline cellulose) as an enabling agent to optimize the systemic delivery of epinephrine through the nasal passages. In some embodiments, microcrystalline cellulose and at least one excipient (e.g., at least one enabling agent, at least one carrier, and/or at least one flow agent) are combined in various ratios and employed to optimize the systemic delivery of epinephrine through the nasal passages. In some embodiments, the concentration of the at least one enabling agent ranges from 0% and 20% w/w. The at least one enabling agent includes, but is not limited to, at least one mucoadhesive, at least one absorption enhancer, at least one permeability enhancer, at least one surfactant, at least one surface modifier, at least one sustained release agent, at least one vasodilator, at least one anticaking agent, and at least one disintegrant. In some embodiments, the formulation includes at least one excipient that impacts the characteristics of the dry powder formulation that directly or indirectly affect the transport of epinephrine across the nasal mucosa. In some embodiments, the formulation includes at least one excipient that facilitates solubilization or processing so as to provide a spray-dried powder having the desired characteristics.

In one embodiment, the invention provides a pharmaceutical formulation including dry powder epinephrine and mannitol (e.g., dry powder mannitol) as an enabling agent to optimize the systemic delivery of epinephrine through the nasal passages. In some embodiments, mannitol and at least one excipient (e.g., at least one enabling agent, at least one carrier, and/or at least one flow agent) are combined in various ratios and employed to optimize the systemic delivery of epinephrine through the nasal passages. In some embodiments, the concentration of the at least one enabling agent ranges from 0% and 20% w/w. The at least one enabling agent includes, but is not limited to, at least one mucoadhesive, at least one absorption enhancer, at least one permeability enhancer, at least one surfactant, at least one surface modifier, at least one sustained release agent, at least one vasodilator, at least one anticaking agent, and at least one disintegrant. In some embodiments, the formulation includes at least one excipient that impacts the characteristics of the dry powder formulation that directly or indirectly affect the transport of epinephrine across the nasal mucosa.

In some embodiments, the formulation includes at least one excipient that facilitates solubilization or processing so as to provide a spray-dried powder having the desired characteristics.

In one embodiment, the invention provides a pharmaceutical formulation including dry powder epinephrine and a mixture of mannitol and hydroxypropyl methylcellulose (HPMC) (e.g., a mixture of dry powder mannitol and HPMC) as an enabling agent to optimize the systemic delivery of epinephrine through the nasal passages. In some embodiments, a mixture of mannitol and HPMC and at least one excipient (e.g., at least one enabling agent, at least one carrier, and/or at least one flow agent) are combined in various ratios and employed to optimize the systemic delivery of epinephrine through the nasal passages. In some embodiments, the concentration of the at least one enabling agent ranges from 0% and 20% w/w. The at least one enabling agent includes, but is not limited to, at least one mucoadhesive, at least one absorption enhancer, at least one permeability enhancer, at least one surfactant, at least one surface modifier, at least one sustained release agent, at least one vasodilator, at least one anticaking agent, and at least one disintegrant. In some embodiments, the formulation includes at least one excipient that impacts the characteristics of the dry powder formulation that directly or indirectly affect the transport of epinephrine across the nasal mucosa. In some embodiments, the formulation includes at least one excipient that facilitates solubilization or processing so as to provide a spray-dried powder having the desired characteristics.

It will be appreciated that, while the categories of pharmaceutical ingredients cited herein (that accompany the primary active ingredient of epinephrine) are used as a means of describing the inventive formulations, some of the ingredients will have multiple mechanisms of action that fall under more than one category. These categories of ingredients, including absorption enhancers, mucoadhesives, vasodilators, surfactants, sustainer release agents, carriers, and manufacturing process enablers, are therefore indicative while not being mutually exclusive. For example, an ingredient may serve as a carrier but also have a mucoadhesive action in the context of nasal delivery of epinephrine, or an ingredient may have actions as both a surfactant and absorption enhancer.

In some embodiments, the devised formulation(s) for delivery to the nasal passages and nasal mucosa are of a particle size having a range of about 3 μm to about 100 μm for epinephrine. In some embodiments, the devised formulation(s) for delivery to the nasal passages and nasal mucosa are of a particle size having a range of about 3 μm to about 50 μm for epinephrine. In another embodiment, the formulation(s) for delivery of epinephrine to the nasal passages and nasal mucosa are a nanoformulation with a particle size having a range of about 30 nm to about 100 nm.

In some embodiments, the devised formulation(s) for delivery of epinephrine to the nasal passages and nasal mucosa have a particle size for the at least one enabling agent, the at least one carrier, the at least one flow agent, and/or other generally accepted as safe (GRAS) pharmaceutical excipients having a range of about 3 μm to about 200 μm . In one embodiment, the formulation(s) for delivery of epinephrine to the nasal passages and nasal mucosa are a nanoformulation with a particle size having a range of about 30 nm to about 100 nm for the at least one enabling agent, the at least one carrier, the at least one flow agent, and/or other generally accepted as safe (GRAS) pharmaceutical excipients.

In another embodiment, provided herein is a product including a nasal delivery device containing the pharmaceutical formulations described herein. The nasal delivery device contains in a body, a cavity, or a reservoir of the device the required quantity of the pharmaceutical preparation to deliver the appropriate amount of epinephrine to the nasal passages and nasal mucosa for systemic delivery. The nasal delivery device preferably includes a pump spray device, a metered dose spray, or a mechanical spray device. In one embodiment, the nasal delivery device is a single dose device. Alternatively, the nasal delivery device is a multiple dose device.

In another embodiment, provided herein is a method of treating anaphylaxis, the method including delivering a dry powder formulation as provided herein intranasally to a subject in need thereof. In some instances, the dry powder formulation is administered using a nasal delivery device as described herein.

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Rat Studies

The plasma pharmacokinetics (PK) following intranasal (IN) delivery of a plurality of dry powder formulations of epinephrine administered at different doses in Sprague Dawley rats was determined. Epinephrine administered by the intramuscular (IM) route was used as a comparator. The dose formulations were one of seven combinations of epinephrine with either lactose or sodium carboxymethylcellulose (CMC) as the carrier. The excipients included sodium chloride, polysorbate (TWEEN 80), niacin, caffeine, hyaluronate, and leucine. Blood was collected at 7 time-points after dose administration using jugular catheterized animals. The nominal dose for the epinephrine IN formulations was 0.5 mg of epinephrine free base. The dose of the reference IM formulation was 0.7 mg/kg.

Each formulation was administered via intranasal instillation once using a nasal device (APTAR) with 5 mg of the formulation per unit, except for group 15 controls via intramuscular injection. Each nasal device was weighed before and after injection.

Blood (250 μ L) was collected from each animal prior to dosing and at 1, 3, 5, 10, 15, 30, and 60 minutes after dosing via a jugular catheter and placed into a K₂EDTA tube. Samples were centrifuged within 30 minutes of collection at 1500 g for 15 min at 4° C. and the plasma removed. For each sample, 5 μ L of SMBS (sodium metabisulfite) was added to each 100 μ L of plasma, and the sample was vortexed. Plasma samples were stored frozen at or below -70° C. in cryotubes for bioanalysis.

FIG. 10 is a table of epinephrine concentration-time data in rats for epinephrine with a lactose carrier. FIG. 11 is a table of epinephrine concentration-time data in rats for epinephrine with a sodium CMC carrier.

FIG. 12 is a table of epinephrine concentration-time data in rats for epinephrine and caffeine (5% w/w) with a lactose carrier. FIG. 13 is a table of epinephrine concentration-time data in rats for epinephrine and caffeine (5% w/w) with a sodium CMC carrier.

FIG. 14 is a table of epinephrine concentration-time data in rats for epinephrine and hyaluronate (0.5% w/w) with a lactose carrier. FIG. 15 is a table of epinephrine concentration-time data in rats for epinephrine and hyaluronate (0.5% w/w) with a sodium CMC carrier.

FIG. 16 is a table of epinephrine concentration-time data in rats for epinephrine and leucine (10% w/w) with a lactose carrier. FIG. 17 is a table of epinephrine concentration-time data in rats for epinephrine and leucine (10% w/w) with a sodium CMC carrier.

FIG. 18 is a table of epinephrine concentration-time data in rats for epinephrine and sodium chloride (2% w/w) with a lactose carrier. FIG. 19 is a table of epinephrine concentration-time data in rats for epinephrine and sodium chloride (2% w/w) with a sodium CMC carrier.

FIG. 20 is a table of epinephrine concentration-time data in rats for epinephrine and niacin (2% w/w) with a lactose carrier. FIG. 21 is a table of epinephrine concentration-time data in rats for epinephrine and niacin (2% w/w) with a sodium CMC carrier.

FIG. 22 is a table of epinephrine concentration-time data in rats for epinephrine and polysorbate (0.5% w/w) with a lactose carrier. FIG. 23 is a table of epinephrine concentration-time data in rats for epinephrine and polysorbate (0.5% w/w) with a sodium CMC carrier.

FIG. 24 is a table of epinephrine concentration-time data in rats for epinephrine intramuscular injection at 0.7 mg/kg.

FIG. 25 is a table of epinephrine pharmacokinetic parameters in rats for epinephrine with a lactose carrier including C_{max} (i.e., maximum observed plasma concentration), T_{max} (i.e., time of maximum concentration, obtained directly from the observed concentration versus time data), AUC_{last} (i.e., area under the concentration-time curve from time 0 to the last measurable concentration calculated by the linear up-log down trapezoidal method), $t_{1/2}$ (i.e., terminal phase half-life, estimated as $\ln(2)/\lambda_z$, where λ_z is the elimination rate constant estimated by linear regression of the log-transformed concentration versus time data), and AUC_{inf} (i.e., the area under the concentration-time curve from time 0 extrapolated to infinity, calculated as $AUC_{last} + C_{last}/\lambda_z$). FIG. 26 is a table of epinephrine pharmacokinetic parameters in rats for epinephrine with a sodium CMC carrier including C_{max} , T_{max} , AUC_{last} , $t_{1/2}$, and AUC_{inf} .

FIG. 27 is a table of epinephrine pharmacokinetic parameters in rats for epinephrine and caffeine (5% w/w) with a lactose carrier including C_{max} , T_{max} , AUC_{last} , $t_{1/2}$, and AUC_{inf} . FIG. 28 is a table of epinephrine pharmacokinetic parameters in rats for epinephrine and caffeine (5% w/w) with a sodium CMC carrier including C_{max} , T_{max} , AUC_{last} , $t_{1/2}$, and AUC_{inf} .

FIG. 29 is a table of epinephrine pharmacokinetic parameters in rats for epinephrine and hyaluronate (0.5% w/w) with a lactose carrier including C_{max} , T_{max} , AUC_{last} , $t_{1/2}$, and AUC_{inf} . FIG. 30 is a table of epinephrine pharmacokinetic parameters in rats for epinephrine and hyaluronate (0.5% w/w) with a sodium CMC carrier including C_{max} , T_{max} , AUC_{last} , $t_{1/2}$, and AUC_{inf} .

FIG. 31 is a table of epinephrine pharmacokinetic parameters in rats for epinephrine and leucine (10% w/w) with a lactose carrier including C_{max} , T_{max} , AUC_{last} , $t_{1/2}$, and AUC_{inf} . FIG. 32 is a table of epinephrine pharmacokinetic parameters in rats for epinephrine and leucine (10% w/w) with a sodium CMC carrier including C_{max} , T_{max} , AUC_{last} , $t_{1/2}$, and AUC_{inf} .

FIG. 33 is a table of epinephrine pharmacokinetic parameters in rats for epinephrine and sodium chloride (2% w/w) with a lactose carrier including C_{max} , T_{max} , AUC_{last} , $t_{1/2}$, and AUC_{inf} . FIG. 34 is a table of epinephrine pharmacokinetic parameters in rats for epinephrine and sodium chloride (2% w/w) with a sodium CMC carrier including C_{max} , T_{max} , AUC_{last} , $t_{1/2}$, and AUC_{inf} .

FIG. 35 is a table of epinephrine pharmacokinetic parameters in rats for epinephrine and niacin (2% w/w) with a lactose carrier including C_{max} , T_{max} , AUC_{last} , $t_{1/2}$, and AUC_{inf} . FIG. 36 is a table of epinephrine pharmacokinetic

parameters in rats for epinephrine and niacin (2% w/w) with a sodium CMC carrier including C_{max} , T_{max} , AUC_{last} , $t_{1/2}$, and AUC_{inf} .

FIG. 37 is a table of epinephrine pharmacokinetic parameters in rats for epinephrine and polysorbate (0.5% w/w) with a lactose carrier including C_{max} , T_{max} , AUC_{last} , $t_{1/2}$, and AUC_{inf} . FIG. 38 is a table of epinephrine pharmacokinetic parameters in rats for epinephrine and polysorbate (0.5% w/w) with a sodium CMC carrier including C_{max} , T_{max} , AUC_{last} , $t_{1/2}$, and AUC_{inf} .

FIG. 39 is a table of epinephrine pharmacokinetic parameters in rats for epinephrine intramuscular injection at 0.7 mg/kg including C_{max} , T_{max} , AUC_{last} , $t_{1/2}$, and AUC_{inf} .

FIG. 40 is a linear graph of epinephrine concentration-time profiles in rats for epinephrine with a lactose carrier. FIG. 41 is a log-linear graph of epinephrine concentration-time profiles in rats for epinephrine with a lactose carrier.

FIG. 42 is a linear graph of epinephrine concentration-time profiles in rats for epinephrine with a sodium CMC carrier. FIG. 43 is a log-linear graph of epinephrine concentration-time profiles in rats for epinephrine with a sodium CMC carrier.

FIG. 44 is a linear graph of epinephrine concentration-time profiles in rats for epinephrine and caffeine (5% w/w) with a lactose carrier. FIG. 45 is a log-linear graph of epinephrine concentration-time profiles in rats for epinephrine and caffeine (5% w/w) with a lactose carrier.

FIG. 46 is a linear graph of epinephrine concentration-time profiles in rats for epinephrine and caffeine (5% w/w) with a sodium CMC carrier. FIG. 47 is a log-linear graph of epinephrine concentration-time profiles in rats for epinephrine and caffeine (5% w/w) with a sodium CMC carrier.

FIG. 48 is a linear graph of epinephrine concentration-time profiles in rats for epinephrine and hyaluronate (0.5% w/w) with a lactose carrier. FIG. 49 is a log-linear graph of epinephrine concentration-time profiles in rats for epinephrine and hyaluronate (0.5% w/w) with a lactose carrier.

FIG. 50 is a linear graph of epinephrine concentration-time profiles in rats for epinephrine and hyaluronate (0.5% w/w) with a sodium CMC carrier. FIG. 51 is a log-linear graph of epinephrine concentration-time profiles in rats for epinephrine and hyaluronate (0.5% w/w) with a sodium CMC carrier.

FIG. 52 is a linear graph of epinephrine concentration-time profiles in rats for epinephrine and leucine (10% w/w) with a lactose carrier. FIG. 53 is a log-linear graph of epinephrine concentration-time profiles in rats for epinephrine and leucine (10% w/w) with a lactose carrier.

FIG. 54 is a linear graph of epinephrine concentration-time profiles in rats for epinephrine and leucine (10% w/w) with a sodium CMC carrier. FIG. 55 is a log-linear graph of epinephrine concentration-time profiles in rats for epinephrine and leucine (10% w/w) with a sodium CMC carrier.

FIG. 56 is a linear graph of epinephrine concentration-time profiles in rats for epinephrine and sodium chloride (2% w/w) with a lactose carrier. FIG. 57 is a log-linear graph of epinephrine concentration-time profiles in rats for epinephrine and sodium chloride (2% w/w) with a lactose carrier.

FIG. 58 is a linear graph of epinephrine concentration-time profiles in rats for epinephrine and sodium chloride (2% w/w) with a sodium CMC carrier. FIG. 59 is a log-linear graph of epinephrine concentration-time profiles in rats for epinephrine and sodium chloride (2% w/w) with a sodium CMC carrier.

FIG. 60 is a linear graph of epinephrine concentration-time profiles in rats for epinephrine and niacin (2% w/w)

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with a lactose carrier. FIG. 61 is a log-linear graph of epinephrine concentration-time profiles in rats for epinephrine and niacin (2% w/w) with a lactose carrier.

FIG. 62 is a linear graph of epinephrine concentration-time profiles in rats for epinephrine and niacin (2% w/w) with a sodium CMC carrier. FIG. 63 is a log-linear graph of epinephrine concentration-time profiles in rats for epinephrine and niacin (2% w/w) with a sodium CMC carrier.

FIG. 64 is a linear graph of epinephrine concentration-time profiles in rats for epinephrine and polysorbate (0.5% w/w) with a lactose carrier. FIG. 65 is a log-linear graph of epinephrine concentration-time profiles in rats for epinephrine and polysorbate (0.5% w/w) with a lactose carrier.

FIG. 66 is a linear graph of epinephrine concentration-time profiles in rats for epinephrine and polysorbate (0.5% w/w) with a sodium CMC carrier. FIG. 67 is a log-linear graph of epinephrine concentration-time profiles in rats for epinephrine and polysorbate (0.5% w/w) with a sodium CMC carrier.

FIG. 68 is a linear graph of epinephrine concentration-time profiles in rats for epinephrine intramuscular injection at 0.7 mg/kg. FIG. 69 is a log-linear graph of epinephrine concentration-time profiles in rats for epinephrine intramuscular injection at 0.7 mg/kg.

FIG. 70 is a graph of mean epinephrine concentration-time profiles following IN administration in rats for epinephrine with a lactose carrier and epinephrine with a sodium CMC carrier.

FIG. 71 is a graph of mean epinephrine concentration-time profiles following IN administration in rats for epinephrine and caffeine (5% w/w) with a lactose carrier and epinephrine and caffeine (5% w/w) with a sodium CMC carrier.

FIG. 72 is a graph of mean epinephrine concentration-time profiles following IN administration in rats for epinephrine and hyaluronate (0.5% w/w) with a lactose carrier and epinephrine and hyaluronate (0.5% w/w) with a sodium CMC carrier.

FIG. 73 is a graph of mean epinephrine concentration-time profiles following IN administration in rats for epinephrine and leucine (10% w/w) with a lactose carrier and epinephrine and leucine (10% w/w) with a sodium CMC carrier.

FIG. 74 is a graph of mean epinephrine concentration-time profiles following IN administration in rats for epinephrine and sodium chloride (2% w/w) with a lactose carrier and epinephrine and sodium chloride (2% w/w) with a sodium CMC carrier.

FIG. 75 is a graph of mean epinephrine concentration-time profiles following IN administration in rats for epinephrine and niacin (2% w/w) with a lactose carrier and epinephrine and niacin (2% w/w) with a sodium CMC carrier.

FIG. 76 is a graph of mean epinephrine concentration-time profiles following IN administration in rats for epinephrine and polysorbate (0.5% w/w) with a lactose carrier and epinephrine and polysorbate (0.5% w/w) with a sodium CMC carrier.

FIG. 77 is a graph of mean epinephrine concentration-time profiles following IN administration in rats grouped by lactose carrier including epinephrine, epinephrine and caffeine (5% w/w), epinephrine and hyaluronate (0.5% w/w), epinephrine and leucine (10% w/w), epinephrine and sodium chloride (2% w/w), epinephrine and niacin (2% w/w), and epinephrine and polysorbate (0.5% w/w).

FIG. 78 is a graph of mean epinephrine concentration-time profiles following IN administration in rats grouped by

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sodium CMC carrier including epinephrine, epinephrine and caffeine (5% w/w), epinephrine and hyaluronate (0.5% w/w), epinephrine and leucine (10% w/w), epinephrine and sodium chloride (2% w/w), epinephrine and niacin (2% w/w), and epinephrine and polysorbate (0.5% w/w).

FIG. 79 is a table of calibration curve data for epinephrine in rat potassium EDTA plasma.

FIG. 80 is a table of quality control evaluation data for epinephrine in rat potassium EDTA plasma.

For epinephrine alone, Cmax ranged from 28.0 to 153 ng/mL, Tmax ranged from 3 to 5 hours, and half-life averaged 8.38 minutes for lactose. Cmax ranged from 13.3 to 39.2 ng/mL, Tmax ranged from 10 to 30 minutes, and a half-life average was unable to be reliably estimated due to slow absorption for sodium CMC. AUClast values were similar, averaging 707 min*ng/mL for lactose and 755 min*ng/mL for sodium CMC.

For epinephrine with 5% caffeine, Cmax averaged 93.9 ng/mL with low variability, Tmax ranged from 3 to 5 minutes, and half-life averaged of 23.3 minutes for lactose. Cmax ranged from 29.5 ng/mL, Tmax ranged from 5 to 30 minutes, and a half-life average was unable to be reliably estimated due to slow absorption for sodium CMC. AUClast values were similar, averaging 902 min*ng/mL for lactose and 909 min*ng/mL for sodium CMC.

For epinephrine with 0.5% hyaluronate, Cmax averaged 20.0 ng/mL and Tmax ranged from 1 to 5 minutes for lactose. Cmax averaged 31.5 ng/mL and Tmax ranged from 15 to 60 minutes for sodium CMC. Half-life values could not be reliably estimated for either formulation. AUClast values averaged 76.1 min*ng/mL for lactose and much higher, 1300 min*ng/mL, for sodium CMC due to protracted absorption throughout the sampling interval.

For epinephrine with 10% leucine, no quantifiable concentrations of epinephrine were observed in any animal. Cmax averaged 19.8 ng/mL and Tmax ranged from 10 to 30 minutes. AUClast values averaged 582 min*ng/mL for sodium CMC due to protracted absorption throughout the sampling interval. Half-life and AUCinf estimates do not appear to be reliable.

For epinephrine with 2% sodium chloride, Cmax averaged 60.1 ng/mL with low variability, Tmax ranged from 3 to 10 minutes, and a half-life average was 27.9 minutes for lactose. Cmax averaged 68.9 ng/mL, Tmax ranged from 1 to 30 minutes, and a half-life average was 65.4 minutes for sodium CMC. AUClast values averaged 889 min*ng/mL for lactose and 1930 min*ng/mL for sodium CMC.

For epinephrine with 2% niacin, Cmax averaged 233 ng/mL, Tmax ranged from 1 to 3 minutes, and a half-life average was 11.9 minutes for lactose. Cmax was much lower, averaging 22.9 ng/mL, Tmax ranged from 5 to 30 minutes, and a half-life average was 56.1 minutes for sodium CMC. AUClast values averaged 934 min*ng/mL for lactose and 929 min*ng/mL for sodium CMC.

For epinephrine with 0.5% polysorbate, Cmax averaged 101 ng/mL, Tmax ranged from 1 to 5 minutes, and an AUClast value was 824 min*ng/mL for lactose. Cmax averaged 21.5 ng/mL, Tmax ranged from 1 to 15 minutes, and an AUClast value was 1580 min*ng/mL for sodium CMC. Half-life values could not be reliably estimated for either formulation.

Variability was high in plasma epinephrine concentrations following IM administration of epinephrine at a dose level of 0.7 mg/kg, precluding the calculation of relative bioavailability values.

Based on the studies, it was determined that rats are not a good species for pharmacokinetic studies for intranasal

epinephrine because the amount loaded in the device (APTAR) is lower than the optimal performance range of the device, which resulted in high variability. Further, administration to the animal required an adapter due to the small size of the rat nostril, which also may have created high variability in dose delivery and concentration-time data.

As shown in FIGS. 70-78, the fastest absorption of epinephrine occurs with lactose formulations (e.g., epinephrine alone, epinephrine with caffeine, niacin, or polysorbate). In contrast, formulations with sodium CMC typically displayed much slower absorption based on later observed Tmax values. The excipients leucine and hyaluronic acid resulted in the lowest absorption of epinephrine. The excipients caffeine and niacin resulted in the highest bioavailability; however, there are safety concerns with use due to animal deaths. Sodium chloride and polysorbate 80 were determined to be the excipients used in future canine pharmacokinetic studies.

Canine Studies

The plasma pharmacokinetics (PK) following intranasal (IN) delivery of four dry powder formulations of epinephrine administered at different doses in four anesthetized Beagle dogs was determined. Epinephrine administered by the intramuscular (IM) route was used as a comparator.

The formulations used in the study include a 40 mg dosage of 10% w/w epinephrine, 2% NaCl, 0.1% w/v polysorbate (TWEEN 80), and lactose as a dry powder pre-loaded into a nasal device (APTAR) for intranasal dosing; a 60 mg dosage of 10% w/w epinephrine, 2% NaCl, and lactose as a dry powder pre-loaded into a nasal device (APTAR) for intranasal dosing; a 60 mg dosage of 10% w/w epinephrine, 2% NaCl, 0.1% w/v polysorbate (TWEEN 80), and lactose as a dry powder pre-loaded into a nasal device (APTAR) for intranasal dosing; a 30 mg dosage of 20% w/w epinephrine, 2% NaCl, and lactose as a dry powder pre-loaded into a nasal device (APTAR) for intranasal dosing; a 30 mg dosage of 20% w/w epinephrine, 2% NaCl, 0.1% w/v polysorbate (TWEEN 80), and lactose as a dry powder pre-loaded into a nasal device (APTAR) for intranasal dosing; and a 0.3 mg IM dosage of epinephrine using an EPIPEN as shown in Table 2.

TABLE 2

Treatment	Route	Formulation Amount (mg)	Epinephrine Dose (mg)	No. of dogs dosed
Lactose + 2% NaCl + 0.1% w/v Tween 80 with 10% w/w Epinephrine	IN	40	4	4
Lactose + 2% NaCl w/ 10% w/w Epinephrine	IN	60	6	3
Lactose + 2% NaCl + 0.1% w/v Tween 80 w/ 10% w/w Epinephrine	IN	60	6	2
Lactose + 2% NaCl w/ 20% w/w Epinephrine	IN	30	6	4
Lactose + 2% NaCl + 0.1% w/v Tween 80 w/ 20% w/w Epinephrine	IN	30	6	3
EPIPEN IM	IM	—	0.3	4

For the studies, the animals received a single treatment of the comparator (EPIPEN) intramuscularly. For the studies, formulations in various amounts were administered intranasally in one nostril, with one spray representing one dose.

Variations in target dose were controlled by formulation and/or amount of powder loaded into the nasal device. At least two days of washout were present between Studies. Blood was collected at a plurality of time points before and after treatment: two samples were collected before dosing, at 10 minutes post anesthesia and at 20 minutes post anesthesia. Post-dose samples were obtained at 2 or 3 minutes, and at 5, 7, 10, 15, 20, 30, 60, 90, and 120 minutes after dosing.

30.0 mL of 200 mg/mL sodium metabisulfite (Na₂S₂O₅, CAS #7681-57-4), pH 3.0 was prepared by adding 6.00 g Na₂S₂O₅ to 24.0 mL of water. The pH was adjusted to 3.0 with hydrochloric acid and the solution was brought to a final volume of 30.0 mL. The 200 mg/mL solution of sodium metabisulfite, pH 3.0 is operable to be stored at 2-8° C. for up to 30 days.

Blood collection tubes containing potassium EDTA (ethylenediaminetetraacetic acid) were pre-spiked with 10 µL of the 200 mg/mL solution of sodium metabisulfite, pH 3.0 for the collection of 1 mL of whole blood. Tubes are operable to be kept on wet ice immediately prior to use. The blood samples collected in the pre-spiked tubes were centrifuged at 1300 g for a minimum of 10 minutes at 2° C. to 8° C. within 30 minutes of collection. The resulting plasma was removed, placed into a cryovial, and frozen at -90° C. to -70° C. until bioanalysis.

Filled devices were weighed prior to dosing and after dosing. The tip of the device was wiped free of obvious secretions prior to weighing the device after dosing.

To dose the animal, the nose was wiped free of any obvious secretions. One spray was delivered intranasally to one nostril per animal for each study. Animals were anesthetized with inhaled isoflurane for dosing and through all blood collection timepoints.

Plasma samples were analyzed for epinephrine concentration using a qualified LC-MS/MS method. The assay was verified with respect to standard bioanalytical methodology including acceptable accuracy and precision based upon quality control sample analysis and results. The quantifiable range of the method is 200 to 10,000 pg/mL.

Pharmacokinetic parameters were derived using noncompartmental methods employing PHOENIX WINNONLIN version 8.3 (Pharsight Corp, St. Louis, MO).

FIG. 81 illustrates a table of plasma epinephrine concentration-time data in dogs following a 40 mg dosage of 10% w/w epinephrine, 2% NaCl, 0.1% w/v polysorbate (TWEEN 80), and lactose. FIG. 82 illustrates a table of plasma epinephrine concentration-time data in dogs following a 60 mg dosage of 10% w/w epinephrine, 2% NaCl, and lactose. FIG. 83 illustrates a table of plasma epinephrine concentration-time data in dogs following a 60 mg dosage of 10% w/w epinephrine, 2% NaCl, 0.1% w/v polysorbate (TWEEN 80), and lactose. FIG. 84 illustrates a table of plasma epinephrine concentration-time data in dogs following a 30 mg dosage of 20% w/w epinephrine, 2% NaCl, and lactose. FIG. 85 illustrates a table of plasma epinephrine concentration-time data in dogs following a 30 mg dosage of 20% w/w epinephrine, 2% NaCl, 0.1% w/v polysorbate (TWEEN 80), and lactose. FIG. 86 illustrates a table of plasma epinephrine concentration-time data in dogs following a 0.3 mg IM dosage of epinephrine using an EPIPEN.

FIG. 87 illustrates a table of epinephrine pharmacokinetic parameters in dogs following a 40 mg dosage of 10% w/w epinephrine, 2% NaCl, 0.1% w/v polysorbate (TWEEN 80), and lactose. The pharmacokinetic parameters include Cmax (pg/mL), Tmax (min), and AUClast (min*pg/mL). FIG. 88 illustrates a table of epinephrine pharmacokinetic parameters in dogs following a 60 mg dosage of 10% w/w

FIG. 102 illustrates a log-linear graph of individual epinephrine concentration-time profiles for a 40 mg dosage of 10% w/w epinephrine, 2% NaCl, 0.1% w/v polysorbate (TWEEN 80), and lactose. FIG. 103 illustrates a log-linear graph of individual epinephrine concentration-time profiles for a 60 mg dosage of 10% w/w epinephrine, 2% NaCl, and

FIG. 115 illustrates a graph of partial AUC for 0-10 minutes by treatment. The treatments shown include a 40 mg dosage of 10% w/w epinephrine, 2% NaCl, 0.1% w/v polysorbate (TWEEN 80), and lactose; a 60 mg dosage of

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10% w/w epinephrine, 2% NaCl, and lactose; a 60 mg dosage of 10% w/w epinephrine, 2% NaCl, 0.1% w/v polysorbate (TWEEN 80), and lactose; a 30 mg dosage of 20% w/w epinephrine, 2% NaCl, and lactose; a 30 mg dosage of 20% w/w epinephrine, 2% NaCl, 0.1% w/v polysorbate (TWEEN 80), and lactose; and a 0.3 mg IM dosage of epinephrine using an EPIPEN.

FIG. 116 illustrates a graph of partial AUC for 0-20 minutes by treatment. The treatments shown include a 40 mg dosage of 10% w/w epinephrine, 2% NaCl, 0.1% w/v polysorbate (TWEEN 80), and lactose; a 60 mg dosage of 10% w/w epinephrine, 2% NaCl, and lactose; a 60 mg dosage of 10% w/w epinephrine, 2% NaCl, 0.1% w/v polysorbate (TWEEN 80), and lactose; a 30 mg dosage of 20% w/w epinephrine, 2% NaCl, and lactose; a 30 mg dosage of 20% w/w epinephrine, 2% NaCl, 0.1% w/v polysorbate (TWEEN 80), and lactose; and a 0.3 mg IM dosage of epinephrine using an EPIPEN.

FIG. 117 illustrates a graph of partial AUC for 0-30 minutes by treatment. The treatments shown include a 40 mg dosage of 10% w/w epinephrine, 2% NaCl, 0.1% w/v polysorbate (TWEEN 80), and lactose; a 60 mg dosage of 10% w/w epinephrine, 2% NaCl, and lactose; a 60 mg dosage of 10% w/w epinephrine, 2% NaCl, 0.1% w/v polysorbate (TWEEN 80), and lactose; a 30 mg dosage of 20% w/w epinephrine, 2% NaCl, and lactose; a 30 mg dosage of 20% w/w epinephrine, 2% NaCl, 0.1% w/v polysorbate (TWEEN 80), and lactose; and a 0.3 mg IM dosage of epinephrine using an EPIPEN.

FIG. 118 illustrates a graph of partial AUC for 0-60 minutes by treatment. The treatments shown include a 40 mg dosage of 10% w/w epinephrine, 2% NaCl, 0.1% w/v polysorbate (TWEEN 80), and lactose; a 60 mg dosage of 10% w/w epinephrine, 2% NaCl, and lactose; a 60 mg dosage of 10% w/w epinephrine, 2% NaCl, 0.1% w/v polysorbate (TWEEN 80), and lactose; a 30 mg dosage of 20% w/w epinephrine, 2% NaCl, and lactose; a 30 mg dosage of 20% w/w epinephrine, 2% NaCl, 0.1% w/v polysorbate (TWEEN 80), and lactose; and a 0.3 mg IM dosage of epinephrine using an EPIPEN.

FIG. 119 illustrates a graph of individual animal plasma epinephrine concentration-time profiles for 0.3 mg IM administration for a pilot study and a main study.

FIG. 120 illustrates a graph of mean plasma epinephrine concentration-time profiles for 0.3 mg IM administration for a pilot study and a main study.

Tabular results with summary statistics are displayed in FIGS. 81-85 for the IN formulations and in FIG. 86 for IM administered EPIPEN. The concentration-time data are plotted for individual animals within each treatment group in linear (FIGS. 96-101) and log-linear (FIGS. 102-107). FIGS. 108-113 contain plots of the data for each treatment received by each individual animal.

Measurable epinephrine concentrations were observed in all dosed animals for all formulations treatments at the first sampling time (e.g., 2 or 3 minutes after dosing). All dosed animals also had measurable concentrations at 120 minutes after dosing. All pre-dose concentrations were below the lower limit of quantification (LLOQ) of the method of 200 pg/mL. Following IN administration, most dogs displayed an initial peak followed by a decay phase that lasted until about 30 minutes after dosing. Concentrations were then relatively flat through to the 120-minute sample. However, for the EPIPEN, all dogs displayed an initial peak followed by a later peak that, in 3 out of 4 dogs was higher than the initial one. Between-dog variability was high for all IN treatments and the EPIPEN formulation. For the IN formu-

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lations, the percent coefficient of variation (% CV) values ranged from 36% to 98.6% for all formulations except the 20%/30 mg formulation where the % CV values were typically <30%. The EPIPEN formulation also displayed high variability with most of % CV values >50%.

Pharmacokinetic parameters for epinephrine following IN formulation administration are summarized in FIGS. 87-91. There was insufficient terminal phase data to permit reliable estimation of half-lives so the noncompartmental pharmacokinetic analysis was restricted to C_{max}, T_{max}, and AUC_{last}. Epinephrine was rapidly absorbed following IN administration as evidenced by T_{max} occurring at either 2 or 3 minutes for the majority of dogs. There were a limited number of T_{max} values that occurred at 5 to 7 minutes. A single dog had a T_{max} at 120 minutes after dosing. Mean C_{max} values ranged from 2840 to 8850 ng/mL with considerable variability between formulation and between-dog variability. Mean AUC_{last} values ranged from 147,000 to 367,000 min*pg/mL, also with high variability.

Pharmacokinetic parameters after EPIPEN dosing are summarized in FIG. 92. T_{max} values tended to occur at later times than those observed following IN dosing. One dog displayed a T_{max} of 3 minutes, but 2 dogs had T_{max} values of 60 minutes and for 1 dog the T_{max} occurred at the last sampling time at 120 minutes. Mean C_{max} was 1710 pg/mL and AUC_{last} averaged 132,000 min*pg/mL. Both were quite variable with % CV of 45% and 36%, respectively.

Partial AUC values were calculated, and box plots of the results are displayed in FIGS. 115-118. The data indicated that much of the AUC following administration of the IN formulations occurs early in the concentration-time profile, whereas the opposite is true for the EPIPEN and the 20%/30 mg formulation with polysorbate.

AUC_{last} values for all dogs and the treatments they received are presented in FIG. 93. These values were used to calculate the relative and comparative bioavailability for the IN formulations in comparison to EPIPEN and the results are summarized in FIG. 94 and FIG. 95, respectively. This study used a crossover design to reduce variability, but since a replacement dog was required, and the results from 3 dogs were excluded from the analysis, all dogs did not receive all treatments as shown in FIG. 93. For dogs who did receive both a IN formulation and an EPIPEN dose, the relative bioavailability ranged from 4.3 to 16.3%. The combination of the IN doses of 4 or 6 mg epinephrine and the relative bioavailability of those formulations resulted in comparative bioavailability values ranging from 65.2 to 325%.

Surprisingly, while polysorbate is a surfactant, formulations with polysorbate are more available than formulations without the polysorbate, which is illustrated in the figures above. See, e.g., FIG. 111. In one example, the presence of polysorbate elevated the whole curve, increasing the AUC, but not affecting the kinetics (e.g., T_{max}, elimination rate). In another example, the presence of polysorbate shifted the T_{max} towards the y-axis (i.e., earlier) and changed the clearance rate (i.e., tail on the right) by making the epinephrine more readily available, thereby shutting down the vasculature (vasoconstriction).

Analytical Test Data

Four batches of epinephrine were studied for shot weight, actuation force, particle distribution, and moisture content.

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TABLE 3

	Batch 1	Batch 2	Batch 3	Batch 4
Lactose (% w/w)	88	88	78	78
Epinephrine (% w/w)	10	10	20	20
NaCl (% w/w)	2	2	2	2
Tween 80 (% w/v)	N/A	0.1	N/A	0.1

Each sample included a fill weight of 30 mg. The shot weight (i.e., the weight of composition metered with actuation) was assessed using a Proveris Actuator (Hudson, MA) with an actuation velocity of 70 mm/s, an actuation acceleration of 5000 m/s², a contact force of 0.3 kg, an end of stroke force of 8 kg, and a hold time of 300 ms. In each case, the shot weight was found to be reproducible, indicating no adverse clogging or blocking of the valve mechanism. As shown in FIG. 121, the results indicate that at least 90% of the filled dose is delivered.

The actuation force was assessed using an actuation velocity of 70 mm/s, an actuation acceleration of 5000 m/s², a contact force of 0.3 kg, an end of stroke force of 8 kg, and a hold time of 300 ms. Two samples were tested for Batch 1, Batch 2, and Batch 4. In each case, the actuation force was found to be reproducible. One sample was tested for Batch 3. FIG. 122 illustrates the actuation force for the detachment center piece and the detachment ball. A force to actuate representative profile is shown in FIG. 123.

Particle size distribution was assessed using an actuation velocity of 70 mm/s, an actuation acceleration of 5000 m/s², a contact force of 0.3 kg, an end of stroke force of 8 kg, and a hold time of 300 ms. FIG. 124 illustrates particle size distribution at 4 cm in complete phase. FIG. 125 illustrates particle distribution at 4 cm in complete phase without outliers. As shown in FIGS. 124-125, the results indicate a large variability for Batch 3 and Batch 4. Additionally, a higher percentage of fine particles are present for Batch 1 and Batch 3.

Moisture content was determined using Karl Fischer as shown in FIG. 126. No moisture content was performed on Batch 1 because the formulation was too sticky. Results for Batches 2-4, which were performed in duplicate, are shown in Table 4 below. The second repetition for Batches 3 and 4 has a higher water content, suggesting a high hygroscopicity of the material, which was visually confirmed. The formulation becomes stickier and more transparent in less than 10 minutes.

TABLE 4

Sample Name	Water Content (%)
Batch 2-1	4.34
Batch 2-2	4.47
Batch 3-1	8.58
Batch 3-2	13.09
Batch 4-1	5.13
Batch 4-2	11.93

Visual appearance of all four batches was noted immediately after opening and after exposure to laboratory conditions as shown in Table 5.

TABLE 5

	Immediately After Opening	After Exposure to Laboratory Conditions
Batch 1	White, agglomerated, and "sticky"	Becomes translucent and "sticky"

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TABLE 5-continued

	Immediately After Opening	After Exposure to Laboratory Conditions
Batch 2	White powder, electrostatic	Stays white, but becomes slightly "sticky"
Batch 3	White, agglomerated, and "sticky"	Becomes slightly translucent and "sticky"
Batch 4	White powder, electrostatic	Stays white, but becomes slightly "sticky"

Impact of device orientation was determined for the UDS-P nasal delivery device. Delivery angle and insertion have an impact on regional deposition. Lower angles (e.g., 30° and 45° result in higher posterior deposition. Higher insertion depth results in higher olfactory region deposition.

FIG. 127 illustrates a delivery angle. FIG. 128 illustrates an angle from the center wall of the nose.

FIG. 129 illustrates a graph of impact of insertion depth and angle using a 5° angle from the center wall. Samples (shown left to right) include a 60° delivery angle, 5° angle from the center wall, and 0.5 cm insertion depth; a 60° delivery angle, 5° angle from the center wall, and 1 cm insertion depth; a 60° delivery angle, 5° angle from the center wall, and 1.5 cm insertion depth; a 45° delivery angle, 5° angle from the center wall, and 0.5 cm insertion depth; a 45° delivery angle, 5° angle from the center wall, and 1 cm insertion depth; a 45° delivery angle, 5° angle from the center wall, and 1.5 cm insertion depth; a 30° delivery angle, 5° angle from the center wall, and 0.5 cm insertion depth; a 30° delivery angle, 5° angle from the center wall, and 1 cm insertion depth; and a 30° delivery angle, 5° angle from the center wall, and 1.5 cm insertion depth. The graph includes % deposition for the nose, nasal valve, floor, turbinates, ethmoids, rhino-pharynx, and lungs.

EXAMPLE COMBINATIONS

The following are examples of embodiments used in combination. However, the present disclosure is not limited to the example embodiments provided below. The intranasal dry powder compositions and/or unit doses are operable to include any combination of the at least one active pharmaceutical ingredient, the at least one enabling agent, and/or the at least one carrier and/or excipient.

Example Composition 1

In one embodiment, the intranasal dry powder compositions and/or unit doses include a vasoactive agent (e.g., epinephrine). In one embodiment, the intranasal dry powder compositions and/or unit doses further include a vasodilator (e.g., phentolamine). In one embodiment, the intranasal dry powder compositions and/or unit doses further include a COMT inhibitor (e.g., entacapone). In one embodiment, the intranasal dry powder compositions and/or unit doses further include at least one enabling agent and/or at least one carrier and/or excipient.

The at least one enabling agent includes, but is not limited to, at least one epinephrine potentiator, at least one mucoadhesive, at least one absorption enhancer, at least one permeability enhancer, at least one surfactant, at least one surface modifier, at least one sustained release agent, at least one anticaking agent, at least one systemic vasodilator, at least one nasal mucosal vasodilator, at least one mucosal permeation enhancer, at least one agent that reduces mucosal transit time, at least one agent that increases mucosal absorption or adhesion or transport, at least one chelator, at least

one steroid, at least one non-sulfite stabilizer, at least one preservative, at least one thickening agent, at least one humectant, at least one antihistamine, at least one solubilizing agent, at least one masking agent (e.g., taste, smell), at least one antioxidant, at least one viscosity enhancing agent, at least one dispersing agent, and/or at least one colorant. In some instances, the enabling agent includes at least one agent that reduces mucosal transit time, at least one agent that increases mucosal absorption and/or adhesion, at least one agent that enhances mucosal transport, or the enantiomers, diastereoisomers, racemates, or salts of such compounds with pharmaceutically acceptable counterions.

In one embodiment, the at least one enabling agent includes, but is not limited to, vasopressin, caffeine, at least one antihistamine (e.g., diphenhydramine, doxylamine, loratadine, desloratadine, cetirizine, levocetirizine, fexofenadine, famotidine, cimetidine, nizatidine), hydrocortisone, EDTA, magnesium stearate, tribasic calcium phosphate, citrate, citric acid, and/or ascorbic acid. In one embodiment, the at least one enabling agent includes vasopressin. In one embodiment, the at least one enabling agent includes caffeine. In one embodiment, the at least one enabling agent includes diphenhydramine. In one embodiment, the at least one enabling agent includes doxylamine. In one embodiment, the at least one enabling agent includes loratadine. In one embodiment, the at least one enabling agent includes desloratadine. In one embodiment, the at least one enabling agent includes cetirizine. In one embodiment, the at least one enabling agent includes levocetirizine. In one embodiment, the at least one enabling agent includes fexofenadine. In one embodiment, the at least one enabling agent includes famotidine. In one embodiment, the at least one enabling agent includes cimetidine. In one embodiment, the at least one enabling agent includes nizatidine. In one embodiment, the at least one enabling agent includes hydrocortisone. In one embodiment, the at least one enabling agent includes EDTA. In one embodiment, the at least one enabling agent includes tribasic calcium phosphate. In one embodiment, the at least one enabling agent includes citrate. In one embodiment, the at least one enabling agent includes citric acid. In one embodiment, the at least one enabling agent includes ascorbic acid.

In one embodiment, the at least one carrier and/or excipient includes at least one cellulose derivative and/or starch. In one embodiment, the at least one carrier and/or excipient includes lactose. In one embodiment, the at least one carrier and/or excipient includes sodium carboxymethylcellulose. In one embodiment, the at least one carrier and/or excipient includes chitosan. In one embodiment, the at least one carrier and/or excipient includes a cyclodextrin (e.g., dimethyl-beta-cyclodextrin). In one embodiment, the at least one carrier and/or excipient includes mannitol. In one embodiment, the at least one carrier and/or excipient includes dodecylphosphocholine. In one embodiment, the at least one carrier and/or excipient includes an alkyl saccharide. In one embodiment, the at least one carrier and/or excipient includes ethanol (e.g., dried ethanol). In one embodiment, the at least one carrier and/or excipient includes lactose and hyaluronic acid (e.g., sodium hyaluronate). In one embodiment, the at least one carrier and/or excipient includes lactose and leucine. In one embodiment, the at least one carrier and/or excipient includes lactose and sodium chloride. In one embodiment, the at least one carrier and/or excipient includes lactose and niacin. In one embodiment, the at least one carrier and/or excipient includes lactose and polysorbate. In one embodiment, the at least one carrier and/or excipient includes sodium carboxymethylcel-

lulose and hyaluronic acid (e.g., sodium hyaluronate). In one embodiment, the at least one carrier and/or excipient includes sodium carboxymethylcellulose and leucine. In one embodiment, the at least one carrier and/or excipient includes sodium carboxymethylcellulose and sodium chloride. In one embodiment, the at least one carrier and/or excipient includes sodium carboxymethylcellulose and niacin. In one embodiment, the at least one carrier and/or excipient includes sodium carboxymethylcellulose and/or polysorbate. In one embodiment, the at least one carrier and/or excipient includes sodium carboxymethylcellulose, lactose, sodium chloride, and/or polysorbate. In one embodiment, the polysorbate is polysorbate 80. In one embodiment, the sodium chloride is present in a range between 0.01% w/w and 5% w/w. In one embodiment, the polysorbate is present in a range between 0.01% w/v and 5% w/v. In another embodiment, the polysorbate is present in a range between 0.01% w/v and 1% w/v.

Example Composition 2

In one embodiment, the intranasal dry powder compositions and/or unit doses include a vasoactive agent (e.g., epinephrine) and at least one antihistamine. In one embodiment, the at least one antihistamine includes, but is not limited to, diphenhydramine, doxylamine, loratadine, desloratadine, cetirizine, levocetirizine, fexofenadine, famotidine, cimetidine, and/or nizatidine. In one embodiment, the intranasal dry powder compositions and/or unit doses further include a vasodilator (e.g., phenolamine). In one embodiment, the intranasal dry powder compositions and/or unit doses further include a COMT inhibitor (e.g., entacapone). In one embodiment, the intranasal dry powder compositions and/or unit doses further include at least one enabling agent and/or at least one carrier and/or excipient.

The at least one enabling agent includes, but is not limited to, at least one epinephrine potentiator, at least one mucoadhesive, at least one absorption enhancer, at least one permeability enhancer, at least one surfactant, at least one surface modifier, at least one sustained release agent, at least one anticaking agent, at least one systemic vasodilator, at least one nasal mucosal vasodilator, at least one mucosal permeation enhancer, at least one agent that reduces mucosal transit time, at least one agent that increases mucosal absorption or adhesion or transport, at least one chelator, at least one steroid, at least one non-sulfite stabilizer, at least one preservative, at least one thickening agent, at least one humectant, at least one solubilizing agent, at least one masking agent (e.g., taste, smell), at least one antioxidant, at least one viscosity enhancing agent, at least one dispersing agent, and/or at least one colorant. In some instances, the enabling agent includes at least one agent that reduces mucosal transit time, at least one agent that increases mucosal absorption and/or adhesion, at least one agent that enhances mucosal transport, or the enantiomers, diastereoisomers, racemates, or salts of such compounds with pharmaceutically acceptable counterions.

In one embodiment, the at least one enabling agent includes, but is not limited to, vasopressin, caffeine, hydrocortisone, EDTA, magnesium stearate, tribasic calcium phosphate, citrate, citric acid, and/or ascorbic acid. In one embodiment, the at least one enabling agent includes vasopressin. In one embodiment, the at least one enabling agent includes caffeine. In one embodiment, the at least one enabling agent includes hydrocortisone. In one embodiment, the at least one enabling agent includes EDTA. In one embodiment, the at least one enabling agent includes mag-

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nesium stearate. In one embodiment, the at least one enabling agent includes tribasic calcium phosphate. In one embodiment, the at least one enabling agent includes citrate. In one embodiment, the at least one enabling agent includes citric acid. In one embodiment, the at least one enabling agent includes ascorbic acid.

In one embodiment, the at least one carrier and/or excipient includes at least one cellulose derivative and/or starch. In one embodiment, the at least one carrier and/or excipient includes lactose. In one embodiment, the at least one carrier and/or excipient includes sodium carboxymethylcellulose. In one embodiment, the at least one carrier and/or excipient includes chitosan. In one embodiment, the at least one carrier and/or excipient includes a cyclodextrin (e.g., dimethyl-beta-cyclodextrin). In one embodiment, the at least one carrier and/or excipient includes mannitol. In one embodiment, the at least one carrier and/or excipient includes dodecylphosphocholine. In one embodiment, the at least one carrier and/or excipient includes an alkyl saccharide. In one embodiment, the at least one carrier and/or excipient includes ethanol (e.g., dried ethanol). In one embodiment, the at least one carrier and/or excipient includes lactose and hyaluronic acid (e.g., sodium hyaluronate). In one embodiment, the at least one carrier and/or excipient includes lactose and leucine. In one embodiment, the at least one carrier and/or excipient includes lactose and sodium chloride. In one embodiment, the at least one carrier and/or excipient includes lactose and niacin. In one embodiment, the at least one carrier and/or excipient includes lactose and polysorbate. In one embodiment, the at least one carrier and/or excipient includes sodium carboxymethylcellulose and hyaluronic acid (e.g., sodium hyaluronate). In one embodiment, the at least one carrier and/or excipient includes sodium carboxymethylcellulose and leucine. In one embodiment, the at least one carrier and/or excipient includes sodium carboxymethylcellulose and sodium chloride. In one embodiment, the at least one carrier and/or excipient includes sodium carboxymethylcellulose and niacin. In one embodiment, the at least one carrier and/or excipient includes sodium carboxymethylcellulose and/or polysorbate. In one embodiment, the at least one carrier and/or excipient includes sodium carboxymethylcellulose, lactose, sodium chloride, and/or polysorbate. In one embodiment, the polysorbate is polysorbate 80. In one embodiment, the sodium chloride is present in a range between 0.01% w/w and 5% w/w. In one embodiment, the polysorbate is present in a range between 0.01% w/v and 5% w/v. In another embodiment, the polysorbate is present in a range between 0.01% w/v and 1% w/v.

Example Composition 3

In one embodiment, the intranasal dry powder compositions and/or unit doses include a vasoactive agent (e.g., epinephrine) and at least one steroid (e.g., hydrocortisone). In one embodiment, the intranasal dry powder compositions and/or unit doses further include a vasodilator (e.g., phenolamine). In one embodiment, the intranasal dry powder compositions and/or unit doses further include a COMT inhibitor (e.g., entacapone). In one embodiment, the intranasal dry powder compositions and/or unit doses further include at least one enabling agent and/or at least one carrier and/or excipient.

The at least one enabling agent includes, but is not limited to, at least one epinephrine potentiator, at least one mucoadhesive, at least one absorption enhancer, at least one permeability enhancer, at least one surfactant, at least one

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surface modifier, at least one sustained release agent, at least one anticaking agent, at least one systemic vasodilator, at least one nasal mucosal vasodilator, at least one mucosal permeation enhancer, at least one agent that reduces mucosal transit time, at least one agent that increases mucosal absorption or adhesion or transport, at least one chelator, at least one non-sulfite stabilizer, at least one preservative, at least one thickening agent, at least one humectant, at least one antihistamine, at least one solubilizing agent, at least one masking agent (e.g., taste, smell), at least one antioxidant, at least one viscosity enhancing agent, at least one dispersing agent, and/or at least one colorant. In some instances, the enabling agent includes at least one agent that reduces mucosal transit time, at least one agent that increases mucosal absorption and/or adhesion, at least one agent that enhances mucosal transport, or the enantiomers, diastereoisomers, racemates, or salts of such compounds with pharmaceutically acceptable counterions.

In one embodiment, the at least one enabling agent includes, but is not limited to, vasopressin, caffeine, at least one antihistamine (e.g., diphenhydramine, doxylamine, loratadine, desloratadine, cetirizine, levocetirizine, fexofenadine, famotidine, cimetidine, nizatidine), hydrocortisone, EDTA, magnesium stearate, tribasic calcium phosphate, citrate, citric acid, and/or ascorbic acid. In one embodiment, the at least one enabling agent includes vasopressin. In one embodiment, the at least one enabling agent includes caffeine. In one embodiment, the at least one enabling agent includes diphenhydramine. In one embodiment, the at least one enabling agent includes doxylamine. In one embodiment, the at least one enabling agent includes loratadine. In one embodiment, the at least one enabling agent includes desloratadine. In one embodiment, the at least one enabling agent includes cetirizine. In one embodiment, the at least one enabling agent includes levocetirizine. In one embodiment, the at least one enabling agent includes fexofenadine. In one embodiment, the at least one enabling agent includes famotidine. In one embodiment, the at least one enabling agent includes cimetidine. In one embodiment, the at least one enabling agent includes nizatidine. In one embodiment, the at least one enabling agent includes EDTA. In one embodiment, the at least one enabling agent includes magnesium stearate. In one embodiment, the at least one enabling agent includes tribasic calcium phosphate. In one embodiment, the at least one enabling agent includes citrate. In one embodiment, the at least one enabling agent includes citric acid. In one embodiment, the at least one enabling agent includes ascorbic acid.

In one embodiment, the at least one carrier and/or excipient includes at least one cellulose derivative and/or starch. In one embodiment, the at least one carrier and/or excipient includes lactose. In one embodiment, the at least one carrier and/or excipient includes sodium carboxymethylcellulose. In one embodiment, the at least one carrier and/or excipient includes chitosan. In one embodiment, the at least one carrier and/or excipient includes a cyclodextrin (e.g., dimethyl-beta-cyclodextrin). In one embodiment, the at least one carrier and/or excipient includes mannitol. In one embodiment, the at least one carrier and/or excipient includes dodecylphosphocholine. In one embodiment, the at least one carrier and/or excipient includes an alkyl saccharide. In one embodiment, the at least one carrier and/or excipient includes ethanol (e.g., dried ethanol). In one embodiment, the at least one carrier and/or excipient includes lactose and hyaluronic acid (e.g., sodium hyaluronate). In one embodiment, the at least one carrier and/or excipient includes lactose and leucine. In one embodiment,

the at least one carrier and/or excipient includes lactose and sodium chloride. In one embodiment, the at least one carrier and/or excipient includes lactose and niacin. In one embodiment, the at least one carrier and/or excipient includes lactose and polysorbate. In one embodiment, the at least one carrier and/or excipient includes sodium carboxymethylcellulose and hyaluronic acid (e.g., sodium hyaluronate). In one embodiment, the at least one carrier and/or excipient includes sodium carboxymethylcellulose and leucine. In one embodiment, the at least one carrier and/or excipient includes sodium carboxymethylcellulose and sodium chloride. In one embodiment, the at least one carrier and/or excipient includes sodium carboxymethylcellulose and niacin. In one embodiment, the at least one carrier and/or excipient includes sodium carboxymethylcellulose and/or polysorbate. In one embodiment, the at least one carrier and/or excipient includes sodium carboxymethylcellulose, lactose, sodium chloride, and/or polysorbate. In one embodiment, the polysorbate is polysorbate 80. In one embodiment, the sodium chloride is present in a range between 0.01% w/w and 5% w/w. In one embodiment, the polysorbate is present in a range between 0.01% w/v and 5% w/v. In another embodiment, the polysorbate is present in a range between 0.01% w/v and 5% w/v.

Example Composition 4

In one embodiment, the intranasal dry powder compositions and/or unit doses include a vasoactive agent (e.g., epinephrine) and a vasodilator (e.g., phenolamine). In one embodiment, the intranasal dry powder compositions and/or unit doses further include a COMT inhibitor (e.g., entacapone). In one embodiment, the intranasal dry powder compositions and/or unit doses further include at least one enabling agent and/or at least one carrier and/or excipient.

The at least one enabling agent includes, but is not limited to, at least one epinephrine potentiator, at least one mucoadhesive, at least one absorption enhancer, at least one permeability enhancer, at least one surfactant, at least one surface modifier, at least one sustained release agent, at least one anticaking agent, at least one systemic vasodilator, at least one nasal mucosal vasodilator, at least one mucosal permeation enhancer, at least one agent that reduces mucosal transit time, at least one agent that increases mucosal absorption or adhesion or transport, at least one chelator, at least one steroid, at least one non-sulfite stabilizer, at least one preservative, at least one thickening agent, at least one humectant, at least one antihistamine, at least one solubilizing agent, at least one masking agent (e.g., taste, smell), at least one antioxidant, at least one viscosity enhancing agent, at least one dispersing agent, and/or at least one colorant. In some instances, the enabling agent includes at least one agent that reduces mucosal transit time, at least one agent that increases mucosal absorption and/or adhesion, at least one agent that enhances mucosal transport, or the enantiomers, diastereoisomers, racemates, or salts of such compounds with pharmaceutically acceptable counterions.

In one embodiment, the at least one enabling agent includes, but is not limited to, vasopressin, caffeine, at least one antihistamine (e.g., diphenhydramine, doxylamine, loratadine, desloratadine, cetirizine, levocetirizine, fexofenadine, famotidine, cimetidine, nizatidine), hydrocortisone, EDTA, magnesium stearate, tribasic calcium phosphate, citrate, citric acid, and/or ascorbic acid. In one embodiment, the at least one enabling agent includes vasopressin. In one embodiment, the at least one enabling agent includes caffeine. In one embodiment, the at least one enabling agent

includes diphenhydramine. In one embodiment, the at least one enabling agent includes doxylamine. In one embodiment, the at least one enabling agent includes loratadine. In one embodiment, the at least one enabling agent includes desloratadine. In one embodiment, the at least one enabling agent includes cetirizine. In one embodiment, the at least one enabling agent includes levocetirizine. In one embodiment, the at least one enabling agent includes fexofenadine. In one embodiment, the at least one enabling agent includes famotidine. In one embodiment, the at least one enabling agent includes cimetidine. In one embodiment, the at least one enabling agent includes nizatidine. In one embodiment, the at least one enabling agent includes hydrocortisone. In one embodiment, the at least one enabling agent includes EDTA. In one embodiment, the at least one enabling agent includes magnesium stearate. In one embodiment, the at least one enabling agent includes tribasic calcium phosphate. In one embodiment, the at least one enabling agent includes citrate. In one embodiment, the at least one enabling agent includes citric acid. In one embodiment, the at least one enabling agent includes ascorbic acid.

In one embodiment, the at least one carrier and/or excipient includes at least one cellulose derivative and/or starch. In one embodiment, the at least one carrier and/or excipient includes lactose. In one embodiment, the at least one carrier and/or excipient includes sodium carboxymethylcellulose. In one embodiment, the at least one carrier and/or excipient includes chitosan. In one embodiment, the at least one carrier and/or excipient includes a cyclodextrin (e.g., dimethyl-beta-cyclodextrin). In one embodiment, the at least one carrier and/or excipient includes mannitol. In one embodiment, the at least one carrier and/or excipient includes dodecylphosphocholine. In one embodiment, the at least one carrier and/or excipient includes an alkyl saccharide. In one embodiment, the at least one carrier and/or excipient includes ethanol (e.g., dried ethanol). In one embodiment, the at least one carrier and/or excipient includes lactose and hyaluronic acid (e.g., sodium hyaluronate). In one embodiment, the at least one carrier and/or excipient includes lactose and leucine. In one embodiment, the at least one carrier and/or excipient includes lactose and sodium chloride. In one embodiment, the at least one carrier and/or excipient includes lactose and niacin. In one embodiment, the at least one carrier and/or excipient includes lactose and polysorbate. In one embodiment, the at least one carrier and/or excipient includes sodium carboxymethylcellulose and hyaluronic acid (e.g., sodium hyaluronate). In one embodiment, the at least one carrier and/or excipient includes sodium carboxymethylcellulose and leucine. In one embodiment, the at least one carrier and/or excipient includes sodium carboxymethylcellulose and sodium chloride. In one embodiment, the at least one carrier and/or excipient includes sodium carboxymethylcellulose and niacin. In one embodiment, the at least one carrier and/or excipient includes sodium carboxymethylcellulose and/or polysorbate. In one embodiment, the at least one carrier and/or excipient includes sodium carboxymethylcellulose, lactose, sodium chloride, and/or polysorbate. In one embodiment, the polysorbate is polysorbate 80. In one embodiment, the sodium chloride is present in a range between 0.01% w/w and 5% w/w. In one embodiment, the polysorbate is present in a range between 0.01% w/v and 5% w/v. In another embodiment, the polysorbate is present in a range between 0.01% w/v and 1% w/v.

Example Composition 5

In one embodiment, the intranasal dry powder compositions and/or unit doses include a vasoactive agent (e.g.,

epinephrine) and sodium chloride. In one embodiment, the intranasal dry powder compositions and/or unit doses further include a vasodilator (e.g., phentolamine). In one embodiment, the intranasal dry powder compositions and/or unit doses further include a COMT inhibitor (e.g., entacapone). In one embodiment, the intranasal dry powder compositions and/or unit doses further include at least one enabling agent and/or at least one carrier and/or excipient.

The at least one enabling agent includes, but is not limited to, at least one epinephrine potentiator, at least one mucoadhesive, at least one absorption enhancer, at least one permeability enhancer, at least one surfactant, at least one surface modifier, at least one sustained release agent, at least one anticaking agent, at least one systemic vasodilator, at least one nasal mucosal vasodilator, at least one mucosal permeation enhancer, at least one agent that reduces mucosal transit time, at least one agent that increases mucosal absorption or adhesion or transport, at least one chelator, at least one steroid, at least one non-sulfite stabilizer, at least one preservative, at least one thickening agent, at least one humectant, at least one antihistamine, at least one solubilizing agent, at least one masking agent (e.g., taste, smell), at least one antioxidant, at least one viscosity enhancing agent, at least one dispersing agent, and/or at least one colorant. In some instances, the enabling agent includes at least one agent that reduces mucosal transit time, at least one agent that increases mucosal absorption and/or adhesion, at least one agent that enhances mucosal transport, or the enantiomers, diastereoisomers, racemates, or salts of such compounds with pharmaceutically acceptable counterions.

In one embodiment, the at least one enabling agent includes, but is not limited to, vasopressin, caffeine, at least one antihistamine (e.g., diphenhydramine, doxylamine, loratadine, desloratadine, cetirizine, levocetirizine, fexofenadine, famotidine, cimetidine, nizatidine), hydrocortisone, EDTA, magnesium stearate, tribasic calcium phosphate, citrate, citric acid, and/or ascorbic acid. In one embodiment, the at least one enabling agent includes vasopressin. In one embodiment, the at least one enabling agent includes caffeine. In one embodiment, the at least one enabling agent includes diphenhydramine. In one embodiment, the at least one enabling agent includes doxylamine. In one embodiment, the at least one enabling agent includes loratadine. In one embodiment, the at least one enabling agent includes desloratadine. In one embodiment, the at least one enabling agent includes cetirizine. In one embodiment, the at least one enabling agent includes levocetirizine. In one embodiment, the at least one enabling agent includes fexofenadine. In one embodiment, the at least one enabling agent includes famotidine. In one embodiment, the at least one enabling agent includes cimetidine. In one embodiment, the at least one enabling agent includes nizatidine. In one embodiment, the at least one enabling agent includes hydrocortisone. In one embodiment, the at least one enabling agent includes EDTA. In one embodiment, the at least one enabling agent includes magnesium stearate. In one embodiment, the at least one enabling agent includes tribasic calcium phosphate. In one embodiment, the at least one enabling agent includes citrate. In one embodiment, the at least one enabling agent includes citric acid. In one embodiment, the at least one enabling agent includes ascorbic acid.

In one embodiment, the at least one carrier and/or excipient includes at least one cellulose derivative and/or starch. In one embodiment, the at least one carrier and/or excipient includes lactose. In one embodiment, the at least one carrier and/or excipient includes sodium carboxymethylcellulose. In one embodiment, the at least one carrier and/or excipient

includes chitosan. In one embodiment, the at least one carrier and/or excipient includes a cyclodextrin (e.g., dimethyl-beta-cyclodextrin). In one embodiment, the at least one carrier and/or excipient includes mannitol. In one embodiment, the at least one carrier and/or excipient includes dodecylphosphocholine. In one embodiment, the at least one carrier and/or excipient includes an alkyl saccharide. In one embodiment, the at least one carrier and/or excipient includes ethanol (e.g., dried ethanol). In one embodiment, the at least one carrier and/or excipient includes lactose and hyaluronic acid (e.g., sodium hyaluronate). In one embodiment, the at least one carrier and/or excipient includes lactose and leucine. In one embodiment, the at least one carrier and/or excipient includes lactose and sodium chloride. In one embodiment, the at least one carrier and/or excipient includes lactose and niacin. In one embodiment, the at least one carrier and/or excipient includes lactose and polysorbate. In one embodiment, the at least one carrier and/or excipient includes sodium carboxymethylcellulose and hyaluronic acid (e.g., sodium hyaluronate). In one embodiment, the at least one carrier and/or excipient includes sodium carboxymethylcellulose and leucine. In one embodiment, the at least one carrier and/or excipient includes sodium carboxymethylcellulose and sodium chloride. In one embodiment, the at least one carrier and/or excipient includes sodium carboxymethylcellulose and niacin. In one embodiment, the at least one carrier and/or excipient includes sodium carboxymethylcellulose and/or polysorbate. In one embodiment, the at least one carrier and/or excipient includes sodium carboxymethylcellulose, lactose, sodium chloride, and/or polysorbate. In one embodiment, the polysorbate is polysorbate 80. In one embodiment, the sodium chloride is present in a range between 0.01% w/w and 5% w/w. In one embodiment, the polysorbate is present in a range between 0.01% w/v and 5% w/v. In another embodiment, the polysorbate is present in a range between 0.01% w/v and 1% w/v.

Example Composition 6

In one embodiment, the intranasal dry powder compositions and/or unit doses include a vasoactive agent (e.g., epinephrine) and polysorbate. In one embodiment, the polysorbate is polysorbate 80. In one embodiment, the intranasal dry powder compositions and/or unit doses further include a vasodilator (e.g., phentolamine). In one embodiment, the intranasal dry powder compositions and/or unit doses further include a COMT inhibitor (e.g., entacapone). In one embodiment, the intranasal dry powder compositions and/or unit doses further include at least one enabling agent and/or at least one carrier and/or excipient.

The at least one enabling agent includes, but is not limited to, at least one epinephrine potentiator, at least one mucoadhesive, at least one absorption enhancer, at least one permeability enhancer, at least one surfactant, at least one surface modifier, at least one sustained release agent, at least one anticaking agent, at least one systemic vasodilator, at least one nasal mucosal vasodilator, at least one mucosal permeation enhancer, at least one agent that reduces mucosal transit time, at least one agent that increases mucosal absorption or adhesion or transport, at least one chelator, at least one steroid, at least one non-sulfite stabilizer, at least one preservative, at least one thickening agent, at least one humectant, at least one antihistamine, at least one solubilizing agent, at least one masking agent (e.g., taste, smell), at least one antioxidant, at least one viscosity enhancing agent, at least one dispersing agent, and/or at least one

colorant. In some instances, the enabling agent includes at least one agent that reduces mucosal transit time, at least one agent that increases mucosal absorption and/or adhesion, at least one agent that enhances mucosal transport, or the enantiomers, diastereoisomers, racemates, or salts of such compounds with pharmaceutically acceptable counterions.

In one embodiment, the at least one enabling agent includes, but is not limited to, vasopressin, caffeine, at least one antihistamine (e.g., diphenhydramine, doxylamine, loratadine, desloratadine, cetirizine, levocetirizine, fexofenadine, famotidine, cimetidine, nizatidine), hydrocortisone, EDTA, magnesium stearate, tribasic calcium phosphate, citrate, citric acid, and/or ascorbic acid. In one embodiment, the at least one enabling agent includes vasopressin. In one embodiment, the at least one enabling agent includes caffeine. In one embodiment, the at least one enabling agent includes diphenhydramine. In one embodiment, the at least one enabling agent includes doxylamine. In one embodiment, the at least one enabling agent includes loratadine. In one embodiment, the at least one enabling agent includes desloratadine. In one embodiment, the at least one enabling agent includes cetirizine. In one embodiment, the at least one enabling agent includes levocetirizine. In one embodiment, the at least one enabling agent includes fexofenadine. In one embodiment, the at least one enabling agent includes famotidine. In one embodiment, the at least one enabling agent includes cimetidine. In one embodiment, the at least one enabling agent includes nizatidine. In one embodiment, the at least one enabling agent includes hydrocortisone. In one embodiment, the at least one enabling agent includes EDTA. In one embodiment, the at least one enabling agent includes magnesium stearate. In one embodiment, the at least one enabling agent includes tribasic calcium phosphate. In one embodiment, the at least one enabling agent includes citrate. In one embodiment, the at least one enabling agent includes citric acid. In one embodiment, the at least one enabling agent includes ascorbic acid.

In one embodiment, the at least one carrier and/or excipient includes at least one cellulose derivative and/or starch. In one embodiment, the at least one carrier and/or excipient includes lactose. In one embodiment, the at least one carrier and/or excipient includes sodium carboxymethylcellulose. In one embodiment, the at least one carrier and/or excipient includes chitosan. In one embodiment, the at least one carrier and/or excipient includes a cyclodextrin (e.g., dimethyl-beta-cyclodextrin). In one embodiment, the at least one carrier and/or excipient includes mannitol. In one embodiment, the at least one carrier and/or excipient includes dodecylphosphocholine. In one embodiment, the at least one carrier and/or excipient includes an alkyl saccharide. In one embodiment, the at least one carrier and/or excipient includes ethanol (e.g., dried ethanol). In one embodiment, the at least one carrier and/or excipient includes lactose and hyaluronic acid (e.g., sodium hyaluronate). In one embodiment, the at least one carrier and/or excipient includes lactose and leucine. In one embodiment, the at least one carrier and/or excipient includes lactose and sodium chloride. In one embodiment, the at least one carrier and/or excipient includes lactose and niacin. In one embodiment, the at least one carrier and/or excipient includes lactose and polysorbate. In one embodiment, the at least one carrier and/or excipient includes sodium carboxymethylcellulose and hyaluronic acid (e.g., sodium hyaluronate). In one embodiment, the at least one carrier and/or excipient includes sodium carboxymethylcellulose and leucine. In one embodiment, the at least one carrier and/or excipient includes sodium carboxymethylcellulose and sodium chlo-

ride. In one embodiment, the at least one carrier and/or excipient includes sodium carboxymethylcellulose and niacin. In one embodiment, the at least one carrier and/or excipient includes sodium carboxymethylcellulose and/or polysorbate. In one embodiment, the at least one carrier and/or excipient includes sodium carboxymethylcellulose, lactose, sodium chloride, and/or polysorbate. In one embodiment, the polysorbate is polysorbate 80. In one embodiment, the sodium chloride is present in a range between 0.01% w/w and 5% w/w. In one embodiment, the polysorbate is present in a range between 0.01% w/v and 5% w/v. In another embodiment, the polysorbate is present in a range between 0.01% w/v and 1% w/v.

Example Composition 7

In one embodiment, the intranasal dry powder compositions and/or unit doses include a vasoactive agent (e.g., epinephrine), sodium chloride, and polysorbate. In one embodiment, the polysorbate is polysorbate 80. In one embodiment, the intranasal dry powder compositions and/or unit doses further include a vasodilator (e.g., phenolamine). In one embodiment, the intranasal dry powder compositions and/or unit doses further include a COMT inhibitor (e.g., entacapone). In one embodiment, the intranasal dry powder compositions and/or unit doses further include at least one enabling agent and/or at least one carrier and/or excipient.

The at least one enabling agent includes, but is not limited to, at least one epinephrine potentiator, at least one mucoadhesive, at least one absorption enhancer, at least one permeability enhancer, at least one surfactant, at least one surface modifier, at least one sustained release agent, at least one anticaking agent, at least one systemic vasodilator, at least one nasal mucosal vasodilator, at least one mucosal permeation enhancer, at least one agent that reduces mucosal transit time, at least one agent that increases mucosal absorption or adhesion or transport, at least one chelator, at least one steroid, at least one non-sulfite stabilizer, at least one preservative, at least one thickening agent, at least one humectant, at least one antihistamine, at least one solubilizing agent, at least one masking agent (e.g., taste, smell), at least one antioxidant, at least one viscosity enhancing agent, at least one dispersing agent, and/or at least one colorant. In some instances, the enabling agent includes at least one agent that reduces mucosal transit time, at least one agent that increases mucosal absorption and/or adhesion, at least one agent that enhances mucosal transport, or the enantiomers, diastereoisomers, racemates, or salts of such compounds with pharmaceutically acceptable counterions.

In one embodiment, the at least one enabling agent includes, but is not limited to, vasopressin, caffeine, at least one antihistamine (e.g., diphenhydramine, doxylamine, loratadine, desloratadine, cetirizine, levocetirizine, fexofenadine, famotidine, cimetidine, nizatidine), hydrocortisone, EDTA, magnesium stearate, tribasic calcium phosphate, citrate, citric acid, and/or ascorbic acid. In one embodiment, the at least one enabling agent includes vasopressin. In one embodiment, the at least one enabling agent includes caffeine. In one embodiment, the at least one enabling agent includes diphenhydramine. In one embodiment, the at least one enabling agent includes doxylamine. In one embodiment, the at least one enabling agent includes loratadine. In one embodiment, the at least one enabling agent includes desloratadine. In one embodiment, the at least one enabling agent includes cetirizine. In one embodiment, the at least one enabling agent includes levocetirizine. In one embodiment, the at least one enabling agent includes fexofenadine.

In one embodiment, the at least one enabling agent includes famotidine. In one embodiment, the at least one enabling agent includes cimetidine. In one embodiment, the at least one enabling agent includes nizatidine. In one embodiment, the at least one enabling agent includes hydrocortisone. In one embodiment, the at least one enabling agent includes EDTA. In one embodiment, the at least one enabling agent includes magnesium stearate. In one embodiment, the at least one enabling agent includes tribasic calcium phosphate. In one embodiment, the at least one enabling agent includes citrate. In one embodiment, the at least one enabling agent includes citric acid. In one embodiment, the at least one enabling agent includes ascorbic acid.

In one embodiment, the at least one carrier and/or excipient includes at least one cellulose derivative and/or starch. In one embodiment, the at least one carrier and/or excipient includes lactose. In one embodiment, the at least one carrier and/or excipient includes sodium carboxymethylcellulose. In one embodiment, the at least one carrier and/or excipient includes chitosan. In one embodiment, the at least one carrier and/or excipient includes a cyclodextrin (e.g., dimethyl-beta-cyclodextrin). In one embodiment, the at least one carrier and/or excipient includes mannitol. In one embodiment, the at least one carrier and/or excipient includes dodecylphosphocholine. In one embodiment, the at least one carrier and/or excipient includes an alkyl saccharide. In one embodiment, the at least one carrier and/or excipient includes ethanol (e.g., dried ethanol). In one embodiment, the at least one carrier and/or excipient includes lactose and hyaluronic acid (e.g., sodium hyaluronate). In one embodiment, the at least one carrier and/or excipient includes lactose and leucine. In one embodiment, the at least one carrier and/or excipient includes lactose and sodium chloride. In one embodiment, the at least one carrier and/or excipient includes lactose and niacin. In one embodiment, the at least one carrier and/or excipient includes lactose and polysorbate. In one embodiment, the at least one carrier and/or excipient includes sodium carboxymethylcellulose and hyaluronic acid (e.g., sodium hyaluronate). In one embodiment, the at least one carrier and/or excipient includes sodium carboxymethylcellulose and leucine. In one embodiment, the at least one carrier and/or excipient includes sodium carboxymethylcellulose and sodium chloride. In one embodiment, the at least one carrier and/or excipient includes sodium carboxymethylcellulose and niacin. In one embodiment, the at least one carrier and/or excipient includes sodium carboxymethylcellulose and/or polysorbate. In one embodiment, the at least one carrier and/or excipient includes sodium carboxymethylcellulose, lactose, sodium chloride, and/or polysorbate. In one embodiment, the polysorbate is polysorbate 80. In one embodiment, the sodium chloride is present in a range between 0.01% w/w and 5% w/w. In one embodiment, the polysorbate is present in a range between 0.01% w/v and 5% w/v. In another embodiment, the polysorbate is present in a range between 0.01% w/v and 1% w/v.

Example Composition 8

In one embodiment, the intranasal dry powder compositions and/or unit doses include a vasoactive agent (e.g., epinephrine), a salt, and a surfactant. In one embodiment, the surfactant is a polysorbate, a fatty acid, a phospholipid, sodium glycocholate, sodium taurocholate, polyoxyethylene lauryl ether, polyacrylic acid gel, sodium lauryl sulfate, and/or sodium deoxycholate. In one embodiment, the salt includes sodium chloride, potassium chloride, sodium phos-

phate, calcium phosphate, calcium sulfate, and/or magnesium sulfate. In one embodiment, the polysorbate is polysorbate 80. In one embodiment, the salt is present in a range between 0.01% w/w and 5% w/w. In one embodiment, the surfactant is present in a range between 0.01% w/v and 5% w/v. In another embodiment, the surfactant is present in a range between 0.01% w/v and 1% w/v. In one embodiment, the intranasal dry powder compositions and/or unit doses further include a vasodilator (e.g., phentolamine). In one embodiment, the intranasal dry powder compositions and/or unit doses further include a COMT inhibitor (e.g., entacapone). In one embodiment, the intranasal dry powder compositions and/or unit doses further include at least one enabling agent and/or at least one carrier and/or excipient.

The at least one enabling agent includes, but is not limited to, at least one epinephrine potentiator, at least one mucoadhesive, at least one absorption enhancer, at least one permeability enhancer, at least one surfactant, at least one surface modifier, at least one sustained release agent, at least one anticaking agent, at least one systemic vasodilator, at least one nasal mucosal vasodilator, at least one mucosal permeation enhancer, at least one agent that reduces mucosal transit time, at least one agent that increases mucosal absorption or adhesion or transport, at least one chelator, at least one steroid, at least one non-sulfite stabilizer, at least one preservative, at least one thickening agent, at least one humectant, at least one antihistamine, at least one solubilizing agent, at least one masking agent (e.g., taste, smell), at least one antioxidant, at least one viscosity enhancing agent, at least one dispersing agent, and/or at least one colorant. In some instances, the enabling agent includes at least one agent that reduces mucosal transit time, at least one agent that increases mucosal absorption and/or adhesion, at least one agent that enhances mucosal transport, or the enantiomers, diastereoisomers, racemates, or salts of such compounds with pharmaceutically acceptable counterions.

In one embodiment, the at least one enabling agent includes, but is not limited to, vasopressin, caffeine, at least one antihistamine (e.g., diphenhydramine, doxylamine, loratadine, desloratadine, cetirizine, levocetirizine, fexofenadine, famotidine, cimetidine, nizatidine), hydrocortisone, EDTA, magnesium stearate, tribasic calcium phosphate, citrate, citric acid, and/or ascorbic acid. In one embodiment, the at least one enabling agent includes vasopressin. In one embodiment, the at least one enabling agent includes caffeine. In one embodiment, the at least one enabling agent includes diphenhydramine. In one embodiment, the at least one enabling agent includes doxylamine. In one embodiment, the at least one enabling agent includes loratadine. In one embodiment, the at least one enabling agent includes desloratadine. In one embodiment, the at least one enabling agent includes cetirizine. In one embodiment, the at least one enabling agent includes levocetirizine. In one embodiment, the at least one enabling agent includes fexofenadine. In one embodiment, the at least one enabling agent includes famotidine. In one embodiment, the at least one enabling agent includes cimetidine. In one embodiment, the at least one enabling agent includes nizatidine. In one embodiment, the at least one enabling agent includes hydrocortisone. In one embodiment, the at least one enabling agent includes EDTA. In one embodiment, the at least one enabling agent includes magnesium stearate. In one embodiment, the at least one enabling agent includes tribasic calcium phosphate. In one embodiment, the at least one enabling agent includes citrate. In one embodiment, the at least one enabling agent includes citric acid. In one embodiment, the at least one enabling agent includes ascorbic acid.

In one embodiment, the at least one carrier and/or excipient includes at least one cellulose derivative and/or starch. In one embodiment, the at least one carrier and/or excipient includes lactose. In one embodiment, the at least one carrier and/or excipient includes sodium carboxymethylcellulose. In one embodiment, the at least one carrier and/or excipient includes chitosan. In one embodiment, the at least one carrier and/or excipient includes a cyclodextrin (e.g., dimethyl-beta-cyclodextrin). In one embodiment, the at least one carrier and/or excipient includes mannitol. In one embodiment, the at least one carrier and/or excipient includes dodecylphosphocholine. In one embodiment, the at least one carrier and/or excipient includes an alkyl saccharide. In one embodiment, the at least one carrier and/or excipient includes ethanol (e.g., dried ethanol). In one embodiment, the at least one carrier and/or excipient includes lactose and hyaluronic acid (e.g., sodium hyaluronate). In one embodiment, the at least one carrier and/or excipient includes lactose and leucine. In one embodiment, the at least one carrier and/or excipient includes lactose and sodium chloride. In one embodiment, the at least one carrier and/or excipient includes lactose and niacin. In one embodiment, the at least one carrier and/or excipient includes lactose and polysorbate. In one embodiment, the at least one carrier and/or excipient includes sodium carboxymethylcellulose and hyaluronic acid (e.g., sodium hyaluronate). In one embodiment, the at least one carrier and/or excipient includes sodium carboxymethylcellulose and leucine. In one embodiment, the at least one carrier and/or excipient includes sodium carboxymethylcellulose and sodium chloride. In one embodiment, the at least one carrier and/or excipient includes sodium carboxymethylcellulose and niacin. In one embodiment, the at least one carrier and/or excipient includes sodium carboxymethylcellulose and/or polysorbate. In one embodiment, the at least one carrier and/or excipient includes sodium carboxymethylcellulose, lactose, sodium chloride, and/or polysorbate. In one embodiment, the polysorbate is polysorbate 80. In one embodiment, the sodium chloride is present in a range between 0.01% w/w and 5% w/w. In one embodiment, the polysorbate is present in a range between 0.01% w/v and 5% w/v. In another embodiment, the polysorbate is present in a range between 0.01% w/v and 1% w/v.

Kits

As previously described, in one embodiment, the present invention includes at least one kit. In one embodiment, the kit includes: (a) a dose of an intranasal dry powder composition disclosed herein and (b) instructions reciting when the dry powder composition in (a) is to be administered to a subject. In some embodiments, the kit further includes at least one intranasal delivery apparatus for dispensing the dry powder composition. In some embodiments, each of the at least one intranasal delivery apparatus is operable to deliver a therapeutically acceptable amount of the dry powder composition. In some embodiments, the apparatus is operable to intranasally deliver a therapeutically acceptable amount of the dry powder composition. In a preferred embodiment, the dry powder composition is delivered intranasally.

In another preferred embodiment, the kit includes a plurality of nasal devices. For example, and not limitation, the plurality of nasal devices includes a first nasal device, a second nasal delivery device, a third nasal delivery device, and/or a fourth nasal delivery device. In one embodiment, the plurality of nasal devices includes at least two nasal delivery devices including the same dry powder composition. For example, and not limitation, the plurality of nasal

devices includes a dry powder composition including epinephrine. A second dose of epinephrine is often needed when anaphylaxis occurs. See, e.g., Patel N, Chong K W, Yip A Y G, Ierodiakonou D, Bartra J, Boyle R J, Turner P J. Use of multiple epinephrine doses in anaphylaxis: A systematic review and meta-analysis. *J Allergy Clin Immunol*. 2021 November; 148(5):1307-1315. doi: 10.1016/j.jaci.2021.03.042. Epub 2021 Apr. 20. PMID: 33862009; PMCID: PMC8588837, which is incorporated herein by reference in its entirety. Advantageously, the kits of the present invention meet the long-standing, unmet need of delivering nasal epinephrine with a plurality of nasal devices.

In one embodiment, the plurality of nasal devices includes a first nasal device having a first dry powder composition and a second nasal device having a second dry powder composition. For example, and not limitation, the first nasal device includes a first dry powder composition (e.g., epinephrine, lactose, polysorbate, and/or sodium chloride), and the second nasal device includes a second dry powder composition (e.g., epinephrine, sodium carboxymethylcellulose, polysorbate, and/or sodium chloride).

In one embodiment, the plurality of nasal devices includes a first nasal device having a first dry powder composition, a second nasal device having a second dry powder composition, and/or a third nasal device having a third dry powder composition. For example, and not limitation, the first nasal device includes a first dry powder composition (e.g., epinephrine, lactose, polysorbate, and/or sodium chloride), the second nasal device includes a second dry powder composition (e.g., epinephrine, sodium carboxymethylcellulose, polysorbate, and/or sodium chloride); and the third nasal device includes a third dry powder composition (e.g., epinephrine, lactose, sodium carboxymethylcellulose, polysorbate, sodium chloride, at least one antihistamine, and/or at least one steroid (e.g., hydrocortisone)).

In yet another embodiment, the kit includes at least one nasal device and at least one beta-2 agonist. In one embodiment, the at least one beta-2 agonist is included in an inhaler (e.g., a meter-dose inhaler). In one embodiment, the at least one beta-2 agonist includes, but is not limited to, salbutamol or albuterol. In one embodiment, the kit further includes a spacer. In another embodiment, the at least one beta-2 agonist is in liquid form. In one embodiment, the kit further includes a nebulizer.

In one embodiment, the kit is included in a pouch. In one embodiment, the pouch includes at least one closeable opening. In one embodiment, the at least one closeable opening is operable to open and/or close via at least one securing mechanism. The at least one securing mechanism includes, but is not limited to, at least one zipper, a hook and loop system (e.g., VELCRO®), at least one button, at least one snap, at least one hook, at least one tie, at least one clip, and/or at least one buckle. The pouch is preferable waterproof or water resistant. In one embodiment, the pouch includes a desiccant. In one embodiment, the desiccant is a plastic desiccant.

In another embodiment, the kit is included in a hard case. In one embodiment, the hard case is formed of polypropylene or acrylonitrile butadiene styrene (ABS). The hard case is preferably waterproof or water resistant. In one embodiment, the hard case includes at least one handle (e.g., for carrying the hard case) and/or at least one loop. In one embodiment, the at least one handle and/or at least one loop is operable to attach a carabiner or other hook. In one embodiment, the hard case includes a desiccant. In one embodiment, the desiccant is a plastic desiccant.

In one embodiment, the pouch and/or the hard case is MOLLE-compatible. In one embodiment, the pouch incorporates a pouch attachment ladder system (PALS), which is a grid of webbing used to attach smaller equipment onto load-bearing platforms, such as vests and backpacks. For example, the PALS grid consists of horizontal rows of 1-inch (2.5 cm) webbing, spaced about one inch apart, and reat-

tached to the backing at 1.5-inch (3.8 cm) intervals. In one embodiment, the webbing is formed of nylon (e.g., cordura nylon webbing, MIL-W-43668 Type III nylon webbing). In one embodiment, an exterior finish of the pouch and/or the hard case is operable to be any color including, but not limited to, white, brown, green, orange (e.g., international orange), yellow, black, red, or blue, or any pattern (e.g., camouflage). In one embodiment, the exterior of the pouch and/or the hard case includes a reflective tape, fabric, or material. Advantageously, the reflective tape, fabric, or material improves visibility of the user in low-light conditions.

Methods of Treatment

Provided herein are methods of treating a patient by intranasally administering the dry powder composition disclosed herein. Also provided herein are methods of treating a patient by using the delivery devices and/or kits disclosed herein.

In one embodiment, the patient has been exposed to at least one organophosphate compound. The at least one organophosphate compound includes, but is not limited to, sarin (GB), tabun (GA), soman (GD), cyclosarin (GF), VX, VR (Russian VX), diisopropyl-fluorophosphate, azinphos-methyl, chlorpyrifos, diazinon, dichlorvos, dimethoate, ethephon, malathion, methamidophos, naled, oxydemeton-methyl, parathion, fenthion, ethion, echothiophate, isofluorophate, trichlorfon, and/or tribufos.

Also provided herein are methods for treating patients exposed to nerve agents including applying to a mucosal surface(s) of the nasal cavity or cavities of an individual (e.g., the mucosal surfaces of the anterior regions of the nose, the frontal sinus, the maxillary sinuses, and/or on each of the mucosal surfaces which overlie the turbinates covering the conchas) any of the pharmaceutical compositions or unit doses herein.

The methods, kits, compositions, doses, or products herein are useful for treating patients. In some instances, the patient has minimal to severe respiratory distress including bronchorrhea and bronchospasms. In some instances, the patient has excess sweating and salivation, seizures, and paralysis. In some embodiments, the composition provides a fast onset time and is suitable for intranasal use.

In some embodiments, the intranasal dry powder composition is sufficient to improve respiratory function and breathing in the patient within 60 minutes, 50 minutes, 40 minutes, 30 minutes, 20 minutes, 10 minutes, 5 minutes, 3 minutes, 2 minutes, or 1 minute after administration. In some embodiments, the intranasal dry powder composition is sufficient to reduce bronchorrhea and bronchospasms in the patient within 60 minutes, 50 minutes, 40 minutes, 30 minutes, 20 minutes, 10 minutes, 5 minutes, 3 minutes, 2 minutes, or 1 minute after administration. In some embodiments, the intranasal dry powder composition is sufficient to decrease excess sweating and salivation, seizures, and paralysis in the patient within 60 minutes, 50 minutes, 40 minutes, 30 minutes, 20 minutes, 10 minutes, 5 minutes, 3 minutes, 2 minutes, or 1 minute after administration. In one embodiment, the intranasal administration of the dry powder composition is used to treat a patient with asthma.

In some embodiments, the method of dilating a bronchus in a subject includes the intranasal administration of the dry powder composition disclosed herein. In some embodiments, the dilation occurs without substantial pulmonary inhalation. In some embodiments, the method of delivering epinephrine in a subject at least one of alpha-adrenergic receptors, beta-adrenergic receptors, or any combination thereof, include the intranasal administration of the dry powder composition disclosed herein. In some embodiments, the alpha-adrenergic receptors consist of the group including alpha-1 and alpha-2 adrenergic receptors. In some embodiments, the beta-adrenergic receptors consist of the group including beta-1, beta-2, and beta-3 adrenergic receptors. In some embodiments, delivery of epinephrine is localized. In some embodiments, delivery of epinephrine is systemic. In some embodiments, the method of treating a subject with asthma includes the intranasal administration of the dry powder composition disclosed herein. In some embodiments, the method of treating a subject with croup includes the intranasal administration of the dry powder composition disclosed herein. In some embodiments, the method of treating a subject by increasing the heart rate of the subject includes the intranasal administration of the dry powder composition disclosed herein. In some embodiments, the method of treating a subject by increasing the respiratory rate of the subject includes the intranasal administration of the dry powder composition disclosed herein. In some embodiments, the method of increasing the blood concentration of epinephrine in a subject includes the intranasal administration of the dry powder composition disclosed herein. In some embodiments, the method of treating pulmonary edema in a subject includes the intranasal administration of the dry powder composition disclosed herein. In some embodiments, the method of treating serum sickness in a subject includes the intranasal administration of the dry powder composition disclosed herein. In some embodiments, the method of counteracting bronchoconstriction effects in a subject following certain chemical exposures includes the intranasal administration of the dry powder composition disclosed herein.

The methods, kits, compositions, doses, or products herein are useful for treating patients. In some embodiments, the patient has bronchoconstriction, hypotension, and/or minimal or no cardiac activity. In some embodiments, the patient has low blood pressure. In some embodiments, the patient has hypotension. In some embodiments, the patient is experiencing hypotensive shock. In some embodiments, the composition also provides a fast onset time and is suitable for intranasal use.

In some embodiments, the intranasal dry powder composition is sufficient to increase arterial pressure in the patient within 60 minutes, 50 minutes, 40 minutes, 30 minutes, 20 minutes, 10 minutes, 5 minutes, 3 minutes, 2 minutes, or 1 minute after administration. In some embodiments, the intranasal dry powder composition is sufficient to increase a mean arterial pressure in the patient within 60 minutes, 50 minutes, 40 minutes, 30 minutes, 20 minutes, 10 minutes, 5 minutes, 3 minutes, 2 minutes, or 1 minute after administration. In some embodiments, the intranasal dry powder composition is sufficient to increase coronary perfusion pressure in the patient within 60 minutes, 50 minutes, 40 minutes, 30 minutes, 20 minutes, 10 minutes, 5 minutes, 3 minutes, 2 minutes, or 1 minute after administration. In some embodiments, the intranasal dry powder composition is sufficient to resume a spontaneous circulation in the patient within 60 minutes, 50 minutes, 40 minutes, 30 minutes, 20 minutes, 10 minutes, 5 minutes, 3 minutes, 2

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minutes, or 1 minute after administration. In some embodiments, the intranasal dry powder composition is sufficient to relieve the bronchoconstriction in the patient within 60 minutes, 50 minutes, 40 minutes, 30 minutes, 20 minutes, 10 minutes, 5 minutes, 3 minutes, 2 minutes, or 1 minute after administration.

Also provided herein are methods for treating bronchospasm, cardiac arrest, hypotensive shock, or other situations requiring the need to implement cardiopulmonary resuscitation (CPR) and/or basic or advanced cardiac life support (ACLS) in an individual, including, applying to a mucosal surface(s) of the nasal cavity or cavities of an individual (e.g., the mucosal surfaces of the anterior regions of the nose, the frontal sinus, the maxillary sinuses, and/or on each of the mucosal surfaces which overlie the turbinates covering the conchas) any of the pharmaceutical compositions or unit doses herein by administering a vasoactive agent (e.g., epinephrine). In one embodiment, a nasal loading dose is an amount of epinephrine administered nasally that results in the systemic blood bioequivalent of intravenously (IV), intramuscularly (IM), or subcutaneously (SQ) administered epinephrine (e.g., 1 mg IV or 0.3 mg doses of EPIPEN®). In a related aspect, the method of treating a patient with cardiac arrest and/or bronchospasm in need of treatment with a composition including the nasal loading dose of about 0.05 mg to about 10 mg of a vasoactive agent (e.g., epinephrine), for example, about 0.5 mg to about 5 mg. In one embodiment, the composition includes about 0.75, 1.5, or 3.0 mg of the vasoactive agent (e.g., epinephrine) and optionally includes about 0.001 mg (or 1 µg) to 10 mg of a vasodilator (e.g., phentolamine), for example, about 0.1 mg to about 5 mg. In one embodiment, the composition includes about 0.1 to about 1 mg, or about 0.5 mg of the vasodilator (e.g., phentolamine). In one embodiment, the composition includes a pharmaceutically acceptable carrier mixture of about 1 to about 50 mg, for example about 10 to about 30 mg, about 15 to about 20 mg, or about 18 mg, and optionally, an agent that reduces mucosal transit time, an agent that increases mucosal absorption and/or adhesion, an agent that enhances mucosal transport, (or the enantiomers, diastereoisomers, racemates, and the salts of such compounds with pharmaceutically acceptable counterions), wherein the amounts are operable to be synergistic for the treatment of bronchospasm and/or cardiac arrest. When used in such low doses, compositions herein are operable to provide a sufficiently high peak blood plasma concentration of the vasoactive agent (e.g., epinephrine) of at least about: 2-fold, 3-5 fold, 5-7 fold, or 7-10 fold more than baseline levels rapidly after administration, within about: 60, 50, 40, 30, 20, 15, 10, 5, or 3 minutes to be effective in the treatment or reducing the symptoms of bronchospasm and/or cardiac arrest.

In some embodiments, the method herein further includes (a) initiating cardiopulmonary resuscitation (CPR), (b) using an automated external defibrillator (AED), or both (a) and (b). In some embodiments, the intranasal dry powder composition is administered if (a), (b), or both fail to increase the arterial pressure in the patient within 60 minutes, 50 minutes, 40 minutes, 30 minutes, 20 minutes, 10 minutes, 5 minutes, 3 minutes, 2 minutes, or 1 minute after administration. In some embodiments, the intranasal dry powder composition is administered if (a), (b), or both fail to increase the mean arterial pressure in the patient within 60 minutes, 50 minutes, 40 minutes, 30 minutes, 20 minutes, 10 minutes, 5 minutes, 3 minutes, 2 minutes, or 1 minute after administration. In some embodiments, the intranasal dry powder composition is administered if (a), (b), or both fail to increase coronary perfusion pressure in the patient within

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60 minutes, 50 minutes, 40 minutes, 30 minutes, 20 minutes, 10 minutes, 5 minutes, 3 minutes, 2 minutes, or 1 minute after administration. In some embodiments, the intranasal dry powder composition is administered if (a), (b), or both fail to resume a spontaneous circulation in the patient within 60 minutes, 50 minutes, 40 minutes, 30 minutes, 20 minutes, 10 minutes, 5 minutes, 3 minutes, 2 minutes, or 1 minute after administration. In some embodiments, the intranasal dry powder composition is administered if (a), (b), or both fail to relieve the allergic reaction in the patient within 60 minutes, 50 minutes, 40 minutes, 30 minutes, 20 minutes, 10 minutes, 5 minutes, 3 minutes, 2 minutes, or 1 minute after administration.

In another aspect, the methods, kits, compositions, doses, or products herein are useful for treating patients in a hospital. In some embodiments, the patient is not in a hospital. In some embodiments, the patient is in a hospital. In some embodiments, the patient is in a combat setting. In some embodiments, the patient is in a civil emergency setting. In some embodiments, the patient has a wound.

Location data is created in the present invention using one or more hardware and/or software components. By way of example and not limitation, location data is created using the Global Positioning System (GPS), low energy BLUETOOTH based systems such as beacons, wireless networks such as WIFI, Radio Frequency (RF) including RF Identification (RFID), Near Field Communication (NFC), magnetic positioning, and/or cellular triangulation. By way of example, location data is determined via an Internet Protocol (IP) address of a device connected to a wireless network. A wireless router is also operable to determine identities of devices connected to the wireless network through the router, and thus is operable to determine the locations of these devices through their presence in the connection range of the wireless router.

FIG. 9 is a schematic diagram of an embodiment of the invention illustrating a computer system, generally described as **800**, having a network **810**, a plurality of computing devices **820**, **830**, **840**, a server **850**, and a database **870**.

The server **850** is constructed, configured, and coupled to enable communication over a network **810** with a plurality of computing devices **820**, **830**, **840**. The server **850** includes a processing unit **851** with an operating system **852**. The operating system **852** enables the server **850** to communicate through network **810** with the remote, distributed user devices. Database **870** is operable to house an operating system **872**, memory **874**, and programs **876**.

In one embodiment of the invention, the system **800** includes a network **810** for distributed communication via a wireless communication antenna **812** and processing by at least one mobile communication computing device **830**. Alternatively, wireless and wired communication and connectivity between devices and components described herein include wireless network communication such as WI-FI, WORLDWIDE INTEROPERABILITY FOR MICRO-WAVE ACCESS (WIMAX), Radio Frequency (RF) communication including RF identification (RFID), NEAR FIELD COMMUNICATION (NFC), BLUETOOTH including BLUETOOTH LOW ENERGY (BLE), ZIGBEE, Infrared (IR) communication, cellular communication, satellite communication, Universal Serial Bus (USB), Ethernet communications, communication via fiber-optic cables, coaxial cables, twisted pair cables, and/or any other type of wireless or wired communication. In another embodiment of the invention, the system **800** is a virtualized computing system capable of executing any or all aspects of software and/or

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application components presented herein on the computing devices **820**, **830**, **840**. In certain aspects, the computer system **800** is operable to be implemented using hardware or a combination of software and hardware, either in a dedicated computing device, or integrated into another entity, or distributed across multiple entities or computing devices.

By way of example, and not limitation, the computing devices **820**, **830**, **840** are intended to represent various forms of electronic devices including at least a processor and a memory, such as a server, blade server, mainframe, mobile phone, personal digital assistant (PDA), smartphone, desktop computer, netbook computer, tablet computer, workstation, laptop, and other similar computing devices. The components shown here, their connections and relationships, and their functions, are meant to be exemplary only, and are not meant to limit implementations of the invention described and/or claimed in the present application.

In one embodiment, the computing device **820** includes components such as a processor **860**, a system memory **862** having a random access memory (RAM) **864** and a read-only memory (ROM) **866**, and a system bus **868** that couples the memory **862** to the processor **860**. In another embodiment, the computing device **830** is operable to additionally include components such as a storage device **890** for storing the operating system **892** and one or more application programs **894**, a network interface unit **896**, and/or an input/output controller **898**. Each of the components is operable to be coupled to each other through at least one bus **868**. The input/output controller **898** is operable to receive and process input from, or provide output to, a number of other devices **899**, including, but not limited to, alphanumeric input devices, mice, electronic styluses, display units, touch screens, signal generation devices (e.g., speakers), or printers.

By way of example, and not limitation, the processor **860** is operable to be a general-purpose microprocessor (e.g., a central processing unit (CPU)), a graphics processing unit (GPU), a microcontroller, a Digital Signal Processor (DSP), an Application Specific Integrated Circuit (ASIC), a Field Programmable Gate Array (FPGA), a Programmable Logic Device (PLD), a controller, a state machine, gated or transistor logic, discrete hardware components, or any other suitable entity or combinations thereof that can perform calculations, process instructions for execution, and/or other manipulations of information.

In another implementation, shown as **840** in FIG. 9, multiple processors **860** and/or multiple buses **868** are operable to be used, as appropriate, along with multiple memories **862** of multiple types (e.g., a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core).

Also, multiple computing devices are operable to be connected, with each device providing portions of the necessary operations (e.g., a server bank, a group of blade servers, or a multi-processor system). Alternatively, some steps or methods are operable to be performed by circuitry that is specific to a given function.

According to various embodiments, the computer system **800** is operable to operate in a networked environment using logical connections to local and/or remote computing devices **820**, **830**, **840** through a network **810**. A computing device **830** is operable to connect to a network **810** through a network interface unit **896** connected to a bus **868**. Computing devices are operable to communicate communication media through wired networks, direct-wired connections or wirelessly, such as acoustic, RF, or infrared,

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through an antenna **897** in communication with the network antenna **812** and the network interface unit **896**, which are operable to include digital signal processing circuitry when necessary. The network interface unit **896** is operable to provide for communications under various modes or protocols.

In one or more exemplary aspects, the instructions are operable to be implemented in hardware, software, firmware, or any combinations thereof. A computer readable medium is operable to provide volatile or non-volatile storage for one or more sets of instructions, such as operating systems, data structures, program modules, applications, or other data embodying any one or more of the methodologies or functions described herein. The computer readable medium is operable to include the memory **862**, the processor **860**, and/or the storage media **890** and is operable to be a single medium or multiple media (e.g., a centralized or distributed computer system) that store the one or more sets of instructions **900**. Non-transitory computer readable media includes all computer readable media, with the sole exception being a transitory, propagating signal per se. The instructions **900** are further operable to be transmitted or received over the network **810** via the network interface unit **896** as communication media, which is operable to include a modulated data signal such as a carrier wave or other transport mechanism and includes any delivery media. The term "modulated data signal" means a signal that has one or more of its characteristics changed or set in a manner as to encode information in the signal.

Storage devices **890** and memory **862** include, but are not limited to, volatile and non-volatile media such as cache, RAM, ROM, EPROM, EEPROM, FLASH memory, or other solid state memory technology; discs (e.g., digital versatile discs (DVD), HD-DVD, BLU-RAY, compact disc (CD), or CD-ROM) or other optical storage; magnetic cassettes, magnetic tape, magnetic disk storage, floppy disks, or other magnetic storage devices; or any other medium that can be used to store the computer readable instructions and which can be accessed by the computer system **800**.

In one embodiment, the computer system **800** is within a cloud-based network. In one embodiment, the server **850** is a designated physical server for distributed computing devices **820**, **830**, and **840**. In one embodiment, the server **850** is a cloud-based server platform. In one embodiment, the cloud-based server platform hosts serverless functions for distributed computing devices **820**, **830**, and **840**.

In another embodiment, the computer system **800** is within an edge computing network. The server **850** is an edge server, and the database **870** is an edge database. The edge server **850** and the edge database **870** are part of an edge computing platform. In one embodiment, the edge server **850** and the edge database **870** are designated to distributed computing devices **820**, **830**, and **840**. In one embodiment, the edge server **850** and the edge database **870** are not designated for distributed computing devices **820**, **830**, and **840**. The distributed computing devices **820**, **830**, and **840** connect to an edge server in the edge computing network based on proximity, availability, latency, bandwidth, and/or other factors.

It is also contemplated that the computer system **800** is operable to not include all of the components shown in FIG. 9, is operable to include other components that are not explicitly shown in FIG. 9, or is operable to utilize an architecture completely different than that shown in FIG. 9. The various illustrative logical blocks, modules, elements, circuits, and algorithms described in connection with the

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embodiments disclosed herein are operable to be implemented as electronic hardware, computer software, or combinations of both. To clearly illustrate this interchangeability of hardware and software, various illustrative components, blocks, modules, circuits, and steps have been described above generally in terms of their functionality. Whether such functionality is implemented as hardware or software depends upon the particular application and design constraints imposed on the overall system. Skilled artisans may implement the described functionality in varying ways for each particular application (e.g., arranged in a different order or partitioned in a different way), but such implementation decisions should not be interpreted as causing a departure from the scope of the present invention.

Disclosed are materials, compositions, and components that can be used for, can be used in conjunction with, can be used in preparation for, or are products of the disclosed methods and compositions. These and other materials are disclosed herein, and it is understood that when combinations, subsets, interactions, groups, etc. of these materials are disclosed that while specific reference of each various individual and collective combinations and permutations of these compounds may not be explicitly disclosed, each is specifically contemplated and described herein. For example, if a method is disclosed and discussed and a number of modifications that can be made to a number of molecules including in the method are discussed, each and every combination and permutation of the method, and the modifications that are possible are specifically contemplated unless specifically indicated to the contrary. Likewise, any subset or combination of these is also specifically contemplated and disclosed. This concept applies to all aspects of this disclosure including, but not limited to, steps in methods using the disclosed compositions. Thus, if there are a variety of additional steps that can be performed, it is understood that each of these additional steps can be performed with any specific method steps or combination of method steps of the disclosed methods, and that each such combination or subset of combinations is specifically contemplated and should be considered disclosed.

Recitation of ranges of values herein are merely intended to serve as a shorthand method of referring individually to each separate value falling within the range, unless otherwise indicated herein, and each separate value is incorporated into the specification as if it were individually recited herein. For example, if a concentration range is stated as 1% to 50%, it is intended that values such as 2% to 40%, 10% to 30%, or 1% to 3%, etc., are expressly enumerated in this specification. These are only examples of what is specifically intended, and all possible combinations of numerical values between and including the lowest value and the highest value enumerated are to be considered to be expressly stated in this disclosure.

The above-mentioned examples are provided to serve the purpose of clarifying the aspects of the invention, and it will be apparent to one skilled in the art that they do not serve to limit the scope of the invention. By nature, this invention is highly adjustable, customizable and adaptable. The above-mentioned examples are just some of the many configurations that the mentioned components can take on. All modifications and improvements have been deleted herein for the sake of conciseness and readability but are properly within the scope of the present invention.

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The invention claimed is:

1. An intranasal device for administration of a pharmaceutical composition comprising:

a reservoir and a means for discharging one or more doses of the pharmaceutical composition, wherein the reservoir contains a quantity of the pharmaceutical composition;

wherein the pharmaceutical composition is a dry powder comprising:

epinephrine or a pharmaceutically acceptable salt thereof, wherein the pharmaceutical composition provides a dose of about 0.01 mg to about 10 mg of the epinephrine or the pharmaceutically acceptable salt thereof;

a carrier; and

a citrate;

wherein the pharmaceutical composition does not include an additional vasodilator; and

wherein the intranasal device further includes a nasal probe and an actuator.

2. The intranasal device of claim 1, wherein a median particle diameter of the epinephrine or the pharmaceutically acceptable salt thereof is about 20 μm to about 75 μm .

3. The intranasal device of claim 1, wherein the pharmaceutical composition further comprises an anticaking agent.

4. The intranasal device of claim 1, wherein the pharmaceutical composition further comprises at least one antihistamine.

5. The intranasal device of claim 1, wherein the pharmaceutical composition further comprises at least one steroid.

6. The intranasal device of claim 1, wherein the pharmaceutical composition further comprises a catechol-o-methyl transferase (COMT) inhibitor.

7. The intranasal device of claim 1, wherein the pharmaceutical composition further comprises one or more agents selected from a group consisting of a mucosal permeation or penetration enhancer, a mucoadhesive, a mucosal transit slowing agent, a mucosal transport enhancer, or any combination thereof.

8. The intranasal device of claim 1, wherein the intranasal device has 360° functionality and is constructed and configured to dispense a dose from any position.

9. The intranasal device of claim 1, wherein the intranasal device further includes an intranasal device global positioning system (GPS) device or is coupled to a GPS device, wherein the intranasal device GPS or the GPS device is constructed and configured to provide location data.

10. The intranasal device of claim 1, wherein the intranasal device further includes a communications interface, wherein the communications interface is constructed and configured to transmit data wirelessly to at least one remote device.

11. The intranasal device of claim 1, wherein the carrier includes mannitol, a cyclodextrin, citric acid, lactose, and/or sodium carboxymethylcellulose.

12. The intranasal device of claim 1, wherein the pharmaceutical composition does not include a preservative.

13. The intranasal device of claim 1, wherein the quantity of the pharmaceutical composition is between about 5 mg and about 75 mg.

14. The intranasal device of claim 1, wherein the intranasal device further includes a sensor that is adapted to detect a displacement or a deformation of a portion of the intranasal device when the dry powder is dispensed from the reservoir.

15. The intranasal device of claim 14, wherein the intranasal device further includes a display and a power supply,

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a timer, a clock, and/or a printed circuit board, wherein the display is constructed and configured to display a time of dose dispensation.

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